Answers

1 (a) The cost of equity capital is derived from the Capital Asset Pricing Model but given this is an unquoted company a proxy must be taken for the company's beta and regeared to reflect the different financial risk exposure of Fly4000.

Ideally, regearing beta requires an estimate of the market gearing for both companies. In the absence of that the book gearing can be used. However, the presence of corporation tax means that we need the values for both the debt and equity in Rover Airways.

$$BV(equity) = \frac{\$3bn}{1.25} = \$2.4bn$$

$$Gearing = \frac{BV(debt)}{BV(equity)}$$

$$BV(debt) = BV(equity) x gearing$$

$$BV(debt) = \$2.4bn x 1.5 = \$3.6bn$$

Using the formula for the asset beta where debt carries zero market risk:

$$\beta_A = \beta_e x(1-W_d)$$

where

$$W_d = \frac{BV_d x(1-T)}{BV_e+BV_d x(1-T)}$$

$$W_d = \frac{3.6x(0.7)}{2.4+3.6x(0.7)}$$

$$W_d = 0.5122$$

$$\beta_A = 2.0x(1-0.5122)$$

$$\beta_A = 0.9756$$

This is the asset beta for Rover Airways. We can now regear the beta to that for FliHI as follows:

Calculate the tax adjusted gearing ratio for FliHi.

$$W_d = \frac{BV_d x(1-T)}{BV_e+BV_d x(1-T)}$$

$$W_d = \frac{150x(0.7)}{120+150x(0.7)}$$

$$W_d = 0.4667$$

$$\beta_e = \frac{\beta_a}{(1-W_d)}$$

$$\beta_e = \frac{0.9756}{(1-0.4667)}$$

$$\beta_e = 1.8294$$

This is the estimated equity beta for FliHi which, when applied to the CAPM, gives an expected rate of return as follows:

$E(r_e) = R_F+\beta_e xERP$
$E(r_e) = 0.045+1.8294x0.035$
$E(r_e) = 0.045+1.8294x0.035$
$E(r_e) = 0.1090(=10.90\%)$

The equity cost of capital for FliHi is therefore approximately 10.90 per cent.

The modelling of the equity cost of capital has embedded within it the assumptions implicit in the CAPM that:
- Investors are mean variance efficient
- Markets are frictionless
- Expectations are homogenous and,
- There is a risk free asset

However, of more practical significance we have also assumed that:

- The underlying exposure to market risk is the same for both companies (this is questionable given the differences in the markets in which they operate).
- That the book gearing ratio is a reasonable approximation to the market gearing ratio. The use of book values, can seriously distort the cost of capital that is calculated converting it into a measure of the average cost of capital in the firm's historical gearing ratio rather than in the ratio of the current capitalised values of the firm's equity and debt.
- That FliHi Ltd does not carry a size and default premium on its cost of capital. Default and size premia can be included through the use of such variants on the standard CAPM as the Fama and French 3 factor model which incorporates these elements of risk.

(b) Gordon's approximation requires a retention ratio which can be derived from the cash flow statement. The free cash flow to equity (before reinvestment) is defined as operating cash flow less interest and tax:

For 2005 the FCFE is as follows:

FCFE = operating cash flow – net interest paid – tax
FCFE ($m) = 210 + 1.5 – 4.1 = $207.4 million

In the current year $120.2m was reinvested. This implies a retention ratio (b) of:

$$b = \frac{reinvestment}{FCFE}$$

$$b = \frac{120.2}{207.4}$$

$$b = 0.58$$

Gordon's approximation was originally developed to measure the growth in earnings assuming a given retention ratio and rate of return. We can apply the same logic to the free cash flow model but here we are looking at the rate of cash generation by the business on new capital investment. The current rate of return is, in principle, the internal rate of return on the current business portfolio. If we assume that the business is highly competitive then the internal rate of return will be close to the company's equity cost of capital. Growth is therefore expected to be:

$g = bxr_e$
$g = 0.58x0.1090$
$g = 0.06322(=6.322\%)$

If we use the company's current rate of return on equity from the accounts we would have:

$$ROE = \frac{\text{net profit}}{\text{equity employed (net assets)}}$$

$$ROE = \frac{50}{120} = 42\%$$

This would suggest a rate of growth of 24.2 per cent which is unlikely in this industry.

On the basis of a growth rate of 6.322 per cent and given that the year six growth rate and forward will be 4 per cent, the pattern of growth we anticipate is therefore:

Year	**1**	**2**	**3**	**4**	**5**	**6**
Growth rate	6.322%	6.322%	6.322%	6.322%	6.322%	4.00%

The assumptions here are embedded within the method of measuring growth. Gordon's growth approximation will give the next year value for the FCFE for the business on the assumption that the cost of capital is achieved and no more. We have also assumed that the current figures for cash generation and reinvestment are typical and likely to be replicated over the near term. In this context we would note the significant increase in operating cash flow from 2004 and question whether this was sustainable.

(c) Using the free cash flow to equity net of reinvestment we have the free cash flow which is, in principle, distributable. We build a valuation model expanding this free cash flow through the next five years. From year six forward the rate of growth is a perpetuity and we use the free cash flow analogue of the dividend growth model to estimate the value at year six.

Step 1: take the growth rates as projected and estimate the future free cash flow to equity taking ($210m – $120.2m=$87.2m) as the starting point.
Step 2 discount these projected values at the cost of equity capital (10.9%) to give a present value of $384.89million.
Step 3 using the formula:

$$V_e = \frac{FCFE_0(1+g)}{r_e - g}$$

Calculate the value of the growing perpetuity at the end of year 6 (note the timing of the year is important) using the expected FCFE in year 5. This gives a value at year 5 of $1,785.73.

Step 4: discount this at 10.9 per cent to give a present value of the residual term of $1,064.53

Step 5: add the two present values to give a valuation of the firm's equity at $1,449.42 million

Year		2006	2007	2008	2009	2010	2011
Growth		6.32%	6.32%	6.32%	6.32%	6.32%	4.00%
FCFE (2005)=$87.2m		92.71	98.57	104.81	111.43	118.48	
Discount at 10.9 per cent		83.60	80.15	76.84	73.67	70.63	
PV of year 1–6		$384.89					
PV of perpetuity at 2011						$1,785.73	
PV of perpetuity at 2005	$1,064.53						
Present value of the firm's equity	$1,449.42						

The limitations of the method is that we assume:

- The current operating cash flows are sustainable
- The constant patterns of growth of operating cash flow will be achieved as specified.
- The rate of return required by investors is constant throughout the life of the business
- The business has an indefinite life beyond 2011.

These assumptions are unlikely to hold in practice and it should be noted that where the rate of growth (4%) is relatively low compared with the cost of equity that we would not expect the perpetuity to be a good approximation of residual value. It might be better to seek a likely break up value from the accounts as the residual value of the business. If we take the company's net asset figure and extrapolate forward at the rate of growth as shown above we obtain a figure of $169.55 million in 2011 which has a present value of $91.14 million. This would put a value on the business of $476.03 million. This incidentally gives a closer approximation to the market valuation if we apply the Rover Airlines P/E ratio to FliHi's net profit figure:

Value = benchmark P/E x FliHi Earnings
= 11.0 x $50million
= $550 million

(d) To whomsoever it may concern:

The proposed acquisition of FliHi represents a substantial capital investment for your airline. However, there are a number of issues which you might wish to consider before making a bid. These issues I have separated into synergies, risk exposure, future options, financing and valuation.

Synergies: From the perspective of your respective markets there would appear to be considerable advantage in integration. From the synergistic perspective these can be categorised as:

Revenue synergies: is there likely to be an enhancement in your ability to capture market share in a way that will add shareholder value. Simply acquiring the business as it stands will not be sufficient as investors can achieve the same at lower cost by diversification. Synergies only arise if a market opportunity presents itself which would not exist if both firms remain independent. One example would be where the domestic and European service can be used as a feeder system for an expanded long haul business from your principal airport and centre of operations.

Cost synergies: are there opportunities to save cost through more efficient operations? Economies of scale and scope are available in the airline business in the areas of in-flight catering, fuel supplies, maintenance and ticketing. The larger fleet size would also present operational opportunities.

Financial synergies: would the company have greater opportunities in the domestic and international capital market to acquire finance at more favourable rates and under better conditions.

Risk exposure: the larger operation would not necessarily improve the firm's exposure to market risk and indeed is likely to leave it unchanged as we would expect the underlying asset beta of both firms to be the same. There are a number of other risk areas that could be improved: operational risk may be mitigated by the firm's increased ability to hedge its operations (see the notes on the real options available below). Other risk effects include: economic, political, transactions and translation. Some of these are minimised (transactions) largely because your business is principally in the domestic market although this may change if you decide, for example, to develop your European network as a feeder system to your hub.

Future options: an acquisition of this type can create real options to expand, redeploy and exchange resources which add value to the proposition not easily captured with conventional valuation procedures. A real option is a claim upon some future course of action which can be exercised at your discretion. The availability of the new Airbus 380 offers potential both within the long haul business but also to medium haul holiday destinations at peak seasons. Access to your fleet of short to medium range aircraft offers the possibility of opening the European market to your long haul business. When and how you exercise these real options depends on circumstances at the time but paradoxically the more uncertain the underlying business the greater the value that attaches to this flexibility.

Financing: an acquisition of this type would require substantial extra financing. FlHi would appear to have a high level of off balance sheet value partly because of the scale of their operational leasing of aircraft but more significantly because of their business name and the quality of their operations. A substantial sum is likely to be paid for the goodwill of this business which suggests that this may not be a proposition that would be attractive to the debt market. Your own substantial cash reserves and the level of retained earnings suggest that this may be the route to financing this acquisition through a cash offer plus shares.

Valuation: we estimate the value of FliHi based upon its current cash flow generation to be of the order of $1.450 billion. Using market multiples a lower figure of $550million is obtained. The key point of this valuation process is to determine the lowest likely figure that the owners of FliHi would be prepared to accept. Our judgement is that the figure is likely to be closer to the upper end of this range. The free cash flow model relies upon our best forecast of future cash flow and reinvestment within the business. We believe that the owners of FliHi would have access to similar advice. The key question now to resolve is what would be the value of FliHi to your company. This has most of the characteristics of a type 2 acquisitions where the financial risk of the business is likely to be disturbed. For this reason we would need to value your current business using available market data and revalue it on the basis that the bid goes ahead using your preferred financing package. The potential increase in the value of your firm will reveal the potential control premium at any proposed offer price that may be decided upon.

2 (a) Sydonics Engineering
Memorandum

Our FOREX risk exposure is primarily driven by our European and US business and the variability of the Sterling/Euro and Sterling/Dollar exchange rates. The volatilities of both rates are of the order of 20 per cent (annualised) which indicates significant Value at Risk in any forward commitment to either currency. There are arguments for and against the use of derivative contracts to manage this transactions risk exposure.

In principal, nothing is added to the value of the firm if we attempt to do something which the investors can easily achieve, possibly, at lower cost. If the capital markets are perfect then investors should be indifferent to firm specific risk (which they can diversify away) and will only be concerned about the market driven risks as reflected in the firm's beta value. Arguably investors can diversify away the forex risk exposure within their portfolio much more efficiently and at lower cost than the company can through hedging. Hedging also brings costs: there are the direct costs of the treasury management function and the indirect costs of developing the in-house expertise required to assess hedging alternatives. There are also compliance costs in that any imperfect hedging agreement must be valued and shown in the accounts. Finally, hedging through derivative contracts can be avoided by the creation of internal hedging arrangements whereby finance is raised in the country of operations and hence borrowing costs are currency matched with the revenue streams.

On the other hand, the market perfection argument can be turned on its head. In as far as hedging is a means of reducing the firm's exposure to exchange rate volatility (a market wide phenomenon), then its impact will be to reduce the firm's beta and thus its cost of capital. This is a general argument for hedging but does not necessarily imply that we should use the derivative markets to manage our exposure to exchange rate risk. The empirical evidence on practice in industry is surprisingly sparse although Geczy, Minton and Schrand, (1997) found that 41 per cent of their sample of 370 US firms actively used derivative instruments to manage their forex risk. The benefits of hedging through the use of derivatives is that, with care, risk exposure can be tightly controlled although the use of exchange traded derivatives still leaves a residual 'basis risk' because of differences between the closeout rates and the underlying rates of exchange. Basis risk can be avoided through the use of OTC agreements wherever possible.

Hedging FOREX risk using derivatives can be expensive especially, as in this case, when the exposure is uncertain. Where the exposure is certain the use of futures or forward contracts can eliminate a large element of uncertainty at much lower cost. Any policy must therefore address the following issues:

- What is the magnitude of the risk exposure (in this case the very high volatility and long term exposure creates a very high hedging cost). Value at Risk (VaR) may be an appropriate method for measuring the likely financial exposure.
- The materiality of the exposure in terms of the magnitude of the sums involved.
- To what extent can the risk be mitigated by matching agreements (borrowing in the counter currency to mitigate CAPEX for example)?
- To what extent has the exposure crystallised? If it is uncertain FOREX options allow the hedging of the downside risk but at a high cost.

A policy would then lay down the principles that should be followed to cover the following:

- Risk assessment – an assessment of the likelihood and impact of any given risk upon the financial position of the firm. Derivative positions can be highly geared and the firm may be exposed to very high liabilities under certain exchange rate and/or interest rate conditions. It is recommended that derivatives should only be used for the management of specific risks and that no speculative or uncovered position should be taken.
- Cost of hedging – measured not only in terms of the direct costs involved but also in the use of scarce management time in establishing the positions and operating the internal control procedures appropriate where derivatives are used as a means of hedging risk. It is recommended that hedging costs should be minimised to an agreed percentage of the Value at Risk through the unhedged position.
- Contract and approval procedures – where OTC products are purchased and specific contracts are raised then the firm needs to establish policies with respect to the legal aspects of the contracting process. With both OTC and exchange traded products the firm should also establish an approval process with clear lines of responsibility and sign off on contracts up to and including board level. It is recommended that the board appoint a risk management committee to review and monitor all hedging contracts where the Value at Risk is in excess of an agreed amount.
- Hedge monitoring – policy needs to be established as to the monitoring of derivative positions and the conditions under which any given position will be reversed. It is recommended that the risk management committee actively monitor all open positions.

(b) This part of the question focuses on the use of options for foreign exchange hedging and deploys the Grabbe (1983) variant of the Black Scholes model.

The analytics for the question are shown in the tables below:

The price of the 'at the money' options are as follows:

	3 months	9 months
Sterling Euro Spot	0.6900	0.6900
Sterling Euro (indirect)	1.4493	1.4493
Euro Libor	2.7656	3.0519
Sterling Libor	4.6231	4.7303
Sterling Euro forward (indirect)	1.4426	1.4317
Sterling Euro forward (direct)	0.6932	0.6985
Annual volatility of £/Euro	0.2200	0.2200
D1	0.0968	0.1595
D2	-0.0132	-0.0310
N(d1)	0.5386	0.4367
N(d2)	0.4948	0.5124
3 month call and 9 month put price (pence per Euro)	3.1581	4.6853
(ie the price of a call or put to buy or sell one Euro at spot		
Contract value (euro equivalent)	3158	4685

The values for d_1 and d_2 have been calculated as follows:

$$d_1 = \frac{\left(\ln\left(\frac{F}{S}\right) + \sigma^2 \frac{T}{2}\right)}{\sigma\sqrt{T}}$$

$$d_1 = \frac{\left(\ln\left(\frac{0.6932}{0.6900}\right) + 0.22^2 \frac{0.25}{2}\right)}{0.22\sqrt{0.25}}$$

$d_1 = 0.0968$

$d_2 = d_1 - \sigma\sqrt{T}$

$d_2 = 0.0968 - 0.22\sqrt{0.25}$

$d_2 = -0.0132$

Where F and S are the forward and spot rates respectively (note the forward must have the same maturity as the option), σ is the volatility, T is the time to maturity.

$N(d_1)$ and $N(d_2)$ are derived from the supplied tables for the normal density function for situations where $N(x) \geq 0$ and $N(x) \leq 0$.

The value of the currency call and the currency put are as follows:

$c = e^{-rT}[FN(d_1) - SN(d_2)]$

$c = e^{-0.046231 \times 0.25}[0.6932 \times 0.5386 - 0.6900 \times 0.4948]$

$c = 3.1581 \textit{pence}$

and

$p = e^{-rT}[SN(-d_2) - FN(-d_1)]$

$p = e^{-0.047303 \times 0.75}[0.6900 \times 0.5124 - 0.6985 \times 0.4367]$

$p = 4.6853 \textit{pence}$

The call should be exercised if the exchange rate rises above 0.6900 (thus making the sterling equivalent more expensive) and the nine month put should be exercised if the spot should be below 0.6900 at the exercise date.

The key points here are to note that the rates into the formula must all be direct, the volatility and interest rates are employed on an annual basis (they have been quoted as such in the question).

Candidates should also be aware of the money market conventions for quoting exchange rates that the base currency is shown first and the counter currency second. They should be sufficiently familiar with the relative values of the principal currencies to identify whether the quote is direct to indirect.

(iii) The hedge ratio (N(d1) and (N(-d1)) reveals the inverse of the number of option contracts we require to hedge a one euro exposure. Therefore the number of contracts we will require to hedge the exposures are as follows:

	3 months	9 months
Number of contracts	2,785	3,435
Cost of hedge	8,795,723	16,094,798
As percentage of value	0.058638	0.214597

(iv) The issues to be borne in mind are:

- Hedging with options eliminates downside risk (unlike futures) and are particularly useful when the exposure is uncertain.
- The cost of this type of hedge can be very high (especially for the long dated put) although the company may wish to reduce the cost by purchasing out of the money options. This will not eliminate the downside risk completely but will allow them to hedge to a known exposure.
- An alternative approach would be to hedge the three month exposure which is where the option is most valuable given the uncertainty over whether the bid will be accepted and then, if it is accepted, to enter into a future contract at that date to lock in the prevailing spot rate. Alternatively, purchasing the put option may be held back until the contract is won.
- Given the set contract sizes it is not possible to create a perfect hedge. The position would need continual monitoring and adjustment to offset gamma risk which is likely to be high for near-the-money options.
- There is timing risk given that currency options are quoted as Europeans. Early sale of the option if the requirement materialises early will create basis risk, if the requirement materialises late the residual time delay will be unhedged.

3 **(a)** Board Paper Presenting Proposed Procedures for Large CAPEX.

This paper proposes revised guidelines for the Board approval of large (in excess of $10,000) capital investment projects. The current two stage process of preliminary and final approval serves an important role in ensuring that any initial concerns of the Board in terms of strategic fit and risk are brought to the attention of the Financial Appraisal Team. The two stage process would consider:

Stage 1:
Business proposal including assessment of strategic requirement, business fit and identified risks.

Outline financial appraisal to include capital requirement, mode of financing, expected net present value, modified internal rate of return and project duration.

It is recommended that conventional payback is dropped because it ignores the cost of finance and the magnitude of post payback cash flows. Duration is recommended as this measures the time required to recover half of the project value.

Stage 2:

- A proposed business plan must be presented giving the business case with an assessment of strategic benefits, risks, finalised capital spend and capital source.
- A value impact assessment giving an NPV calculation supported by a calculation of the project value at risk. The net present value of the project represents our best estimate of the likely impact of the investment on the value of the firm. This is the key statistic from the capital market perspective in that, unless we are assured that the project NPV is positive, the investment will reduce and not enhance the value of the firm. This net present value calculation should be supported by a modified internal rate of return which measures the additional economic return of the project over the firm's cost of capital where intermediate cash flows are reinvested at that cost of capital. In a highly competitive business the reinvestment assumption implicit in the MIRR is more realistic that that assumed with IRR where intermediate cash flows are assumed to be reinvested at the IRR. This may be satisfactory for near-the-money projects but is far less satisfactory for projects which offer high levels of value addition to the firm.
- An accounting impact assessment including the differential rate of return on capital employed and a short term liquidity assessment. Although positive NPV projects are value enhancing they may not do so in ways that are readily apparent in the financial reports. To manage investor expectations effectively the firm needs to be aware of the impact of the project on the firm's reported profitability and this is most accurately reflected by the differential rate of return measure. Accounting rate of return as normally calculated does not examine the impact of the project on the financial position of the firm but is restricted to the rate of return the investment offers on the average capital employed.
- An assessment of the project duration. This project, for example, reveals a duration of 4.46 years which is the mean time over which half of the project value is recovered. This is more useful than the other liquidity based measures especially when used as a relative as opposed to an absolute measure of the cash recovery. Cash recovery assumes that the future project cash flows are achieved at a constant rate over the life of the project.

(b) The proposed business case concludes that this is a key strategic investment for the firm to maintain operating capacity at the Gulf Plant. The financial assessment is as detailed above (excluding an assessment of the impact of the project on the financial reports of the firm).

(i) The net present value of this project is calculated using a discount rate of 8 per cent and gives a value of \$1.964 million. The volatility attaching to the net present value of \$1.02 million indicates that there is (z) standard deviations between the expected net present value and zero as follows:

$$z = \frac{1.964-0}{1.02} = 1.9255$$

This suggests that this project has a 97.3 per cent probability that it will have a positive net present value or conversely a 2.7 per cent probability of a negative net present value (these probabilities are taken from the normal density function tables supplied).

Unlike the assessment of the probability of the project failing to generate a positive net present value given above, VaR measures the value which could potentially be lost by the business, over the project life time, given the volatility of the net present value. The project value at risk is based upon an assessment of the number of years that the project cash flow is at risk (10), the annual volatility and the confidence level required by the firm. VaR assumes that the net present value of the project is normally distributed and thus there is the potential, in theory at least, for those losses to be unlimited. The application of the confidence level of 95% places a level of potential loss that the firm will consider in its project evaluation processes whilst recognising that there is a one in twenty chance that the loss could turn out to be greater. The formula for project VaR is:

$$\text{projectVaR} = N(0.95)s\sqrt{T}$$
$$\text{projectVaR} = 1.645\text{x}1.02\text{x}3.162 = \$5.3\text{million}$$

This assumes a 95 per cent confidence level, at 99 per cent the project VaR is \$7.51 million. This value reflects the fact that the capital invested is at risk for ten years and assumes that the volatility of the project is fairly represented by the volatility of its net present value.

(ii) Project Return

The internal rate of return is shown as 11.01 per cent. The Modified Internal Rate of Return is calculated by (i) projecting forward the cash flows in the recovery stage of the project at 8 per cent to future value of \$41.798 million and (ii) discounting back the investment phase cash flows to give a present value of the investment of \$17.396 million.

The Modified Internal Rate of Return is therefore:

$$\text{MIRR} = \sqrt[10]{\frac{41.798}{17.396}}-1$$
$$\text{MIRR} = 9.16\%$$

This rate suggests that the margin on the cost of capital is rather small with only a 1.16 per cent premium for the strategic and competitive advantage implied by this project.

(iii) Project Liquidity

With a present value of the recovery phase of \$19.6931 million and of the investment phase of \$17.3961 million this suggests that the project will have a recovery period of:

$$recovery = 2 + \frac{17.396}{19.361} \text{ x } 8 = 9.188 \; years$$

In practice the actual recovery is shorter than this because the expected cash in flows occur earlier rather than later during the recovery phase of the project.

The project duration is calculated by multiplying the proportion of cash recovered in each year (discounted recovery cash flow/present value of the recovery phase) by the relevant year number from project commencement. The sum of the weighted years gives the project duration.

Year		3	4	5	6	7	8	9	10
Discounted cash flow (recovery) (\$m)		3.5722	4.7042	4.9342	4.0961	3.2092	2.1611	-1.0005	-2.316
Present value of recovery phase	19.3606								
duration of recovery phase									
proportion of CF recovered		0.1845	0.2430	0.2549	0.2116	0.1658	0.1116	-0.0517	-0.1196
weighted years		0.5535	0.9719	1.2743	1.2694	1.1603	0.8930	-0.4651	-1.1962
project duration (Years)	4.46								

The project duration reveals that the project is more highly cash generative in the early years notwithstanding the two year investment phase.

In summary, the analysis confirms that this project if financially viable, it will be value adding to the firm although there is substantial value at risk given the volatility of the net present value quoted. In terms of return the premium over the firm's hurdle rate is small at 1.16 per cent and any significant deterioration in the firms cost of capital would be very damaging to the value of this project. The liquidity statistics reveal that the bulk of the project's cash returns are promised in the early part of the recovery phase and that half the value invested in the project should be recovered by year five. Taking this into account acceptance is recommended to the board.

4 (a) Calculation of the expected WACC

A preliminary view suggests that the increased cost of debt is more than offset by the impact of the tax shield. The steps in the calculation are to estimate the cost of debt before and after the new issue. The existing cost of equity is calculated from the WACC and the ungeared equity cost calculated using M&M. The equity is then regeared to the new level assuming the higher level of gearing but not assuming any increase in the cost of debt through default or term structure. We then use the increased cost of debt to calculate the revised WACC. The analytics are as below:

Current WACC	**6.80%**			
	D/E	**D/(D+E)**		
Current gearing	0.1667	0.1429		
New gearing	0.2667	0.2105		
		risk free	**spread**	**Total**
Current yield on debt		0.03450	0.00400	0.03850
New yield on debt	3 years	0.03450	0.00440	0.03890
spread = 0.6x70 basis points (bp) + 0.4x107bp = 84.8bp	10 years	0.04200	0.00848	0.05048
Combined rate	**MVd**	**Rd**		
market value of 3 year debt	0.5	0.03890	0.02431	
market value of new debt	0.3	0.05048	0.01893	
Total market value of debt and combined rate	0.8		0.04324	
Existing cost of equity	7.48%			
Calculate ungeared cost of equity	7.11%			
Calculate regeared cost of equity (keeping cost of debt constant)	7.72%			
Calculate new weighted average cost of capital	6.73%			

The existing cost of equity is calculated by rearranging the WACC as follows:

$$r_e = \frac{\text{wacc-}w_d r_d(1\text{-}T)}{(1\text{-}w_d)}$$

$$r_e = \frac{6.8\%\text{-}0.1429\text{x}3.85\%\text{x}0.7}{(1\text{-}0.1429)}$$

$$r_e = 7.48\%$$

The ungeared equity cost of capital (r_e^1) is discovered from Modigliani and Miller Proposition 2:

$$r_e = r_e^1 + (r_e^1 \text{-} r_d)\frac{D}{E}(1\text{-}T)$$

Rearranging and substituting:

$$r_e = r_e^1 + r_e^1\frac{D}{E}(1\text{-}T)\text{-}r_d\frac{D}{E}(1\text{-}T)$$

$$r_e^1 = \frac{r_e + r_d\frac{D}{E}(1\text{-}T)}{1+\frac{D}{E}(1\text{-}T)}$$

$$r_e^1 = \frac{7.48\%+3.85\%\text{x}0.1667\text{x}0.7}{1+0.1667\text{x}0.7}$$

$$r_e^1 = 7.11\%$$

The regeared cost of equity capital assuming that the cost of debt remains unchanged is given by again using proposition 2:

$$r_e = r_e^1 + (r_e^1 - r_d)\frac{D}{E}(1-T)$$

$$r_e = 7.11\% + (7.11\% - 3.850\%) x 0.2667 x 0.7$$

$$r_e = 7.72\%$$

This assumes that the equity cost of capital is unaffected by the change in the level of default risk. This is a reasonable assumption at modest levels of gearing but would not be expected to hold at very high gearing levels.

Finally the new weighted average costs of capital is calculated using the revised equity cost and the revised debt cost as follows:

$$wacc = (1-w_d)r_e + w_d r_d (1-T)$$

$$wacc = (1-0.2105) x 7.72\% + 0.2105 x 4.324\% x 0.7$$

$$wacc = 6.73\%$$

This demonstrates a less than 10 basis point fall in the weighted average cost of capital.

(b) Paper on the alternative sources of capital

For a company in this position the following sources of finance suggest themselves:

Sale and lease back of existing assets
US debt financed through a further bond issue
Debt raised on the Chinese market
Equity finance by rights or new issue

The choice of financing is partly down to the cost and availability of the various sources and partly down to the method the company chooses to hedge its FOREX exposure.

If we ignore foreign exchange considerations for a moment, pecking order theory suggests that debt should be preferred to equity and the weighted average cost of capital calculation suggests that the firm should increase its gearing to capture further tax shield effects which are not currently being offset by increased default risk (static trade off theory). However, issue costs may be expensive and the company may seek to raise finance by sale and lease back of existing assets. There are implications for reporting under FASB 13 depending upon whether the leases are financing or operating leases. Raising $300 million of debt by a bond issue is at the low end of the scale for new debt issues of this type although it may be possible that a syndicated issue where a number of companies of similar credit rating are joined by a lead bank could be arranged. It is to be expected that the costs of the issue will be high in terms of commissions and underwriting fees.

Raising finance directly in China has been eased considerably with recent changes in the rules of the Chinese Securities Regulatory Commission opening better access of foreign firms to the Chinese bond and equity markets. However, the entry of China into the WTO in 2002 is still feeding through the economy as tariff barriers and other constraints are removed. This process of liberalisation is likely to continue accelerating although, as with any emerging market, there are risks associated with inward investment and capital entry. These risks may be sufficient to raise the risk assessment against this company and as a result the benign implications of increased gearing outlined above may not be realised in practice.

The problem of hedging the foreign exchange exposure can be partly solved by borrowing directly in China and using the income flows from the new venture to finance the interest charges and capital repayments. Because the borrowing and the income flows are in the same currency transactions exposure is largely eliminated although any appreciation in the Chinese currency would increase the dollar value of the translated debt in the firm's balance sheet. If the borrowing is used to purchase matching assets in China then the translation risk is mitigated along with the transaction risk. However, if the assets are not owned but leased or rented then translation effects will impact upon the balance sheet and may be misread by the market. A second alternative would be to raise finance in the US and then engage in a currency swap for a ten year term. The effect of this would be to lock in the current exchange rate for the duration of the borrowing. However, finding a swap of this type would entail the services of a financial institution specialising in bringing appropriate counter-parties together. Such derivative arrangements have been mis-sold in the past with disastrous leveraging effects built into the contract.

5 **(a)** Normally, obtaining a listing consists of three steps: legal, regulatory and compliance. In the UK, as in many other jurisdictions a company must ensure that it is entitled to issue shares to the public as opposed to an issue by private treaty. In the UK a firm seeking listing must register as a public limited company. This entails a change in its memorandum and articles agreed by the existing members at a special meeting of the company.

The company must then meet the regulatory requirements of the Listing Agency which, in the UK, is part of the Financial Services Authority (FSA). These requirements impose a minimum size restriction on the company and other conditions concerning length of time trading. Once these requirements are satisfied the company is then placed on an official list and is allowed to make a public offering of its shares.

Once the company is on the official list it must then seek the approval of the Stock Exchange for its shares to be traded. In principal it is open to any company to seek a listing on any exchange where shares are traded. The London Exchange imposes strict requirements and invariably the applicant company will need the services of a sponsoring firm that specialises in this type of work.

(b) The advantages of seeking a public listing are that it opens the capital market to the firm. It offers the company access to equity capital from both institutional and private investors and the sums that can be raised are usually much greater than can be obtained through private equity sources. The presence of the firm as a listed company on a major exchange also enhances its credibility as investors and the general public are aware that by doing so it has opened itself to a much higher degree of public scrutiny than is the case for a firm that is privately financed. The disadvantages are significant. A distributed shareholding does place the firm in the market for corporate control increasing the likelihood that the firm will be subject to a takeover bid. There is also a much more public level of scrutiny with a range of disclosure requirements. Financial accounts must be prepared in accordance with IFRS or FASB and with the relevant GAAP as well as the Companies Acts. Under the rules of the London Stock Exchange companies must also comply with the governance requirements of the Combined Code and also have in place an effective and ongoing business planning process. Much of this may be regarded as desirable within a privately owned company but the requirements to comply or explain imposed on a public company can impose a significant regulatory burden and exposure to critical comment.

(c) There is an ethical dimension to the request made by Martin Pickles. He is of course entitled to acquire or dispose of his equity claim as he sees fit but his position as a large shareholder does impose on him certain duties with respect to the other shareholders. He should not undertake any action which is prejudicial to them and in this case making any move towards listed status would require their consent. It may well be that the private equity firm involved has in mind its own exit strategy and that his proposed course of action would be acceptable to them. If the decision was made to go public his own intentions would be a material factor in the valuation of the firm and the offer price made to subscribers. An immediate intention to divest would need to be disclosed. There would appear to be nothing wrong at this stage with asking the CFO to investigate the matter on a confidential basis although the request that he or she should seek to enhance the earnings of the business should be resisted in as far as it represents an instruction to engage in earnings manipulation beyond that required to present a true and fair view of the affairs of the business.

Earnings management techniques whereby revenues and costs are accelerated/ decelerated to achieve desired earnings figures are severely limited by GAAP and in the US for example could lead to arraignment under the Sarbanes Oxley Act. The proposal that the CFO would be offered share options adds a veneer of impropriatery to this discussion. He or she is in a difficult position but does need to make clear that he or she must act to preserve the interests of all the shareholders and all the directors. An invitation to participate in a share option scheme is a fairly crude attempt to win support for the proposed course of action (as the shares are not yet quoted) and should be resisted.

1 30 marks distributed over four sections. Key numbers presented in the analysis must be contextualised and justified.

(a) Estimate the current cost of equity capital for FliHi using the Capital Asset Pricing Model making notes on any assumptions that you have made. (9 marks)

Calculation of the asset beta for Rover Airways at 0.9756	2
Regear for FliHi to 1.8294.	1
Calculate FliHI's cost of equity at 10.90 per cent	2
Notes on the assumptions relating to the CAPM	2
Notes on the assumptions of more practical significance	2

(b) Estimate the expected growth rate of this company using the current rate of retention of Free Cash Flow and your estimate of the required rate of return on equity for each of the next seven years. Make notes on any assumptions you have made. (6 marks)

Calculation of the retention ratio as specified in the question as 0.58	2
Estimate of the growth for the next six years (justifying the return on equity chosen)	2
Assumptions embedded within the calculation of growth and Gordon's approximation	2

(c) Value this company on the basis of its expected Free Cash Flow to Equity explaining the limitations of the methods you have used. (7 marks)

Calculation and projection of the free cash flow to equity with explanations	2
Discount these values at the rate of return on equity	2
Calculate the present value of the perpetuity at 2011 with explanations	2
Calculate the present value of the firm's equity	1

(d) Write a brief report outlining the considerations your colleagues on the Board of Fly4000 might bear in mind when contemplating this acquisition. (8 marks)

Discussion of synergies and their capture	2
Examination of the risk exposure and the potential real options	3
Review of the financing options	2
Summary of valuation and recommendation of the next steps to the board of Fly4000	1

2 30 marks distributed over four sections.

(a) Prepare a memorandum, to be considered at the next board meeting, which summarises the arguments for and against forex risk hedging and recommends a general policy concerning the hedging of foreign exchange risk. (10 marks)

Summary of the principle arguments for and against hedging with specific reference to the market perfection and reduction of cost of capital arguments.	4
Clear policy recommendations focusing risk assessment, cost management, contracts and approvals, and monitoring.	4
Quality of the memorandum assessed in terms of clarity, argumentation and persuasiveness.	2

(b) Prepare a short report justifying your use of derivatives to minimise the firm's exposure to foreign exchange risk. Your report should contain:

(iv) The likely option price for an at the money option stating the circumstances in which the option would be exercised (you should use the Grabbe variant of the Black-Scholes model for both transactions adjusted on the basis that deposits generate a rate of return of LIBOR). (10 marks)

Correct calculation of d1 and d2 using the forward and spot rates as specified	5
Calculation of N(x) from the tables given	1
Correct calculation of the currency call	2
Correct calculation of the currency put	2

(v) A calculation of the number of contracts that would be required to eliminate the exchange rate risk and the cost of establishing a hedge to cover the likely FOREX exposure. (4 marks)

Calculation of the number of contracts for the 3 month hedge	2
Calculation of the number of contracts for the 9 month hedge	2

(vi) A summary of the issues the board should bear in mind when reviewing a hedging proposal such as this taking into account the limitations of the modelling methods employed and the balance of risk to which the firm will still be exposed when the position is hedged. (6 marks)

Cost of the hedge and identification of alternatives	2
Problems of establishing a perfect hedge (contract size)	2
Understanding demonstrated of the nature of basis risk and how it is influenced by timing.	2

3	**(a)**	**Prepare a paper for the next Board meeting recommending procedures for the assessment of capital investment projects. Your paper should make proposals about the involvement of the Board at a preliminary stage and the information that should be provided to inform their decision. You should also provide an assessment of the alternative appraisal methods.**	**(8 marks)**
		Clear definition of a two stage process for Board involvement in capital expenditure decisions	2
		Recommendation for the stage 1 appraisal procedure and metrics focusing on the role of payback and viable alternatives	3
		Stage 2 appraisal focusing on the business plan, value and accounting impact and cash recovery.	3
	(b)	**Using the appraisal methods you have recommended in (a) prepare a paper outlining the case for acceptance of the project to build a distillation facility at the Gulf plant with an assessment of the company's likely value at risk. You are not required to undertake an assessment of the impact of the project upon the firm's financial accounts.**	**(12 marks)**
		Calculation of the project VAR and assessment of its significance	4
		Estimation of the potential value impact using MIRR and the assumptions that underpin it.	3
		Estimation of the potential cash recovery using procedures recommended in a)	3
		Quality and persuasiveness of the written report	2
4	**(a)**	**Estimate the expected cost of capital for this project on the assumption that the additional finance is raised through a bond issue in the US market.**	**(10 marks)**
		Estimation of the current cost of debt capital at 4.324 per cent	3
		Calculation of the existing cost of equity capital at 7.48 per cent	2
		Ungearing and regearing the cost of equity capital using M&M proposition 2	3
		Calculation of the revised weighted average cost of capital and comment thereon	2
	(b)	**Draft a brief report for the board outlining the alternative sources of finance that are potentially available for this project including a brief discussion of their advantages and disadvantages and the likely impact upon the firm's cost of capital of each of the alternatives that you consider.**	**(10 marks)**
		Note of the theoretical issues in the selection of finance source (M&M, pecking order theory)	2
		Clear description of the potential sources of capital within the context of the Chinese economy, regulation and capital market	2
		Evaluation of the likely advantages and disadvantages of the principal sources of finance	2
		Discussion of the risk management strategies available	2
		Professional marks	2
5	**(a)**	**Prepare a board paper describing the procedure for obtaining a listing on an international stock exchange such as the London Stock Exchange.**	**(6 marks)**
		Outline of the three step procedure: registration, listing and admission to trading (2 marks for each step)	6
	(b)	**Prepare a briefing note itemising the advantages and disadvantages of such a step for a medium sized company.**	**(6 marks)**
		Note of the advantages: capital market access, reputation effects	2
		Note of the disadvantages: compliance costs, vulnerability to takeover, public scrutiny	2
		Well written briefing note weighing the advantages and disadvantages and focusing on the judgements that the board would be required to make	2
	(c)	**Discuss any ethical considerations or concerns you may have concerning this proposed course of action.**	**(8 marks)**
		Identification of the principal ethical issues involved	2
		Note on the issue of transparency and the protection of minority rights	2
		Discussion of alternative ways that the CFO could proceed and the ethical implications of each	2
		Commentary on the ethical issues involved in earnings management	2

Answers

Tutorial note: *These model answers are considerably longer and more detailed than would be expected from any candidate in the examination. They should be used as a guide to the form, style and technical standard (but not in length) of answer that candidates should aim to achieve. However, these answers may not include all valid points mentioned by a candidate – credit will be given to candidates mentioning such points.*

1 **(a)** A short form cash flow statement is as follows (note: a more detailed analysis of operating cash flow is not required)

Operating profit	108·8
Add depreciation	28·0
	136·8
Add change in:	
trade payables	1·1
trade receivables	3·5
inventories	0·5
Operating cash flow	141·9
less interest	–2·3
less taxation	–25·6
Free cash flow before reinvestment	114·0
CAPEX	–80·0
Dividends	–28·0
Financing	–10·0
Net cash change	–4·0

(b) Dividend capacity is determined by the free cash flow available to equity investors after net reinvestment. Reinvestment is that which is required to maintain the operating capacity of the business at the planned rate of growth specified by management as necessary to achieve the firm's strategic goals. Where new capital has been introduced this is deducted from the capital expenditure during the year to give the amount of investment financed from the free cash flow.

The free cash flow before reinvestment is $114 million which with CAPEX of $80 million implies a reinvestment rate of 0·70. The maximum dividend capacity is $24 million given the intention to repay $10 million of debt and excluding any further expansion in the company's working capital requirement. However, given that 18 days of cost of sales is unfunded this implies that (18/365 × (143·2 – 28·0)) = $5·68 million of extra capital would be required to finance the working capital cycle reducing the dividend capacity to $24m – $5·68m = $18·32 million.

Given the actual dividend is $28 million this implies that the firm has over distributed given its free cash generation during the year and its necessary reinvestment. The shortfall has been taken from the firm's cash reserves. Whether the firm has over distributed in terms of its capital account depends upon the firm's level of earnings for the year and its target rate of retention.

(c) The necessary calculations for this report are as follows:

Cost of equity 2007 using the capital asset pricing model

$E(R_e) = R_F + \beta_{i,m} \times ERP$
$E(R_e) = 3\% + 1{\cdot}4 \times 5\%$
$E(R_e) = 10\%$

The firm's market gearing ratio is:

$$\text{market gearing (2007)} = \frac{45}{45 + 25 \times 4 \times 16{\cdot}2} = 0{\cdot}027$$

and the weighted average cost of capital is:

WACC (2007) = (1 – 0·027) × 10% + 0·027 * 5% * 0·7 = 9·82%

On the basis of this the EVA for both years is:

EVA = *NOPAT* – *WACC* × Capital employed
EVA (2007) = (102·3 × ·7) – 0·0982 × (179·0 + 45) = $49·6 million

Or 22·1% on capital employed and

EVA (2008) = 108·8 * ·7 – 0·0982 × (253·9 + 35) = $47·79m
Or 16·54% of the capital employed in the business

The return on capital employed (defined as profit before interest and tax over capital employed) is:

ROCE (2007) = 51·2%
ROCE (2008) = 42·4%

Report to Management

On the basis of the forecast for the next 12 months we estimate that profit before tax should rise from the current level of \$102·3 million per annum to \$108·6 million an increase of 6·16%. An estimate of the Economic Value Added for 2007 and 2008 (projected) shows a small decrease from \$49·60million to \$47·79 million which is accounted for by an increase in the expected NOPAT figure from \$71·61 million to \$76·2 million offset by an increased capital charge incurred by the expected increase in the book value of capital employed and the increase in the firm's cost of capital. Overall, the EVA return on capital employed is expected to fall from 22·1% to 16·54% and the EVA margin from 22·54% to 18·24%. This deterioration in the EVA figure is attributable to an increase to \$36·1 million in the other operating costs of the business and an increase in the equity capital of the firm which on the current projections will not lead to a commensurate increase in the firm's value generation. A similar picture is presented by the expected decline in the firm's return on capital employed from 51·2% to 42·4%.

It is likely that the market will regard an outcome in line with our projections as a deterioration in the economic performance of the firm and this will inevitably have a negative impact upon both price of the firm's equity and the company's cost of capital. In order to surmount this difficulty the board should consider why the increased capital base of the firm (up by a projected 19·43%) is not being fully reflected in terms of increased turnover, profitability and value generation. Part of the explanation lies in the expected rate of increase of turnover which is expected to grow at a slower rate than the level of capital invested. However there is also an anticipated deterioration in margins (operating profit margin deteriorates from 39% to 37·7%) suggesting that further measures to control costs is necessary.

The underlying difficulty we face is the speed with which new capital is generating returns. This is largely governed by our actual reinvestment. Cash generation is well ahead of our planned rate of reinvestment and servicing of finance. In the medium term we need to review the use of this cash either in terms of new capital projects, acquisitions or repayment of capital.

2 (a) The first step in the valuation is to calculate each company's weighted average cost of capital. The cost of capital is calculated post tax and using the relevant market values to calculate the market gearing ratio.

	Burcolene	**PetroFrancais**
Cost of equity (using the CAPM, 3% risk free and 4% equity risk premium)	$= \cdot 03 + 1\cdot 85 \times 0\cdot 04$ $= 0\cdot 104$	$= \cdot 03 + 0\cdot 95 \times 0\cdot 04$ $= 0\cdot 068$
Market gearing	$= 3\cdot 3/(3\cdot 3 + 9\cdot 9)$ $= 0\cdot 25$	$5\cdot 8/(5\cdot 8 + 6\cdot 7)$ $= 0\cdot 464$
Cost of debt	$= 0\cdot 03 + 0\cdot 016$ $= 0\cdot 046$	$= 0\cdot 03 + 0\cdot 03$ $= 0\cdot 06$
WACC	$= 0\cdot 75 \times \cdot 104 + \cdot 25 \times \cdot 046 \times 0\cdot 7$ $= 0\cdot 0861$	$= 0\cdot 536 \times \cdot 068 + \cdot 464 \times \cdot 06 \times 0\cdot 75$ $= 0\cdot 0573$

Note: there may be rounding differences.

The core valuation formula is:

$$V_0 = \frac{FCF_1}{WACC - g}$$

As free cash flow is NOPAT – net reinvestment then:

$$V_0 = \frac{FCF_1}{WACC - g} = \frac{(\text{NOPAT} - \text{net reinvestment})_1}{WACC - g}$$

The figures quoted are for NOPAT – reinvestment for the current year. For the two companies the value before either pension or share option scheme adjustments is therefore (in \$):

$$V_B = \frac{450 \times (1\cdot 05)}{0\cdot 0861 - 0\cdot 05} = \$13\cdot 107\text{bn}$$

and

$$V_P = \frac{205 \times (1\cdot 04)}{0\cdot 0573 - 0\cdot 04} = \$12\cdot 303\text{bn}$$

However, both company values will need to be reduced by the relevant charge for the outstanding options and the pension deficit. Using the fair value approach the value of each option outstanding is given by:

Option value = intrinsic value + time value
Option value = (actual price – exercise price) + time value
Option value = 29·12 – 22·00 + 7·31 = \$14·43

(the actual price is given by the total value of equity \$9·9 billion/340 million = \$29·12 per share)

The number of options likely to be exercised:

options $= 25{\cdot}4\text{m} \times (1 - 0{\cdot}05)^3 \times 0{\cdot}8 = 17{\cdot}42$ million

Which gives a value of options outstanding = \$251·4m

And an estimated market valuation of Burcolene of:

$V_B = \$13{\cdot}107\text{bn} - \$251{\cdot}35\text{m} = \$12{\cdot}855\text{bn}$

The value of PetroFrancais is much more straightforward being:

$V_P = \$12{\cdot}303\text{bn} - \$430\text{m} = \$11{\cdot}873\text{bn}$

(b) Burcolene

Report to management

Subject: Valuation and Financial Implications of an Acquisition of PetroFrancais

This is potentially a type three acquisition where both the firm's exposure to business risk and financial risk change. As a consequence the value of the combined entity will depend upon the post acquisition values of the component cash streams: (i) the cash flow from the existing business; (ii) the cash flow from the acquired business and (iii) any synergistic cash flows less the cost of acquisition. However, estimating the value of these cash flows relies upon an estimate of the post acquistion required rate of return – which cannot be estimated until we know the value of the component cash flows. This problem requires an iterative solution and which can be solved using a spreadsheet package.

Validity of the Free Cash Flow to Equity Model

Our estimates of the value using NOPAT as a proxy for free cash flow produces values that are reasonably close to the current market valuation of both companies. The models value Burcolene at \$12·855 billion and PetroFrancais at \$11·873 billion compared with current market valuations of \$13·1 billion and \$12·5 billion respectively. The estimation error is 1·9% and 5·3% respectively. Although minor the differences can be explained by any of the following:

- The model used may mis-specify the market valuation process. In either case NOPAT may not be a sufficiently close approximation to each firm's free cash flow.
- The underpinning models in the cost of capital calculations may not be valid. The capital asset pricing model, for examples, does not capture fully all the risk elements that are priced in competitive markets.
- The estimates of growth may be overoptimistic (both valuations are highly sensitive to variation in the implied level of growth).
- The markets may have reacted positively to rumours of an acquisition.
- The capital markets may be inefficient.

However, on the basis of this preliminary analysis, the low levels of modelling error suggest that the NOPAT based model should form the basis for valuing a combined business.

Deriving a bid price:

In preparation of an offer, a due diligence process should, as part of its brief, consider the likely growth of each cash stream within the context of the combined business and the variability associated with the future growth rates of each cash stream. This information could then be used to estimate the firm's future cash flows (i) to (iii) above using a cost of capital derived from the current required rates of return and market values. An iterative procedure can then be employed to bring the derived values into agreement with those used to estimate the firm's cost of capital. This valuation less the cost of acquisition and the firm's current debt gives the post acquisition equity value. The maximum price that should be paid for PetroFrancais is that which leaves the equity value of Burcolene unchanged. This estimation process whilst procedurally complex does reinforce a key point with type III acquisitions that the sum of the equity valuation of both parties is not a good indication of the value of the combined business.

Providing the management of Burcolene can come to a reliable valuation of the combined business then, providing they remain within the bid-price parameters, the acquisition should increase shareholder value. Good valuation methods should capture the benefits and the consequential costs of combined operation. It is important that management recognises this point and do not double count strategic opportunities when negotiating a bid price. In this case, improving equity value for the Burcolene investors depends upon a number of factors. A simulation of the most important parameters in the valuation model: forward growth, the cost of equity, default premiums and the cost of debt should allow Burcolene to estimate the likely equity value at risk given any chosen bid price. A simulation would also provide an estimate of the probability of a loss of equity value for Burcolene's investors at the chosen bid price.

Implications for gearing and cost of capital:

Financing an acquisition of this magnitude through debt will raise the book gearing of the business although its impact upon the market gearing of the firm is less easy to predict. Much depends on the magnitude of any surplus shareholder value generated by the combination and how it is distributed. An acquisition such as this will increase market gearing if the benefits accrue to the target shareholders. The reverse may occur if the bulk of the acquisition value accrues to the Burcolene investors. Similarly the impact upon the firm's overall cost of capital, the impact of the tax shield and the exposure to default risk again all depend upon the agreed bid price and the distribution of acquisition value between the two groups of investors.

3 **(a)** This is a straightforward application of the Black and Scholes option pricing model. Each of the input components is stated in the question:

Current price = Present Value of the Project = \$28 million
Exercise price = capital expenditure = \$24 million
Exercise date = 2 years (or 500 trading days)
Risk free rate = 5%
Volatility = 25%

Using the formula as specified:

$$d_1 = \frac{\ln\left(\frac{28}{24}\right) + \left(0{\cdot}05 + 0{\cdot}5 \times 0{\cdot}25^2\right) \times 2}{0{\cdot}25 \times \sqrt{2}} = 0{\cdot}8956$$

$$d_2 = 0{\cdot}8956 - 0{\cdot}25 \times \sqrt{2} = 0{\cdot}5421$$

The areas under the normal curves for these two values are $N(d_1) = 0{\cdot}8147$ and $N(d_2) = 0{\cdot}7061$.

Using the derived values for $N(d_1)$ and $N(d_2)$ the value of the call option on the value represented by this project is as follows:

$c = 0{\cdot}8147 \times 28 - 0{\cdot}7061 \times 24 \times e^{-{\cdot}05 \times 2} = \$7{\cdot}48$ million

This implies that at the current time the project has a value equal to its net present value plus the value of the call option to delay, i.e., \$11·48 million. The additional value arises because the delay option allows the company to avoid the downside element of risk.

(b) Digunder Ltd

Housing Development at Newtown

This project has a net present value of \$4million on a capital expenditure of \$24 million which whilst significant has a volatility estimate of 25% of the present value per annum. This volatility is brought about by uncertainties about Government's intentions with respect to the Bigcity–Newtown transport link and the consequential impact upon property values. Currently, the project presents substantial value at risk and there is a high likelihood that the project will not be value generating. To surmount this, an estimate is provided of the value of the option to delay construction for two years until the Government's transport plans will be made known.

The option to delay

The option to delay construction is particularly valuable in this case. It eliminates much of the downside risk that the project does not generate the cash flows expected and it gives us the ability to proceed at a point in time most favourable to us. The nature of the delay option is that it is more valuable the greater the volatility of the underlying cash flows and the greater the time period before we are required to exercise.

The valuation of the option to delay has been undertaken using the Black and Scholes model which members have been briefed about with respect to fair value accounting practices under the International Financial Reporting Standards. The model has some limiting assumptions relating to the underlying nature of the cash flows and our ability to adjust our exposure to risk as time passes. In reality, the use of this type of modelling is more appropriate for financial securities that are actively traded. Our use of the model is an approximation of the value of the flexibility inherent in this project and although the model will not have the precision found in its security market applications it does indicate the order of magnitude of the real option available. A positive value of \$11·5 million is suggested by the model underlying the considerable benefit in delay.

In interpreting this valuation it is important to note that the actual project present value at commencement could be significantly larger than currently estimated and will certainly not be less than zero (otherwise we will not exercise the option to build). The additional value reflects the fact that downside risk is eliminated by our ability to delay the decision to proceed.

On the basis of our valuation the option to delay commencement of the project should be taken and investment delayed until the Government's intention with respect to transport links becomes clearer. On this basis we would place a value of \$11·5 million on the project including the delay option.

(c) The Black and Scholes model makes a **number of assumptions** about the underlying nature of the pricing and return distributions which may not be valid with this type of project. More problematically it assumes that continuous adjustment of the hedged position is possible and that the option is European style. Where the option to delay can be exercised over any set period of time up to the exercise date the Black and Scholes model will cease to be accurate. For a call option, such as the option to delay, then the level of inaccuracy is likely to be quite low especially for options that are close to the money. Given that an option always has time value it will invariably be in the option holder's interest to wait until exercise date before exercising his or her option. However, in those situations where the level of accuracy is particularly important, or where it is suspected that the Black and Scholes assumptions do not hold, then the binomial option pricing approach is necessary.

4 (a) Agenda for change

Over recent years, the competitiveness of our business has been reduced by a number of factors not least the significant reductions in defence spending in our constituent markets. Over this time we have introduced new technologies through acquisition and have not engaged in significant research and development on our own account. Our current position is one of significant strength: we have substantial cash reserves and cash flow generation is still strong. Our gearing at 12% of capital is very low and this combined with our earnings history and liquidity gives us a high credit rating and hence a relatively low cost of capital. Our weakness is our lack of investment in new projects and our lack of R&D to support such investment. In this position we are exposed to the risk of a hostile bid by one of the many companies which do have substantial R&D but are chronically short of liquidity for future developments. I propose the following alternatives for discussion. These alternatives are not mutually exclusive:

Alternative 1:

Given that building a viable R&D ability would take many years of investment and development that would not appear to be a route to follow. However, we do have the financial resources to acquire a competitor who does have strong R&D in relevant technologies. This would achieve two ends, it would reduce the risk of predatory attack – there is substantial evidence that companies who acquire are less likely to be acquired themselves – it would eliminate one part of the competition and it would give us the capability for development that we do not currently have. The downside is if the perceived benefits do not materialise and shareholder value is lost. We may also lose shareholder value if we do not get the level of our bid right and in that regard much will depend on our investigation of potential targets.

Alternative 2:

We work harder at our current strategy seeking key technology targets at the best price and redoubling efforts to manage our own cost base. In our industry cost has become a strategic tool and we have not invested in advanced manufacturing technology to the extent of some of our competitors. In a contract build environment it is important to achieve high levels of efficiency within our matrix structure and to minimise the dysfunctional aspects of this type of operational management. To this end we should consider making projects responsible for the labour they utilise at current market rates and to establish a coherent project based budgeting and cost control system.

Alternative 3:

We recognise that our future cash generation is based upon a set of current projects of finite life and that we are moving into a phase of the company's life where new positive net present value projects are unlikely to be found. Our recent history suggests that we do have agency problems in that there are lower level managers who have championed projects which have not generated the promised returns and this is indicative of a situation where the market opportunities are very limited and net present values have been competed down to zero. In this situation one option is to return cash to shareholders through enhanced dividends or share repurchase schemes. The latter has the advantage that it does not tie us to payout commitments in the future.

Alternative 4:

We recognise that we have a shortage of managerial talent at different levels and that this has been brought about by relatively low levels of remuneration compared with the rest of the sector. Executive remuneration is not just about salary levels but we may wish to consider a stock option scheme where managers are rewarded for the delivery of positive net present value investments through the resulting increase in share price. This may well have the desirable effect of reducing the agency loss through overselling the merits of projects and overstating potential returns.

(b) The key element of this case is that this company is no longer able to find positive net present value projects and as a result its rate of growth is slowing and may, very soon, start to decline. The engine of growth would appear to be new technology and superior management practice in the management and control of projects and their costs. However, the company has no effective R&D expertise and its scope for technology led acquisitions appears to be very limited.

The ethical issue here is that if a company is no longer able to use its owner's cash then it should return money to its investors and not use it to enhance managerial rewards and perks. There is only one justification for increasing levels of executive remuneration and that is that those managers are better motivated to create the high levels of growth that lead to increased shareholder value. There are some who would argue that maximising shareholder value is a constrained objective and that the firm owes a duty to other stakeholders such as employees, managers, suppliers and customers. However, overriding this is the efficiency argument. By returning cash to shareholders, the effective operation of the capital based system ensures that they have at their disposal those cash resources and can make their own judgements about the most efficient use of their resources – to the greater benefit of the stakeholders of those businesses in which they choose to invest. The ethical arguments are therefore based upon both social policy and property rights. Social policy is involved in that the efficient operation of market economies and the maximisation of social welfare and property rights in that the surplus value within a company belongs to its investors both legally and morally.

Depending upon the situation options that increase shareholder return either through the maximisation of the firm's value, or by returning cash to them for new investment elsewhere are to be preferred.

The case also raises a question mark concerning the firm's accounting practice and the use of defensive accounting policies. If the ratio of EBITDA to operating cash flow is consistently less than one for a growing firm this would suggest that the company is deliberately hiding earnings. There are a number of reasons for this: it may be that the firm is attempting to smooth its earnings figures in order to present more consistent performance measures over the years or it is hiding earnings in order to suppress pressure from various other stakeholders for higher wages or other forms of compensation. It may be that the company is also trying to present a relatively low earnings history as part of its pricing negotiations. However, this type of earnings management strategy is self correcting in the longer run and it is doubtful to what extent the market is fooled. The ethical dimension arises if this represents an intention to deceive rather than a function of the firm's type of business and the constraints of the GAAP.

Finally, given the ethical requirement to act responsibly it is also important to consider the environmental issues presented by the case. There is an argument that operational economy and the efficient use of resources has an environmental dimension in as far as a given level of growth can be achieved with a given level of inputs. However, the case also reveals that the company has been fined for allowing untreated discharge into a local river. As social concern about the environmental impact of industry increases, the regulation of waste and the punishments for breaches of environmental security are likely to become more and more severe. If for no other reason than the protection of the shareholders' interest the company should make the necessary investment to control its effluent discharge. It should also seek to minimise energy consumption across all its operations.

5 (a) The coupon rate on the new debt

The coupon rate should be the same as the yield for four-year debt at 6%. If the firm's bankers have overestimated the credit risk and set the spread too high, then a coupon of 6% will result in the debt being issued at a premium in the market. If they have set it too low then the debt will not be fully taken up and the underwriters will have to issue it at a discount. The investment banks suggest that at a yield and hence a coupon of 6% that this would guarantee that the issue would be taken up by their institutional clients. On this basis the firm may wish to ask for an underwriting agreement to that effect although there would inevitably be a charge for this.

(b) Impact of the new debt upon the company's market valuation and its cost of debt

The issue of the new debt can only be achieved at the cost of a reduction in our company's credit rating and/or a consequent increase in its cost of debt capital.

Using our current market gearing ratio the current amount of debt in issue is calculated as follows:

$$gearing = \frac{V_d}{V_e + V_d}$$

$$0{\cdot}25 = \frac{V_d}{1{\cdot}2 + V_d}$$

$$V_d = \frac{0{\cdot}25 \times 1{\cdot}2}{0{\cdot}75} = \$0{\cdot}4 \text{ billion}$$

Thus, the existing market value of our company debt is \$400 million. Given that the coupon (4%) and the current market yield (3·5% plus 50 basis points) are the same then the current market value is also its par value.

The yield on the new debt would be 5·1% plus 90 basis points to give 6%. If the new debt is issued at par at this yield of 6% then the market value of the existing debt will fall in line with the decrease in the company's credit rating and the consequential increase in yield to 4·4%:

$$MV_d = \frac{4}{1{\cdot}044} + \frac{4}{1{\cdot}044^2} + \frac{104}{1{\cdot}044^3}$$

$$MV_d = \$98{\cdot}90\%$$

Which when applied to the \$400 million par value gives a market value of \$395·90 million. On the assumption that the new debt is taken up at par then the new market value of debt in issue will rise to \$795·90 million.

The firm's effective cost of debt capital is calculated by weighting the yields of the two components of debt and then adjusting for tax:

$$r_d = \left[\frac{V_d^n}{V_d^n + V_d^o} r_d^n + \frac{V_d^o}{V_d^n + V_d^o} r_d^o\right] \times \left(1 - T\right)$$

$$r_d = \left[\frac{400}{400 + 395{\cdot}90} \times 6\% + \frac{395{\cdot}90}{400 + 395{\cdot}90} \times 4{\cdot}4\%\right] \times 0{\cdot}7$$

$$r_d = 3{\cdot}64\%$$

The firm's current cost of debt capital is 4% × (1 – 0·3) = 2·8% so the increase in gearing will raise the firm's cost of debt capital (after tax) by 84 basis points. However, this increase is in part due to the longer term to maturity on the new borrowing rather than the increase in the credit spread and the firm might wish to consider extending the term depending upon the yield curve and rates beyond four years.

(c) The advantages and disadvantages of this mode of capital financing.

Debt finance is a relatively low cost method of raising long term finance. Under static trade off theory we would expect higher gearing to generate improvements in the firm's cost of capital given the benefit of the tax shield. However, the cost of debt capital consists of three components: the pure risk free rate, the term premium and the credit spread. In this case we are proposing to alter our capital structure by taking on longer term debt and thus the advantages of higher gearing are to a certain extent obscured. Pecking order theory suggests that debt finance should be preferred to new equity finance and is normally taken by the market as a signal that management believe that the company is undervalued. In the context of an efficient market this is doubtful but it is certainly the case that there are strong agency effects through debt. Debt will exert a greater discipline over our action than equity finance and tends to suppress opportunistic investment and over consumption of perks.

From a transactions costs perspective, debt tends to be preferred for the acquisition of general assets with high marketability and equity for intangibles and highly specific assets. In the airline business finance of this level is normally for aeroplane acquisitions which do have a reasonably active second hand market.

The marking strategy for this paper is to construct a mark ramp such that the average student can attain a pass mark of 50% through the demonstration of a satisfactory level of knowledge and skills. Much of this ramp effect will be achieved through q1 and q2 but with each of the latter questions giving a substantial core of easily gained marks. Part A questions will carry a stated number of marks assigned to the professional quality of the answer. At this level, professional quality covers both presentational aspects and the ability to integrate both numerical and written material into a single coherent piece of work.

			Marks
1	**(a)**	For the correct calculation and presentation of the following:	
		Cash flow statement: operating cash flow	3
		Other items	3
		Total	**6**
	(b)	Calculation of the free cash flow before reinvestment	1
		Reinvestment deduction	2
		Working capital adjustment	2
		Summary of dividend capacity	1
		Total	**6**
	(c)	For the correct calculation of:	
		Calculation of the 2007 cost of equity using the CAPM	2
		Calculation of the WACC	3
		Calculation of the 2007/2008 EVA	4
		Report to management to include other metrics and commentary*	9
		Total	**18**

**Professional marks awarded for the quality of the layout, clarity and persuasiveness of the presentation and integration of analytical data with the written text.*

2	**(a)**	Calculation of the WACC	5
		Calculation of the value of both companies	6
		Deduction for option value	3
		Deduction for pension liability	2
		Total	**16**
	(b)	Identification, using market risk measures, of type three acquisition	1
		Comparison of FCF model with market capitalisation	2
		Note on the estimation error and likely causes	3
		Outline of iteration method for type three acquisition and the method of identifying a bid price	2
		The indeterminate outcome for market gearing following acquisition	2
		The likely impact upon the firm's WACC	2
		Report to management*	2
		Total	**14**

**Professional marks awarded for the quality of the layout, clarity and persuasiveness of the presentation and integration of analytical data with the written text.*

			Marks
3	**(a)**	Identification of inputs into BS model	4
		Calculation of d1 and d2	4
		Calculation of real option value	3
		Conclusion on the value of the option to delay	1
		Total	**12**
	(b)	Estimation of overall project value at $11·48 million	2
		Justification for the use of BS model	2
		Total	**4**
	(c)	Outline of the limitations of the BS model	2
		Identification of the American style real option in the given circumstances	1
		Note on the appropriate technique for solving the American style option	1
		Total	**4**
4	**(a)**	Setting the scene and identification of core problem and its source	2
		Layout principal alternatives	
		Acquisition strategy	2
		Reorganisation, organic growth with cost minimisation	2
		Return cash to investors	2
		Incentives for management with share option scheme	2
		Total	**10**
	(b)	Identification of core ethical issue	2
		Social policy and property rights arguments underpinning ethical dimension	2
		Resolution and advice	4
		Note on ethics of earnings management	2
		Total	**10**
5	**(a)**	Advice on the appropriate coupon rate	4
		Total	**4**
	(b)	Estimate of current market value of debt	2
		Estimate of market value of debt following new issue	3
		Calculation of the revised cost of debt capital	3
		Total	**8**
	(c)	Relative advantages and disadvantages	
		Asset specificity and matching	2
		Agency effects	2
		Static tradeoff arguments	2
		Pecking order	2
		Total	**8**

Answers

1 Mercury Training

(a) The solution strategy for this part of the question is to determine the proxy beta by weighting the individual asset betas and then gearing them to the level appropriate for Mercury Training Ltd. The difficulty is that the gearing should be tax adjusted. The required conversion formula is shown in the formula sheet. The equity and weighted average costs of capital are then straightforward.

The tax adjusted gearing for Jupiter and for the financial services sector, given a tax rate of 40%, is as follows:

Jupiter: $$w_d' = \frac{1-T}{w_d^{-1}-T} = \frac{0{\cdot}6}{0{\cdot}12^{-1}-0{\cdot}4} = 7{\cdot}56\%$$

Financial Services Sector: $$w_d' = \frac{1-T}{w_d^{-1}-T} = \frac{0{\cdot}6}{0{\cdot}25^{-1}-0{\cdot}4} = 16{\cdot}67\%$$

Note: other approaches to calculating the tax adjusted gearing for Jupiter are acceptable.
Calculating the amount of debt for Jupiter can be found as follows:

$$1-w_d = \frac{E}{D+E}$$

$$D = \frac{E}{1-w_d} - E = \frac{29}{0{\cdot}88} - 29 = \$3{\cdot}96\ million$$

Using this debt value and the stated equity value of $29 million the tax adjusted gearing for Jupiter is as follows:

$$w_d = \frac{D(1-T)}{E+D(1-T)} = \frac{3{\cdot}96\times 0{\cdot}6}{29+3{\cdot}96\times 0{\cdot}6} = 7{\cdot}56\%$$

The asset beta for Jupiter is as follows:

$$\beta_A = \beta_e(1-w_d^1)$$
$$\beta_A = 1{\cdot}5\times(1-0{\cdot}0756)$$
$$\beta_A = 1{\cdot}387$$

Repeating the same exercise for the financial services sector using the gearing ratio of 16·66% and an equity beta of 0·9 gives an asset beta as follows:

$$\beta_A = \beta_e(1-w_d^1)$$
$$\beta_A = 0{\cdot}9\times(1-0{\cdot}1667)$$
$$\beta_A = 0{\cdot}75$$

The proxy asset beta for Mercury is as follows:

$$\beta_{A(mercury)} = (0{\cdot}67\times 1{\cdot}387) + (0{\cdot}33\times 0{\cdot}75) = 1{\cdot}177$$

The proxy equity beta for Mercury is calculated by converting the quoted market gearing of 30% to the tax adjusted rate and then using the formula above expressing the relationship between the equity and asset beta as follows:

$$w_d^1 = \frac{1-T}{w_d^{-1}-T} = \frac{0{\cdot}6}{0{\cdot}30^{-1}-0{\cdot}4} = 20{\cdot}45\%$$

$$\beta_E = \frac{\beta_A}{(1-w_d^1)}$$

$$\beta_E = \frac{1{\cdot}177}{(1-0{\cdot}2045)}$$

$$\beta_E = 1{\cdot}480$$

The equity and weighted average costs of capital are then as follows:

$$r_e = R_F + \beta_i(R_m - R_F)$$

$$r_e = 0{\cdot}045 + 1{\cdot}480\times 0{\cdot}035 = 9{\cdot}68\%$$

And WACC:

$$WACC = w_e r_e + w_d r_d(1-T)$$

$$WACC = 0{\cdot}7\times 0{\cdot}0968 + 0{\cdot}3\times(0{\cdot}045+0{\cdot}025)\times 0{\cdot}60 = 8{\cdot}04\%$$

The equity cost of capital is used for valuing income flows (such as dividends or free cash flow to equity) which go directly to the equity investor. The weighted average cost of capital is for valuing flows attributable to the business entity such as project cash flows or NOPAT.

(b) This part of the question requires advice on the likely range of prices.

At the low end the firm's net assets at fair value would be the realisable value of the equity between a willing buyer and seller. This is 650c per share which would represent the lower end of any negotiating range.

Using the dividend valuation model we estimate the share price at the upper end using the latest DPS of 25¢ per share and the cost of equity capital of 9·68%. Three potential growth rates present themselves: the historic earnings growth which at 12% is greater than the firm's equity cost of capital and is therefore not sustainable over the very long run, the anticipated growth rate of the two sectors weighted according to the firm's revenue from each (0·67 × 6% + 0·33 × 4% = 5·33%) and the rate implied from the firm's reinvestment:

$$g = br_e = \frac{(100 - 25)}{100} \times 0{\cdot}0968 = 7{\cdot}26\%$$

The value of the firm using the growth model and the higher of the two feasible growth rates is:

$$P_0 = \frac{D_0(1+g)}{r_e - g}$$

$$P_0 = \frac{25 \times (1{\cdot}0726)}{(0{\cdot}0968 - 0{\cdot}0726)} = \$11{\cdot}08 \text{ per share}$$

In addition, the share price gives a spot estimate of the value of a dividend stream in the hands of a minority investor. If the option to float is taken then a share price of $11·08 could be achieved especially if a portion of the equity and effective control are retained. However, if a sale is made to a private equity investor then it may be appropriate to value the firm taking into account the benefits of control which can be substantial if the purchaser is able to generate significant synergistic benefits either in terms of revenue enhancement, cost efficiency or more favourable access to the capital market. Control premiums can be as much as 30–50% of the spot price of the equity. In this case an opening negotiation may start with a share price of $16·62.

(c) **The Directors**
Mercury Training

The two principal sources of large-scale equity finance are either through a public listing on a recognised stock exchange or through the private equity market. The former represents the traditional approach for firms who have grown beyond a certain size and where the owners wish to release, in whole or in part, their equity stake within the firm, or where they wish to gain access to new, large scale equity finance. The procedure for gaining a public listing is lengthy and invariably requires professional sponsorship from a company that specialises in this type of work. Depending upon the jurisdiction there are three stages that may have to be fulfilled before a firm can raise capital on a stock exchange:

1 Formalise the company's status as a public limited company with rights to issue its shares to the public. In some jurisdictions this requires re-registration and in others it is implicit in the conferment of limited liability.

2 Seek regulatory approval for admission to a public list of companies who have met the basic criteria required for entry to a stock exchange (in the UK this process is under the jurisdiction of the Financial Services Authority).

3 Fulfil the requirements of the exchange concerned which may entail the publication of a prospectus which is an audited document containing, among other things, projections of future earnings and profitability.

The disadvantage of public listing is that a company will be exposed to stake building by other companies, regulatory oversight by the stock exchange and greater public scrutiny. Stock exchanges require that quoted companies comply with company law as a matter of course but also that they adhere to various codes of practice associated with good governance and takeovers. They must also comply with stock exchange rules with respect to the provision of information and dealing with shareholders.

Private equity finance is the name given to finance raised from investors organised through the mediation of a venture capital company or a private equity business. As the name suggests these investors do not operate through the formal equity market but they operate within the context of the wider capital market for high risk finance. Because of its position, PEF does not impose the same regulatory regime as the public market. Transaction costs tend to be lower and there is evidence to suggest that private equity finance offers companies the ability to restructure and take long term decisions which have adverse short term consequences. In some jurisdictions there are favourable tax advantages to private equity investors.

2 Venus Systems

This question focuses on the problems of a tied supplier which has experienced a downturn in its business because of factors largely beyond its control. The candidates' report should focus on the problems of managing business cycles, the nature of financial distress, working capital management and turnaround.

The Board of Directors
Venus Systems Ltd (VSL)

From: Molly Quelle, Financial Advisor

(a) Analysis of the current financial situation and advice on future business prospects.

Since 2006 your company (VSL) has experienced a sudden drop in business turnover with falls of 2% in 2007 and 17·7% in 2008. Bottom line performance has been more mixed for reasons noted in your briefing but the position is now that VSL is only just in profit. The company is in distress and it is worth reflecting for a moment on the five core causes of financial distress in any business:

– Revenue failure caused by either internal or external factors. Revenue failure may be through a loss of orders (market failure) or through the acceptance of business which does not contribute to the growth of shareholder value.

– Cost failure caused by weak cost control, changes in technology, inappropriate accounting policies, inadvertent or exceptional cost burdens, poor financial management or failure of effective governance.

– Failure in asset management through failure to invest in appropriate technology, poor working capital management, inappropriate write off and reinvestment or poor organisation of the available assets.

– Failure in liability management through failure to manage the company's relationship with the money markets, weak control of interest rate risk and currency risk or unsustainable credit policies.

– Failure of capital management through either over or under-capitalisation or poor management of the company's relationship with the capital markets and in particular the company's debt portfolio and the optimisation of its cost of capital.

In practice problems rarely occur in isolation. A business is an internal and external network of relationships of assets and individuals and problems in one area invariably have consequences elsewhere. To structure my investigation I have focused on the four areas of performance, efficiency, risk and liquidity and used a variety of accounting measures under each heading to identify the principal problem areas and the potential remedies.

(b) Performance

VSL has experienced problems in its two principal areas of activity through failure in its market. In the case of the military business sustained global hostilities have come to an end and military spending, as it did in the early 1990s, has begun to slow down to a maintenance level. However, on the civil side there is no reason to doubt that once production against advance orders for the new European Aircoach commences orders for VSL components will rapidly accelerate. There has been significant rebalancing of the firm's production, new investment took place in 2006 and is planned for 2009, and steps have been taken to reskill and redeploy the workforce.

On the cost management side there are some general issues and steps that can be taken. The company's gross margin is very consistent and typical of manufacturing at 60% before depreciation is charged. It may be that the company can secure greater cost efficiency through focusing on the reduction of change, logistic and other non-value adding costs. The firm's depreciation policy should also be reconsidered. The average rate of 10% per annum is appropriate for fixed equipment but not for buildings and land.

In 2007 the company's operating profit margin was 12·5% but has since deteriorated sharply because of the fall in revenues. Other operating costs have risen by more than general inflation at 7% for 2007/8 and 5% for 2006/7. This is a significant cause for concern and should be investigated further. As a consequence of the deterioration of revenues the company's return on capital employed, return on equity and other return measures have all deteriorated rapidly. These all reflect the general failure of profitability. However, one pair of ratios is worthy of comment and that is the relationship between return on capital employed and return on fixed capital employed.

Ratios	2008	2007	2006
	$m	$m	$m
Earnings before interest and tax/Total capital employed	1·53%	12·28%	7·49%
Earnings before interest and tax/Fixed capital employed	9·18%	29·72%	9·52%

The key point to note here is not the sudden change in 2007 due to the commissioning and decommissioning of capital equipment but the loss in return because of the company's working capital management. If a company has a neutral working capital management policy, with a current asset ratio equal to unity, then these two ratios should be approximately the same. The higher the current asset ratio, the lower the return on capital employed versus the return on fixed capital employed and vice versa. In manufacturing, it is rarely possible to operate with current asset ratios less than one and so this points to working capital management as a priority area for the company.

Finally, it is clear from the collapse in the share price that the market recognises the difficulties that the company is facing. However the company's P/E ratio clearly indicates that the market expects the business to recover.

Efficiency
The company's asset efficiency has rapidly increased from 107% in 2006 to 427% in 2009. This has been mostly due to the rationalisation of the firm's capital assets and the sale of the non-productive refuelling gear plant. With respect to cost efficiency the company's production costs appear to be well controlled with a 5% improvement in 2007 but a 5% deterioration in 2008. Given the problems of redeployment and restructuring cost management at this level does not appear to be of concern. However, when we turn our attention to the efficiency with which the VSL converts its activities into cash, a significant problem does emerge:

Ratios	2008 $m	2007 $m	2006 $m
operating cycle analysis			
receivables*365/turnover	183	91	71
inventory*365/cost of sales (ex depn)	146	73	82
trade and other payables*365/cost of sales (ex depn)	34	32	36
scale adjustment (see below)	0·77	0·67	0·67
adjusted receivables age (days)	141	61	47
adjusted inventory days	113	49	54
funded/(unfunded) days	–220	–77	–66
cash requirement ($ millions)	34	12	11

Because of the difference in the magnitude of the cash flow recovered from customers compared to that required to settle outstanding payables a scale adjustment is required to estimate the average number of days that the operation cycle is unfunded. For example, in 2008:

Scale adjustment = (operating costs – depreciation)/revenue = (44 + 27 – 15)/72·6 = 0·77

When comparing with a given payables age we effectively reduce the receivables period from 183 to 141 days and the inventory holding period from 146 to 113 days. The unfunded period is the sum of the adjusted receivable and inventory periods (254 days) less the payables period of 34 days to give 220 days of cash outflow not covered by the cash recovered from customers. As noted above the payment period to creditors has remained constant at around 34 days. However, the number of unfunded days has risen from 66 days in 2006 to 220 days in 2008 committing an additional $21 million of cash compared with 2006 to the maintenance of working capital. As a matter of urgency, the company needs to bring the figures for receivables back down to the 2006 levels although inventories may be more difficult to reduce in the short run.

Risk
Of key concern is the prospect of default. Our assessment is that in the short run the risk is moderate but not imminent. The Z score for the business (which predicts failure within five years with 80% accuracy) has fallen to 2·54 in 2008 from 3·33 in 2007. At this level it lies within an indeterminate region where many companies survive.

Depending on how candidates treated 'total debt' minor variations in the Z score can arise. The examiner accepted as correct all reasonable interpretations of the inputs to the score. Using total liabilities the following ratios and Z score are obtained:

Z score analysis	factor	2008	2007	2006
X_1 = working capital/total assets	1·2	0·75	0·56	0·20
X_2 = retained earnings/total assets	1·4	0·33	0·32	0·23
X_3 = earnings before interest and tax/total assets	3·3	0·01	0·10	0·06
X_4 = market value of equity/book value of total debt	0·6	0·90	1·98	1·41
X_5 = sales/total assets	1·0	0·60	0·70	0·66
Z score		**2·54**	**3·33**	**2·27**

An alternative approach attributable to William Beaver's (1966) work on corporate failure gives further cause for concern. The company's operating cash flow to total debt ratio has fallen below the survival threshold of 0·15. Currently the ratio is at –0·16 reflecting the fall in operating cash flows largely due to the increase in working capital.

The firm's operational gearing is 37% compared with 26% in 2006 which reflects the increase in the firm's fixed operating costs noted earlier. However, the principal concern is the firm's ability to pay its immediate interest payments and it is to this that the Beaver ratio points as a key indicator of the firm's short run survival.

Liquidity
As we have already noted the firm's cash generation has fallen rapidly, the sell-off of assets in 2007 and 2008 has released significant cash into the business and the company has $56·1 million in hand. The current asset and acid test ratios confirm what we already know, but of more significance is the cash exhaustion ratio which tells us that the available cash could sustain over a year of operations without any additional revenue. In our judgement, there is enough cash in the business to sustain the reinvestment planned of $35 million and to repurchase 10% of the firm's equity at a cost of $5 million. However, it is not clear at this stage that an increase in the firm's gearing would offer a significant benefit to the firm's cost of capital nor is it clear, in advance of a possible and sudden expansion of the firm's civil business, that the cash resources of the business should be returned to the investors in a share repurchase.

(c) Ethical and Governance

When a company is in financial distress it is particularly important that the company maintains the highest standards of corporate governance and ethical responsibility. As non-executive directors, it is clear that in commissioning this report you are exercising your responsibilities effectively. It is also clear that the executive directors are being both open and conscientious in supplying you with relevant financial information and details of the plans that they have in order to help the business recover from this downturn in its business activities. At this point in the company's financial fortunes it is worth re-emphasising the importance of sound internal control and of maintaining good relations with investors. The company has historically relied upon the diversification of its aerospace business between military and civil markets. This has been a source of the company's strength and it is particularly unfortunate that the substantial delays in the European Aircoach business have led to a collapse in the company's performance. The capital markets have appreciated the problem, as we would expect, but at this sensitive time it is important that the company makes careful profit forecasts in line with what it can reasonably expect and does not try to inject a sense of false optimism into its future prospects.

At this stage, and given the company's substantial cash holdings, it is to be expected that other companies may see Venus Systems Limited as an attractive target for a takeover bid. It is important from a shareholder perspective that all such bids are reviewed carefully and that any decisions you make are made as a whole board without any sectional interests. Although there does not appear to be a potential acquirer at the moment the company should be on its guard for any untoward stake building and regular monitoring of changes in the company's share register should be undertaken.

Ethically, the company has performed as would be expected of a high quality, long established business of this type. There is a body of opinion that any company that engages in the military market cannot be classified as 'ethical' for investment purposes.

However, leaving that on one side the company has attempted to minimise the impact of the downturn in its business upon its key stakeholder groups and in particular through redeployment and reskilling of staff, the maintenance of good payment policies with respect to its suppliers, and the transparency with which it has conducted its business.

Recommendations:

The directors' proposed recovery plan is based upon the assumption of a rapid return to growth of the civil aviation business. There is no doubt that the company has suffered significantly from delays in production of the European Aircoach. Contingency plans should be put in place for further delays in the start of that project. The commitment of management to the reduction of the company's working capital to its 2006 levels is commendable and will be the source of short-term cash flow required to offset any further operating losses caused by further delay in the Aircoach business. In our view, the capital reinvestment programme of $35 million should proceed but further consideration of the advisability of the share buyback should be given based upon an assessment of the impact upon the firm's cost of capital and market value.

3 Asteroid Systems

Currency hedging.

The objective is to fix (Euro/SFr) currency exchange rates for two months for an expected remittance of SFr 2·4299 million. This is achieved with a money market hedge for the two month exposure.

(a) Hedging the two month Swiss franc exposure

Forward contracts in the money market are the most straightforward way of eliminating transaction risk. The exposure to movements in the Swiss Franc can be eliminated by entering into a forward contract to purchase the currency at Euro SFr 1·6199 (see note 1). Alternatively, given access to fixed rate finance in the Swiss market a reverse money market hedge can be established by borrowing in SFr and depositing in Euros. Given that the interest rates can be locked in this would offer a better forward rate at two months at any borrowing rate less than SFr LIBOR + 7 (note 2).

Supporting notes

Note (1)

The current expectation of remittances is based upon an estimate of the two month forward rate of $(1{\cdot}6223 + 1{\cdot}6176)/2 = 1{\cdot}6199$ for the Swiss Franc

Swiss Francs = $1{\cdot}6199 \times 1{\cdot}5$m = SFr 2·4299 million in two months using the forward rate above.

Note (2)

The money market hedge is based upon interest rate parity and using the IRP formula we can calculate the maximum rate of interest that can be borne for the money market hedge to be worthwhile.

The technique for a reverse money market hedge is identical to that for a conventional hedge except that the counter currency is borrowed in the foreign market, converted at spot and deposited in the domestic market. However, for a money market hedge to work, the company must be able to secure short term money market finance in both the base and the counter currency area. With the SFr exposure it is possible to borrow at fixed in the Swiss market. Using the no arbitrage condition of the interest rate parity formula we can determine the maximum rate of interest the company should agree in creating a money market hedge. The interest rate in the base currency to use is the best rate for depositing in the Euro market:

$$1{\cdot}6199 = 1{\cdot}6244 \times \frac{\left(1 + i_c \times \frac{2}{12}\right)}{\left(1 + 0{\cdot}03725 \times \frac{2}{12}\right)}$$

$i_c = 2{\cdot}13\%$

If the company can borrow at less than spot Swiss 2·13% in the Swiss market then the money market hedge will be preferred to a forward sale of SFr 1·5 million.

(b) The relative advantages and disadvantages of the use of a money market hedge versus exchange traded derivatives

A money market hedge is a mechanism for the delivery of foreign currency, at a future date, at a specified rate without recourse to the forward FOREX market. If a company is able to achieve preferential access to the short term money markets in the base and counter currency zones then it can be a cost effective substitute for a forward agreement. However, it is difficult to reverse quickly and is cumbersome to establish as it requires borrowing/lending agreements to be established denominated in the two currencies.

Exchange traded derivatives such as futures and foreign exchange options offer a rapid way of creating a hedge and are easily closed out. For example, currency futures are normally closed out and the profit/loss on the derivative position used to offset the gain or loss in the underlying. The fixed contract sizes for exchange traded products mean that it is often impossible to achieve a perfect hedge and some gain or loss on the unhedged element of the underlying or the derivative will be carried. Also, given that exchange traded derivatives are priced in a separate market to the underlying there may be discrepancies in the movements of each and the observed delta may not equal one. This basis risk is minimised by choosing short maturity derivatives but cannot be completely eliminated unless maturity coincides exactly with the end of the exposure. Furthermore less than perfectly hedged positions require disclosure under IFRS 39. Although rapid to establish, currency hedging using the derivatives market may also involve significant cash flows in meeting and maintaining the margin requirements of the exchange. Unlike futures, currency options will entail the payment of a premium which may be an expensive way of eliminating the risk of an adverse currency movement.

With relatively small amounts, the OTC market represents the most convenient means of locking in exchange rates. Where cross border flows are common and business is well diversified across different currency areas then currency hedging is of questionable benefit. Where, as in this case, relatively infrequent flows occur then the simplest solution is to engage in the forward market for hedging risk. The use of a money market hedge as described may generate a more favourable forward rate than direct recourse to the forex market. However the administrative and management costs in setting up the necessary loans and deposits are a significant consideration.

(c) The only risk which will impact on a firm's cost of capital is that risk which is priced in either the equity or the debt markets. Considering these two markets in turn:

Currency risk forms part of a firm's exposure to market risk and will impact on a firm's cost of capital through its beta value and, as a result, its equity cost of capital. The extent to which this is significant depends on the exposed volume of currency transactions conducted in a given period, the average duration of the exposure and the correlation of the currency with the market. If a given currency has the same correlation with the market as the company, removing currency risk will have no impact on the firm's overall exposure to market risk and as a result no impact on the firm's cost of capital. The greater the difference in the relative correlations the higher the potential improvement in shareholder value from hedging.

The impact on the cost of debt is more complex. The most significant impact is through the firm's exposure to default risk. If currency transactions are significant and the foreign currency is highly correlated with the domestic currency then the impact is unlikely to be significant and the gains from hedging modest. Where the degree of correlation is low or indeed negative then eliminating currency risk may significantly alter the firm's default risk. However, unlike market risk default risk is related to the overall volatility of a firm's underlying value and its ability to finance its debt. The elimination of currency risk is therefore likely to have at least some impact on the volatility of the firm's cash flows and therefore its cost of capital.

4 Saturn Systems

Note: regulatory regimes with respect to takeovers vary between different jurisdictions and in answering this question the examiners will wish to see evidence that candidates are either (a) aware of the regulations or codes within their own countries or (b) are familiar with the UK's City Code for Takeovers and Mergers.

Memorandum
To: John Moon
From: Conny Date

Potential Bid for Pluto Ltd

The remarks made at the dinner last night whilst general in nature and non-specific about this firm's intentions could be construed as an intention to bid for Pluto Ltd. Below are the principal regulatory, financial and ethical issues we currently face:

Regulatory issues

Most regulatory regimes around the world impose strict rules concerning the release of price sensitive information such as an intention to bid. Furthermore, as directors of a publicly quoted company we are obliged to take all reasonable care that any information we put into the public domain does not mislead investors. Within the UK for example, the City Code stresses the vital importance of absolute secrecy before any announcement is made. The Code places the burden of secrecy upon anybody in possession of confidential information, particularly where the information is price sensitive, and it stresses that they should conduct themselves so as to minimise the risk of any accidental disclosure.

When a target company is a particular subject of rumour, speculation or untoward movement in its share price, and where it is reasonable to assume that the source was the offeror, then the Code stipulates that an announcement of intention should be made.

An announcement that we do not intend to bid, or are withdrawing a bid, normally means that we are restricted from making another bid within a specified time period (normally six months) unless:

– an offer to be made by us is recommended by the board of Pluto to its shareholders,
– another offer is made by a third party,
– a whitewash proposal is made by the board of Pluto or,
– there is a significant change of circumstances such as the regulator is disposed to waive the requirement.

The underpinning requirement here is that in our public announcements we have a duty to be as clear as possible, not to be seen as creating a false market in the shares of Pluto Ltd and providing all shareholders (both our own and those of Pluto) with equal access to information about our intentions.

Financial issues

The comments made by you might not have been interpreted as significant by the market under different circumstances. However, the 15% price movement strongly suggests that the market now views Pluto as a target and Saturn as one of a number of potential predators. Evidence accumulated on the reaction of the market to information entering the public domain – the so called reaction studies of the semi-strong form of the Efficient Markets Hypothesis – confirms that the market has a powerful ability to anticipate announcements and it is no surprise that the market recognises the potential of the situation with respect to Pluto. The sudden price increase strongly suggests that we are now seen as potential bidders. The problems that we face are three fold: first, we are not yet in a position to announce a formal bid in that we have undertaken neither a due diligence study of Pluto nor have we undertaken a valuation of the company, second the price range over which we can negotiate has now been raised and third, if we withdraw we cannot make a further bid within six months.

The valuation of Pluto will not be straightforward. Our preliminary view is that although the company is part of our supply chain it may not carry our exposure to business risk. We have entered into preliminary discussions with our investment bankers about raising the necessary debt finance. The size of the acquisition means that it is most likely that we will alter both our exposure to business and financial risk. This means that the value of Pluto to us cannot be determined independently of a revaluation of our existing business on the presumption that the acquisition proceeds. The increase in the valuation of Saturn would determine the maximum that we should be willing to pay without impacting adversely upon our own shareholder value.

Ethical issues

One possibility is to deny that we are considering a bid for Pluto. The distinction we need to note here is whether our current investigations could be classed as 'strategic scanning' or we are actively considering this company as a bid target. The major difficulty we have is that Pluto is one of four companies discussed at the May board meeting as a potential target and although that discussion was very speculative investors will be aware that we have consistently sought growth through acquisition rather than organically. Given these circumstances and the commitment of this company to adopt the highest standards of ethical conduct with respect to transparency and the treatment of investors I recommend that we clarify our position.

Recommendation

I therefore recommend that following consultation with the regulator we make an announcement as follows:

Following recent speculation, the Board of Saturn Ltd confirms that it is considering a possible combination with Pluto Ltd. However, the Board has decided not to make an offer at this time. The Board reserves its right to make an offer or take any other action which might otherwise be restricted under the six month rule in the event of (a) an agreement from the Board of Pluto, (b) an announcement by any other party of a possible or actual offer, (c) a whitewash proposal from the board of Pluto Ltd or (d) significant change in circumstances.

Given this announcement we will still have the opportunity to develop a bid proposal and to enter into substantive negotiations with the board of Pluto. However, we will not be able to make a hostile bid in the absence of another company making a bid for Pluto.

5 Neptune

(a) The adjusted present value technique separates the value created by the Galileo project into two components: (a) the value of the project cash flows at the firm's pure rate of equity (i.e., the unlevered WACC), and (b) the gain or loss of value associated with the costs and benefits of the new finance.

The adjusted present value method is only appropriate where the project does not affect the firm's exposure to business risk. With the Galileo project we can estimate the alteration to the firm's cash flows as a whole if the project were to proceed. This adjustment entails a substantial tax benefit because of writing down allowances but also a delayed tax charge attributable to the increase in the firm's taxable earnings generated by the project.

The project cash flows exclude all non-relevant costs but include the opportunity costs associated with the redeployment of labour. The projected relevant cash flows and the calculation on the eventual profit on the sale of the capital equipment are as follows:

	0	1	2	3	4	5
Sales revenue		680·00	900·00	900·00	750·00	320·00
Direct costs		–408·00	–540·00	–540·00	–450·00	–192·00
Redeployment of labour		–150·00	–150·00			
operating cash flow		122·00	210·00	360·00	300·00	128·00
capital payments and receipts	–800·00					40·00
less written down value						31·00
profit on sale of equipment						8·90

On the basis of this the net of tax operating cash flows and the tax savings attributable to the write-off for tax purposes of the capital equipment are:

	0	1	2	3	4	5	6
operating cash flow		122·00	210·00	360·00	300·00	128·00	
tax on operating cash flows			–36·00	–63·00	–108·00	–90·00	–38·40
		122·00	173·40	297·00	192·00	38·00	–38·40
capital cash flow	–800·00					40·00	
capital allowance saving		120·00	48·00	28·80	17·28	10·37	6·22
sale of capital equipment							–2·67
nominal project cash flow	–800·00	242·00	221·40	325·80	209·28	88·37	–34·85

The valuation of this future cash flow involves the firm's cost of capital on the basis that it is ungeared (the pure equity rate). The calculation of the ungeared rate is shown in note 1.

Using the ungeared rate of 8·97% to discount the project cash flows and LIBOR plus the firm's credit spread (7·2%) to discount the tax benefit associated with the project an adjusted present value is estimated at $100·69 million as follows:

	0	1	2	3	4	5	6
nominal project cash flow	–800·00	242·00	221·40	325·80	209·28	88·37	–34·85
discount at pure equity rate	–800·00	222·07	186·44	251·76	148·41	57·50	–20·81
Net present value of post tax operating flow	45·38						
new capital introduced	–816·32						
tax saving on annual interest		17·63	17·63	17·63	17·63	17·63	
discounted value of tax shield on interest	71·63	15·87	15·06	14·29	13·56	12·86	
Cost of financing	–16·32						
Adjusted present value	100·69						

Note: it is assumed that there is no tax relief on debt.

This suggests that the new project will add substantial value to the firm although $71·63 million is attributable to the tax benefit associated with the new financing less 2% transaction cost associated with the new debt finance. With this project the clear advice on the basis of the financial analysis is to proceed although it is worth bearing in mind that more marginal projects may be solely justified through the financing effect.

(b) The modified internal rate of return (MIRR) is calculated using the rate of return required to discount those cash flows as follows:

One procedure for calculating the MIRR is to calculate the terminal value of the cash flows from the recovery phase of the project using the company's cost of capital of 8·97%.

	0	1	2	3	4	5	6
nominal project cash flow	–800·00	242·00	221·40	325·80	209·28	88·37	–34·85
compound factor using 8·97%		1·5365	1·4100	1·2940	1·1874	1·0897	1·0000
terminal cash flow		371·83	312·18	421·57	248·51	96·29	–34·85
future value of recovery cash flows	1,415·54						

The modified internal rate of return is found by calculating the internal rate of return of the modified project cash flow:

$$-800 + \frac{1{,}415{\cdot}54}{(1+MIRR)^6} = 0$$

Therefore:

$$MIRR = \sqrt[6]{\frac{1{,}415{\cdot}54}{800}} - 1 = 9{\cdot}98\%$$

An alternative approach is as follows:

$$MIRR = [1 + PI]^{\frac{1}{n}} \times (1 + rate) - 1$$

where PI is the profitability index or the net present value of the project's operating cash flows divided by the capital invested.

In this case, because we are concerned solely with the project's nominal cash flow excluding financing costs we use the ungeared cost of equity as the appropriate discount and reinvestment rate.

$$MIRR = \left[1 + \frac{45{\cdot}38}{800}\right]^{\frac{1}{6}} \times \left(1{\cdot}0897\right) - 1 = 9{\cdot}98\%$$

It is to be expected that in a highly competitive business new business opportunities with significant net present values are hard to find. This project, if successful, will add to the value of the firm and offers a rate of return just 1% higher than the current reinvestment rate.

Note 1

Below we show the required elements of the estimate of the pure equity rate deducted from the firm's current, tax adjusted market gearing.

The current market gearing w_d is:

$$w_d = \frac{2,500}{7,500 + 2,500} = 0{\cdot}25$$

The tax adjusted market gearing using a company tax rate of 30% is:

$$w_d^{'} = \frac{2,500 \times 0{\cdot}7}{7,500 + 2,500 \times 0{\cdot}7} = 0{\cdot}1892$$

Given a beta of 1·4 this implies an ungeared (or asset) beta of:

$\beta_A = \beta_E(1 - w_d') = 1{\cdot}4 \times (1 - 0{\cdot}1892) = 1{\cdot}1351$

The cost of equity capital (ungeared) is found as follows:

$r_e = 0{\cdot}05 + 1{\cdot}1351 \times 0{\cdot}035 = 0{\cdot}0897$ (8·97%)

Tutorial note: *These model answers are considerably longer and more detailed than would be expected from any candidate in the examination. They should be used as a guide to the form, style and technical standard (but not in length) of answer that candidates should aim to achieve. However, these answers may not include all valid points mentioned by a candidate – credit will be given to candidates mentioning such points.*

Professional marks are awarded for the quality of the layout, clarity and persuasiveness of the presentation and integration of analytical data with the written text.

			Marks
1	(a)	Calculate the tax adjusted gearing for Jupiter and the FS sector	2
		Calculate the proxy asset beta for Mercury and its estimated equity beta	4
		Estimate Mercury's equity cost of capital and WACC	2
		Note of circumstances under which each is used	2
			Total 10
	(b)	Lowest rate set at liquidated value of the business	2
		Calculation of reinvestment rate and growth	3
		Estimation of upper boundary price per share	3
		Identification of control premium	2
			Total 10
	(c)	Advantages and disadvantages of listing (regulatory, takeover, process)	4
		Nature of PE finance and its advantages and disadvantages	4
			Total 8

2 Note candidates may answer this question in a variety of ways and in different order to that set.

Up to two marks are awarded for the professional quality of the report.

			Marks
	(a)	Identification of the principal causes of failure (revenue, cost, asset, liability or capital management (one mark for each))	5
			Total 5
	(b)	Assessment of the performance of the busines and its prospects	2
		Assessment of the efficiency of the business	2
		Assessment of the general risk exposure	2
		Assessment of the risk of default	4
		Assessment of the liquidity of the business	2
		Calculation of operating cash cycle	3
		Conclusion with respect to proposed plan of action	2
			Total 17
	(c)	Role of the board and duty to stakeholders	2
		Role of good internal control and risk management	2
		Response of the board to potential acquisitions	2
			Total 6
	(d)	Professional marks	4
			Total 4

			Marks
3	(a)	Calculation of forward rates	4
		Calculation of minimum rate at LIBOR + 7 through reverse money market hedge	5
		Conclusion	1
			Total 10
	(b)	Advantages and disadvantages of OTC versus ET derivatives	
		Basis risk	2
		Under/over hedging	1
		Counterparty risk	1
		Flexibility	1
		Margin	1
			Total 6
	(c)	Impact upon the cost of equity capital	2
		Impact upon cost of debt capital	2
			Total 4
4	(a)	Identification of the problems created by Mr Moon's remarks	3
		Significance of the price reaction	3
		The firm's current position and identification of the holding option on the PR	4
		Ethical problems of insider information, fairness to stakeholders, avoidance of dissembling	5
			Total 15
	(b)	Issue of immediate press release	2
		Draft of statement reserving the company's position under six month rule	3
			Total 5
5	(a)	Note of the two approaches to Adjusted Present Value	2
		Projection of relevant cash flows	4
		Identification and calculation of appropriate discount rates	4
		Calculation of value of project and tax shield using APV	4
			Total 14
	(b)	Calculation of MIRR using 8·97% as reinvestment rate	6
			Total 6

Answers

Tutorial note: *These model answers are considerably longer and more detailed than would be expected from any candidate in the examination. They should be used as a guide to the form, style and technical standard (but not in length) of answer that candidates should aim to achieve. However, these answers may not include all valid points mentioned by a candidate – credit will be given to candidates mentioning such points.*

1 Blipton International Entertainment Group

Management Report: Blipton International Entertainment Group
400 bed Olympic Hotel, London
Completion: 31 December 2009

(a) Projection of $value cash flows for both the project investment and the project return.

In projecting the cash flow for this project we have created a forecast of the capital requirement, the six year operating cash flow and the residual value of the property net of repairs and renewals at the end of the project. On the basis of the specified occupancy rates and a target nightly rental of £60 we have projected the revenues for the hotel and the expected costs. These are projected at current prices to give a real cash flow before conversion to nominal at the UK rate of inflation. Tax is calculated both in terms of the offset available against the construction costs but also at 30% of the operating surplus from the project. We assume that the benefit of the capital allowances will be recovered irrespective of the success of the operating phase of the project. They are therefore considered a credit to the investment phase (candidates who assume that they are part of the recovery phase will not be penalised).

Finally, using purchasing power parity, future spot rates are estimated. The rate specified is indirect with respect to the dollar and declines as sterling strengthens.

We have separated the calculation of the present value of the investment phase from that of the return phase as follows:

Investment phase (values in £)	**01 Jan 2009**	**31 Dec 2009**	**31 Dec 2010**	**31 Dec 2011**	**31 Dec 2012**	**31 Dec 2013**	**31 Dec 2014**
Nominal project cash flow		–6,200,000					
Capital allowance (tax saving)		930,000	310,000	310,000	310,000		
Nominal project cash flow after tax (investment phase)		–5,270,000	310,000	310,000	310,000		
Rate of exchange	0·6700	0·6552	0·6409	0·6268	0·6130	0·5996	0·5864
$value of investment phase		–8,043,346	483,695	494,576	505,710	0	0
Return phase (value in £)							
Occupancy rate		0	0·4	0·5	0·9	0·6	0·6
Terminal value of property							8,915,309
Rooms let (400 × occ. Rate × 365)			58,400	73,000	131,400	87,600	87,600
Revenue (rooms let × £60)			3,504,000	4,380,000	7,884,000	5,256,000	5,256,000
Variable operating costs (rooms let × £30)			–1,752,000	–2,190,000	–3,942,000	–2,628,000	–2,628,000
Fixed costs			–1,700,000	–1,700,000	–1,700,000	–1,700,000	–1,700,000
Project operating cash flow (real)			52,000	490,000	2,242,000	928,000	928,000
Project operating cash flow (nominal)			54,633	527,676	2,474,749	1,049,947	1,076,195
Tax on operating cash flows (at 30%)			–16,390	–158,303	–742,425	–314,984	–322,859
Nominal project cash flow after tax (return phase)			38,243	369,373	1,732,324	734,963	9,668,646
Rate of exchange	0·6700	0·6552	0·6409	0·6268	0·6130	0·5996	0·5864
$value of return phase			59,670	589,300	2,825,977	1,225,755	16,488,141

(b) Project evaluation

Net present value
Given that the Dubai rate of inflation is 4·8% per annum and the company's real cost of capital is 4·2% per annum the nominal cost of capital is estimated using the Fisher formula:

$$i_{nom} = (1 + \text{inf})(1 + i_{real}) - 1$$
$$i_{nom} = (1{\cdot}048)(1{\cdot}042) - 1 = 9{\cdot}2016\%$$

Discounting the project cash flows (investment plus return) at this nominal cost of capital gives a project net present value as follows:

	01 Jan 2009	31 Dec 2009	31 Dec 2010	31 Dec 2011	31 Dec 2012	31 Dec 2013	31 Dec 2014
Nominal project cash flow (investment plus return)		–8,043,346	543,365	1,083,876	3,331,687	1,225,755	16,488,141
Nominal cost of capital (Dubai)	0·092016						
Discounted cash flow	0	–7,365,593	455,653	832,324	2,342,870	789,330	9,722,942
Net present value	6,777,525						

A net present value of $6,777,525 strongly suggests that this project is viable and will add to shareholder value.

Modified internal rate of return

The modified internal rate of return can be estimated by calculating the internal rate of return of the sum of the return cash flows compounded at the cost of capital to give a year six terminal value. The discount rate which equates the present value of this terminal value of return cash flows with the present value of the investment cash flows is the modified internal rate of return.

$$MIRR = \left[\frac{PV_R}{PV_I}\right]^{\frac{1}{n}}(1 + r_e) - 1$$

Where PV_R is the present value of the return phase of the project, PV_I is the present value of the investment phase and r_e is the firm's cost of capital.

	01 Jan 2009	31 Dec 2009	31 Dec 2010	31 Dec 2011	31 Dec 2012	31 Dec 2013	31 Dec 2014
Modified internal rate of return							
Present value of return phase	13,002,093		50,038	452,532	1,987,251	789,330	9,722,942
Present value of investment phase	–6,224,568	–7,365,593	405,614	379,792	355,619		
Present value per $ investment	2·0888						
Sixth root of present value of PV_R/PV_I	1·1306						
MIRR	23·47%						

The calculation of the MIRR is as follows:

$$MIRR = \left[\frac{13{,}002{,}093}{6{,}224{,}568}\right]^{\frac{1}{6}}(1{\cdot}092016) - 1 = 23{\cdot}47\%$$

Alternatively the modified internal rate of return can be found by compounding forward the return phase cash flows at the firm's cost of capital and then calculating the internal rate of return using the terminal value of the return phase and the present value of the investment phase as follows:

	01 Jan 2009	31 Dec 2009	31 Dec 2010	31 Dec 2011	31 Dec 2012	31 Dec 2013	31 Dec 2014
Modified internal rate of return (Method 2)							
year 6 cash flow							16,488,141
year 5 cash flow							1,338,544
year 4 cash flow							3,369,975
year 3 cash flow							767,403
year 2 cash flow							84,854
Future value of return phase							22,048,918
Present value of investment phase	–6,224,568						

The modified internal rate of return is the discount rate which solves the following equation:

$$6{,}224{,}568 = \frac{22{,}048{,}918}{(1 + MIRR)^6}$$

$$MIRR = \sqrt[6]{\frac{22{,}048{,}918}{6{,}224{,}568}} - 1 = 23{\cdot}47\%$$

(c) Recommendation and discussion of method

I have examined the project plan for the proposed project and referring to the appendices (see above) report that this project is expected to deliver an increase in shareholder value of $6·78 million, at the firm's current cost of finance. I have estimated the increase in shareholder value using the net present value (NPV) method. Net present value focuses on the current equivalent monetary value associated with capital expenditure leading to future cash flows arising from investment. The conversion to present value is achieved by discounting the future cash flows at the firm's cost of capital – a rate designed to reflect the scarcity of capital finance, inflation and risk.

Although the net present value technique is subject to a number of assumptions about the perfection and efficiency of the capital market it does generate an absolute measure of increase in shareholder value and as such avoids scale and other effects associated with percentage performance measures. Given the magnitude of the net present value of the project it is safe to assume that it is value-adding assuming that the underlying cash projections can be relied upon.

However, in certain circumstances it can be useful to have a 'headroom' percentage which reliably measures the rate of return on an investment such as this. In this case the modified internal rate of return of 23·47% is 14·26% greater than the firm's cost of capital. MIRR measures the economic yield of the investment (i.e. the discount rate which delivers a zero net present value) under the assumption that any cash surpluses are reinvested at the firm's current cost of capital. The standard IRR assumes that reinvestment will occur at the IRR which may not, in practice, be achievable. MIRR does not suffer from the multiple root problem when calculating IRR on complex cash flows.

Although MIRR, like IRR, cannot replace net present value as the principle evaluation technique it does give a measure of the maximum cost of finance that the firm could sustain and allow the project to remain worthwhile. For this reason it gives a useful insight into the margin of error, or room for negotiation, when considering the financing of particular investment projects.

2 Jupiter Co

To: Rosa Nelson
From: An accountant
Briefing Note:

The impact of the proposed financing package requires an estimation of the firm's cost of capital before and after the event and a calculation of the likely impact of the refinancing scheme upon the value of the firm:

(a) The current cost of debt, equity and the weighted average cost of capital

The current debt has a cost of finance of 4·65% (4·2% + 45bp) for a yield to maturity of four years in the Euro market.

The current cost of equity is as follows:

$$r_e = r_f + \beta_i(r_m - r_f)$$
$$r_e = 4\% + 1{\cdot}5 \times 3\% = 8{\cdot}5\%$$

The weighted average cost of capital relies upon a valuation of the current debt. This is achieved by discounting the current average coupon rate applied to a nominal $100 of borrowing at the current cost of debt finance:

$$MV_d = \frac{5{\cdot}6}{(1{\cdot}0465)} + \frac{5{\cdot}6}{(1{\cdot}0465)^2} + \frac{5{\cdot}6}{(1{\cdot}0465)^3} + \frac{105{\cdot}6}{(1{\cdot}0465)^4} = £103{\cdot}4\ percent$$

This gives a total market value of debt of 103·4% × $800 million = $827·17 million

Given the current share price, the market value of equity is ($13·80 × 500 million shares in issue) $6,900 million and the market gearing ratio w_d is therefore 10·70%.

From this the weighted average cost of capital is as follows:

$$WACC = (1 - w_d)r_e + w_d r_d(1 - T)$$
$$WACC = (0{\cdot}8930 \times 8{\cdot}5\%) + (0{\cdot}1070 \times 4{\cdot}65\% \times 0{\cdot}75) = 7{\cdot}96\%$$

(b) The revised cost of debt, equity and weighted average cost of capital

The revised cost of debt is calculated as a weighted average of the 10 year risk free rate plus the credit premium in both the Euro and the Yen markets:

Cost of debt = (0·5 × 2·30%) + (0·5 × 5·45%) = 3·875%

The market value of the debt (on the assumption that the fixed rate on the bond is set at the current weighted average cost of debt) will be $2,400 million.

The increased gearing will impact upon the cost of equity for the company. The procedure for calculating the revised cost of equity is to ungear the current beta and revise it to the new gearing level using the market value of debt following the new issue. In practice this would entail an iterative calculation as the revised market gearing ratio will be dependent upon the value of the equity after the issue of the new debt.

Using the formula for the asset beta assuming that the debt beta is zero:

$$\beta_a = \left[\frac{V_e}{\left(V_e + V_d\left(1-T\right)\right)}\beta_e\right] = \left[\frac{6,900}{6,900 + 827{\cdot}17 \times 0{\cdot}75} \times 1{\cdot}5\right] = 1{\cdot}3763$$

Assuming that the value of equity does not change the revised equity beta will be:

$$\beta_e = \left[\frac{\beta_a}{\left(\frac{V_e}{\left(V_e + V_d\left(1-T\right)\right)}\right)}\right] = \left[\frac{1{\cdot}3763}{\left(\frac{6,900}{6,900 + 2,400 \times 0{\cdot}75}\right)}\right] = 1{\cdot}735$$

The cost of equity capital to the new firm is:

$r_e = r_f + \beta_i(r_m - r_f)$
$r_e = 4\% + 1{\cdot}7353 \times 3\% = 9{\cdot}21\%$

Assuming no alteration in the market value of the firm's equity the revised gearing and WACC after the issue of the new debt will be:

$$w_d = \frac{2,400}{2,400 + 6,900} = 0{\cdot}258$$

$WACC = (1 - w_d)r_e + w_d \times r_d(1 - T)$
$WACC = (0{\cdot}742 \times 9{\cdot}21\%) + (0{\cdot}258 \times 3{\cdot}875\% \times 0{\cdot}75) = 7{\cdot}58\%$

(c) Estimation of the minimum rate of return on the additional debt financing required to maintain shareholder value

The free cash flow to equity model appears to satisfactorily predict the value of the firm. The model gives a share price as follows:

$$V_0 = \frac{FCFE_0(1 + br_e)}{r_e - br_e}$$

$$V_0(\text{million}) = \frac{400 \times (1 + 0{\cdot}3 \times 0{\cdot}085)}{0{\cdot}085 - 0{\cdot}3 \times 0{\cdot}085} = \$6,894 \text{ million}$$

Or \$13·79 per share

Given the refinancing of the business and the revised cost of equity capital we can rearrange the valuation formula to find the free cash flow required to maintain shareholder value as follows:

$$FCFE_0 = \frac{V_0 \times \left(r_e - br_e\right)}{(1 + br_e)}$$

Using 9·21% as the revised cost of equity following refinancing the free cash flow to equity required to maintain shareholder value is as follows:

$$FCFE(million) = \$6,894 \times \frac{(0{\cdot}0921 - 0{\cdot}3 \times 0{\cdot}0921)}{(1 + 0{\cdot}3 \times 0{\cdot}0921)} = \$432{\cdot}3 \text{ million}$$

or an increase of \$32·3 million.

The addition to the operating cash flow that this implies is calculated by estimating the free cash flow before tax and adding back the interest charge on the new debt:

$$\Delta OCF(\$million) = \frac{32{\cdot}3}{0{\cdot}75} + \left[(2,400 \times 0{\cdot}03875) - (800 \times 0{\cdot}056)\right] = \$91{\cdot}30 \text{ million}$$

This means that the company needs to generate \$91·30 million on the new debt investment of \$2,400 million or 3·80%.

(d) Comparison of the proposed method of raising finance for investment compared with the alternatives

The proposed method of financing through a bond issue is an attractive means of raising large scale debt. There are significant issue costs and there is the risk that the issue will not be fully subscribed – although much of that risk can be mitigated through an underwriting agreement. A more popular means of financing an issue of this size is through a syndicated loan.

Global lending through this means is now believed to exceed US$3 trillion with an average loan size in excess of US$400 million. Syndication entails the creation of a banking syndicate led by an 'arranging' bank whose function it is to bring together other banks who are willing to participate in the loan. The arranging bank may also act in an on-going agency relationship with the client company once the deal has been established. The advantages of syndication are:

(i) Loans can be arranged that are considerably greater than could be managed by any single bank without unbalancing its lending portfolio or indeed breaching its capital requirements under the Basle agreements.

(ii) Banks in different currency jurisdictions combine to create mixed lending packages to suit the needs of companies requiring finance for investment in different countries.

(iii) Speed of creation and relatively low transaction costs.

Excluding the FOREX risk arising with any overseas borrowing, the disadvantages of syndication from the borrower's point of view are relatively small: there is a risk of bank default and the rates offered may be somewhat above the spreads available in the bond market.

3 Aston Co

(a) Default is defined as that point at which the firm is unable to discharge its interest and/or principal payments when they fall due. From the monthly average cash flow we deduct the monthly interest payment. The monthly payment is based upon the effective monthly rate as follows:

$$i_m = \sqrt[1/12]{(1+i_a)} - 1 = \sqrt[1/12]{1{\cdot}08} - 1 = 0{\cdot}6434\%$$

The expected monthly cash flow will then be:

c_m = \$14,400 – 0·006434 × \$1,500,000 = \$4,748·4.

To give an annual expected cash flow after interest of $56,987·4.

We now need to determine the probability over the course of 12 months that this figure will fall below zero. For this we need to calculate the annual volatility of cash flows. Given that the monthly volatility before interest is 13%, we must translate this to volatility after fixed interest in two stages. First, calculate the proportion of fixed interest to monthly cash flow (G):

$$G = \frac{(0{\cdot}006434 \times \$1{,}500{,}000)}{14{,}400} = 0{\cdot}6702$$

And second, calculate the monthly volatility after interest (σ_m) as follows:

$$\sigma_m = \frac{\sigma'_m}{1-G} = \frac{13\%}{1-0{\cdot}6702} = 39{\cdot}42\%$$

The annualised volatility is as follows:

Annualised volatility = $0{\cdot}3942 \times \sqrt{12}$ = 136·55%

Standard deviation of annual cash flows = 136·55% × \$56,987 = \$77,818

Given a critical cash value of zero (i.e. if the cash flow less interest falls below zero) then the expected cash difference to default including the \$8,500 of cash in hand is \$65,487. This figure represents 0·842 standard deviations.

Using the standard normal tables this shows a value of 0·3 or a cumulative probability that the cash flow plus reserve will be above the default probability of 0·8. This tells us that there is a 20% chance of failure within 12 months.

Alternatively this can be expressed as a 'distance to default' of 12 months × 0·8 = 9·6 months.

(b) Making a loan of this type resolves down to an estimate of the probability of default, the potential default loss, the rate the lender needs to recover to cover the cost of finance in the inter-bank market plus an additional charge for bearing the risk attaching to the loan.

The probability of default on interest payments is determined by the expected annual cash flow after interest payments and the volatility of that residual cash flow. If, on an annual basis, that cash flow falls below zero then default is deemed to occur.

The next most important issue for a lender is the potential recoverability of its borrowing. This will be governed by a number of factors: the percentage of the loan covered by the firm's net assets excluding the loan, the liquidity of those assets under forced sale, the market demand for these assets (and hence their price) and any director's or other guarantees that may be in place. If the net assets exceed the value of the outstanding loan then default will not occur as it would be possible for the firm to liquidate some of the surplus asset value to service the debt.

In this case the probability of recovery is 90% which implies a loss of 10% of the loan in the event of default. When estimating its potential loss the bank will calculate the present value of the outcomes over a year of each £1 invested discounting at the risk free rate plus the additional rate of return it requires as compensation for bearing the risk of default. Using the following decision tree:

$$1 \begin{cases} \xrightarrow{1 - p_r} & 1 \times \dfrac{(1 + r_d)}{(1 + r_f + r_c)} \\[2ex] \xrightarrow{p_r} & 1 \times \text{Recovery} \times \dfrac{(1 + r_d)}{(1 + r_f + r_c)} \end{cases}$$

Where: p_r is the probability of default, r_d is the rate of return required to compensate for the loss on default, r_f is the risk free rate and r_c is the additional return required by the bank for bearing the risk attaching to this loan.

The logic of this calculation is that the bank has put at risk, over 12 months, the value of its original loan (£1) and the rate of interest it decides to charge (r_d). If the bank was indifferent to the outcomes, i.e. it is risk neutral with respect to those outcomes, then it would discount each of the possibilities at the risk free rate (r_f). However, it is not risk neutral and r_c is the premium it requires to offset its risk aversion.

Putting the information given in the question into an equation and solving:

$$1 = (1 - p_r) \times \frac{(1 + r_d)}{(1 + r_f + r_c)} + p_r \times \text{Recovery} \times \frac{(1 + r_d)}{(1 + r_f + r_c)}$$

$$r_d = \frac{1 + r_f + 0{\cdot}034}{(1 - p_r) + p_r \times \text{Recovery}} - 1 = \frac{1{\cdot}0584}{(1 - 0{\cdot}2) + 0{\cdot}2 \times 0{\cdot}9} - 1 = 8{\cdot}00\%$$

To summarise, the bank requires 5·5% to cover its own cost of finance on the money market, 2·16% to cover the expected loss on the loan given the probability of default and the potential for recovery and 0·34% as compensation for carrying the risk attaching to the loan.

In practice the estimation of default probability and recoverability will depend upon a number of judgements about the credit-worthiness of the business, an assessment of its business plan and the willingness of the lender to finance a high risk business of this type.

4 Solar Supermarkets

The directors of Solar Supermarkets face a number of issues which may reflect a lack of understanding of the consequences:

(i) The company's more hostile competitive environment with pressure on prices and costs;

(ii) The need to offer management compensation that will provide an incentive to seek value adding business opportunities;

(iii) The investor pressure to release the value in property assets with the implied concern that if they do not acquiesce a private equity team may well do the job for them; and

(iv) The willingness of the firm to support the potential liabilities of a final salary pension scheme for its employees.

In reviewing these issues it is important to bear in mind the overarching duty of the directors which is to maximise the value of the firm and to act in its (i.e. the company's) best interests. Traditionally this concept of duty to the firm has been taken to mean the same as duty to existing shareholders. However, this is neither the legal nor arguably the moral duty of directors. The threat of a private equity acquisition may be real or it may simply be a ploy by institutional investors to liquidate a part of the value of the firm on the assumption that the firm will be able to operate just as effectively without owning its premises. Leaving aside the issue of whether such leases would be financial or operational, the investors do not appear to recognise that the market has valued the firm currently on the basis that its value generation is both retail and property driven. Disinvestment of the property portfolio will simply skew the risk of the business towards retail and given the increasing international competition in the sector this may result in the firm's value falling even if the firm manages to maintain its current levels of profitability and growth. As things stand the investors (and other stakeholders) in Solar Supermarkets benefit from its diversified value generation with the added benefit that management can focus on the difficult job of retailing and leave the property market to look after itself.

Disinvestment of the property portfolio could increase the risk of the business and in this context it is worth reviewing the share option proposal. Currently, senior management and directors are compensated by a mix of salary, perks and profit related bonuses. To this extent there is a degree of risk sharing between owners and managers. The problem with share options schemes is that they tend to increase the risk appetite of managers in that the holder is no longer so concerned about downside risk in the company's performance which is shifted to the writer (effectively the shareholders). However, to the extent that the firm is financed by debt, the shareholders in their turn hold a call option written by the lenders on the underlying assets of the firm and so the lenders are the ones who bear the ultimate risk. The combination of limited liability and equity options in the hands of directors may well create a wholly unacceptable appetite for risk and a willingness to take on new projects that they would otherwise have rejected.

Finally, the move to a money purchase pension scheme may be suggestive of a less than generous approach to the firm's future employees who, in the retail sector, tend to be poorly paid with incomes of shop floor workers close (in the UK and Europe) to the minimum hourly wage. It is safe to assume that the pension fund is principally designed for the various management grades within the firm. The move from a final salary scheme to a money purchase scheme brings benefits in terms of fund management and financing but passes risk to the beneficiaries of the fund particularly in terms of their exposure to future investment returns and annuity rates. The maintenance of the final salary scheme for existing staff will no doubt be welcomed by them but the firm should recognise that it may be more difficult to appoint staff of the same quality as currently at current rates of pay. In so far as the labour market is efficient we would anticipate wage rates to rise to compensate for the reduction of pension benefit and so the gains from this move may well be illusory.

From an ethical perspective, the directors are in the position of attempting to balance the interests of a range of different stakeholders as well as satisfying their own compensation requirements. It is clear that all four options involve the transfer of risk from one stakeholder group to another in ways which are not immediately obvious. The duties of directors in this case can be summarised as ones of transparency, effective communication and integrity in the choices that they make. It is important that the directors should not be seen as taking a more advantageous position with respect to other groups, without their consent, either in terms of the return they take or the risk they bear.

5 Phobos Co

(a) (i) This is a straightforward question on hedging interest rate exposure using interest rate futures:

Step (1) Calculate the current interest:

Current interest = (LIBOR + 50) × exposure time × principal

$$\text{Current interest} = 6{\cdot}50\% \times \frac{4}{12} \times £30{,}000{,}000 = £650{,}000$$

Step (2) Select the shortest available future with maturity following the commencement of exposure and choose the appropriate hedging strategy.

Sell March with an open of 93·8 and a settlement of 93·88.

Step (3) Calculate the number of contracts:

$$\text{Contracts} = \frac{\text{principal}}{\text{contract size}} \times \frac{\text{Exposure period}}{\text{Contract period}}$$

$$\text{Contracts} = \frac{£30{,}000{,}000}{£500{,}000} \times \frac{4}{3} = 80$$

Step (4) Calculate the basis:

Basis = spot price – futures price = 94·00 – 93·88 = 12 basis points or 'ticks'

Assuming linear convergence then movement between closure and maturity is four ticks given the contracts will have one month to run.

Step (5) Estimate close-out price if interest rates (a) increase by 100 basis points (b) decrease by 100 basis points.

(a) Close out will be 93·00 – 0·04 = 92·96
(b) Close out will be 95·00 – 0·04 = 94·96

Step (6) Calculate gain and/or loss in the futures market and the equivalent cost:

Interest rate at close out	**7·00%**	**5·00%**
Current open price	93·88	93·88
Futures price at close out	92·96	94·96
Ticks	92	–108
On 80 contracts at £12·50 per tick	92,000	–108,000
Cost of loan in spot market	750,000	550,000
less profit/(loss) on futures	92,000	–108,000
Net cost of loan	658,000	658,000
Annual equivalent	6·58%	6·58%

(ii) Traded options allow the management of this type of risk, but the hedge carries a premium. Given the current LIBOR of 6% and an exposure commencing in March the end of March puts at 94,000 are best suited for this type of exposure. A put option allows the holder, at exercise, the right to short the futures at the stated price. These options are exercised (or sold back to the market) if the March futures rate is less than the stated exercise price.

Step (1) Choose the most effective option strategy to minimise basis risk from the point of exposure to contract exercise date on the underlying and to minimise time value.

March puts on three month futures at 94000

Step (2) Calculate the required numbers of contracts:

The calculation as for futures = 80 contracts

Step (3) Calculate premium payable:

Premium = $80 \times 16{\cdot}8 \times £12{\cdot}50 = £16,800$

Step (4) Calculate basis on the underlying (as before) = 4 ticks

Step (5) Test outcomes against expected movements in interest rates:

Interest rate at close out	**7·00%**	**5·00%**
Futures price at close out	92·96	94·96
Exercise price	94·00	94·00
Option payoff	104·00	0
Position payoff on 80 contracts at £12·50 per tick	104,000	0
Cost of loan in spot market	750,000	550,000
less option payoff	–104,000	0
less premium	16,800	16,800
Net cost of loan	662,800	566,800
Annual equivalent	6·63%	5·67%
Expected payoff assuming equal likelihoods	6·15%	

In this calculation we have ignored the time value of the option at close out but have assumed that it will only be the intrinsic value. With one month before close out with the volatilities implied in this example the time value of the in-the-money options could be significant and should be calculated.

At 6·63% the effective cost is just above the required threshold of 6·6% but with an expected payoff of 6·15% (given equal likelihoods of a rise or a fall in interest rates). Given the absence of a time value estimate on close out, and the possibility of capturing the benefit of a fall in rates, the use of options should be the preferred alternative.

(b) Derivatives offer an opportunity for a firm to vary its exposure to interest rate risk at a given rate of interest on the underlying principal (hedging) or to decrease the rate of interest on its principal at an increased level of risk exposure. For hedging purposes derivatives permit the management of exposure either for the long term (swaps) or for the short term (Forward Rate Agreements (FRAs), Interest Rate Futures (IRFs), Interest Rate Options (IROs) and hybrids). With forward and futures contracts, the mechanism of hedging is the same in that an offsetting position is struck such that both parties forego the possibility of upside in order to eliminate the risk of downside in the underlying rate movements. Where the option to benefit from favourable rate movements is required or in situations where there is uncertainty whether a hedge will be required, then an IRO may be the more appropriate but higher cost alternative. Such hedging can be more or less efficient depending upon the ability to set up perfectly matched exposures with zero default risk. Matching depends upon the nature of the contract. With OTC agreements the efficiency of the match may be perfect but the risk of default remains. With traded derivatives, the efficiency of the match may be less than perfect either through size effects or because of the lack of a perfect match on the underlying (for example the use of a LIBOR derivative against an underlying reference rate which is not LIBOR). There will also be basis risk where the maturity of the derivative does not coincide exactly with the underlying exposure.

Where a company forms a view that future spot rates will be lower than those specified by the forward yield curve they may decide to alter their exposure to interest rate risk in order to capture the benefit of the reduced rate. This can be achieved through the use of IROs. Alternatively, leveraged swap or leveraged FRA positions can be taken to avoid the upfront cost of an IRO. For example, taking multiples of the variable leg of a swap (i.e. agreeing to swap fixed for variable) where a higher than market fixed rate is swapped for 'n' multiples of the variable rate. However, as a number of cases have demonstrated it may be very difficult with these types of arrangement to gauge the degree of risk exposure and to ensure that they are effectively managed by the firm. In the 1990s a number of companies in the US and elsewhere took leveraged positions, without recognising the degree of their exposure and took losses that threatened the survival of the firm.

Professional marks are awarded for the quality of the layout, clarity and persuasiveness of the presentation and integration of analytical data with the written text.

			Marks
1	**(a)**	Identification of construction cost and estimation of terminal value	1
		Estimation of the number of room/nights let	2
		Projection of real cash flow on the return phase	2
		Conversion to nominal using the UK inflation rate	2
		Estimation of investment phase including the savings attaching to the capital allowances	2
		Tax charge and capital gain	1
		Conversion to dollars	2
		Total	**12**
	(b)	Calculation of the nominal $ rate of discount using the Fisher formula	2
		Calculation of the net present value	2
		Calculation of the MIRR	4
		Total	**8**
	(c)	Definitive conclusion on the project	1
		NPV as absolute as opposed to relative measure of increase in shareholder value	2
		Problems with underlying assumptions of the NPV model (efficiency arguments)	2
		Weaknesses of return measures	2
		Advantage of MIRR to IRR (reinvestment rate and single root arguments)	2
		MIRR gives headroom in cost of finance negotiations	1
		Total	**(maximum) 8**
	(d)	Professional marks	**2**
2	**(a)**	Calculation of current debt cost	1
		Calculation of cost of equity	1
		Calculation of market value of current debt	2
		Calculation of current WACC	2
		Total	**6**
	(b)	Calculation of revised cost of debt	2
		Ungearing of current beta to give the asset beta	2
		Regearing the beta	1
		Calculation of the new cost of equity	1
		Calculation of the new WACC	2
		Total	**8**
	(c)	Estimation of the firm's market value using the FCFE model	2
		Calculation of the FCFE required to maintain SH value	2
		Required change to the firm's OCF	1
		Calculation of required rate of return on new investment	1
		Total	**6**
	(d)	Outline of mechanism and rationale of a bond issue	2
		Note of importance as a financing method	2
		Advantages	2
		Alternatives: syndication or single source loan	2
		Total	**8**
		Professional marks	**2**

				Marks
3	**(a)**	Definition of default		1
		Calculation of effective monthly rate		2
		Calculation of expected annual cash flow		1
		Calculation of leverage		1
		Calculation of monthly volatility after interest		1
		Calculation of annual volatility		1
		Calculation of standard deviation of cash flows		1
		Calculation of Z		1
		Calculation of probability of default		1
		Total		**10**
	(b)	Bank issue: probability of default and determinants		3
		Bank issue: estimation of recoverability and determinants		3
		Rationale for the risk premium		1
		Justification of interest charge as 5·5% + 2·16% + 0·34%		3
		Total		**10**
4	Essay			
	Overarching duty of directors			3–4
	Commentary on the threat of acquisition by private equity			3–4
	Merits or issues of sale/leaseback (cost of breakup and risk effects)			3–4
	Commentary on share options schemes and redistribution of risk			3–4
	Labour market reactions to move to a money purchase scheme			3–4
	Ethical commentary on distribution of risk between stakeholders			5–6
	Total			**(max) 20**
5	**(a)**	Calculate current interest rate		1
		Identification of appropriate future and hedge strategy		1
		Calculation of number of contracts		3
		(i)	Calculation of basis for IRF	2
			Calculate gain or loss on alternative closeouts	2
		(ii)	Identification of most appropriate option strategy	1
			Calculate premium payable	1
			Calculate loan cost under alternative payoffs	2
			Estimate expected payoff given equal likelihoods	1
		Total		**14**
	(b)	Discussion of the use of derivatives for interest rate risk management		
		Problems of making an efficient match		1
		Hedge efficiency issues		1
		Default and basis risk		1
		Use of derivatives to reduce interest rate		
		Leveraged swaps and FRAs		2
		Dangers of leveraging		1
		Total		**6**

Answers

1 (a) The project cash flow contains a number of errors of principle which should be corrected. As the project cash flows are shown after tax, the corrections should be made net of tax by either adding back or deducting the change required.

– Interest has been deducted and should be added back as this finance charge is properly charged through the application of the discount rate.
– Depreciation should be added back as this is not a cash flow.
– The indirect cost charge should be added back as this does not appear to be a decision relevant cost.
– Infrastructure costs should be deducted as these have not been included in the original projection.
– Site clearance and reinstatement costs of $5 million have been included net of tax.
– The unclaimed capital allowance is calculated as follows:

	0	1	2	3	4	5	6
Capital investment	150·00	50·00					
Deduct FYA at 50%	–75·00	–25·00					
Deduct WDA at 25% of residual		–18·75	–20·31	–15·23	–11·43	–8·57	–6·43
Pool	75·00	81·25	60·94	45·71	34·28	25·71	19·28
Proceeds of sale							7·00
Unclaimed CA							12·28

This will generate a positive tax benefit in year six of $3·68 million at the tax rate of 30%.

The adjusted project cash flow and net present value calculation for this project are as follows:

	0	1	2	3	4	5	6
Project after tax cash flow	–127·50	–36·88	44·00	68·00	60·00	35·00	20·00
Add back net interest			2·80	2·80	2·80	2·80	2·80
Add back depreciation (net of tax)			2·80	2·80	2·80	2·80	2·80
Add back ABC charge (net of tax)			5·60	5·60	5·60	5·60	5·60
Less corporate infrastructure costs			–2·80	–2·80	–2·80	–2·80	–2·80
Estimate for site clearance							–3·50
Tax benefit of unrecovered capital allowances							3·68
Adjusted project cash flow	–127·50	–36·88	52·40	76·40	68·40	43·40	28·58
Discount factor	1·0000	0·9091	0·8264	0·7513	0·6830	0·6209	0·5645
Discounted cash flow at 10%	–127·50	–33·52	43·31	57·40	46·72	26·95	16·14
Net present value	29·48						

The sensitivity of the project to a 1% increase in capital expenditure is as follows:

Sensitivity to a $1 million increase in CAPEX at year 0	0	1	2	3	4	5	6
Equipment purchase/written down value	1	0·50	0·37	0·28	0·21	0·16	0·12
FYA	–0·5						
WDA		–0·13	–0·09	–0·07	–0·05	–0·04	–0·03
Balance	0·5	0·37	0·28	0·21	0·16	0·12	0·09
Impact upon CAPEX	–1						
Tax saving due to WDA and FYA	0·15	0·039	0·028	0·021	0·015	0·012	0·009
Unrecovered allowance							0·027
Net impact	–0·85	0·039	0·028	0·021	0·015	0·012	0·036
Discount factor	1·0000	0·9091	0·8264	0·7513	0·6830	0·6209	0·5645
Discounted cash flow	–0·85	0·0355	0·0223	0·0158	0·0102	0·0075	0·0203
Net present value	–0·63						

Thus an increase in CAPEX by $1million results in a loss of NPV of $0·63 million due to the benefit of the capital allowances available discounted over the life of the project.

(b) The discounted payback is estimated as follows:

	0	1	2	3	4	5	6
Discounted cash flows from project	–127·50	–33·52	43·31	57·40	46·72	26·95	16·14
Cumulative discounted cash flow	–127·50	–161·02	–117·72	–60·32	–13·60	13·35	29·48
Payback (discounted)	4·50						

The duration of a project is the average number of years required to recover the present value of the project.

Duration	0	1	2	3	4	5	6
Discounted cash flow at 10%			43·31	57·40	46·72	26·95	16·14
Present value of return phase	190·52						
Proportion of present value in each year			0·2273	0·3013	0·2452	0·1415	0·0847
Weighted years			0·4546	0·9039	0·9809	0·7073	0·5082
Duration (= sum of the weighted years)	3·55						

Payback, discounted payback and duration are three techniques that measure the return to liquidity offered by a capital project. In theory, a firm that has ready access to the capital markets should not be concerned about the time taken to recapture the investment in a project. However, in practice managers prefer projects to appear to be successful as quickly as possible. Payback as a technique fails to take into account the time value of money and any cash flows beyond the project date. It is used by many firms as a coarse filter of projects and it has been suggested to be a proxy for the redeployment real option. Discounted payback does surmount the first difficulty but not the second in that it is still possible for projects with highly negative terminal cash flows to appear attractive because of their initial favourable cash flows. Conversely, discounted payback may lead a project to be discarded that has highly favourable cash flows after the payback date.

Duration measures either the average time to recover the initial investment (if discounted at the project's internal rate of return) of a project, or to recover the present value of the project if discounted at the cost of capital as is the case in this question. Duration captures both the time value of money and the whole of the cash flows of a project. It is also a measure which can be used across projects to indicate when the bulk of the project value will be captured. Its disadvantage is that it is more difficult to conceptualise than payback and may not be employed for that reason.

(c) Report to Management
Prepared by: D Obbin, ACCA

Project acceptance and criteria for acceptability

I have reviewed the proposed capital investment and after making a number of adjustments have estimated that the project will increase the value of the firm by approximately $29·48 million. The project is highly sensitive to changes in the level of capital investment. Increases in immediate capital spending on this project will lead to a concomitant loss in the overall project value less the tax saving resulting from the increased capital allowances. However, given the size of the net present value of the project, it is unlikely that an adverse movement in this variable would lead to a significant reduction in the value of the firm.

The analysis of the payback on this project using discounted cash flows suggests that the value of the capital invested will be wholly recovered within four years of commencement. The bulk of the cash flow recovery occurs early within the life cycle of the project with an average recovery of the total present value occurring 3·55 years from commencement.

On the basis of the figures presented and the sensitivity analysis conducted, I recommend the Board approves this project for investment.

For many years the Board has used payback as one technique for evaluating investment projects. The Board has noted concerns that (i) the method chosen does not reflect the cost of finance either in the cash flows or in the discount rate applied and (ii) it fails to reflect cash flows beyond the payback date. Discounted payback surmounts the first but not the second difficulty. I would recommend that the Board considers the use of 'duration' which measures the time to recover either half the value invested in a project or, by alternative measurement, half the project net present value. Because this measure captures both the full value and the time value of a project it is recommended as a superior measure to either payback or discounted payback when comparing the time taken by different projects to recover the investment involved.

As part of its review process the Board has asked for sensitivities of the project to key variables. Sensitivity analysis demonstrates the likely gain or loss of project value as a result of small changes in the value of the variables chosen. Unfortunately, some variables such as, for example, price changes and the cost of finance, are highly correlated with one another and focusing upon the movement in a single variable may well ignore significant changes in another variable. To deal with this and given our background information about the volatility of input variables and their correlation, I would recommend that a simulation is conducted taking these component risks into account. Simulation works by randomly drawing a possible value for each variable on a repeated basis until a distribution of net present value outcomes can be established and the priority of each variable in determining the overall net present value obtained. Furthermore, the Board will be in a position to review the potential 'value at risk' in a given project.

I recommend that the Board reviews a simulation of project net present values in future and that this forms part of its continuing review process.

2 BBS Stores
Report to Management
From: N Erd, Financial Consultant

(a) Impact of Property Unbundling on the Statement of Financial Position and the Reported Earnings per Share

The unbundling of the buildings component entails a sale value of 50% of the land and buildings and 50% of the assets under construction to yield a sale value of $1,231 million. Under option 1 $360 million would be used to repay the outstanding medium-term loan notes and the balance as reinvestment within the business of $871 million. Option 2 would entail repayment of the loan and a share buyback. The value released would buy back $871 million/$4 = 217·75 million shares with a nominal value of $54·44 million and a charge to reserves of $817 million. The comparative balance sheets under each option are as follows:

	As at year end 2008	Sale proceeds	Reinvestment option 1		Share buyback option 2		As at year end 2007
	$m	$m	$m	$m	$m	$m	$m
ASSETS							
Non-current assets							
Intangible assets	190			190		190	160
Property, plant and equipment	4,050	–1,231	871	3,690	–1,231	2,819	3,600
Other assets	500			500		500	530
	4,740			4,380		3,509	4,290
Current assets	840	1,231	–1,231	840		840	1,160
Total assets	5,580			5,220		4,349	5,450
LIABILITIES							
Current liabilities	1,600			1,600		1,600	2,020
Non-current liabilities							
Borrowings and other financial liabilities	1,130		–360	770	–360	770	1,130
Other liabilities	890			890		890	900
	2,020			1,660		1,660	2,030
Total liabilities	3,620			3,260		3,260	4,050
Net assets	1,960			1,960		1,089	1,400
EQUITY							
Called up share capital – equity	425			425	–54	371	420
Retained earnings	1,535			1,535	–817	718	980
Total equity	1,960			1,960		1,089	1,400

The first option has the effect of reducing the company's book gearing from 36·6% (borrowing and other financial liabilities to total capital employed) to 28·2% with option 1 or increasing it to 41·4% with option 2. The net impact upon the earnings of the business is less straightforward. Under options 1 and 2 the company would benefit from a reduction in interest payable but would be required to pay an open market rent at 8% per annum on the property released. In addition, under option 1, the reduction in gearing would lead to a 30 basis points saving in interest on the variable component of the swap. Under option 1 the company would be able to earn a rate of return of 13% on the funds reinvested. The adjustment to the current earnings to show these effects is as follows:

	Current	Option 1	Option 2
	$m	$m	$m
Earnings for the year	670·00	670·00	670·00
add back interest saved (net of tax) $360 million × 6·2% × 0·65		14·51	14·51
reduction in credit spread on six-year debt $770 million × 0·003 × 0·65		1·50	
deduct additional property rent (net of tax) $1,231 × 8% × 0·65		–64·01	–64·01
add additional return on equity $871 million × 13% × 0·65		74·00	
Revised earnings	670·00	696·00	620·50
Number of shares in issue	1,700	1,700	1,484
Revised EPS (c per share)	39·41	40·94	41·81

Given that the company has swapped out its variable rate liability there will be no change in the interest charge to earnings for the six-year debt from the current 5·5% per annum unless the lender has the ability to vary the variable rate on current borrowing for changes in credit rating. Because the fixed rate and the floating rate are the same at 6·2% (given the current credit spread is 70 basis points over LIBOR, or in addition to the swap rate of 5·5%) the nominal value of the company's debt is equal to its current market value.

(b) Impact of unbundling on the firm's overall cost of finance

Because the current firm is a combination of both retail and property it is necessary to estimate the asset beta for the retail business alone. Once this is achieved it is then straightforward to estimate the asset beta of a firm with a given element of property removed.

The current equity cost of capital is given as follows:

$$E\left(r_e\right) = R_F + \beta_i \times ERP$$
$$E\left(r_e\right) = 5\% + 1{\cdot}82 \times 3\% = 10{\cdot}47\%$$

The current weighted average cost of capital is:

$$WACC = w_e r_e + w_d r_d \left(1 - T\right)$$
$$WACC = \frac{6{,}800}{6{,}800 + 1{,}130} \times 10{\cdot}47\% + \frac{1{,}130}{6{,}800 + 1{,}130} \times 6{\cdot}2\% \times {\cdot}65 = 9{\cdot}56\%$$

To calculate the unbundled cost of equity capital we first ungear the current company beta as follows:

$$\beta_A = \beta_e \times \left(1 - w'_d\right)$$

where

$$w'_d = \frac{V_d(1-T)}{V_e + V_d(1-T)} = \frac{1{,}130 \times 0{\cdot}65}{6{,}800 + 1{,}130 \times 0{\cdot}65} = 9{\cdot}748\%$$

and

$$\beta_A = 1{\cdot}824 \times \left(1 - 0{\cdot}09748\right) = 1{\cdot}646$$

The retail asset beta can then be calculated from the weighted average of the component betas as:

$$\beta_A = \frac{V_R}{V_T}\beta_R + \frac{V_P}{V_T}\beta_P$$
$$1{\cdot}646 = \frac{4{,}338}{6{,}800} \times \beta_R + \frac{2{,}462}{6{,}800} \times 0{\cdot}625$$
$$\beta_R = \left[1{\cdot}646 - \frac{2{,}462}{6{,}800} \times 0{\cdot}625\right] \times \frac{6{,}800}{4{,}338} = 2{\cdot}225$$

However, the beta of the continuing firm will be a combination of this retail beta and the property beta of the remaining firm. On the assumption that the share price does not change under either option the cost of equity capital is estimated as follows:

	Option 1	**Option 2**
Value of the equity	$= 425 \times 4 \times 4 = \$6{,}800$ million	$= 371 \times 4 \times 4 = \$5{,}936$ million
Asset beta of reconstructed firm	$\beta_A = \frac{V_R}{V_T}\beta_R + \frac{V_P}{V_T}\beta_P$	$\beta_A = \frac{V_R}{V_T}\beta_R + \frac{V_P}{V_T}\beta_P$
	$\beta_A = \frac{5{,}569}{6{,}800} \times 2{\cdot}225 + \frac{1{,}231}{6{,}800} \times 0{\cdot}625 = 1{\cdot}935$	$\beta_A = \frac{4{,}705}{5{,}936} \times 2{\cdot}225 + \frac{1{,}231}{5{,}936} \times 0{\cdot}625 = 1{\cdot}893$
Tax adjusted market gearing	$= \frac{770 \times 0{\cdot}65}{6{,}800 + 770 \times 0{\cdot}65} = 6{\cdot}86\%$	$= \frac{770 \times 0{\cdot}65}{5{,}936 + 770 \times 0{\cdot}65} = 7{\cdot}78\%$
Equity beta of reconstructed firm	$\beta_e = \frac{1{\cdot}935}{\left(1 - 0{\cdot}0686\right)} = 2{\cdot}0775$	$\beta_e = \frac{1{\cdot}893}{\left(1 - 0{\cdot}0778\right)} = 2{\cdot}053$
Cost of equity	$= 5\% + 2{\cdot}0775 \times 3\% = 11{\cdot}23\%$	$= 5\% + 2{\cdot}053 \times 3\% = 11{\cdot}16\%$

$$WACC(1) = \frac{6{,}800}{\left(6{,}800 + 770\right)} \times 11{\cdot}23\% + \frac{770}{\left(6{,}800 + 770\right)} \times 5{\cdot}9\% \times 0{\cdot}65 = 10{\cdot}48\%$$

And

$$WACC(2) = \frac{5{,}936}{\left(5{,}936 + 770\right)} \times 11{\cdot}16\% + \frac{770}{\left(5{,}936 + 770\right)} \times 6{\cdot}2\% \times 0{\cdot}65 = 10{\cdot}34\%$$

Note that under option 1, the variable component of the swap would be reduced by 30 basis points. However, the market value of the debt would remain unchanged because given LIBOR and the fixed component of the swap are the same at 5·5%, the reduction in basis points will reduce the effective coupon and the yield to 5·9%.

Both options will significantly increase the cost of capital for the company from 9·56% to 10·48% in the case of option 1 and 10·34% in the case of option 2.

(c) The potential impact upon the value of the firm

The value of the firm for a low geared business such as this is represented by the present value of the firm's future earnings discounted at the company's cost of capital. The ownership of property does not add value to the business providing that the company can enjoy a continuing and unencumbered use of the asset concerned.

On the assumption that an independent property company can be established and an arm's length rental agreement concluded then it is possible that the ownership of the property assets could be taken off balance sheet. However, the ease with which this can be done depends upon the local accounting regulations and accounting standards.

As it stands option 1 appears to increase the potential earnings more than currently or than option 2. However option 2 offers the highest EPS. With an unbundling exercise such as this it is difficult to predict with precision the likely impact upon the value of the firm. The removal of part of the firm's property portfolio will increase the equity beta but this will be offset by the reduction in the firm's gearing. Much also depends upon the ability of the business to generate a return of 13% on the reinvested proceeds of the property sale. If this is not achieved then a significant loss in shareholder value would result. For this reason the shareholders might prefer the lower risk option of a repurchase of their equity at 400c leaving the firm's EPS largely unchanged.

The analysis of the impact upon the firm's equity cost of capital assumes that the value of the firm's equity at 400c per share will remain unchanged. In practice that is unrealistic and a model of the firm's value would have to be constructed to test the full impact of either option on shareholder value. Given the problem is recursive in that the output value determines the estimation of the equity beta, which is also an input variable in the calculation, computer modelling would be required.

3 (a) The requirement is to discover the fixed exchange rate for an agreed monthly delivery of Euros over six months to settle a contract for 10,000 tonnes of aggregate at an agreed price of €220 per tonne.

The calculation required is to equate the present value of a single agreed forward rate with the present value of the six forward rates specified in the question.

The procedure is to calculate the present value of the variable forward rates using discount rates derived from the yield curve data. The sum of the discounted forward prices is then divided by the sum of the discount rates to obtain an equivalent fixed forward rate on the swap. The calculations are as follows:

	1	2	3	4	5	6
Forward Rates (Euros per $)	0·8326	0·8314	0·8302	0·8289	0·8278	0·8267
$ short zero coupon yield curve	3·25%	3·45%	3·50%	3·52%	3·52%	3·52%
Discount factor	0·9973	0·9943	0·9913	0·9883	0·9854	0·9826
Prepaid forward price	0·8303	0·8267	0·8229	0·8192	0·8157	0·8123
Present value of prepaid forward prices	4·9272					
Sum of discount factors	5·9392					
Fixed forward rate on swap	0·8296					

As rates change continuously, the monthly discount factor should be calculated on a continuous time basis:

$$\text{Discount Factor}_1 = e^{-yxt} = e^{-0{\cdot}0325 * \frac{1}{12}} = 0{\cdot}9973$$

The use of discrete time discounting whilst less accurate would be accepted.

(b) Currency swaps can be for the exchange of different currencies at an agreed rate, or for the swap of interest rate liabilities on borrowing in different currencies. This currency swap entails an agreement to swap a constant dollar sum in exchange for a sequence of currency payments in Euros at the agreed amount each month. The attraction of a currency swap such as this is that it avoids having to enter into a sequence of forward contracts (a forward strip) with a currency dealer with the associated charges and budget variability entailed. The two rates quoted assume a zero arbitrage swap rate and no commission on the swap. Inevitably, a higher rate will be quoted on the swap to cover the required commission. A disadvantage of swap agreements is the relatively complex contract procedure which must be pursued to ensure that the counter-parties to the swap are in agreement as to the terms. Forward contracts tend to be less cumbersome to both negotiate and contract with the currency dealer.

4 To: Chief Financial Officer
From: Deputy Financial Officer

Four issues of concern

The four issues you have raised with me touch upon our ethical and commercial responsibilities.

(i) The payment of a commission to an official in any country can be justified if it was in recognition of a service performed that (a) she or he was legitimately entitled to perform for payment, (b) that the payment was duly authorised within this company and (c) a service was received for which the payment was fair and reasonable. Clearly, such a payment should not have been made if it contravened the ruling law in either this or the official's country. Given this, a payment for consultancy, legal or lobbying services to an independent consultant would be legitimate. However, given that the individual concerned was an official of the agency concerned then the payment should not have been authorised. As the payment was for a substantial amount the matter should be taken up with the company's chief executive officer with a view to an internal investigation being mounted. Disciplinary action should be considered when a more detailed understanding of the circumstances is known. It may also be appropriate to raise this with the health department in the government of the country concerned with a view to full public disclosure of the facts once the situation has been clarified.

(ii) The actions of the agent in using price sensitive information for personal gain would be classed as insider dealing irrespective of whether the transactions took place on the shares, or on options on the shares of our company. Much depends on what could be established in the circumstances. Did the agent know that the licensing agreement would be forthcoming or was it a speculative trade on the anticipation that it would be granted? If it were the former then it would be classed as insider trading. The more difficult issue relates to speculative trading in that if the attempt to gain the licence was in the public domain then the dealing would not be an issue. If however, the agent was only aware of the possibility through his or her relationship with this company then it would be insider dealing.

(iii) This problem raises a number of issues. First, hedging is not an efficient means of reducing translation risk. Translation risk arises because of the conversion of assets and liabilities held in dollars into the domestic currency for accounting purposes. Translation risk will impact upon the residual earnings of the business but does not impact upon the firm's cash flow as no transaction has occurred. There would appear to be an absence of risk management policy in this area and even though a senior member of the treasury team has been making the trades, policy of this type should be set at board level through a risk management committee. Second, substantial trades of this type should be authorised and again, it would appear, that there is an absence of policy in this area. Disciplinary action against the treasury manager concerned would only be appropriate if trading of this type lay outside his or her role description or if there was an explicit policy in place requiring authorisation for trades of greater than a given size. In deciding what action to take, making a gain or a loss is irrelevant. Third the current position suggests a $1·25 million loss is likely against a dollar position of $12·5 million. This may not be material from the point of view of the company's overall financial position but the potential for further loss on an uncovered position such as this should be immediately reversed by shortening the contract concerned. If the loss is deemed material then a brief statement to shareholders should be made specifying the magnitude of the loss and the action taken.

(iv) This problem appears to be an abuse of copyright and as such is against the WTO's Trade Related Aspects of Intellectual Property Rights 'TRIPS' agreement. However, to gain protection under TRIPS we have to make sure that we have made supply available of the drugs concerned. If a member country takes the view that we have abused our patent position then they can issue a 'compulsory licence' which would allow a competitor to produce the product under licence. At the Doha ministerial conference in 2001 it was agreed that TRIPS should not prevent a country adopting measures for the protection of its population's health. In this case it would appear that the dispute is one of piracy and gaining protection through the local courts. Whilst this could in principle be resolved through the WTO and the dispute resolution procedure, this may be an issue which would be most satisfactorily resolved through intergovernmental mediation. Any bilateral concession would need to be multilateralised through the WTO in light of the member's most favoured-nation obligations. It would be worthwhile attempting to discover exactly why the government concerned has blocked access to the courts to ensure that there are no public health policy issues involved and from there attempting to secure the involvement of our own government in helping to resolve the issue.

5 **(a)** Net present value is not a sufficient criterion for choosing between projects when capital is in short supply. On the assumption that the priority of the firm is to maximise net present value overall then the optimal ranking of projects is achieved through the profitability index as measured by the net present value per $ of invested capital at year zero. The ranking of the projects using the net present value index is as follows:

	Investment	NPV	IRR	PI	CumInv
P0805	–120	19	17%	0·1573	120
P0802	–640	69	13%	0·1085	760
P0801	–620	55	16%	0·0892	1,380
P0803	–240	20	15%	0·0841	1,620
P0806	–400	29	15%	0·0733	2,020
P0804	–1,000	72	13%	0·0719	3,020

The first project, PO801 is now the marginal project given the available capital of $1,200,000. However, this ordering of projects is not viable as PO801 cannot be varied and is either promoted in the ranking or is not produced as the plan as it stands requires an investment of $1,380 million to satisfy the supermarket contract. The investment structure can be specified in one of two ways therefore:

Acceptance of PO801 ahead of PO802 (which can be scaled):

	Investment	NPV	IRR	PI	CumInv	proportion	NPV
P0805	–120	19	17%	0·1573	120	1	19
P0801	–620	55	16%	0·0892	740	1	55
P0802	–640	69	13%	0·1085	1,380	0·71875	50
P0803	–240	20	15%	0·0841	1,620		
P0806	–400	29	15%	0·0733	2,020		
P0804	–1,000	72	13%	0·0719	3,020		
						Plan NPV	124

Removal of PO801 from the plan:

	Investment	NPV	IRR	PI	CumInv	proportion	NPV
P0805	–120	19	17%	0·1573	120	1	19
P0802	–640	69	13%	0·1085	760	1	69
P0803	–240	20	15%	0·0841	1,000	1	20
P0806	–400	29	15%	0·0733	1,400	0·5	15
P0804	–1,000	72	13%	0·0719	2,400		
P0801	–620	55	16%	0·0892	740		
						Plan NPV	123

(i) The revised plan should be to produce all of PO805, PO801 and a reduced scale of production on PO802 as shown in the revised schedule.

(ii) The net present value of the plan is $124 million

The internal rate of return cannot be calculated using the proportions of projects invested because of scale effects but must be calculated for the overall plan. Using the interpolation method and calculating the net present value of the optimum plan at 14% and 18% the IRR can be estimated by interpolation:

Discount	NPV
14%	12
18%	–85

$$IRR = 14\% + \frac{12}{12+85} \times 4\% = 14{\cdot}5\%$$

(iii) The profitability index for the plan = 124/1200 = $0·1033 per dollar invested.

(b) When calculating the rate for short-term financing the maximum rate which should be offered is that which generates a zero net present value on those projects which do not qualify for the current plan. The internal rate of return is not appropriate as that is the rate that would be the maximum rate for investment over the life of the projects concerned. This is however, a short-term capital rationing problem. The profitability index gives the net present value of each pound invested.

Project	Now	2009	2010	2011	2012	2013	2014
P0802 (balance of the marginal project)	–180	23	34	56	59	118	–8
P0803	–240	120	120	60	10		
P0806	–400	245	250				
P0804	–1,000	300	500	250	290		
Cash flows of rejected projects	–1,820	688	904	366	359	118	–8
Discount at 10%	–1,820	625	747	275	245	73	–5
Net present value of rejected projects	141						
Profitability index	0·07742						

Therefore these projects could support a maximum additional finance charge of the following:

Additional finance = $1,820,000 * 0·0774 = $141,000

Given that 10% is the rate assuming no short-term market failure for finance for this company, the maximum rate for the one year over which capital rationing is expected to hold is 17·74%.

Professional marks are awarded for the quality of the layout, clarity and persuasiveness of the presentation and integration of analytical data with the written text.

			Marks
1	(a)	Add back net interest	1
		Add back depreciation	1
		Add back indirect cost charge	1
		Deduct infrastructure costs	1
		Estimation of site clearance	1
		Calculation of tax on unrecovered capital allowances	3
		Deduct site clearance and reinstatement	1
		Calculation of NPV	1
		Sensitivity to change in CAPEX	4
		Total	**14**
	(b)	Calculation of the discounted payback	2
		Calculation of duration	2
		Comparative assessment of the techniques	1
		Total	**5**
	(c)	Review of techniques and recommendation	2
		Definition and interpretation of duration measure	2
		Review of sensitivity and its deficiencies	2
		Comment on role of Monte Carlo simulation	2
		Total	**8**
	(d)	Professional marks	**1**
2	(a)	Comparative statement of financial position under the two alternatives (2 each)	4
		Revision of earnings figure for each alternative and calculation of EPS (3 each)	6
		Discussion of the unbundling impact upon the statements of financial position and earnings	3
		Total	**13**
	(b)	Estimation of the current cost of equity	1
		Estimation of the current weighted average cost of capital	1
		Ungearing of current beta and estimation of retail beta	2
		Estimation of the cost of equity under each alternative	4
		Estimation of the WACC under each alternative	2
		Total	**10**
	(c)	Note on problems of taking property assets off balance sheet	2
		Difficulty of predicting net impact of unbundling on shareholder value	2
		Conclusion	2
		Total	**6**
	(d)	Professional mark	**3**

			Marks
3	**(a)**	Calculation of discounted prepaid forward prices	4
		Calculation of present value of forward prices	4
		Calculation of fixed forward rate on swap	4
		Total	**12**
	(b)	Avoidance of commission on strip elements	2
		Avoidance of budget variability	2
		Method shown is for zero arbitrage valuation – extra for commission will be required	2
		Differences in cost of contracting	2
		Total	**8**
4	Essay		
	Discussion of problems of paying commission, principles, ethics and recommendation		5
	Discussion of scope of insider trading and application to agents and options markets		5
	Discussion of relevance of hedging, authority for trading and action to be taken		5
	Advice on position with respect to abuse of patent with reference to TRIPS		5
	Total		**20**
5	**(a)**	Comment on the rationale for the use of PI under capital rationing	2
		Rank ordering of projects	2
		Alternative treatment of non-divisible project	4
		Calculation of overall NPV	2
		Calculation of project IRR graphically or by interpolation	2
		Total	**12**
	(b)	Calculation of the proportions of investment in each project	2
		Calculation of the PI for the rejected projects	2
		Combination of result with current cost of capital to give overall opportunity cost	2
		Summary comments and advice	2
		Total	**8**

Answers

1 (a) Given the details supplied, a forward forecast of the income statement and of the statement of financial position is a precursor to the cash flow forecast. On the assumptions (as stated in the question but not reproduced here) the following projection is obtained (all figures in $'000):

Projected income statement	**Year 1**	**Year 2**	**Year 3**
Revenue (9% growth)	5,450	5,941	6,475
Cost of Sales (9% growth)	3,270	3,564	3,885
Gross Profit	2,180	2,377	2,590
Operating costs (w1)	2,012	2,159	2,317
Operating profit	168	218	273

Projected cash flows			
Operating profit	168	218	273
Add depreciation (w2)	134	144	155
Less incremental working capital (w3)	(20)	(21)	(24)
Less taxation (w4)	(15)	(28)	(43)
Less interest	(74)	(74)	(74)
Free cash flow to equity	193	239	287
Less investment in non-current assets (w2)	(79)	(95)	(114)
Free cash flow	114	144	173

Workings

w1: Operating costs

Variable costs (9% growth)	818	891	971
Fixed costs (6% growth)	1,060	1,124	1,191
Depreciation (w2)	134	144	155
Total operating costs	2,012	2,159	2,317

w2: Non-current assets and depreciation

Non-current assets at beginning	1,266	1,345	1,440
Additions (20% growth)	79	95	114
Non-current assets	1,345	1,440	1,554
Depreciation (10%)	134	144	155

w3: Working Capital

Working capital (9% growth)	240	261	285
Incremental WC	240 – 220 = 20	261 – 240 = 21	285 – 261 = 24

Initial working capital equals net current assets less cash. Alternatively, the full 270 can be used as working capital as well. Credit will be given for either assumption.

w4: Taxation

One year in arrears (30%)	15 (given)	30% x (168 – 74) = 28	30% x (218 – 74) = 43

(b) Our estimate of the value of this business on a going concern basis assumes that cash will be generated and reinvestment made according to the above projection. The free cash flow after reinvestment which is potentially distributable and the terminal value of the business assuming a constant rate of reinvestment of 3% forward is as follows:

		0	1	2	3
Free cash flow after reinvestment			114	144	173
Terminal value					2,546
Required rate of return		10%			
Present value of cash flows (discounted at 10%)			104	119	2,043
Value of the firm	2,266				

The terminal value is calculated as follows:

$$Value_3 = \frac{FCFE_3(1{\cdot}03)}{R_e - 3\%}$$

$$Value_3(\$'000) = \frac{173 \times (1{\cdot}03)}{10\% - 3\%} = 2{,}546$$

Where R_e is the required rate of return of 10% per annum.

The value of the firm, on the basis of the above projections, is $2,266,000.

(c) This valuation is based upon a number of assumptions which you should consider when reviewing this analysis. We have taken your judgement that 10% fairly reflects the market rate of return required for an investment of this type. This rate should compensate you for the business risk to which your firm is exposed. For an investment held within the context of a widely diversified portfolio the rate of return you should expect will only be conditioned by your exposure to market risk. However, in the context of a sole equity investment then the rate of return you may require could be more than that which would be available from the market for an investment of this type.

In generating our projections we have assumed the estimates are certain and that the firm is a going concern. In considering this investment further you may wish to explicitly consider the variability attaching to the underlying variables in the projection and the possible range of values that may result. Of particular importance is the assumption of a three-year forecast. In practice the period chosen does depend upon the nature of the business and in particular the uncertainties to which it is exposed.

Finally we have assumed a terminal value based upon future cash flows from year three forward growing at a compound rate of 3% into the indefinite future. The resulting value will be particularly sensitive to this figure and it may be that you may wish to consider a different rate depending upon what you regard as sustainable in the long term for a business of this type.

(d) In considering the value of this business you should note that the company has a very high level of gearing and that this imparts additional value to you through your possession of limited liability. The value quoted above is a fair value of the business in your hands based upon the expected value of the future cash flows of the enterprise accruing to you. This is the equivalent of the fair value of the firm's assets less its outstanding liabilities including its debt.

However, the possession of limited liability gives you an effective call option on the value of the business against the lenders in that *in extremis*, if the value of your equity fell to zero, you could liquidate and the lenders would bear any loss. As shareholders you have the potential of 'upside' gains but are not exposed to downside losses. This advantage on the upside means that your limited liability represents an asymmetric claim on the business and this is a valuable right which is particularly important during the growth stage of a business.

This additional value is termed its 'structural value' and it is at its maximum when the value of the assets of the firm approximate to the value of the outstanding debt but it diminishes as that differential widens. A structural valuation of your firm could be undertaken but would need estimates of the volatility of the returns generated by the firm's assets and the length of time which the debt will be held to maturity.

2 To Polar Finance

Report on Anchorage Retail Limited

Following your terms of reference we report as follows:

(a) There are many risks in acquisitions which have both a high likelihood of occurring and potentially a significant impact upon your business if they do occur. The principal risks to which Polar Finance might be exposed in an acquisition of this type are as follows:

1. Disclosure risk: in making an acquisition of this type it is important to ensure that the information upon which the acquisition is made is reliable and fairly represents the potential earning power, financial position and cash generation of the business. As part of a due diligence exercise it would be necessary to ensure that the financial accounts have not been unduly manipulated to give a more attractive view of the business than the underlying reality would support. To this end it is important to ensure that the income statement can be supported by the reported cash flow. In addition to this all other company documents should, in due course, be scrutinised as part of a full due diligence exercise.

2. Valuation risk: a substantial acquisition has the potential to alter the risk of the acquirer either because of an alteration in the fund's exposure to financial risk or in its exposure to market risk. Ultimately the value of Polar Finance to its equity investors depends upon the potential returns it offers to them and the risk of those returns. A substantial acquisition of this type can impact upon the perceived risk attaching to the equity which investors have already subscribed and hence the value they place upon the fund. As a consequence the post acquisition value of Polar Finance may not be a simple sum of the fund's current value and the existing equity valuation of Anchorage Retail.

3. Regulatory risk: an acquisition of the size proposed may raise concern with the government or with other regulatory agencies if it is seen to be against the public interest. As Polar Finance is not another retail company then the acquisition is unlikely to be seen as against the public interest on competition grounds. However, concern has been raised in a number of jurisdictions about the lack of accountability of private equity funds and given the reputation of Anchorage Retail this acquisition may result in adverse regulatory scrutiny and pressure.

Each of these risks would need to be explicitly considered as part of your due diligence investigation to ensure that they are either mitigated or avoided. Credit may be given for alternative, relevant risks.

(b) In examining the reported return we have considered the following measures (only two are required in addition to EVA):

Return on capital employed which can be measured as either net operating profit before tax or net operating profit after tax (NOPAT) as a percentage of capital employed.

Economic Value Added which is the difference in a firm's reported return on capital employed and its weighted average cost of capital. EVA measures that rate of value creation within a business reflecting the degree of economic 'super normal profit' or, in other terms, the residual income of the business. EVA return is also reported as the difference between return on capital employed and the weighted average cost of capital for the business.

Return on fixed capital employed which in comparison with return on capital employed indicates the return advantage associated with the company's management of its working capital.

Return on equity which measures the rate of return earned on the total equity funds employed.

Each of the above return measures is presented using the year end figures (although mid-year averages for capital employed are more appropriate in each case). Credit may be given for alternative, relevant suggestions and calculations.

Return on Capital Employed

Return on capital employed focuses attention upon the return generated by all classes of asset. Its measurement in post tax form as NOPAT/(Equity + Debt) gives a measure of the return excluding the benefit of the tax shield on debt. Alternatives, not given here, are (i) to measure the gross return using net operating profit before tax or (ii) the distributable profit plus interest payable. ROCE, using our preferred method and year end capital figures is:

ROCE (2009) = $1{,}250 \times (1 - 30\%)/(2{,}030 + 1{,}900) = 22{\cdot}26\%$

ROCE (2008) = $1{,}030 \times (1 - 30\%)/(1{,}555 + 1{,}865) = 21{\cdot}08\%$

ROCE has shown a modest improvement but a further investigation of the accounting information would be required to establish whether the improvement is a genuine improvement in the underlying performance of the business. A thorough accounting investigation as part of a due diligence exercise should answer the question as to whether the ROCE measure can be trusted.

Marginal ROCE

Marginal ratios can be calculated for one or all of the return ratios. They measure the return generated by the latest capital reinvested within the business. Marginal return on capital employed measures the return generated on capital introduced or reinvested over the last period of account. It is measured as:

$$\Delta ROCE = \frac{\Delta NOPAT}{\Delta CE} = \frac{(1{,}250 - 1{,}030) \times 0{\cdot}7}{3{,}930 - 3{,}420} = 30{\cdot}20\%$$

Movements in this ratio year-on-year give a clearer indication of the improvement or deterioration in the company's performance. In the case of Anchorage, the performance on new capital is considerably better than that on the accumulated capital employed in the business.

Economic Value Added and Economic Value Added Return

EVA is derived from the difference between accounting return on capital employed and the weighted average cost of capital:

$EVA\% = ROCE - WACC$

Where ROCE is measured as net operating profit after tax (NOPAT) over total capital employed.

In absolute terms EVA can be expressed as a financial surplus by multiplying throughout by the denominator of the ROCE ratio:

EVA = NOPAT – WACC × Capital Employed

Proponents of the EVA measure offer a variety of procedures for 'cleaning' the NOPAT and capital employed figures to increase their economic significance and to reduce distortion through accounting manipulation. Insufficient detail is available at this stage to make any such adjustments. The calculation of the weighted average cost of capital is shown in Annex 1 to this report.

On the stated assumptions and on the basis that WACC has not changed EVA, and the percentage EVA return, for the two years of account is as follows:

EVA (2009) = 1,250 × (1 – 30%) – 6·12% × (2,030 + 1,900) = \$634 million (16·14%)

EVA (2008) = 1,030 × (1 – 30%) – 6·12% × (1,555 + 1,865) = \$512 million (14·96%)

This suggests that there has been a significant improvement in EVA for the business over the two years both in absolute terms and in terms of EVA return. As with ROCE, reliance should only be placed on these figures once a full accounting investigation has been conducted.

Return on equity

This ratio measures in both its absolute and marginal form the return exclusively for the equity investor within the business:

ROE (2009) = 860/2,030 = 42·36%

ROE (2008) = 650/1,555 = 41·80%

As with ROCE this return measure has shown a modest improvement over the year.

Return on Fixed Capital Employed

This measure reflects the return on the capital employed in investment in non-current assets. When compared with ROCE it indicates the return leverage through the firm's working capital policy.

ROFCE (2009) = 1,250 × (1 – 30%)/4,980 = 17·57%

ROFCE (2008) = 1,030 × (1 – 30%)/4,540 = 15·88%

If the current asset ratio of a company is one then ROFCE and ROCE should be identical. If the CAR is greater than one, capital on the form of owner's equity and long-term liabilities is being diverted into financing working as opposed to fixed capital employed in the business and *vice versa*. The comparison of ROFCE with ROCE for Anchorage suggests that, on the basis of the accumulated capital of the business, the company is leveraging additional return through its working capital management policy.

(c) The impact upon the equity cost of capital of this acquisition is an approximation at this stage, pending a more intensive valuation exercise focusing on the impact it would have upon the equity valuation of the fund overall. Unravelling the gearing effect upon the betas of both Anchorage and Polar finance we estimate that the asset beta of each is as follows (assuming that debt beta is zero):

Where there is no tax benefit attaching to debt:

$$\beta_a = \beta_e\left[\frac{V_e}{V_e + V_d}\right]$$

Where there is a tax benefit attaching to debt

$$\beta_a = \beta_e\left[\frac{V_e}{V_e + V_d\left(1 - T\right)}\right]$$

Note that Polar Finance does not pay tax on its income and thus there is no benefit attributable to the tax shield on debt. In this case the market value of debt will not be tax adjusted. However, with Anchorage this is not the case as the current equity beta will reflect a tax shield on its debt.

Polar asset beta = 0·285 (given)

Anchorage asset beta can be calculated as follows:

Anchorage equity beta = 0·75
Anchorage debt proportion = 24%
Therefore equity proportion = 76%
1 – tax rate = 1 – 0·3 = 0·7

Anchorage asset beta = 0·75 × 76/(76 + 24 × 0·7) = 0·614

Asset beta of combined entity (Polar + Anchorage) = 0·8 × 0·285 + 0·2 × 0·614 = 0·351
(this is based on the information that if Anchorage's proportion of cash flows is 20%, then Polar's cash flows without Anchorage would be 80%).

Polar debt before acquisition is 85%, therefore equity is 15% and \$1·125bn. Therefore, debt is \$1·125/0·15 × 0·85 = \$6·375bn.

Level of debt after acquisition is \$6·375bn + \$2·5bn = \$8·875bn.

Equity beta of the combined entity = 0·351 × (8·875 + 1·125)/1·125 = 3·12
Equity beta of Polar before acquisition = 0·285 × (6·375 + 1·125)/1·125 = 1·9

Required rate of return of equity investors before acquisition =
Cost of equity = 5% + 1·9 × 2·224% (see annex) = 9·23%

Required rate of return of equity investors after acquisition =
Cost of equity = 5% + 3·12 × 2·224% = 11·94%

Hence Polar's shareholders will require an increase of 2·71% return as a result of the acquisition.

(d) The view that Anchorage Retail has been undervalued by the market implies that the market is inefficient in pricing the equity of firms of this size. The evidence from tests of the efficient market hypothesis suggests that this is unlikely and given the number of investors involved in an active market for a company of this type we should not assume that, on average, they have mispriced this business. It maybe that investors have irrational as well as rational expectations about this company but we would expect that in the aggregate the irrational component would be unsystematic with respect to the underlying value of the business and thus diversified away in the pricing process.

Assuming that the market has priced in zero growth for Anchorage, capitalising the current dividend payments of $270 million at an equity cost of 6·668% suggests an equity market capitalisation of $4·049bn. The current share price of $2·6 per share, with 1·6 bn shares in issue implies an equity value of $4·16 bn which suggests that the market expects little growth in the business over the longer term. This probably reflects both perceptions of weakness in Anchorage Retail and in its ability to compete in a weakening market. Given what the price reveals about expectations, it is likely that a bid of $3·20 would be attractive to Anchorage shareholders. Whether it is an attractive purchase for Polar depends on how Polar's management deploy their strategies to improve Anchorage's performance.

Yours sincerely

A. Consultant

Annex 1

Estimation of the Weighted Average Cost of Capital (WACC) for Anchorage Retail

To estimate the cost of equity capital for the company we first measure the expected return on the market using the rearranged dividend valuation model on the broadly based market index:

$$r_e = \frac{D_0}{P_0}\left(1+g\right)+g$$

$$r_e = 3{\cdot}1\% \times \left(1{\cdot}04\right) + 4\% = 7{\cdot}224\%$$

Given the market index should have a beta of 1 and with a risk-free rate of interest of 5% this suggests that the rate of return expected on the market is 7·224% and the equity risk premium is 2·224%

Putting this result into the Capital Asset Pricing Model:

$$r_e = R_F + \beta_i(r_m - R_F)$$
$$r_e = 5\% + 0{\cdot}75 \times 2{\cdot}224\% = 6{\cdot}668\%$$

This gives a weighted average cost of capital as follows:

$WACC = 76\% \times 6{\cdot}668\% + 24\% \times 6{\cdot}2\% \times (1 - 30\%) = 6{\cdot}12\%$

3 **(a)** A warrant is an option attached to another financial instrument on issue which can be detached and negotiated independently of the underlying issue. Warrants are usually exercised over a longer term than traded options but can be valued in exactly the same way using the Black Scholes Option Pricing Model by inserting into the standard formula.

The calculation has been performed as follows:

$$c = 85_a N(d_1) - 90N(d_2)e^{-0\cdot05\times5}$$

Where:

$$d_1 = \frac{\ln(85/90) + (0\cdot05 + 0\cdot5 \times 0\cdot2^2) \times 5}{0\cdot2 \times \sqrt{5}} = 0\cdot6548$$

$$d_2 = d_1 - 0\cdot2 \times \sqrt{5} = 0\cdot2076$$

From normal tables we calculate the area under the normal curve represented by d_1 and d_2:

$$c = 85 \times 0\cdot7437 - 90 \times 0\cdot5822 \times e^{-0\cdot05 \times 5} = 22\cdot41$$

Given that each warrant represents an option on 100 equity shares the value of each warrant is $2,241.

The Black Scholes model makes a number of restrictive assumptions:

1. The warrant is a 'European' style option.
2. The share price follows a log-normal distribution and is continuously traded.
3. Unrestricted short selling of the underlying security is permitted.
4. There are no market frictions such as taxes or transaction costs.
5. No dividends are paid during the life of the warrant.

These assumptions are less realistic with a company such as Alaska Salvage than with a large enterprise with a full listing. It is unlikely, for example, that the company's shares will be actively traded or that the share market is efficient in its pricing of the equity.

(b) The coupon rate is derived from the cash flow to the lender as follows:

1. Lay out the cash flow to the lender showing the value of the warrant as a benefit accruing immediately to the lender.

	0	1	2	3	4	5
Coupon	(10,000)	100 × c%	100 × c%	100 × c%	100 × c%	100 × c%
Repayment						10,000
Call value	2,241					
Cash flow to lender	(7,759)	100 × c%	100 × c%	100 × c%	100 × c%	10,000 + 100 × c%

2. Solve the following equation where c% is the coupon rate and A and V are the five-year annuity and discount factors at 13% respectively:

   $7{,}759 = 100 \times c\% \times A + 10{,}000 \times V$

   Therefore:

   $7{,}759 = 100 \times c\% \times 3\cdot517 + 10{,}000 \times 0\cdot543$

   By rearrangement:

   $$c\% = \frac{7{,}759 - 10{,}000 \times 0\cdot543}{100 \times 3\cdot517} = 6\cdot62\%$$

Therefore a 6·62% coupon rate will give an effective rate of return on the investment to the lender of 13%.

(c) Mezzanine debt such as this is one mechanism by which a small, high growth firm such as Alaska Salvage can raise debt finance where the risk of default is high and/or there is a low level of asset coverage for the loan. In this case raising a loan of $1·6 million would raise the market gearing of the firm from zero (assuming there is no current outstanding debt) to 13·6% (debt to total capitalisation). This increase in borrowing against what might be presumed to be specialised salvaging equipment and the forward cost of operation may not be attractive to the commercial banking sector and may need specialised venture finance. The issue of warrants gives the lender the opportunity to participate in the success of the venture but with a reasonable level of coupon assured. However, the disadvantage for the current equity investors is that the value of their investment will be reduced by the value of the warrants issued. The extent to which this will be worthwhile depends upon the value of the firm on the assumption that the project proceeds and is financed in the way described. This should ultimately decide the maximum value that they would be prepared to pay to finance the new project.

4 **(a)** The credit crunch has led to an unwillingness by the banks to lend, particularly to one another, resulting in a drain in the liquidity across the capital markets. Interest rates are now low as central banks attempt to stimulate their economies. The business of banks is to earn profits by borrowing short and lending long and they are still willing to lend to high class corporate customers. And probably at competitive interest rates. The banks will be concerned about the following:

The risk of default: although there has been a slackening of demand for the company's product our relatively high credit rating would still make us an attractive prospect for lenders. They will need to undertake a credit risk assessment which will include a thorough examination of our asset strength, existing capitalisation, operating strength and income gearing. An important measure will be the firm's current cash flow/debt obligations ratio.

Recovery: the assessment of our asset strength will form a part of their assessment of the potential recoverability of the debt in the event of default.

(b) Syndication is where a group of banks combine with one bank taking the lead in the arrangement. Syndication allows banks to offer much larger loans in combination than would be feasible singly, and given the range of banks involved can tailor loans (perhaps across different currencies) to more exactly match our requirements. The management of the syndicate lies with the arranging bank but the effective cost will be somewhat higher than with a conventional loan but usually much lower than the cost of raising the necessary finance through a bond issue.

A bond issue is where the debt is securitised and floated onto the capital market normally with a fixed interest coupon and a set redemption date. Initial set up costs can be high especially if the issue is underwritten. A loan of the size envisaged is towards the low end of what would normally be raised through this means. Some bond issues can be syndicated in that a number of borrowers of similar risk are combined by the investment bank chosen to manage the issue.

The advantage of syndication is that it reduces the costs of issue. However, it may be that the best offer would entail accepting a variable rate based on LIBOR which would have to be swapped out if we wished to minimise interest rate risk.

(c) In assessing this capital investment we have to make some assumptions about the immediate future in terms of the general economic conditions and to what extent we have a delay option on the project concerned. Where there is a positive delay option then from a financial perspective the best advice may be to delay investment dependent on the likely actions by the competitors and how markets for the product develop. Where there are significant competitive reasons for proceeding we should only proceed if the net present value of the project is worthwhile. Given the magnitudes of the uncertainties involved at this stage of the economic cycle the decision to proceed should only be made when we are sure that we have estimated the potential magnitude of the risks and taken them into account in our analysis.

For all parts to question 4, credit will be given for alternative and relevant points.

5 **(a)** Possible strategies available to the Katmai company may include:

(i) Do nothing: with this strategy the company is taking a gamble on the yield curve that future interest rates will remain unchanged or indeed may fall. Much would depend on the degree of interest rate diversification the company may have on overseas debt and on the magnitude of any interest payable.

(ii) Retire and reissue fixed rate debt: with this strategy the company will issue fixed rate bonds and use the proceeds to retire the floating rate notes. This will eliminate any downside risk associated with increasing interest rates but will be an expensive option to pursue as commissions, arrangement fees and underwriting costs can be between 2–3% of the value of the loan.

(iii) Enter a fixed for variable interest rate swap: with this strategy the company will enter into a swap agreement with a market maker for the term of the loan. Normally this will entail swapping the liability for the LIBOR component of the swap for a fixed rate. The advantage of a swap agreement is that it is easy to establish through the highly organised OTC swap market. The disadvantage is that the company will be committed for the term of the swap which means that the company would have to reverse out of the swap if it found itself in a position to retire the loan notes earlier than planned.

Moving from variable to fixed interest rates is unlikely to have a significant impact upon the company's weighted average cost of capital or its value but it will stabilise the firm's financing flows which should make forward planning and budgeting easier. Therefore, in deciding between the three alternatives the directors should assess whether taking risk of this sort is a part of its core business or whether managing interest rate volatility in the budgeting process is a diversionary activity. Either (ii) or (iii) will reduce managerial and labour risk exposure by reducing the volatility of the surplus from which their remuneration and other compensation is drawn. From an equity investor point of view therefore, given the costs of hedging, (i) may be the preferred strategy. From the perspective of other stakeholders (iii) is likely to be the least costly alternative for achieving stability in future financing flows. From a managerial perspective (iii) is the recommended course of action.

(b) The six monthly interest rate under a vanilla swap given the quoted spread would be as follows:

Payments	LIBOR/2 + 0·6%
Receipt under a vanilla swap	LIBOR/2
Payment on fixed leg	5·4%/2 = 2·7%
Net payment	3·30%

The six-monthly rate is therefore 3·30% or an effective annual rate:

$EAR = 1{\cdot}033^2 - 1 = 6{\cdot}71\%$

(c) Given that the annual interest rate volatility is 1·5%, the standard deviation of six-monthly rates is:

$$\sigma_6 = \sigma_a \times \sqrt{\frac{1}{2}} = 1{\cdot}5\% \times 0{\cdot}7071 = 1{\cdot}061$$

The Value at Risk (VAR) is given by:

$\text{VAR} = \text{Loan} \times \sigma_6 \times \text{CL} = \$150 \text{ million} \times 1{\cdot}061\% \times 1{\cdot}645 = \$2{\cdot}62 \text{ million}$

Where the confidence level of 95% is taken from the normal tables supplied and is, assuming a single tail, 1·645 standard deviations away from the mean.

VAR is defined (Jorion, 2007) as the 'worst loss over a target horizon such that there is a low, prespecified probability that the actual loss will be larger'. Currently the six-monthly interest rate is LIBOR plus 120 basis points. Let us assume for the moment that LIBOR is 5% per annum. The six-monthly interest on this loan will be 3·1% or $4·65 million. There is a 5% chance that the actual interest paid would be greater than ($4·65 million + $2·.62 million) $7·27 million or, to put it another way, there is a 95% likelihood that the actual interest payable will be less than this figure. A number of assumptions constrain our interpretation of VAR. The first is that interest rates are assumed to follow a 'random walk' in that the current rate is the mean of the distribution of future possible rates, second that the volatility of future rates remains unchanged and third, that the distribution of rates is normal. In practice, although the 'normality' assumption may be useful in simplifying the mathematics of VAR, calculation shows that actual rate distributions exhibit significant skew and that the likelihood of extreme outcomes is somewhat higher than would be expected. However, VAR does simplify the representation of risk by placing a monetary value on the exposure arising from different sources.

				Marks
1	**(a)**	Estimation of depreciation		1
		Estimation of taxation		1
		Estimation of changes in working capital		2
		Projection of income statements		4
		Projection of cash flows		4
			Total	**12**
	(b)	Calculation/identification of free cash flow		1
		Calculation of terminal value		2
		Calculation of present values and business value		3
			Total	**6**
	(c)	Commentary on required rate of return		2–3
		Assumptions about the growth rates		2–3
		Other relevant points		1–2
			Max	**6**
	(d)	Discussion of equity as a call option on business assets, issues relating to gearing and time value of the limited liability option		4
			Total	**28**

2 Note candidates may answer this question in a variety of ways and in different order to that set.

	(a)	Identification of principal risks (2 each to a maximum of 6) Disclosure Valuation Regulatory Other		
			Total	**6**
	(b)	Calculation of the equity risk premium and return on equity		2
		Estimation of the WACC		1
		Calculation of the EVA for each year		3
		Other performance ratios and associated commentary	up to	6
			Total	**12**
	(c)	Ungearing the component betas		1
		Calculation of the combined beta		2
		Estimation of impact upon required equity rate of return		3
			Total	**6**
	(d)	Undervaluation and the implication of inefficiency		2
		Review of current price and whether the acquisition may be attractive		2
			Total	**4**
	(e)	Professional marks	**Total**	**4**
			Total	**32**

				Marks
3	(a)	Calculation of:		
		d1		3
		d2		1
		value of the warrant		2
		assumptions (one each to a maximum of 4)		4
			Total	**10**
	(b)	Estimation of the coupon rate (2 marks for deducting option value from face value of warrant and 2 marks for calculation of coupon using annuity and discount factors)		4
			Total	**4**
	(c)	Identification of mezzanine debt as a source of high risk finance		2
		Disadvantage for equity investors (reduction in equity value on exercise)		2
		Advantages: low coupon, additional equity participation		2
			Total	**6**
			Total	**20**

4 The mark ranges reflect the flexibility the examiners use in assessing this question. The following points are indicative only:

	(a)	Importance of asset strength and default assessment		3–4
		Other relevant comments		3–4
			Max	**7**
	(b)	Loan syndication as a means of spreading load and risk		1–2
		Flexibility of syndication and possibility of cross-border finance		2–3
		Cost of loan syndication versus bond issues		1–2
		Flexibility of bonds in terms of repayment and size of issue		1–2
			Max	**7**
	(c)	Note on the significance of the real option to delay and the factors that might influence that decision. And other relevant points such as strategic and operational benefits of not delaying.	up to	**6**
			Total	**20**
5	(a)	Enumeration and evaluation of the three choices (credit will be given for reasonable alternative choices)		
		Do nothing		1–2
		Retire and reissue		1–2
		Other choices		1–2
		Swap		2–3
		Recommendation of appropriate course of action in the case with justification		3
			Max	**9**
	(b)	Calculation of 3·3%		4
		Calculation of annual rate		1
			Total	**5**
	(c)	Calculation of six-month volatility		1
		Calculation of VAR		3
		Comments on the results		2
			Total	**6**
			Total	**20**

Answers

1 The answer should be presented as a briefing note.

The standard financial appraisal is supplemented by a sensitivity analysis. A discussion of the principal uncertainties and recommendation of a further process of evaluation to assess the volatility of the project is presented.

(i) Estimation of the net present value (NPV) of the project

Estimated Project NPV (all values quoted in $million):

30-year annuity factor, growth rate 4% and discount rate 10%

$$A_n = \left[\frac{1-\left(\frac{1+g}{1+i}\right)^n}{i-g}\right](1+g) = \left[\frac{1-\left(\frac{1{\cdot}04}{1{\cdot}1}\right)^{30}}{0{\cdot}1-0{\cdot}04}\right]1{\cdot}04 = 14{\cdot}11$$

Year/Type	Cash Flows ($m)	DF (10%)	Present value ($m)
2012 Construction	(300)	$1{\cdot}1^{-1}$	(272·7)
2013 Construction	(600)	$1{\cdot}1^{-2}$	(495·9)
2014 Construction	(100)	$1{\cdot}1^{-3}$	(75·1)
2015–2044 Annual operating surplus	100	$14{\cdot}11 \times 1{\cdot}1^{-3} = 10{\cdot}601$	1060·1
2044 Decommissioning	600	$1{\cdot}04^{33} \times 1{\cdot}1^{-33} = 0{\cdot}1571$	(94·3)
NPV			122·1

On the basis of the above, the net present value of the project as at 1 January 2012 is $122·1 million which indicates that the project will add value to the business.

(ii) An analysis of the principal uncertainties associated with this project

Uncertainty in capital investment projects can be categorised as misestimation of:

(i) the timing and the level of capital expenditure over the investment phase of the project;
(ii) the operating surpluses from the project and their timing;
(iii) the timing and amount of closure costs; and
(iv) the discount rate.

Capital Expenditure: with large scale investment projects uncertainty will attach to the timing and costs of the capital construction. Where building is undertaken by external contractors there may be legal issues in forming a complete contract and in monitoring performance against it. In addition to engineering difficulties causing delay in construction, problems can also occur on the costs of labour, raw material supplies and other costs.

Operating Surplus: This will depend upon a number of capacity estimates (theoretical and actual), market costs for labour, materials and operating overheads. In addition, the supply of electricity is subject to national demand and unit prices will depend upon alternative capacity, alternative energy supplies (fossil and renewable) and whether they can contribute to marginal or to base load.

Closure Costs: These can be high in this industry although their remoteness diminishes their significance in present value terms. The timing and estimates of the magnitude of such costs will have a small impact upon the viability of the project as a capital investment.

The discount rate: This can be difficult to estimate for a project of this scale. In practice such rates are taken from a mixture of models and sources, all of which have uncertainties attached. The gearing of the business will determine the importance of uncertainty attaching to each capital source. There is also likely to be significant social externalities with a project of this type affecting its valuation and hence the discount rate which should be applied. Such social externalities range from the reduction in dependence upon fossil fuels and associated carbon emissions, the relatively low pollution costs compared with fossil fuels, and the stability and security of supply.

In addition there may be a range of **real options** attaching to a project of this type: (i) the option to delay, (ii) the option to expand or contract capacity, (iii) the option to withdraw early or extend the operating life of the project. Each of these will add value to the firm by helping to eliminate the downside exposure and focusing the expected value calculation on the upside of the possible distribution of outcomes.

(iii) Project sensitivity

In order for the net present value to reduce to nil, the NPV must reduce by $122·1m.

Construction Cost
Cost increase per $100m x ($3 \times 1{\cdot}1^{-1} + 6 \times 1{\cdot}1^{-2} + 1 \times 1{\cdot}1^{-3}$) = $122·1m
Cost increase per $100m x (3 x 0·9091 + 6 x 0·8264 + 1 x 0·7513) = $122·1m

Cost increase per \$100m = \$122·1m/8·4370 = \$14·47m
Year 1 cost will increase to \$343·41m (\$300m + \$14·47m x 3), Year 2 cost will increase to \$686·82m and year 3 cost will increase to \$114·47m. An increase of 14·47% in each year before NPV becomes zero.

Annual Operating Surplus
\$122·1m/10·601 = \$11·52m
Surplus needs to reduce to \$88·48m (11·52%) before NPV becomes zero.

Decommissioning Costs
\$122·1m/0·1571 = \$777·2m
Decommissioning costs need to increase by \$777·2m (129·5%) in 1 January 2012 value before NPV becomes zero.

The annual operating surplus is the most sensitive to changes.

(iv) Volatility Assessment

The assessment of the volatility (or standard deviation) of the net present value of a project entails the simulation of the financial model using estimates of the distributions of the key input parameters and an assessment of the correlations between variables. Some of these variables are normally distributed but some (such as the decommissioning cost) are assumed to have limit values and a most likely value. Given the shape of the input distributions, simulation employs random numbers to select specimen value for each variable in order to estimate a 'trial value' for the project NPV. This is repeated a large number of times until a distribution of net present values emerge. By the central limit theorem the resulting distribution will approximate normality and from which project volatility can be estimated.

In its simplest form, Monte Carlo simulation assumes that the input variables are uncorrelated. However, more sophisticated modelling can incorporate estimates of the correlation between variables. Other refinements such as the Latin Hypercube technique can reduce the likelihood of spurious results occurring through chance in the random number generation process. The output from a simulation will give the expected net present value for the project and a range of other statistics including the standard deviation of the output distribution. In addition, the model can rank order the significance of each variable in determining the project net present value.

2 The answer should be presented in a report format.

Due to the size of the acquisition it is necessary to establish the potential group value on the presupposition that shareholder value is not reduced in AggroChem. This will give the maximum premium that should be paid.

(i) Using the free cash flow valuation model for AggroChem (which is wholly financed by equity) we can estimate the market value:

Estimated market value =
Free cash flows (FCF) x (1 + growth rate)/(cost of capital (Ke) – growth rate (g))

Note: Since AggroChem is financed wholly by equity, its cost of capital is equal to its cost of equity.

Free cash flow \$'000) = NOPAT – net reinvestment = 580 – 180 = 400

Assume: g = retention rate x cost of capital (Ke)

Taking the net reinvestment as a ratio of NOPAT we obtain the retention ratio (b):

$$b = \frac{180}{580} = 31{\cdot}03\%$$

Ke = 5% + 1·26 x 6% = 12·56%

g = 0·3103 x 0·1256 = 0·0390

Market value = \$400,000 (1·0390)/(0·1256 – 0·0390) = \$4,799,000

This modelling method assumes that the free cash flow model fairly represents the fair value of the business, that the company is a going concern, that the discount rate has a flat term structure, and that future growth is constant and the growth rate is based on the assumption that retained earnings can be invested at the cost of capital. In practice one or more of these assumptions may be violated.

(ii) Using the BSOP model in company valuation rests upon the idea that equity is a call option, written by the lenders, on the underlying assets of the business. If the value of the company declines substantially then the shareholders can simply walk away, losing the maximum of their investment. On the other hand, the upside potential is unlimited once the interest on debt has been paid.

BSOP model can be helpful in circumstances where the conventional methods of valuation do not reflect the risks fully or where they can not be used. Given the gearing of the two companies, the low levels of trading in each company's equity, and their future growth potential, including its volatility, it is appropriate to handle the valuation by focusing upon the real option value attributable to the post-acquisition business.

There are five variables which are input into the BSOP model to determine the value of the option. Proxies need to be established for each variable when using the BSOP model to value a company. The five variables are: the value of the underlying asset, the exercise price, the time to expiry, the volatility of the underlying asset value and the risk free rate of return.

For the exercise price, the debt of the company is taken. In its simplest form, the assumption is that the borrowing is in the form of zero coupon debt, i.e., a discount bond. In practice such debt is not used as a primary source of company finance and so we calculate the value of an equivalent bond with the same yield and term to maturity as the company's existing debt. The exercise price in valuing the business as a call option is the value of the outstanding debt calculated as the present value of a zero coupon bond offering the same yield as the current debt.

The proxy for the value of the underlying asset is the fair value of both the companies' assets less current liabilities on the basis that if the company is broken up and sold, then that is what the assets would be worth to the long-term debt holders and the equity holders.

The time to expiry is the period of time before the debt is due for redemption. The owners of the company have that time before the option needs to be exercised, that is when the debt holders need to be repaid. The proxy for the volatility of the underlying asset is the volatility of the business' assets. The risk-free rate is usually the rate on a riskless investment such as a short-term government bond.

(iii) The exercise price is determined by an equivalent zero coupon bond. LeverChem's debt is a variable rate loan; the yield is the current rate of 8%. The equivalent zero coupon debt with a term to maturity of five years is as follows:

$\$3$ million x $1{\cdot}08^{-5} = \$2{\cdot}04175$ million

To value the equity of the combined business as a call option we take the following as the inputs into the model:

The underlying assets of the business = $8·6 million assuming fair values

The exercise price = $2·04175 million

Risk free rate = 5%

Time to exercise = 5 years

Volatility = 0·35

Applying the BSOP Model:

$d_1 = [\ln(8{\cdot}6/2{\cdot}04175) + (0{\cdot}05 + 0{\cdot}5 \times 0{\cdot}35^2) \times 5]/(0{\cdot}35 \times 5^{1/2})$
$d_1 = 2{\cdot}548$

$d_2 = 2{\cdot}548 - (0{\cdot}35 \times 5^{1/2})$
$d_2 = 1{\cdot}765$

$N(d_1) = 0{\cdot}9946$

$N(d_2) = 0{\cdot}9612$

Call value = $\$8{\cdot}6m \times 0{\cdot}9946 - \$2{\cdot}04175 \times 0{\cdot}9612 \times e^{-0{\cdot}05 \times 5} = 8{\cdot}55356 - 1{\cdot}52842 = 7{\cdot}02514$

The value of the call on the combined firm's assets is approximately $7·025 million.

Deducting AggroChem's current equity

$7·025m – $4·799m = $2·226m

This suggests that the maximum price which should be paid to the shareholders in LeverChem is $2·226m.

Market value of LeverChem = $\$1,200,000 \times 1{\cdot}33^1/_3 = 1,600,000$

Maximum premium payable $626,000 or just over 39%.

In part (iii), the risk free rate could be taken as ln(1·05) = 0·0488 or 4.88%. If this rate is used, the answer changes as follows:

$N(d_1) = 0{\cdot}9945$

$N(d_2) = 0{\cdot}9605$

Call value = $7·017 million

Maximum price = $2·218 million

Premium = $618,000 or 38·6%.

(iv) Although the BSOP model is very effective in valuing continuously traded securities in active markets, its validity is more questionable where the basic assumptions supporting the model do not hold. It is assumed that the value of the assets of the business change randomly around a rising trend and are thus log-normally distributed. It is also assumed that they are traded in frictionless markets and the holding can be continuously adjusted. Finally, the model is only appropriate for European style options. Given these assumptions, the modelling can only be expected to give an indicative rather than a definitive value, but in the absence of any alternative is a useful adjunct to commercial judgement.

3 **(a)** In order to estimate the returns an annual cash account should be created showing the cash flow receivable from the pool of assets and the cash payments against the various liabilities created by the securitisation process. In this securitisation a degree of leverage has been introduced by the swap giving a return of 19·05% to the holders of the subordinate certificates but carrying a high degree of risk.

Cash flow receivable	$million	Cash flow payable		$million
$200 million x 10·5%	21·00	A-rated bonds	LIBOR	2·13
less service charge	0·24	$152 million at 1·4%		
		B-rated bonds		
		$19 million at 11%		2·09
		SWAP		
		Receive LIBOR	–LIBOR	
		Pay 8·5% on $152m		12·92
	20·76			17·14
		Balance to the subordinated certificates		3·62

The return to the holders of the certificates is $3·62 million on $19 million or 19·05%. A reduction of 1% in the cash flow receivable brings about a fall in the annual receivable to $20·79 million, reducing the balance available for the subordinated certificates to $3·41 million. A reduction of 1% in the cash flow receivable, results in a fall of 5·80% ([3·62 – 3·41]/3·62) in revenue of the subordinated certificates.

(b) A securitisation of this type is a common method of refinancing in a large business. After the securitisation of mortgages, car loan securitisation is the most important source of refinancing in the US and in Europe. Structured finance arrangements such as this are also used in a variety of other industries from banking to entertainment. The process of credit enhancement is the process whereby a relatively high risk cash flow (car loans) can be converted into a range of collateralised loan obligations satisfying the varying risk appetites of different investors. In the securitisation process a rating agency would normally advise on the structure of the liabilities created such that the AAA tranche will attract investors such as banks and other financial institutions who demand a low level of risk exposure. This reduction in risk for the senior and intermediate level notes is balanced by a significant transfer of risk to the subordinated certificate holders. Tranching the issue rather than creating a single issue of an asset backed security is the most important mechanism for credit enhancement.

Other approaches can entail insuring the risk of the issue through the use of credit default swaps or by transferring a greater asset pool than is securitised (over collateralisation).

(c) There are a number of risks inherent in the securitisation process faced by the investor:

Correlation risk: it is often assumed that defaults on the asset side of the securitisation process are uncorrelated. However, if a degree of positive correlation is present (such as defaulting car loans and repossessions) being positively associated with rising unemployment) then this can create higher than anticipated volatility in the receivables.

Timing and liquidity risk: the question only refers to average returns which presumably consist of a mixture of repayments, interest and possibly the anticipated recovery from repossessions. Modelling the cash flow 'waterfall' is a difficult issue where the timing of the cash receipts is crucial in fulfilling the commitments to the various tranches.

Default and collateral risk: the success of the securitisation will be dependent upon the assessment of the quality of the loans made to car purchasers. Dealers selling cars are responsible for the primary credit assessment and tight controls are necessary for ensuring that the loans are properly negotiated. Risk arises both in terms of default but also in the value of the vehicle on repossession.

From GoSlo's perspective the risks attaching to the process are more straightforward and to a certain extent depend on the motivation for the securitisation. If it is simply to refinance activity then the risk is that the issues will be undersubscribed. If it is also to remove the assets from the balance sheet then much will depend on whether loans of this type can be transferred to a special purpose vehicle such that full legal ownership passes. **(Note: this paragraph is not required to answer part (c), but has been included for tutorial purposes.)**

4 Corporate disposal of this type raises a number of issues. In deciding whether disposal is the appropriate course of action MandM should follow, the company needs to clarify its motives for disposal. The question suggests that there is no business case for retaining the LunarMint operation. However, the concept of what is core business and what is not, is a matter of judgement – and where that business is profitable, senior management need to be very careful in making that judgement. The question arises why did they diversify into this rather odd line of business for a money mint manufacturing company in the first place? Was it because printing money was not generating the returns required for the level of risk being carried?

The steps involved in the procedure for preparing a business such as this for sale are as follows:

1. A decision would need to be made about the nature of the assets being transferred and the process for resolving whether and how any joint assets might be sold. Fair value for all of the assets to be disposed of should then be established on the basis of an orderly sale as a going concern.
2. Checking the status of all intellectual property, that all patents are established and where necessary further valuable corporate knowledge, brand symbols and proprietary processes should be patented or protected by copyright.

3. Identification and valuation of the LunarMint contribution to the business. This may involve improving the business by a thorough operating and business process review, implementing any changes that might improve reported profitability and removing any impediments to a sale. It may be that the whole business is to be sold as a going concern or that only elements of the business will be disposed of. The valuation must assess the impact of the disposal upon the shareholder value in MandM both before and after a potential sale to identify a threshold value which will lead to no loss of shareholder value. Where the impact upon the firm's exposure to systematic risk is significant the valuation process will be recursive entailing sophisticated modelling of the potential loss of cash-flow to the group and the impact of the uncoupling on the firm's asset beta.
4. Any regulatory issues need to be clarified. Would sale to any of the potential purchasers conflict with the public interest and present problems gaining approval for the acquisition?
5. Potential buyers will need to be sought through open tender or through an intermediary. Depending upon the nature of the assets being sold a single bidder may be sought or preparations made for an auction of the business as a going concern. Members of the LunarMint supply chain and distribution channels may be interested, as may be competitors in the confectionary business. High levels of discretion are required in the search process to protect the value of the business from adverse competitive action. An interested and dominant competitor may open a price war in order to force down prices and hence the value of LunarMint prior to a bid.
6. Once a potential buyer has been found, access should be given so that they can conduct their own due diligence. Up-to-date accounts should be made available and all legal documentation relating to assets to be transferred made available.
7. The company should undertake its own due diligence to check the ability of the potential purchaser to complete a transaction of this size. Before proceeding it would be necessary to establish how the purchaser intends to finance the purchase, the timescale involved in their raising the necessary finance and any other issues that may impede a clean sale.
8. If not already involved the firm's legal team will need to assess any contractual issues on sale, the transfer of employment rights, the transfer of intellectual property and any residual rights and responsibilities to MandM. Normally, a clean separation should be sought unless an agreement is required concerning the use of joint assets.
9. In the light of the above and (3) in particular a sale price will be negotiated which will increase the shareholder value of LunarMint. The negotiation process should be conducted by professional negotiators who have been thoroughly briefed on the terms of the sale, the conditions attached and all of the legal requirements. The consideration for the sale, the deeds for the assignment of assets and terms for the transfer of staff and their accrued pension rights will also all be subject to agreement.
10. Once the sale has been agreed in principle it is important to address all of the employment issues which will include communicating with staff the reasons for the sale, the protection of their rights on transfer, the handling of any incentive payments including share options and the transfer of their pension rights. This step may involve discussion with unions and other employee representatives.
11. Given the size of the business being sold shareholder agreement may also be required and if so the process should be put in place to gain their approval.
12. Finally the contracts for sale and the completion documents can be exchanged and the sale completed.

5 (a) Determination of the currency transfers required

There are a number of ways this problem can be handled. Two methods are shown here, the first uses a transactions matrix and the second is a route minimisation algorithm.

Given that all balances are to be cleared through the European office, proceed as follows.

Convert all indebtedness between parties to Euros using the specified exchange rates:

	million	€ million
US$	6·40	4·67
S$	16·00	7·75
US$	5·40	3·94
Euros	8·20	8·20
US$	5·00	3·65
Rm	25·00	5·01
£	2·20	2·34
S$	4·00	1·94
Rm	8·30	1·66

Transactions Matrix

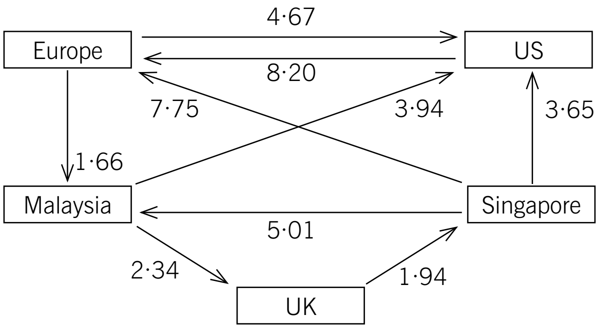
		Owed to					
		Europe	US	Malaysia	Singapore	UK	Owed by
Owed by	Europe		4·67	1·66			6·33
	US	8·20					8·20
	Malaysia		3·94			2·34	6·28
	Singapore	7·75	3·65	5·01			16·41
	UK				1·94		1·94
	Owed to	15·95	12·26	6·67	1·94	2·34	
	Owed by	6·33	8·20	6·28	16·41	1·94	
	Net	9·62	4·06	0·39	–14·47	0·40	

(All amounts are in € million)

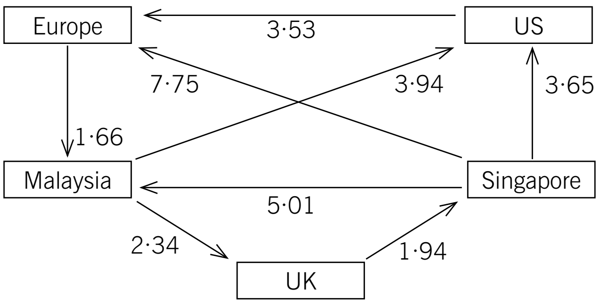
Multidrop (Europe) pays the US, UK and Malaysian business €4·06 million, €0·40 million, and €0·39 million respectively, and receives from Singapore €14·47 million. The net income is €9·62 million to Multidrop (Europe).

Route Minimisation Algorithm

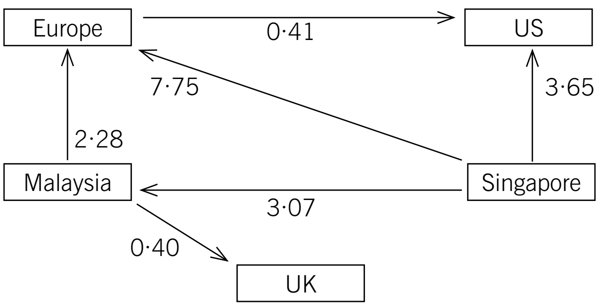
Step 1: Network of Indebtness



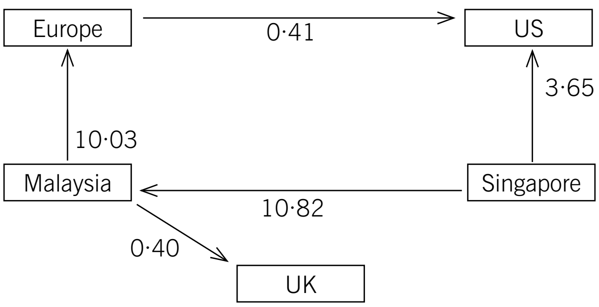
Step 2: Resolve any bilaterals



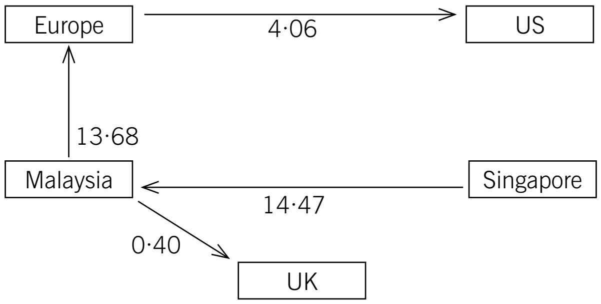
Step 3: Identify and clear circuits



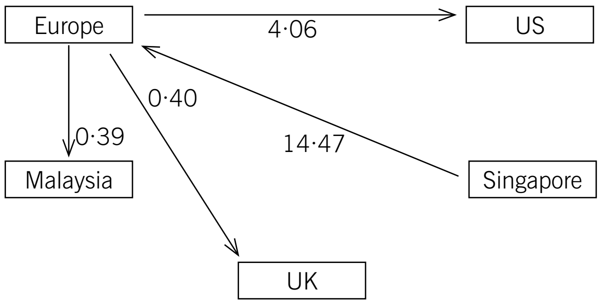
Step 4: Identify and clear cross indebtedness



And also:



Step 5: Finally resolve all indebtedness in favour of Europe



(All amounts are in € million)

(b) Netting Arrangements with the global business and trading partners

Netting is a mechanism whereby mutual indebtedness between group members or between group members and other parties can be reduced. The advantages of such an arrangement is that the number of currency transactions can be minimised, saving transaction costs and focusing the transaction risk onto a smaller set of transactions that can be more effectively hedged. It may also be the case, if exchange controls are in place limiting currency flows across borders, that balances can be offset, minimising overall exposure. Where group transactions occur with other companies the benefit of netting is that the exposure is limited to the net amount reducing hedging costs and counterparty risk.

The disadvantages: some jurisdictions do not allow netting arrangements, and there may be taxation and other cross border issues to resolve. It also relies upon all liabilities being accepted – and this is particularly important where external parties are involved. There will be costs in establishing the netting agreement and where third parties are involved this may lead to re-invoicing or, in some cases, re-contracting.

			Marks	
1	**(i)**	NPV calculation		
		Calculation of annuity factor with growth	2	
		Calculation of PV of construction costs	2	
		Calculation of PV of income	2	
		Calculation of PV of decommissioning costs	3	
		NPV and conclusion	2	
				11
	(ii)	Principal uncertainties		
		Uncertainty associated with CAPEX	2	
		Uncertainty associated with operating surplus	2	
		Uncertainty associated with decommissioning costs	2	
		Uncertainty associated with the discount rate	2	
		Uncertainty associated with real options	2	
		(Credit will be given for alternative, relevant points)	Max	7
	(iii)	Sensitivity to construction costs	3	
		Sensitivity to annual operating surplus	1	
		Sensitivity to decommissioning cost	1	
		Conclusion	1	
				6
	(iv)	Volatility assessment		
		Role of simulation	1–2	
		Process of simulation and its limitations	1–2	
		Outputs (distribution and sensitivities)	1–2	
		(Credit will be given for alternative, relevant points)	Max	4
			Total	**28**

				Marks
2	(i)	Annual free cash flow	1	
		Retention rate	1	
		Cost of equity and growth rate	2	
		Calculation of market value	2	
		Assumptions (1 mark per point, max 3)	3	
				9
	(ii)	Explanation of when it is appropriate to use BSOP to value a company	1–2	
		Explanation of the proxy for the exercise price, the PV of a zero coupon bond	1–2	
		Explanation of the underlying assets	1–2	
		Explanation of the other inputs for the BSOP model	1–2	
			Max	5
	(iii)	Calculation of the PV of the zero coupon bond	2	
		Calculation of $N(d_1)$	3	
		Calculation of $N(d_2)$	1	
		Calculation of call value/value of company	1	
		Maximum price	1	
		Maximum premium	1	
				9
	(iv)	Introductory explanation why BSOP may be limited in determining the value of a company	1–2	
		Explanation of the assumptions made for BSOP	2–4	
		Conclusion about indicative rather than definitive value	1–2	
			Max	5
		Professional marks		4
			Total	**32**
3	(a)	Calculation of expected return from pool (inc service charge and over-collateralisation)	2	
		Impact of swap on A-tranche	2	
		Calculation of annual cash flows payable on interest for tranches A-rated and B-rated	2	
		Estimation of return to subordinated certificates	1	
		Impact of marginal change and calculation of sensitivity	3	
				10
	(b)	Explanation and purpose of securitisation	1–2	
		Methods of credit enhancement:		
		Tranching		
		Ratings agency involvement		
		Over-collateralisation		
		Others	2–3	
			Max	4
	(c)	Correlation risk	2–3	
		Timing and liquidity risk	1–2	
		Collateral risk	2–3	
		(Credit will be given for alternative, relevant points)	Max	6
			Total	**20**

			Marks	
4		Clarification of assets under sale and purpose	1–2	
		Status of IP and extent and remedy of defects in protection	1–2	
		Valuation and note on recursion problem	2–3	
		Identification of regulatory issues	1–2	
		Search for, and resolving issues involving, potential buyers	2–3	
		Due diligence: Both seller and buyer	3–4	
		Contracting and legal issues	1–2	
		Price negotiation	2–3	
		Employment and transfer of rights and pension	2–3	
		Investor agreement	1	
		Contract and completion	1	
		(Credit will be given for alternative, relevant points)	**Max**	**20**

5	**(a)**	**EITHER: Using Transactions Matrix Method**		
		Set principal as European business	1	
		Conversion to Euros	3	
		Initial amounts owed and owing	2	
		Totals owed and owing	2	
		Net amounts owed	2	
		Conclusion: Payments and receipts to Multidrop (Europe)	2	
				12
		OR: Using Route Minimisation Algorithm Method		
		Set principal as European business	1	
		Conversion to Euros	3	
		Step 1: Network of Indebtness	2	
		Steps 2 and 3: Resolve bilaterals and resolve circuits	2	
		Steps 4 and 5: Clear cross Indebtness and resolve in favour of Multidrop (Europe)	2	
		Conclusion: Payments and receipts to Multidrop (Europe)	2	
				12
	(b)	Minimisation of number of transactions	1–2	
		Minimisation of cost of transacting	1–2	
		Avoidance of exchange controls	1–2	
		Minimisation of hedging costs	1–2	
		Limiting exposure to net	1–2	
		Taxation issues	1–2	
		Acceptance of liability	1–2	
		Re-invoicing and re-contracting	1–2	
			Max	8
			Total	**20**

Answers

1 Up to 4 professional marks are available for the presentation of the answer, which should be in a report style.

The decision should be taken in the best interests of the shareholders and other stakeholders of the company.

Obviously all the input numbers should be considered on basis of their reasonableness and accuracy.

Hence the positions of the interested parties will be considered on this basis.

1. **Cease Trading and Liquidate the Company**

This is probably not in the best interest of any party. Debt holders only receive 55·7c for every $1 invested and the shareholders receive nothing (see appendix, proposal 1). Furthermore, the parts division is continuing to make a profit and should possibly continue.

The Board may want to consider closing just the fridge division, and focusing on the parts manufacturing division, with the possibility of pursuing the option of the mobile refrigeration business. However, in this case, the problem of the lack of funding might continue.

2. **Corporate Restructuring and Management Buy-Out**

Shareholders
The shareholders would benefit from either proposal two or three, as opposed to the first proposal, as they stand to gain some funds. The restructuring proposal requires them to pay $40m cash for new shares but lose their control of the company (the shareholding falls to just under 13%). On the other hand the statement of financial position looks robust with a $20m cash float and bank overdraft facilities probably available at previous levels (see appendix, proposal 2). This may make the company more successful in the future, as directors are less restricted by covenants. The value at $256·3m currently only gives existing shareholders a share (or stake) of about $33·3m (13% x $256·3m), which is less than the amount they would inject.

The shareholders would benefit immediately if the management buy-out option is taken because they will receive a premium on the share price, although this may still be lower than when the company's share price was at its height.

Therefore the shareholders need to weigh up whether they would like to possibly benefit from future company prospects (not evident at the moment) or whether they would like to sell their shares for 60c per share. They would probably opt for the management buy-out.

Unsecured Bond Holders
The unsecured bond holders' position is not dissimilar to the shareholders' position in that with the restructuring, their financial position depends on the future success of the company, but with the management buy-out they benefit from receiving the full repayment of their initial investment. However, their preferred proposal is probably more difficult to judge.

With the restructuring option they would become the majority shareholder with just over 87% of the company for a total investment of $210m. They would be able to play a major part in influencing the management's decision possibly with representation on the Board. However, they would be exposed to additional risk as equity holders, as opposed to being debt holders.

The value attributable to them based on perpetuity cash flows is $223m ($256·3 x 87%) approximately, which is more than their investment. Like the shareholders they would benefit from any future projects that the re-structured business undertakes. They would also receive the shares at a significant discount, 270m shares for $210m which is 77·7c per $1 par value. The ability to influence the Board and the possibility of obtaining a higher return than their investment from almost the start may sway them to accept the restructuring option. On the other hand, the management buy-out pays them what is due immediately, but they cannot participate in future benefits.

Bond holders may therefore be more tempted to opt for the restructuring when compared to shareholders.

Directors and Management Participating in the Management Buy-Out
If the restructuring is considered as opposed to liquidation then clearly a significant benefit to the management and directors is that they would retain their employment, unless the new shareholder owners decide to terminate some of their contracts.

The possibility of the offer of the share options is interesting. At first the $1·10 exercise price may seem generous as the directors would be able to exercise when the price of shares increase by just 10%. However, it is unlikely that the share price will start at $1 per share. If the estimate of the value to perpetuity of $256·3m is taken against the total number of shares of 310m, this gives a theoretical share price of 82·7c per share. This means that the share price needs to increase by over 33% before the option will become in-the-money. The option is currently well out-of-money and would have a low value. Given the asymmetric payoff of the option and the need to increase the price dramatically, directors and managers may be tempted to act in an excessively risky manner, to the detriment of the shareholders and other stakeholders. Indeed research has shown that the presence of share options in individual pay structure do make option holders behave in a more risky manner compared to pay structures which do not contain options.

The management buy-out may influence different classes of managers and directors very differently. For those who participate in the management buy-out, the calculations seem to indicate a clear benefit (see appendix, proposal 3) of a gain net of the cost of the buy-out. It would seem that by having a 5% growth, the value has increased to more than 50% of the initial cost of the buy-out, although some unreasonable assumptions have been made. For managers and directors not participating in the buy-out, it is likely that they will lose their employment once the fridge division is sold.

Conclusion
Given that the shareholders will probably prefer the management buy-out, and as it appears to have a significant advantage for the participating managers, it is likely that this will be the option that is preferred. Although some parties may not approve of the option, it is unlikely that their 'voice' will be strong enough to alter the decision.

APPENDIX: Calculatons for each option

Proposal 1: Cease Trading

Estimated Break-up Values of Assets	**$m**
Land and Buildings	60
Machinery and Equipment	40
Inventory	90
Receivables	20
Total	210
Less redundancy and other costs	(54)
	156
Current and non-current liabilities	
Payables	70
Bank overdraft	60
Unsecured loan stock	120
Other unsecured loans	30
	280

The current and non-current liabilities will receive 55·7c per $1 owing to them (156/280). Shareholders will receive nothing.

Proposal 2: Restructure

Financial Position	**$m**
Non-Current Assets	
Land and buildings	70
Machinery and equipment (assuming all new investment is here)	130
	200
Current Assets	
Inventory	180
Receivables	40
Cash ($40m + $90m – $80m – $30m)	20
	240
Total Assets	440
Current Liabilities	
Payables	70
	70
Non-Current Liabilities	
Bank loan (7% interest)	60
	60
Share capital ($1 per share par value, 40m + 270m)	310
	310
Total Liabilities and capital	440

Income position	**$m**
Sales revenue (510 x 1·07)	545·7
Costs prior to depreciation, interest payments and tax	(490)
Tax allowable depreciation (15% x 200)	(30)
Finance cost (interest) (7% x 60)	(4·2)
Tax	(4·3)
Profit	17·2

Note: Even after the 7% growth, the fridge division is still loss-making. Revenue = $340 x 1·07 = $363·8m, costs (unchanged) = $370m.

Cash position	**$m**
Profit before interest and tax	25·7
Tax (20%)	(5·14)
Net cash flow	20·56

Note: depreciation is not added back because it is the same amount as needed to maintain operations.

Value of company after re-structuring

Cost of capital = 6·5%

Estimated value based on cash flows to perpetuity
$20·56m/0·065 = $316·3m

Value attributable to shareholders (net of bank loan) = $316·3 – $60m = $256·3m

Proposal 3: Management Buy-out

	$m
Value of selling fridge division (2/3 x 210 (proposal 1))	140
Redundancy and other costs (2/3 x 54)	(36)
Funds available from sale of division	104
Amount of current and non-current liabilities (proposal 1)	280
Amount of management buy-out funds needed to pay current and non-current liabilities	176
Amount of management buy-out funds needed to pay shareholders	60
Equipment needed to increase sales by 7% ($^1/_3$ of 80m)	26.7
Investment needed for new venture	50
Total funds needed for management buy-out	312·7

Estimating value of new company after buy-out	**$m**
Sales revenue (70%* x 170m x 1·07)	127·3
Costs (70%* x 120)	(84·0)
Profits before depreciation	43·3
Growth in profits due to new venture (5%)	2·2
Depreciation (1/3 x 200 x 15%)	(10)
Tax (20%)	(7·1)
Cash flows before interest payment	28·4

Note: It is assumed that, as before, the depreciation and the amount of capital investment needed are roughly equal. It is assumed that no additional investment in non-current assets or working capital is needed, even though sales revenue is increasing. It is assumed that additional initial working capital requirement is part of the new venture investment of $50m.

*It is assumed that 30% of the sales revenue lost due to the closure of the fridge division is not recovered.

Cost of capital = 11%

Estimated value based on cash flows to perpetuity = 28·4/(0·11 – 0·05) = $473·3m

This is over 50% in excess of the funds invested in the new venture.

2 (a) Base Case Net Present Value

Fubuki Co: Project Evaluation

Base Case	$'000 Unit Price/cost	Inflation	Now	Year 1	Year 2	Year 3	Year 4
Units Produced and sold				1,300	1,820	2,548	2,675
Sales revenue	2·5	3%		3,250	4,687	6,758	7,308
Direct costs	1·2	8%		1,560	2,359	3,566	4,044
Attributable fixed costs	1,000	5%		1,000	1,050	1,103	1,158
Profits				690	1,278	2,089	2,106
Working capital	15%		(488)	(215)	(311)	(82)	1,096
Taxation (w1)				(10)	(157)	(360)	(364)
Incremental cash flows							
Investment/sale			(14,000)				16,000
Net cash flows			(14,488)	465	810	1,647	18,838
Present Value (10%) (w2)			(14,488)	422	670	1,237	12,867
Base case NPV			708				

Working (w1)

	Year 1	Year 2	Year 3	Year 4
Profits	690	1,278	2,089	2,106
Less: allowances	650	650	650	650
Taxable profits	40	628	1,439	1,456
Tax	10	157	360	364

Note: Full credit will be given where the assumption is made that allowances are 750 in the first three years and 350 in the final year.

Working (w2)

Discount rate (Haizum's ungeared Ke)
ke(g) = ke(u) + (1– t)(ke(u) – kd)Vd/Ve
Ve = 2·53 x 15 = 37·95
Vd = 40 x 0·9488 = 37·952
Assume Vd/Ve = 1

14 = ke(u) + 0·72 x (ke(u) – 4·5) x 1
14 = 1·72ke(u) – 3·24
ke(u) = 10·02 assume 10%

Note: The discount rate can be estimated by calculating the asset beta and then using that to estimate the cost of equity.

The base case net present value is calculated as approximately $708,000. This is positive but marginal.

The following financing side effects apply

	$'000
Issue costs 4/96 x $14,488	(604)
Tax Shield	
Annual tax relief = (14,488 x 80% x 0·055 x 25%)	
+ (14,488 x 20% x 0·075 x 25%)	
= 159·4 + 54·3 = 213·7	
213·7 x 3·588	766
Subsidy benefit	
14,488 x 80% x 0·02 x 75% x 3·588	624
Total benefit of financing side effects	786
Adjusted present value (708 + 786)	1,494

Note: Full credit will be given if the issue costs are included in the funds borrowed.

The addition of the financing side effects gives an increased present value and probably the project would not be considered marginal. Once these are taken into account Fubuki Co would probably undertake the project.

Note: In calculating the present values of the tax shield and subsidy benefits, the annuity factor used is based on 4·5% debt yield rate for four years. It could be argued that 7·5% may also be used as this reflects the normal borrowing/default risk of the company.

Credit will be given where this assumption is made to estimate the annuity factor.

(b) The adjusted present value can be used where the impact of using debt financing is significant. Here the impact of each of the financing side effects from debt is shown separately rather than being imputed into the weighted average cost of capital. The project is initially evaluated by only taking into account the business risk element of the new venture. This shows that although the project results in a positive net present value, it is fairly marginal and volatility in the input factors could turn the project. Sensitivity analysis can be used to examine the sensitivity of the factors. The financing side effects show that almost 110% value is added when the positive impact of the tax shields and subsidy benefits are taken into account even after the issue costs.

Assumptions (Credit given for alternative, valid assumptions)

1. Haizum Co's ungeared cost of equity is used because it is assumed that this represents the business risk attributable to the new line of business.
2. The ungeared cost of equity is calculated on the assumption that Modigliani and Miller's (MM) proposition 2 applies.
3. It is assumed that initial working capital requirement will form part of the funds borrowed but the subsequent requirements will be available from the funds generated from the project.
4. The feasibility study is ignored as a past cost.
5. It is assumed that the five-year debt yield is equivalent to the risk-free rate.
6. It is assumed that the annual reinvestment needed on plant and machinery is equivalent to the tax allowable depreciation.
7. It is assumed that all cash flows occur at the end of the year unless specified otherwise.
8. All amounts are given in $'000 to the nearest $'000. When calculating the units produced and sold, the nearest approximation for each year is taken.

Assumptions 4, 5, 6, 7 and 8 are standard assumptions made for a question of this nature. Assumptions 1, 2 and 3 warrant further discussion. Taking assumption 3 first, it is reasonable to assume that before the project starts, the company would need to borrow the initial working capital as it may not have access to the working capital needed. In subsequent years, the cash flows generated from the operation of the project may be sufficient to fund the extra working capital required. In the case of Fubuki Co, because of an expected rapid growth in sales in years 2 and 3, the working capital requirement remains high and the management need to assess how to make sufficient funds available.

Considering assumptions 1 and 2, the adjusted present values methodology assumes that MM proposition 2 applies and the equivalent ungeared cost of equity does not take into account the cost of financial distress. This may be an unreasonable assumption. The ungeared cost of equity is based on another company which is in a similar line of business to the new project, but it is not exactly the same. It can be difficult to determine an accurate ungeared cost of equity in practice. However, generally the discount rate (cost of funds) tends to be the least sensitive factor in investment appraisal and therefore some latitude can be allowed.

3 **(a)** The number of put contracts to purchase depends on the hedge ratio, which in turn depends on the delta of the option. This measures the change in the option price over the change in the price of the share, and therefore helps determine how many option contracts are needed to protect against a fall in the share price. For put options an estimate of the delta is given by $N(-d_1)$.

$d_1 = [\ln(P_a/P_e) + (r + 0{\cdot}5s^2)t]/(st^{1/2})$ (from formulae sheet given in examination)

$d_1 = [\ln(340/350) + ((0{\cdot}04 + 0{\cdot}5 \times 0{\cdot}4^2) \times 1/6)]/(0{\cdot}4 \times 1/6^{0{\cdot}5}) = -0{\cdot}055$

$-d_1 = 0{\cdot}055$

$N(-d_1) = 0{\cdot}5 + (0{\cdot}0199 + (0{\cdot}0239 - 0{\cdot}0199)/2) = 0{\cdot}5219$

Put contracts to be bought for a delta hedge = 200,000/(0·5219 x 1,000) = 383·2 rounding to 383 contracts.

(b) Wenyu's position is based on the theoretical case put forward for not managing corporate risk. In a situation of market efficiency where information is known and securities are priced correctly, holding well diversified portfolios will eliminate (or at least significantly reduce) unsystematic risk. The position against hedging states that in such cases companies would not increase shareholder value by hedging or eliminating risk because there will be no further reduction in unsystematic risk. In a situation of perfect markets, the cost of reducing any systematic risk will exactly equal the benefit derived from such a reduction. Shareholders would not gain from risk management or hedging, in fact if the costs exceed the benefits due to transactional costs then hedging may result in a reduction in shareholder value.

However, hedging or the management of risk may result in increasing corporate (and therefore shareholder) value if market imperfections exist, and in these situations reducing the volatility of a company's earnings will result in higher cash inflows. Proponents of hedging cite three main situations where reduction in volatility may increase cash flows – in situations: where the rate of tax is increasing; where a firm could face significant financial distress costs due to high volatility in earnings; and where stable earnings increases certainty and the ability to plan for the future and therefore resulting in stable investment policies by the firm.

It would appear that none of the reasons for hedging explains the situation described in the scenario of the question. Given that Marengo Co is a large company with a variety of investments, it is unlikely that reducing the volatility of one investment will significantly alter the cash flows of the company. There could be other reasons for reducing risk through hedging and these are explored further on.

Lola's proposal of selling Arion Co shares is based on the fact that such a move would eliminate the risk of a reduction in price altogether. This proposal has a number of limitations which the managers need to consider carefully. Presumably the purpose of Marengo Co's strategy of investing in companies is to generate value from such activity. It is likely that it generates higher value than the risk-free rate of return (which would be obtained if funds were invested in 'no-risk' investments). The funds generated from selling Arion Co shares will need to be invested elsewhere to generate the target returns. Unless investments are available which seem to be better than Arion Co, there may not be a case for selling Arion shares. In addition to this, selling such a large quantity of shares can potentially make the share price reduce significantly and the managers would need to investigate whether there is sufficient liquidity in the markets for such a large sale. Finally, the sale may unbalance the investment portfolio.

Sam's proposal would entail purchasing OTC put options from the bank. This can be an expensive alternative as purchasing a right would entail having to pay premiums on the options, which can be substantial. If the option is not exercised then the cost will be the full amount of the premium. On the other hand, options will allow Marengo Co to be protected against downside movements, whilst still benefiting from positive movements in the share price. However, options bring additional risks with them. For example, delta is not stable and the rate of change in an option is measured by the gamma. The option value also changes as time to expiry reduces (measured by theta) and as volatility of the underlying asset changes (measured by vega).

Active hedging may reduce agency costs. For example, unlike shareholders, managers and employees of the company may not be diversified. Hedging allows the risks exposed to them to be reduced. Additionally hedging may allow managers to not be concerned about market movements which are not within their control and instead allow them to focus on business issues on which they can exercise control. A consistent hedging strategy or policy may be used as a signalling tool to reduce the conflict of interest between bondholders and shareholders, and thus reduce restrictive covenants. Although a single transaction, like this scenario, may have little impact on this, it should be part of the overall risk management policy of the company.

The case for hedging or not is not clear cut and should not be taken on an individual or piecemeal basis. Instead the company should consider its overall risk management strategy and the resultant value creation opportunities. Subsequent hedging decisions should be based on the overall strategy.

(Note: Credit will be given for alternative relevant approaches)

4 (a) Dividend Capacity Prior to TE Proposal Implementation

	$000
Operating profit (30% x $80,000,000)	24,000
Less interest (8% x $35,000,000)	(2,800)
Less taxation (28% x (24,000 – 2,800))	(5,936)
Less investment in working capital (15% x (20/120 x 80,000))	(2,000)
Less investment in additional non-current assets (25% x (20/120 x 80,000))	(3,333)
Less investment in project	(4,500)
Cash flows from domestic operations	5,431
Cash flows from overseas subsidiary dividend remittances (W1)	3,159
Additional tax payable on Magnolia profits (6% x 5,400)	(324)
Dividend capacity	8,266

Dividend Capacity After TE Proposal Implementation

Cash flows from domestic operations (as above)	5,431
Cash flows from overseas subsidiaries dividend remittances (W2)	2,718
Additional tax payable on Magnolia profits (6% x 3,120)	(187)
Dividend capacity	7,962

Estimate of actual dividend for coming year
(7,500 x 1·08) 8,100

Note: The impact of depreciation is neutral, as this amount will be spent to retain assets at their current productive capability.

Workings

W1: Prior to Implementation of TE Proposal

	$000 Strymon	$000 Magnolia
Sales revenue	5,700	15,000
Cost		
Variable	(3,600)	(2,400)
Fixed	(2,100)	(1,500)
Transfer		(5,700)
Profit before tax	Nil	5,400
Tax	Nil	1,188
Profit after tax	Nil	4,212
Remitted	Nil	3,159
Retained	Nil	1,053

W2: After Implementation of TE Proposal

	$000 Strymon	$000 Magnolia
Sales revenue	7,980	15,000
Cost		
Variable	(3,600)	(2,400)
Fixed	(2,100)	(1,500)
Transfer		(7,980)
Profit before tax	2,280	3,120
Tax (42%, 22%)	958	686
Profit after tax	1,322	2,434
Remitted (75% x 1,322 x 90%)	892	
Remitted		1,826
Retained	331	608
Total remitted	2,718	

(b) Lamri's dividend capacity before implementing TE's proposal ($8,266,000) is more than the dividend required for next year ($8,100,000). If the recommendation from TE is implemented as policy for next year then there is a possibility that Lamri will not have sufficient dividend capacity to make the required dividend payments. It requires $8,100,000 but will have $7,962,000 available. The reason is due to the additional tax that will be paid in the country in which Strymon operates, for which credit can not be obtained. Effectively 14% additional tax and 10% withholding tax will be paid. Some of this amount is recovered because lower additional tax is paid on Magnolia's profits but not enough.

The difference between what is required and available is small and possible ways of making up the shortfall are as follows. Lamri could lower its growth rate in dividends to approximately 6·2% (7962/7500 – 1 x 100%) and have enough capacity to make the payment. However, if the reasons for the lower growth rate are not explained to the shareholders and accepted by them, the share price may fall.

An alternative could be to borrow the small amount needed possibly through increased overdraft facilities. However, Lamri may not want to increase its borrowings and may be reluctant to take this option. In addition to this, there is a possibility that because of the change of policy this shortfall may occur more often than just once, and Lamri may not want to increase borrowing regularly.

Lamri may consider postponing the project or part of the project, if that option were available. However, this must be considered in the context of the business. From the question narrative, the suggestion is that Lamri have a number of projects in the pipeline for the future. The option to delay may not be possible or feasible.

Perhaps the most obvious way to get the extra funds required is to ask the subsidiary companies (most probably Strymon) to remit a higher proportion of their profits as dividends. In the past Strymon did not make profits and none were retained hence there may be a case for a higher level of remittance from there. However, this may have a negative impact on the possible benefits, especially manager morale.

(Note: credit will be given for alternative relevant suggestions)

5 (**Solution note:** Question 5 can be answered in a variety of ways and the suggested answer below is indicative. Credit will be given for reasonable answers that take a different approach. The question asks for a discussion of whether or not PMU should undertake a joint venture. It does not require candidates to discuss whether or not PMU should venture into Kantaka. Credit will not be given for answers which discuss this.)

There are a number of benefits that PMU can potentially take advantage of if it went into a joint venture with a Kantaka academic institution, such as sharing of risks, possible lower running costs and the partner's existing experience of the local market and lower capital investment costs. These are balanced against the potential disadvantages of going into a joint venture such as loss of reputation, product quality and staffing, government restrictions, cultural differences, managerial issues, contractual issues and loss of tax concessions. Each of these issues will be discussed, together with strategies for mitigation of the disadvantages and other information that is required.

PMU does not have the experience of doing business overseas and in particular in Kantaka. Having a partner to guide PMU on the local market and its expectations would be beneficial. It may be able to assist PMU in determining how to market the degree programmes effectively. The partner may also be able to advise PMU on pricing decisions and on how to minimise costs. A well constructed contract could be instrumental in effective risk sharing in case the demand and revenues do not grow as expected.

The capital investment cost is lower if PMU enters into a joint venture. Presumably PMU would have access to the partner's infrastructure and systems. They may also be able to utilise the expertise of the local academic and administrative staff. Training costs may also be lower as the academic staff may have the required level of expertise.

PMU may have an easier access to the local capital markets. This may be particularly beneficial where revenues are matched with the costs of capital. If the fee income is in the Kantaka currency but interest is payable in the Rosinante currency, then there is a potential for long-term currency exposure. This may be serious if the Kantaka currency continues to weaken. Although some economic data suggests that this may not be the case, it is not very persuasive. It may therefore be beneficial to borrow money using the local currency markets.

Perhaps the most significant disadvantage of progressing with a joint venture is the loss of reputation. It seems that due to historical reasons the Kantaka government may not treat the joint venture favourably by not recognising the degrees issued and not allowing the graduates to seek employment in government controlled organisations. In addition to this, joint ventures have a low reputation with the local population. PMU needs to consider its wider reputation as well. For example, there may be negative publicity if the institution chosen has a poor reputation. In order to mitigate this, PMU needs to choose a partner carefully through detailed due diligence. It may consider linking with an institution with high reputation and perhaps meet with and petition the government to give public and official backing to its degree programmes. However, this may take considerable time and effort. PMU also needs to consider whether the government will recognise its degrees if it sets up its own site. It is not clear from the question whether this is the case.

Linked to this is the quality of the product. Students would have certain expectations of the institution and the quality of the degrees needs to match this. If PMU enters into a joint venture with the expectation of using its partner's infrastructure, systems and staff, it needs to ensure that these match the expectations of its customers. For example, if it uses local staff to deliver its degrees then these staff must be properly trained to ensure that the correct standards are achieved. It may be the case that training costs make it more expensive than getting staff to come from Rosinante. It may be the case that initially a higher proportion of the staff are from Rosinante and then the numbers are lowered as the local staff gain the required skills.

PMU also needs to consider government restrictions other than the ones described above. For example, would the government issue visas for staff from Rosinante and would the government allow repatriation of funds as easily from a joint venture as from PMU's own investment? PMU would need to meet with government officials or seek legal advice to clarify situations such as these.

PMU needs to ensure that cultural differences between expatriate staff from Rosinante and local staff in Kantaka are minimised and they all share common corporate values. The two organisational cultures may be very different. It would require time and resources to get a common ground between the two organisations and a shared identity. Strategies need to be developed and enacted to ensure that this happens, perhaps through implementation of staff exchange programmes and secondments. Human resources need to develop techniques to help expatriate staff settle into the country. The costs related to all these need to be compared with PMU setting up its own site.

Managers' actions may be restricted in the case of joint ventures because the opinions of both the partners need to be taken into account. Managers of the joint venture may feel that they are not being listened to by the management at PMU. And the managers at PMU may feel that the actions of the managers at the partner are incongruent with aims of PMU. Clear guidelines need to be developed and communicated to limit dysfunctional behaviour between the managers, perhaps asking all the parties concerned to be involved in the development of the joint venture.

There may be difficulties in agreeing to the terms and conditions of the contract between the parties involved. Legal representation and clear communication between the senior management would be needed from both parties to agree: clear terms, conditions and boundaries; the roles and responsibilities of both sides; and, the ownership percentage and profit sharing arrangements.

Finally, if PMU proceeds with financing the joint venture using local funding, it may lose the tax concessions attached to any FDI. The financial consequences of this need to be assessed.

Additional Information

1. Financial assessment of both positions to include for example the consequences of loss of tax concessions. Sensitivity analysis of different projections and an assessment of the likelihood of their occurrence.
2. Possibility of hiring experts to advise PMU to set up in Kantaka, as opposed to going for a joint venture.
3. Outcomes from the government discussions on whether or not PMU degrees will get recognition in Kantaka.
4. Assessment of whether the partner's infrastructure and systems meet PMU's requirements.
5. Likely movement in the Kantaka currency and instruments such as swaps to hedge currency fluctuations.
6. The reputation of a PMU degree with Kantaka's people.
7. Quality of local staff, their ability to teach to the methods required and the need for expatriate staff from Rosinante.
8. An assessment of government restrictions on for example, issuing visas, repatriation of funds, etc.

				Marks
1	**Calculations**			
	(i)	Value if company is broken up and liquidated		3
	(ii)	Restructure Option		
		Income position		3
		(Note: the financial position is not required but helps the discussion)		
		Calculation of cash flows and valuation of business		5
		(Assumption(s) should be included)		
			Total	**8**
	(iii)	Management Buy-out		
		Funds required		4
		Calculation of value		4
		(Assumption(s) should be included)		
			Total	**8**
	(iv)	Discussion		
		Initial comment		1–2
		Break up comment and suggestions for alternatives		1–2
		Shareholders' position		2–3
		Unsecured loan stock holders' position		2–3
		Directors' and managers' position		3–4
		Conclusion		1
			Max	**12**
	Professional Marks			
	Report format			1
	Layout, presentation and structure			3
			Total	**4**
			Total	**35**
2	**(a)**	Sales revenue, direct costs and additional fixed costs		4
		Incremental working capital		1
		Taxation		2
		Estimation of Ke (ungeared)		2
		Net cash flows, present value and base case NPV		2
		Issue costs		1
		Calculation of tax shield impact		2
		Calculation of subsidy impact		1
		Adjusted present value and conclusion		2
			Total	**17**
	(b)	Discussion of using APV		2–3
		Assumption about Haizum as proxy and MM proposition 2		3–4
		Other assumptions		2–3
			Max	**8**
			Total	**25**

			Marks
3	**(a)**	Identifying the need to calculate $N(d_1)$ for hedge ratio	2
		Calculation of d_1	2
		Calculation of $N(-d_1)$	2
		Calculation of number of put options required	1
		Total	**7**
	(b)	Discussing the theoretical argument for not hedging	3–4
		Discussing the limitations/risks/costs of selling the shares	3–4
		Discussing the risks/costs of using OTC options to hedge	2–3
		Discussing the potential benefits of hedging and application to scenario	2–3
		Relevant concluding remarks	1–2
		Max	**13**
		Total	**20**
4	**(a)**	Calculation of operating profit, interest and domestic tax	3
		Calculation of investments in working capital and non-current assets (including correct treatment of depreciation)	3
		Calculation of dividend remittance before new policy implementation	2
		Calculation of additional tax payable on Magnolia profits before new policy implementation	1
		Calculation of dividend remittance after new policy implementation	3
		Calculation of additional tax payable on Magnolia profits after new policy implementation	1
		Dividend capacity	1
		Total	**14**
	(b)	Concluding comments and explanation of reason	2
		Possible actions (1 mark per suggestion)	4
		Total	**6**
		Total	**20**
5		Benefits of joint venture (1 to 2 marks per well explained point)	4–6
		Disadvantages of joint venture, including ways of mitigating disadvantages (2 to 3 marks per well explained point)	10–12
		Additional information (1 mark per point)	3–5
		Max	**20**

Answers

1 Up to 4 professional marks are available for the presentation of the answer, which should be in a report style.

(i) The calculations and estimations for part (i) are given in the appendix. To assess whether or not the acquisition would be beneficial to Pursuit's shareholders, the additional synergy benefits after the acquisition has been paid for need to be ascertained.

The estimated synergy benefit from the acquisition is approximately $9,074,000 (see appendix), which is the post-acquisition value of the combined company less the values of the individual companies. However, once Fodder Co's debt obligations and the equity shareholders have been paid, the benefit to Pursuit Co's shareholders reduces to approximately $52,000 (see appendix), which is minimal. Even a small change in the variables and assumptions could negate it. It is therefore doubtful that the shareholders would view the acquisition as beneficial to themselves or the company.

(ii) The limitations of the estimates stem from the fact that although the model used is theoretically sound, it is difficult to apply it in practice for the following reasons.

The calculations in part (i) are based on a number of assumptions such as the growth rate in the next four years, the perpetual growth rate after the four years, additional investment in assets, stable tax rates, discount rates and profit margins, assumption that debt is risk free when computing the asset beta. All these assumptions would be subject to varying margins of error.

It may be difficult for Pursuit Co to assess the variables of the combined company to any degree of accuracy, and therefore the synergy benefits may be hard to predict.

No information is provided about the pre-acquisition and post-acquisition costs.

Although it may be possible to estimate the equity beta of Pursuit Co, being a listed company, to a high level of accuracy, estimating Fodder Co's equity beta may be more problematic, because it is a private company.

Given the above, it is probably more accurate to present a range of possible values for the combined company depending on different scenarios and the likelihood of their occurrence, before a decision is made.

(iii) The current value of Pursuit Co is $140,000,000, of which the market value of equity and debt are $70,000,000 each. The value of the combined company before paying Fodder Co shareholders is approximately $189,169,000, and if the capital structure is maintained, the market values of debt and equity will be approximately $94,584,500 each. This is an increase of approximately $24,584,500 in the debt capacity.

The amount payable for Fodder Co's debt obligations and to the shareholders including the premium is approximately $49,116,500 [4,009 + 36,086 x 1·25]. If $24,584,500 is paid using the extra debt capacity and $20,000,000 using cash reserves, an additional amount of approximately $4,532,000 will need to be raised. Hence, if only debt finance and cash reserves are used, the capital structure cannot be maintained.

(iv) If Pursuit Co aims to acquire Fodder Co using debt finance and cash reserves, then the capital structure of the combined company will change. It will also change if they adopt the Chief Financial Officer's recommendation and acquire Fodder Co using only debt finance.

Both these options will cause the cost of capital of the combined company to change. This in turn will cause the value of the company to change. This will cause the proportion of market value of equity to market value of debt to change, and thus change the cost of capital. Therefore the changes in the market value of the company and the cost of capital are interrelated.

To resolve this problem, an iterative procedure needs to be adopted where the beta and the cost of capital are recalculated to take account of the changes in the capital structure, and then the company is re-valued. This procedure is repeated until the assumed capital structure is closely aligned to the capital structure that has been re-calculated. This process is normally done using a spreadsheet package such as Excel. This method is used when both the business risk and the financial risk of the acquiring company change as a result of an acquisition (referred to as a type III acquisition).

Alternatively an adjusted present value approach may be undertaken.

(v) The Chief Financial Officer's suggestion appears to be a disposal of 'crown jewels'. Without the cash reserves, Pursuit Co may become less valuable to SGF Co. Also, the reason for the depressed share price may be because Pursuit Co's shareholders do not agree with the policy to retain large cash reserves. Therefore returning the cash reserves to the shareholders may lead to an increase in the share price and make a bid from SGF Co more unlikely. This would not initially contravene the regulatory framework as no formal bid has been made. However, Pursuit Co must investigate further whether the reason for a possible bid from SGF Co might be to gain access to the large amount of cash or it might have other reasons. Pursuit Co should also try to establish whether remitting the cash to the shareholders would be viewed positively by them.

Whether this is a viable option for Pursuit Co depends on the bid for Fodder Co. In part (iii) it was established that more than the expected debt finance would be needed even if the cash reserves are used to pay for some of the acquisition cost. If the cash is remitted, a further $20,000,000 would be needed, and if this was all raised by debt finance then a significant proportion of the value of the combined company would be debt financed. The increased gearing may have significant implications on Pursuit Co's future investment plans and may result in increased restrictive covenants. Ultimately gearing might have to increase to such a level that this method of financing might not be possible. Pursuit Co should investigate the full implications further and assess whether the acquisition is worthwhile given the marginal value it provides for the shareholders (see part (i)).

APPENDIX TO QUESTION 1

Part (i)
Interest is ignored as its impact is included in the companies' discount rates

Fodder cost of capital
Ke = 4·5% + 1·53 x 6% = 13·68%
Cost of capital = 13·68% x 0·9 + 9% x (1 – 0·28) x 0·1 = 12·96% assume 13%

Fodder
Sales revenue growth rate = $(16{,}146/13{,}559)^{1/3}$ – 1 x 100% = 5·99% assume 6%
Operating profit margin = approx. 32% of sales revenue

Fodder Co cash flow and value computation ($000)

Year	1	2	3	4
Sales revenue	17,115	18,142	19,231	20,385
Operating profit	5,477	5,805	6,154	6,523
Less tax (28%)	(1,534)	(1,625)	(1,723)	(1,826)
Less additional investment (22c/$1 of sales revenue increase)	(213)	(226)	(240)	(254)
Free cash flows	3,730	3,954	4,191	4,443
PV (13%)	3,301	3,097	2,905	2,725

	$(000)
PV (first 4 years)	12,028
PV (after 4 years) [4,443 x 1·03/(0·13 – 0·03)] x $1{\cdot}13^{-4}$	28,067
Firm value	40,095

Combined Company: Cost of capital calculation
Asset beta (Pursuit Co) = 1·18 x 0·5/(0·5 + 0·5 x 0·72) = 0·686
Asset beta (Fodder Co) = 1·53 x 0·9/(0·9 + 0·1 x 0·72) = 1·417
Asset beta of combined co. = (0·686 x 140,000 + 1·417 x 40,095)/(140,000 + 40,095) = 0·849
Equity beta of combined company = 0·849 x (0·5 + 0·5 x 0·72)/0·5 = 1·46

Ke = 4·5% + 1·46 x 6% = 13·26%
Cost of capital = 13·26% x 0·5 + 6·4% x 0·5 x 0·72 = 8·93%, assume 9%

Combined Co cash flow and value computation ($000)
Sales revenue growth rate = 5·8%, operating profit margin = 30% of sales revenue

Year	1	2	3	4
Sales revenue	51,952	54,965	58,153	61,526
Operating profit	15,586	16,490	17,446	18,458
Less tax (28%)	(4,364)	(4,617)	(4,885)	(5,168)
Less additional investment (18c/$1 of sales revenue increase)	(513)	(542)	(574)	(607)
Free cash flows	10,709	11,331	11,987	12,683
PV (9%)	9,825	9,537	9,256	8,985

	$(000)
PV (first 4 years)	37,603
PV (after 4 years) [12,683 x 1·029/(0·09 – 0·029)] x $1{\cdot}09^{-4}$	151,566
Firm value	189,169

Synergy benefits = 189,169,000 – (140,000,000 + 40,095,000) = $9,074,000
Estimated premium required to acquire Fodder Co = 0·25 x 36,086,000 = $9,022,000
Net benefit to Pursuit Co shareholders = $52,000

2 **(a)** The information provided enables Casasophia Co to hedge its US$ income using forward contracts, future contracts or option contracts.

Forward contracts
Since it is a dollar receipt, the 1·3623 rate will be used.
Locked in receipt = US$20,000,000/1·3623 = €14,681,054

The hedge fixes the rate at 1·3623 and is legally binding.

Futures Contracts
This hedging strategy needs to show a gain when the Dollar exchange rate depreciates against the Euro and the underlying market shows a loss. Hence, for a US$ receipt, the five-month futures contracts (two-month is too short for the required hedge

period) need be bought, that is a long position needs to be adopted. It is assumed that the basis differential will narrow proportionally to the time expired. However, when the contract is closed out before expiry, this may not be the case due to basis risk, and a better or worse outcome may result.

Predicted futures rate = 1·3698 – (1/3 x (1·3698 – 1·3633)) = 1·3676 (when the five-month contract is closed out in four months' time)

Expected receipt = US$20,000,000/1·3676 = €14,624,159

Number of contracts to be bought = €14,624,159/€125,000 = 117 contracts

[OR: Futures lock-in rate may be estimated from the spot and five-month futures rate:
1·3698 – (1/5 x (1·3698 – 1·3618)) = 1·3682
US$20,000,000/1·3682 = €14,617,746
€14,617,746/€125,000 = 116·9 or 117 contracts (a slight over-hedge)]

This is worse than the forward rate. In addition to this, futures contracts require margin payments and are marked-to-market on a daily basis, although any gain is not realised until the contracts are closed out.

Like the forward contracts, futures contracts fix the rate and are legally binding.

Option Contracts
Options have an advantage over forwards and futures because the prices are not fixed and the option buyer can let the option lapse if the rates move favourably. Hence options have an unlimited upside but a limited downside. However, a premium is payable for this benefit.

Casasophia Co would purchase Euro call options to protect itself against a weakening Dollar to the Euro.

Exercise Price: $1·36/€1
€ receipts = 20,000,000/1·36 = 14,705,882 or 117·6 contracts
117 call options purchased

€ receipts = 117 x 125,000 = €14,625,000
Premium payable = 117 x 0·0280 x 125,000 = US$409,500
Premium in € = 409,500/1·3585 = €301,435
Amount not hedged = US$20,000,000 – (117 x €125,000 x 1·36) = US$110,000
Use forwards to hedge amount not hedged = US$110,000/1·3623 = €80,746

Total receipts = 14,625,000 – 301,435 + 80,746 = €14,404,311

Exercise Price: $1·38/€1
€ receipts = 20,000,000/1·38 = 14,492,754 or 115·9 contracts
115 call options purchased

€ receipts = 115 x 125,000 = €14,375,000
Premium payable = 115 x 0·0223 x 125,000 = US$320,563
Premium in € = 320,563/1·3585 = €235,968
Amount not hedged = US$20,000,000 – (115 x €125,000 x 1·38) = US$162,500
Use forwards to hedge amount not hedged = US$162,500/1·3623 = €119,284

Total receipts = 14,375,000 – 235,968 + 119,284 = €14,258,316

Both these hedges are significantly worse than the forward or futures contracts hedges. This is due to the high premiums payable to let the option lapse if the prices move in Casasophia Co's favour. With futures and forwards, Casasophia Co cannot take advantage of the Dollar strengthening against the Euro. However, this needs to be significant before the cost of the premium is covered.

Conclusion
It is recommended that Casasophia Co use the forward markets to hedge against the Dollar depreciating in four months time against the Euro in order to maximise receipts. However, Casasophia Co needs to be aware that forward contracts are not traded on a formal exchange and therefore default risk exists. And the exchange rate is fixed once the contract is agreed.

(b) Amount expected from US$ receipts is €14,681,054 assuming that forward contracts used.
Invested for two months = €14,681,054 x (1 + (60/360 x 0·0180)) = €14,725,097 say €14,725,000 approximately.

Expected spot rate (E(s)) in 12 months (using purchasing power parity) =
E(s) = 116 x 1·097/1·012 = 125·7
Expected spot rate in 6 months
116 + (125·7 – 116)/2 = 120·9

Investment Amount required = MShs 2,640,000,000/120·9 = €21·84m

Loan finance required = 21·84 – 14·73 = €7·11m

Casasophia Co will need to raise just over €7 million in loans in addition to the receipts from the USA to finance the project in Mazabia. This is on the assumption that the future spot rate follows the purchasing power parity conditions.

[Tutorial Note: Casasophia Co could also consider whether it may be more beneficial to transfer funds directly from the USA to Mazabia instead of converting them into Euros first. This would save transaction costs of converting first into Euros and then into MShs, and also the costs related to using the forward markets. The rates for investing the funds in the USA for two months and the exchange rate between US$ and MShs are not given, but if these were available a comparative analysis could be conducted. In these circumstances the amount of loan finance required would possibly be lower.]

(c) Calculate expected forward rates

Year	Interest Rate Parity Forward rate [MShs/1€]
½ year	128 x 1·108/1·022 = 138·77
	128 + (138·77 – 128)/2 = 133·4
1·5 years	133·4 x 1·108/1·022 = 144·6
2·5 years	144·6 x 1·108/1·022 = 156·8
3·5 years	156·8 x 1·108/1·022 = 170·0

Present Value calculations (Present values in six months)

	Year 1	Year 2	Year 3	Total
Income (MShs, million)	1,500	1,500	1,500	
Income (€ million, based on forward rates)	10·37	9·57	8·82	
Discounted Income (€ million at 12%)	9·26	7·63	6·28	23·17

Net present value = €23·17m – €21·84m = €1·33m

The calculation of the forward rates based on the interest rate parity indicates that the MShs rates are depreciating against the Euro because the Mazabia base rates at 10·8% are higher than the European country's local base rates at 2·2%. However, even where the forward rates are fixed, based on interest rate parity, the project is worthwhile for Casasophia Co.

According to the purchasing power parity, future spot currency rates will change in proportion to the inflation level differentials between two countries. Hence if Mazabia's inflation level is higher than the European Union, its currency will depreciate against the Euro.

Given that the inflation level in Mazabia is expected to range from 5% to 15% over the next few years, there is uncertainty over the NPV of the project in Euros if the swap is not accepted. The swap fixes the future exchange rates, although Casasophia Co will lose out if the inflation rate is lower than 9·7%, since the future spot rate will depreciate by less than what is predicted by the forward rates. The situation will be opposite if the level of inflation is higher than 9·7%.

Casasophia Co will also need to consider the risk of default by the local bank. Casasophia Co may ask Mazabia's government to act as guarantor to reduce this risk. Overall, if such an agreement could be reached, it would probably be beneficial to agree to the swap to ensure a certain level of income.

Casasophia Co may also want to explore whether it is possible for the grant funding from the European Union being paid to it directly, to reduce its exposure to the likely depreciation of MShs.

3 (a) In order to calculate the duration of the two bonds, the present value of the annual cash flows and the price or value at which the bonds are trading at need to be determined. To determine the present value of the annual cash flows, they need to be discounted by the gross redemption yield (i).

Gross Redemption Yield

Try 5%
$60 \times 1{\cdot}05^{-1} + 60 \times 1{\cdot}05^{-2} + 60 \times 1{\cdot}05^{-3} + 60 \times 1{\cdot}05^{-4} + 1{,}060 \times 1{\cdot}05^{-5} =$
$60 \times 4{\cdot}3295 + 1{,}000 \times 0{\cdot}7835 = 1{,}043{\cdot}27$

Try 4%
$60 \times 4{\cdot}4518 + 1{,}000 \times 0{\cdot}8219 = 1{,}089{\cdot}01$

$i = 4 + [(1{,}089{\cdot}01 - 1{,}079{\cdot}68)/(1{,}089{\cdot}01 - 1{,}043{\cdot}27)] = 4{\cdot}2\%$

Bond 1 (PV of cash flows)

$60 \times 1{\cdot}042^{-1} + 60 \times 1{\cdot}042^{-2} + 60 \times 1{\cdot}042^{-3} + 60 \times 1{\cdot}042^{-4} + 1{,}060 \times 1{\cdot}042^{-5}$

PV of cash flows (years 1 to 5) = 57·58 + 55·26 + 53·03 + 50·90 + 862·91 = 1,079·68

Market price = $1,079·68

Duration = [57·58 x 1 + 55·26 x 2 + 53·03 x 3 + 50·90 x 4 + 862·91 x 5]/1,079·68 = 4·49 years

Bond 2 (PV of Coupons and Bond Price)

Price = $40 \times 1{\cdot}042^{-1} + 40 \times 1{\cdot}042^{-2} + 40 \times 1{\cdot}042^{-3} + 40 \times 1{\cdot}042^{-4} + 1{,}040 \times 1{\cdot}042^{-5}$

PV of cash flows (years 1 to 5) = 38·39 + 36·84 + 35·36 + 33·93 + 846·63 = 991·15

Market Price = $991·15

Duration = [38·39 x 1 + 36·84 x 2 + 35·36 x 3 + 33·93 x 4 + 846·63 x 5]/991·15 = 4·63 years

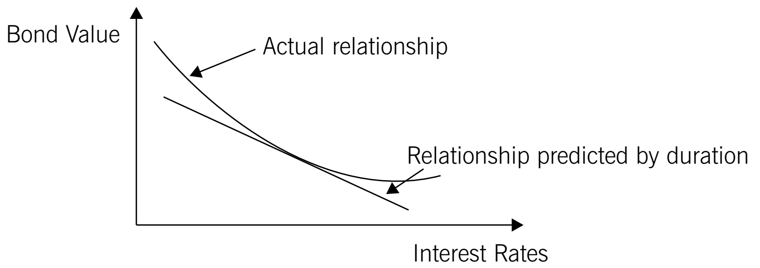
(b) The sensitivity of bond prices to changes in interest rates is dependent on their redemption dates. Bonds which are due to be redeemed at a later date are more price-sensitive to interest rate changes, and therefore are riskier.

Duration measures the average time it takes for a bond to pay its coupons and principal and therefore measures the redemption period of a bond. It recognises that bonds which pay higher coupons effectively mature 'sooner' compared to bonds which pay lower coupons, even if the redemption dates of the bonds are the same. This is because a higher proportion of the higher coupon bonds' income is received sooner. Therefore these bonds are less sensitive to interest rate changes and will have a lower duration.

Duration can be used to assess the change in the value of a bond when interest rates change using the following formula:

$\Delta P = [-D \times \Delta i \times P]/[1 + i]$, where P is the price of the bond, D is the duration and i is the redemption yield.

However, duration is only useful in assessing small changes in interest rates because of convexity. As interest rates increase, the price of a bond decreases and vice versa, but this decrease is not proportional for coupon paying bonds, the relationship is non-linear. In fact, the relationship between the changes in bond value to changes in interest rates is in the shape of a convex curve to origin, see below.



Duration, on the other hand, assumes that the relationship between changes in interest rates and the resultant bond is linear. Therefore duration will predict a lower price than the actual price and for large changes in interest rates this difference can be significant.

Duration can only be applied to measure the approximate change in a bond price due to interest changes, only if changes in interest rates do not lead to a change in the shape of the yield curve. This is because it is an average measure based on the gross redemption yield (yield to maturity). However, if the shape of the yield curve changes, duration can no longer be used to assess the change in bond value due to interest rate changes.

(Note: Credit will be given for alternative benefits/limitations of duration)

4 (a) Net Present Value without the option to delay the decision

Year	**Current**	**1**	**2**	**3**	**4**	**5**	**6**
Cash flows ($)	–7m	–7m	–35m	25m	18m	10m	5m
PV (11%) ($)	–7m	–6·31m	–28·42m	18·28m	11·86m	5·93m	2·68m

Net Present Value = $(2·98 million)
On this basis the project would be rejected.

Value of option to delay the decision until the film is released and its popularity established. Black-Scholes Option Pricing model is used to value the call option.

Present value of project's positive cash flows discounted to current day:

$18·28m + $11·86m + $5·93m + $2·68m = $38·75m

Variables:
Current price (P_a) = $38·75m
Exercise price (P_e) = $35m
Exercise date = 2 years
Risk free rate = 3·5%
Volatility = 30%

$d_1 = [\ln(38{\cdot}75/35) + (0{\cdot}035 + 0{\cdot}5 \times 0{\cdot}30^2) \times 2]/(0{\cdot}30 \times 2^{1/2}) = 0{\cdot}6170$
$d_2 = 0{\cdot}6170 - (0{\cdot}30 \times 2^{1/2}) = 0{\cdot}1927$

Using the Normal Distribution Table provided
$N(d_1) = 0{\cdot}5 + 0{\cdot}2291 + 0{\cdot}7 \times (0{\cdot}2324 - 0{\cdot}2291) = 0{\cdot}7314$
$N(d_2) = 0{\cdot}5 + 0{\cdot}0753 + 0{\cdot}3 \times (0{\cdot}0793 - 0{\cdot}0753) = 0{\cdot}5765$

Value of option to delay the decision $= 38{\cdot}75 \times 0{\cdot}7314 - 35 \times 0{\cdot}5765 \times e^{-0{\cdot}035 \times 2} = 28{\cdot}34 - 18{\cdot}81 = \$9{\cdot}53\text{m}$

Overall value of the project = \$9·53m – \$2·98m = \$6·55m
Since the project yields a positive net present value it would be accepted.

Hence by taking into account the option to delay the decision, the project should be accepted for investment.

(b) The option to delay the decision has given MMC's managers the opportunity to monitor and respond to changing circumstances before committing to the project, such as a rise in popularity of this type of genre of films in the next two years or increased competition from similar new releases or a sustained marketing campaign launched by the film's producers before its launch. Although the project looks unattractive at present, it may not be the case if the film on which it is based is successful. The option pricing formula requires numerous assumptions to be made about the variables, the primary one being the assumption of volatility. It therefore does not provide a correct value but an indication of the value of the option to delay the decision. Hence it indicates that the management should consider the project further and not dismiss it, even though current conventional net present value is negative.

The option to delay the decision may not be the only option within the project. For example, the gaming platform that the company needs to develop for this game may have general programmes which may be used in future projects and MMC should take account of these. Or if the film is successful, it may lead to follow-on projects involving games based on film sequels.

(Note: credit will be given for alternative relevant comments)

5 **(Solution note: Question 5 can be answered in a variety of ways and the suggested answer below is indicative. Credit will be given for reasonable answers considering alternatives or additions to the two issues discussed below.)**

The directors' overarching aim should be to maximise Mezza Co's long-term value and thereby maximise the value to its shareholders. Hence any decision should be made with this aim as the primary objective. However, the directors should also try to minimise the negative consequences resulting from the implementation of the project, taking into account the company's responsibility to its stakeholders.

The first key issue to consider is whether the new project would add value to the company. Initially it would appear that the investment into the new venture may be beneficial to the company. The product would be meeting market needs for a substantial period of time, as a tool in tackling climate change. It would possibly enhance the company's corporate reputation in helping to tackle the negative impact of climate change. Furthermore, it may enable the research subsidiary company to undertake future research and development projects in similar products.

However, whether the positive factors described above lead to an increase in the value of the company warrants further discussion and investigation. The company needs to assess the likely income the investment will generate and take account of the inherent risk of the venture. Presumably this is a new product and therefore it is likely that the uncertainty and risk to income flows will be significant. The directors should also take account of the fact that their remuneration package contains share options and these may induce them to act in an overly risky manner, where they would benefit from increasing share prices but not lose if the share price falls. This may not be beneficial to the shareholders or other stakeholders who do not hold such options.

Due diligence procedures for the project need to be undertaken before the decision is made. The company's directors need to undertake a full assessment of how realistic the estimates of revenues and income are likely to be. They would also need to assess the likelihood of competitors and alternative products which may affect the future sales of the product. A full investigation of the uncertainties and risks needs to be undertaken, possibly using techniques such as sensitivity, probability and project duration analysis. Risks need to be accounted for in the assessment of the likely value added. This would be of particular importance if the directors are to convince the shareholders and other stakeholders that they are not taking unacceptable levels of risk. Realistic time scales need to be determined of how long it would take to commercialise the product, perhaps by considering how other companies undertook similar projects. The adequacy of the expertise and infrastructure required by the company needs to be assessed.

The second key issue for the directors to consider is the location of the plant product. There are a number of factors which would make the location ideal for Mezza Co. The location provides the ideal conditions for the plant to grow in the quantity required for commercialisation. The relationship with the government is strong and the government wants to develop new industries, hence the project is likely to be seen in a positive light. It is possible therefore that many legal and administrative barriers would be reduced to enable production to commence quickly. Finally, Mezza Co has the infrastructure it needs in place and therefore set-up costs are likely to be significantly lower. These factors would provide financial benefits for Mezza Co and may make the investment viable.

However, there are ethical and environmental concerns in using this area for the project. It may be perceived that the relationship with the government is too close and this will prevent proper scrutiny by the government. The livelihood of the affected fishermen needs to be considered, as well as the impact on the wildlife and the environment. Going ahead with the project may result in a significant negative impact on Mezza Co's reputation and possibly contradicts with the company's (and the directors') values. Therefore, the dilemma that the directors face is that the project would be perceived as helping the global environment but damaging the local environment.

The directors could take a number of steps to reduce or eliminate this negative impact. Given that the fishermen do not have a significant 'voice' or power, Mezza's board could try to hide the issue, but it is unlikely that their personal values would allow such a situation. The directors could speak with the leaders of the fishermen's community to explain the benefits and consequences on the fishermen, possibly offering the fishermen priority to the new jobs that the project would create. They could influence, and work with the government, to part-develop the area for tourists and also leave areas for the fishermen to continue their activity. This may be possible if the whole area is not needed for plant cultivation at once. These additional wealth enhancing opportunities may convince the fishermen of the merits of the project. The company could continue looking for alternative areas to cultivate the crop and possibly engage in research and development to create crops which are not harmful to the fish stock and the wildlife. However, these steps would cost money and Mezza Co needs to balance revenues it is likely to receive against the additional costs.

In terms of the relationship with the government, Mezza Co may be able to demonstrate that it worked with the government to improve the livelihood of the fishermen. It could also ensure that it follows due process in terms of legal and administrative requirements, even though this would possibly delay the product's launch.

Mezza Co needs to consider the likely positive benefits against the costs, both direct and to the wider community, before taking on the project. It needs to consider the impact on long term value creation, and corporate reputation would be a major factor in determining this. Although Maienar's government may try to approve the project quickly, Mezza Co should consider the full impact of the proposed project, alternatives and consequences, and try to manage the entire process to ensure that there isn't an overall negative impact on the company's reputation.

				Marks
1	**(i)**	Ignore interest in calculations		1
		Estimate of cost of capital of Fodder Co		1
		Estimates of growth rates and profit margins for Fodder Co		2
		Estimate of intrinsic value of Fodder Co		3
		Equity beta of combined co		3
		Cost of capital of combined company		1
		Estimate of value of combined company		3
		Synergy benefits, value to Pursuit Co shareholders and conclusion		2–3
			Max	16
	(ii)	1 to 2 marks per each point discussed		
		Credit will be given for alternative, relevant points	Max	4
	(iii)	Estimate of the increase in debt capacity after acquisition		1
		Estimate of the funds required to acquire Fodder Co		1
		Conclusion		1
			Total	3
	(iv)	Explanation of the problem of the changing capital structure		2
		Explanation of the resolution of the problem using the iterative process		2
			Total	4
	(v)	Assessment of suitable defence		2–3
		Assessment of viability		2–3
		Credit will be given for alternative, relevant points	Max	5
	Professional Marks			
	Report format			1
	Layout, presentation and structure			3
			Total	4
			Total	**36**
2	**(a)**	Forward contract calculation		1
		Forward contract comment		1
		Futures contracts calculations		3
		Futures contracts comments		2–3
		Option contracts calculations		4
		Option contracts comments		2–3
		Conclusion		1
			Max	15
	(b)	Income from US$ after six months		1
		Expected spot rate in six months		2
		Investment amount required		1
		Loan finance required		1
		Comments		1
			Max	5
	(c)	Estimates of forward rates		3
		Estimates of present values and net present value in Euros		3
		Discussion		4–5
			Max	10
			Total	**30**

			Marks
3	**(a)**	Calculation of gross redemption yield	2
		PV of cash flows and duration of bond 1	3
		PV of cash flows, price and duration of bond 2	4
		Total	9
	(b)	Duration as a single measure of sensitivity of interest rates	3–4
		Explanation of convexity	2–3
		Explanation of change in shape of yield curve and other limitations	2–3
		Max	**8**
		Total	**17**
4	**(a)**	Value of project without considering option to delay decision and conclusion	2
		Current price variable (P_a) for BSOP formula	1
		Additional cost (P_e) for BSOP formula	1
		Other variables for BSOP formula	1
		Calculation of $N(d_1)$	2
		Calculation of $N(d_2)$	2
		Value of the option to delay decision	1
		Revised value of project and conclusion	2
		Total	12
	(b)	1 to 2 marks per well explained point	
		Max	5
		Total	**17**
5		Over-arching corporate aim	1–2
		Discussion of the project adding value and issues relating to return and risk	3–4
		Possible suggestions for mitigating the negative issues to above discussion	3–4
		Discussion of the ethical and environmental issues	3–4
		Possible suggestions for mitigating the ethical and environmental issues	3–4
		Other relevant key issues and suggestions for mitigation	2–3
		Max	**17**

Answers

1 Up to 4 professional marks are available for the presentation of the answer, which should be in a report style.

(i) Initial Evaluation

The information provided has been used to assess whether the production of the *X-IT* should be moved to Gamala from the USA. Initially a base case net present value calculation is conducted to assess the impact of the production in Gamala. This is then adjusted to show the impact of cash flows in the USA as a result of the move, the immediate impact of ceasing production and the impact of the subsidy and the tax shield benefits from the loan borrowing.

The calculations presented in the appendix show that the move will result in a positive adjusted present value of just over $2·4 million. On this basis, the production of *X-IT* should cease in the USA and the production moved to Gamala instead.

Assumptions

It is assumed that the borrowing rate of 5% is used to calculate the benefits from the tax shield. It could be argued that the risk free rate of 3% could be used as the discount rate instead of 5% to calculate the present value of benefits from the tax shields and the subsidies.

In adjusted present value calculations, the tax shield benefit is normally related to the debt capacity of the investment, not the actual amount of debt finance used. Since this is not given, it is assumed that the increase in debt capacity is equal to the debt finance used.

It has been assumed that many of the input variables, such as for example the tax and capital allowances rates, the various costs and prices, units produced and sold, the rate of inflation and the prediction of future exchange rates based on the purchasing power parity, are accurate and will change as stated over the four-year period of the project. In reality any of these estimates could be subject to change to a greater or lesser degree and it would be appropriate for Tramont Co to conduct uncertainty assessments like sensitivity analysis to assess the impact of the changes to the initial predictions.

(Note: credit will be given for alternative relevant assumptions)

(ii) Government Change

From the preamble it would seem that a change of government could have a significant impact on whether or not the project is beneficial to Tramont Co. The threat to raise taxes may not be too significant as the tax rates would need to increase to more than 30% before Tramont Co would lose money. However, the threat by the opposition party to review 'commercial benefits' may be more significant.

Just over 40% of the present value comes from the tax shield and subsidy benefits. If these were reneged then Tramont Co would lose a significant amount of the value attached to the project. Also the new government may not allow remittances every year, as is assumed in part (i). However, this may not be significant since the largest present value amount comes from the final year of operation.

Other Business Factors

Tramont Co should consider the possibility of becoming established in Gamala, and this may lead to follow-on projects. The real options linked to this should be included in the analysis.

Tramont Co's overall corporate strategy should be considered. Does the project fit within this strategy? Even if the decision is made to close the operation in the USA, there may be other alternatives and these need to be assessed.

The amount of experience Tramont Co has in international ventures needs to be considered. For example, will it be able to match its systems to the Gamalan culture? It will need to develop strategies to deal with cultural differences. This may include additional costs such as training which may not have been taken into account.

Tramont Co needs to consider if the project can be delayed at all. From part (i), it can be seen that a large proportion of the opportunity cost relates to lost contribution in years 1 and 2. A delay in the commencement of the project may increase the overall value of the project.

Tramont Co needs to consider the impact on its reputation due to possible redundancies. Since the production of the *X-IT* is probably going to be stopped in any case, Tramont Co needs to communicate its strategy to the employees and possibly other stakeholders clearly so as to retain its reputation. This may make the need to consider alternatives even more important.

(Note: credit will be given for alternative relevant comments)

Appendix
Gamalan Project Operating Cash Flows
(All amounts in GR/$ 000's)

Year	Now	1	2	3	4
Sales revenue (w2)		48,888	94,849	214,442	289,716
Local variable costs (w3)		(16,200)	(32,373)	(75,385)	(104,897)
Imported component (w4)		(4,889)	(9,769)	(22,750)	(31,658)
Fixed costs		(30,000)	(32,700)	(35,643)	(38,851)
Profits before tax		(2,201)	20,007	80,664	114,310
Taxation (w5)		0	0	(7,694)	(18,862)
Investment	(230,000)				450,000
Working capital	(40,000)	(3,600)	(3,924)	(4,277)	51,801
Cash flows (GR)	(270,000)	(5,801)	16,083	68,693	597,249
Exchange rate (w1)	55·00	58·20	61·59	65·18	68·98
Cash flows ($)	(4,909)	(100)	261	1,054	8,658
Discount factor for 9·6% (w6) **(Full credit given if 10% is used as the discount rate)**		0·912	0·832	0·760	0·693
Present values ($)	(4,909)	(91)	217	801	6,000

Net present value (NPV) of the cash flows from the project is approx. $2,018,000.

Adjusted present value (APV)	**$000's**
NPV of cash flows	2,018
Additional USA tax	
Opportunity cost (revenues foregone from current operations)	
Additional contribution from component exported to project (net of tax) (w7)	(1,237)
Closure revenues and costs ($2,300,000 – $1,700,000)	600
Tax shield	
Benefit of subsidy (w8)	1,033
Total APV	2,414

Workings

1. Exchange rates

Year	**1**	**2**	**3**	**4**
GR/$1	55 x 1·09/1·03 = 58·20	58·20 x 1·09/1·03 = 61·59	61·59 x 1·09/1·03 = 65·18	65·18 x 1·09/1·03 = 68·98

2. Sales revenue (GR 000's)

Year	**1**	**2**	**3**	**4**
Price x units x exchange rate	70 x 12,000 x 58·20 = 48,888	70 x 22,000 x 61·59 = 94,849	70 x 47,000 x 65·18 = 214,442	70 x 60,000 x 68·98 = 289,716

3. Local variable costs (GR 000's)

Year	**1**	**2**	**3**	**4**
Cost x units x inflation after yr 1	1,350 x 12,000 = 16,200	1,350 x 22,000 x 1·09 = 32,373	1,350 x 47,000 x $1{\cdot}09^2$ = 75,385	1,350 x 60,000 x $1{\cdot}09^3$ = 104,897

4. Imported Component (GR 000's)

Year	**1**	**2**	**3**	**4**
Price x units x inflation after year 1 x exchange rate	7 x 12,000 x 58·20 = 4,889	7 x 22,000 x 1·03 x 61·59 = 9,769	7 x 47,000 x $1{\cdot}03^2$ x 65·18 = 22,750	7 x 60,000 x $1{\cdot}03^3$ x 68·98 = 31,658

5. Taxation

Year	**1**	**2**	**3**	**4**
Profits before tax	(2,201)	20,007	80,664	114,310
Tax allowable depreciation	(20,000)	(20,000)	(20,000)	(20,000)
Profit/(loss) after depreciation	(22,201)	7	60,664	94,310
Taxable profits	0	0	38,470	94,310
Taxation (20%)	0	0	(7,694)	(18,862)

6. **Gamala project all-equity financed discount rate**
Tramont Co equity beta = 1·17
MVe = $2·40 x 25m shares = $60m
MVd = $40m x $1,428/$1,000 = $57·12m

Tramont Co asset beta (assuming debt is risk free)
1·17 x 60m/(60m + 57·12m x 0·7) = 0·70

Project asset beta = 0·70 + 0·40 = 1·10
Project all-equity financed discount rate = 3% + 6% x 1·1 = 9·6%

7. **Additional tax, additional contribution and opportunity cost ($000's)**

Year	**1**	**2**	**3**	**4**
Additional tax				
Taxable profits x 1/exchange rate x 10%	0	0	38,470 x 1/65·18 x 10% = (59)	94,310 x 1/68·98 x 10% = (137)
Opportunity cost				
Units x contribution x (1 – tax)	40 x $20 x 0·7 = (560)	32 x $20 x 0·7 = (448)	25·6 x $20 x 0·7 = (358)	20·48 x $20 x 0·7 = (287)
Additional Contribution				
Units x contribution x inflation x (1 – tax)	12 x $4 x 0·7 = 34	22 x $4 x 1·03 x 0·7 = 63	47 x $4 x $1·03^2$ x 0·7= 140	60 x $4 x $1·03^3$ x 0·7 = 184
Total cash flows	**(526)**	**(385)**	**(277)**	**(240)**
PV of cash flows				
Discount at 7%	**(492)**	**(336)**	**(226)**	**(183)**

NPV is approx. $(1,237,000)

8. **Tax shield and subsidy benefits ($/GR 000's)**

Year	**1**	**2**	**3**	**4**
Annual tax shield (GR)				
Interest x loan x tax rate	6% x 270m x 20% = 3,240	3,240	3,240	3,240
Annual subsidy benefit (GR)				
Interest gain x loan x (1 – tax rate)	7% x 270m x 0·8 = 15,120	15,120	15,120	15,120
Total tax shield + subsidy benefits (GR)	18,360	18,360	18,360	18,360
Exchange rate (GR/$1)	58·20	61·59	65·18	68·98
Cash flows ($)	**315**	**298**	**282**	**266**
PV of cash flows				
Discount at 5%	**300**	**270**	**244**	**219**

NPV of tax shield and subsidy benefit is approx. $1,033,000

2 **(a)** The main advantage of using a collar instead of options to hedge interest rate risk is lower cost. A collar involves the simultaneous purchase and sale of both call and put options at different exercise prices. The option purchased has a higher premium when compared to the premium of the option sold, but the lower premium income will reduce the higher premium payable. With a normal uncovered option, the full premium is payable.

However, the main disadvantage is that, whereas with a hedge using options the buyer can get full benefit of any upside movement in the price of the underlying asset, with a collar hedge the benefit of the upside movement is limited or capped as well.

(b) Using Futures

Need to hedge against a rise in interest rates, therefore go short in the futures market. Alecto Co needs June contracts as the loan will be required on 1 May.

No. of contracts needed = €22,000,000/€1,000,000 x 5 months/3 months = 36·67 say 37 contracts.

Basis
Current price (on 1/1) – futures price = total basis
(100 – 3·3) – 96·16 = 0·54
Unexpired basis = 2/6 x 0·54 = 0·18

If interest rates increase by 0·5% to 3·8%

Cost of borrowing funds = 4·6% x 5/12 x €22,000,000 =	€421,667
Expected futures price = 100 – 3·8 – 0·18 = 96·02	
Gain on the futures market = (9,616 – 9,602) x €25 x 37 =	€12,950
Net cost =	€408,717
Effective interest rate = 408,717/22,000,000 x 12/5 =	4·46%

If interest rates decrease by 0·5% to 2·8%

Cost of borrowing funds = 3·6% x 5/12 x €22,000,000 =	€330,000
Expected futures price = 100 – 2·8 – 0·18 = 97·02	
Loss on the futures market = (9,616 – 9,702) x €25 x 37 =	€79,550
Net cost =	€409,550
Effective interest rate = 409,550/22,000,000 x 12/5 =	4·47%

(Note: Net cost should be the same. Difference is due to rounding the number of contracts)

Using Options on Futures

Need to hedge against a rise in interest rates, therefore buy put options. As before, Alecto Co needs 37 June put option contracts (€22,000,000/€1,000,000 x 5 months/3 months).

If interest rates increase by 0·5% to 3·8%

Exercise Price	96·00	96·50
Futures Price	96·02	96·02
Exercise ?	No	Yes
Gain in basis points	0	48
Underlying cost of borrowing (from above)	€421,667	€421,667
Gain on options (0 and €25 x 48 x 37)	€0	€44,400
Premium		
16·3 x €25 x 37	€15,078	
58·1 x €25 x 37		€53,743
Net cost	€436,745	€431,010
Effective interest rate	4·76%	4·70%

If interest rates decrease by 0·5% to 2·8%

Exercise Price	96·00	96·50
Futures Price	97·02	97·02
Exercise ?	No	No
Gain in basis points	0	0
Underlying cost of borrowing (from above)	€330,000	€330,000
Gain on options	€0	€0
Premium		
16·3 x €25 x 37	€15,078	
58·1 x €25 x 37		€53,743
Net cost	€345,078	€383,743
Effective interest rate	3·76%	4·19%

Using a collar

Buy June put at 96·00 for 0·163 and sell June call at 96·50 for 0·090.
Premium payable = 0·073

If interest rates increase by 0·5% to 3·8%

	Buy put	**Sell Call**
Exercise Price	96·00	96·50
Futures Price	96·02	96·02
Exercise ?	No	No
Underlying cost of borrowing (from above)	€421,667	
Premium		
7·3 x €25 x 37	€6,753	
Net cost	€428,420	
Effective interest rate	4·67%	

If interest rates decrease by 0·5% to 2·8%

	Buy put	Sell Call
Exercise Price	96·00	96·50
Futures Price	97·02	97·02
Exercise ?	No	Yes
Underlying cost of borrowing (from above)	€330,000	
Premium		
7·3 x €25 x 37	€6,753	
Loss on exercise (52 x €25 x 37)	€48,100	
Net cost	€384,853	
Effective interest rate	4·20%	

Hedging using the interest rate futures market fixes the rate at 4·47%, whereas with options on futures or a collar hedge, the net cost changes. If interest rates fall in the future then a hedge using options gives the most favourable rate. However, if interest rates increase then a hedge using futures gives the lowest interest payment cost and hedging with options give the highest cost, with the cost of the collar hedge being in between the two. If Alecto Co's aim is to fix its interest rate whatever happens to future rates then the preferred instrument would be futures.

This recommendation is made without considering margin and other transactional costs, and basis risk, which is discussed below. These need to be taken into account before a final decision is made.

(Note: credit will be given for alternative approaches to the calculations in part (b))

(c) Basis risk occurs when the basis does not diminish at a constant rate. In this case, if a futures contract is held until it matures then there is no basis risk because at maturity the derivative price will equal the underlying asset's price. However, if a contract is closed out before maturity (here the June futures contracts will be closed two months prior to expiry) there is no guarantee that the price of the futures contract will equal the predicted price based on basis at that date. For example, in part (b) above, the predicted futures price in four months assumes that the basis remaining is 0·18, but it could be more or less. Therefore the actual price of the futures contract could be more or less.

This creates a problem in that the effective interest rate for the futures contract above may not be fixed at 4·47%, but may vary and therefore the amount of interest that Alecto Co pays may not be fixed or predictable. On the other hand, it could be argued that the basis risk will probably be smaller than the risk exposure to interest rates without hedging and therefore, although some risk will exist, its impact will be smaller.

3 **(a)** Spot yield rates applicable to Levante Co (based on A credit rating)

1 year	3·85%
2 year	4·46%
3 year	5·07%
4 year	5·80%
5 year	6·12%

Bond value based on A rating =
$\$4 \times 1{\cdot}0385^{-1} + \$4 \times 1{\cdot}0446^{-2} + \$104 \times 1{\cdot}0507^{-3} = \$97{\cdot}18$ per \$100

Current price based on AA rating = \$98·71

Fall in value = (97·18 – 98·71)/98·71 x 100% = 1·55%

(b) Spot rates applicable to Levante Co (based on A credit rating) [from above]

1 year	3·85%
2 year	4·46%
3 year	5·07%
4 year	5·80%
5 year	6·12%

(i) **Value of 5% coupon bond**
$\$5 \times 1{\cdot}0385^{-1} + \$5 \times 1{\cdot}0446^{-2} + \$5 \times 1{\cdot}0507^{-3} + \$5 \times 1{\cdot}0580^{-4} + \$105 \times 1{\cdot}0612^{-5} = \$95{\cdot}72$
Hence the bond will need to be issued at a discount if only a 5% coupon is offered.

(ii) **New coupon rate for bond valued at \$100 by the markets**
Since the 5% coupon bond is only valued at \$95·72, a higher coupon needs to be offered. This coupon amount can be calculated by finding the yield to maturity of the 5% coupon bond discounted at the above yield curve. This yield to maturity will be the coupon amount for the new bond such that its face value will be \$100.

Therefore, if the yield to maturity is denoted by YTM then
$\$5 \times (1 + YTM)^{-1} + \$5 \times (1 + YTM)^{-2} + \$5 \times (1 + YTM)^{-3} + \$5 \times (1 + YTM)^{-4} + \$105 \times (1 + YTM)^{-5} = \$95{\cdot}72$

Solve by trial and error, assume YTM is 5·5%. This gives the bond value as \$97·86.

Assume YTM is 6%; this gives the bond value as \$95·78, which is close enough to \$95·72
$\$5 \times (1{\cdot}06)^{-1} + \$5 \times (1{\cdot}06)^{-2} + \$5 \times (1{\cdot}06)^{-3} + \$5 \times (1{\cdot}06)^{-4} + \$105 \times (1{\cdot}06)^{-5} = \$95{\cdot}78$

Hence if the coupon payment is 6% or \$6 per \$100 bond unit then the bond market value will equal the par value at \$100.
$\$6 \times (1{\cdot}06)^{-1} + \$6 \times (1{\cdot}06)^{-2} + \$6 \times (1{\cdot}06)^{-3} + \$6 \times (1{\cdot}06)^{-4} + \$106 \times (1{\cdot}06)^{-5} = \$100$

Alternatively:
Take R as the coupon rate, such that:

$(R \times 1{\cdot}0385^{-1}) + (R \times 1{\cdot}0446^{-2}) + (R \times 1{\cdot}0507^{-3}) + (R \times 1{\cdot}0580^{-4}) + (R \times 1{\cdot}0612^{-5}) + (100 \times 1{\cdot}0612^{-5}) = \$100$

$4{\cdot}2826R + 74{\cdot}30 = \$100$
$R = 6\%$ or \$6 per \$100

Advice:
If only a 5% coupon is offered, the bonds will have to be issued at just under a 4·3% discount. To raise the full \$150 million, if the bonds are issued at a 4·3% discount, then 1,567,398 \$100 bond units need to be issued, as opposed to 1,500,000. This is an extra 67,398 bond units for which Levante Co will need to pay an extra \$6,739,800 when the bonds are redeemed in five years.

On the other hand, paying a higher coupon every year of 6% instead of 5% will mean that an extra \$1,500,000 is needed for each of the next five years.

If the directors feel that the drain in resources of \$1,500,000 every year is substantial and that the project's profits will cover the extra \$6,739,800 in five years' time, then they should issue the bond at a discount and at a lower coupon rate. On the other hand, if the directors feel that they would like to spread the amount payable then they should opt for the higher coupon alternative.

(c) Industry risk measures the resilience of the company's industrial sector to changes in the economy. In order to measure or assess this, the following factors could be used:

Impact of economic changes on the industry in terms of how successfully the firms in the industry operate under differing economic outcomes;
How cyclical the industry is and how large the peaks and troughs are;
How the demand shifts in the industry as the economy changes.

Earnings protection measures how well the company will be able to maintain or protect its earnings in changing circumstances. In order to assess this, the following factors could be used:

Differing range of sources of earnings growth;
Diversity of customer base;
Profit margins and return on capital.

Financial flexibility measures how easily the company is able to raise the finance it needs to pursue its investment goals. In order to assess this, the following factors could be used:

Evaluation of plans for financing needs and range of alternatives available;
Relationships with finance providers, e.g. banks;
Operating restrictions that currently exist as debt covenants.

Evaluation of the company's management considers how well the managers are managing and planning for the future of the company. In order to assess this, the following factors could be used:

The company's planning and control policies, and its financial strategies;
Management succession planning;
The qualifications and experience of the managers;
Performance in achieving financial and non-financial targets.

(Note: credit will be given for alternative relevant comments and suggestions)

4 (a) Possible benefits of disposing Tyche Co through a management buy-out may include:

Management buy-out costs may be less for Proteus Co compared with other forms of disposal such as selling the assets of the company or selling the company to a third party.

It may be the quickest method in raising funds for Proteus Co compared to the other methods.

There would be less resistance from the managers and employees, making the process smoother and easier to accomplish.

Proteus Co may retain a better relationship and beneficial links with Tyche Co and may be able to purchase or sell goods and services to it, as seems to have happened with the management service.

It may be able to get a better price for the company. The current management and employees possibly have the best knowledge of the company and are able to make it successful. Therefore they may be willing to pay more for it.

It may increase Proteus Co's reputation among its internal stakeholders such as the management and employees. It may also increase its reputation with external stakeholders and the markets if it manages the disposal successfully and efficiently.

(Note: credit will be given for alternative relevant comments)

(b) In order to calculate whether or not the covenant is breached every year, the proportion of debt to equity needs to be calculated each year. The debt will reduce by $3 million every year and the equity will increase by reserves every year. In order to calculate the increase in reserves every year, the forecast income statements need to be determined.

Forecast Income Statement ($000's)

Year	1	2	3	4	5
Operating income before mgmt. fee (W1)	23,760	25,661	27,714	29,931	32,325
Management service fee	12,000	12,960	13,997	15,116	16,326
Interest payable (W2)	5,850	5,580	5,310	5,040	4,770
Profit before tax	5,910	7,121	8,407	9,775	11,229
Tax payable (25%)	1,478	1,780	2,102	2,444	2,807
Profit after tax	4,432	5,341	6,305	7,331	8,422
Dividend payable (25%)	1,108	1,335	1,576	1,833	2,106
Balance transferred to reserves	3,324	4,006	4,729	5,498	6,316

Book value of equity

Year	1	2	3	4	5
Opening equity	16,000	19,324	23,330	28,059	33,557
Reserves	3,324	4,006	4,729	5,498	6,316
Closing equity	19,324	23,330	28,059	33,557	39,873

Debt/Equity Computations

Year	1	2	3	4	5
Debt Outstanding at year end ($000's)	62,000	59,000	56,000	53,000	50,000
Equity value at year end	19,324	23,330	28,059	33,557	39,873
Debt/Equity	321%	253%	200%	158%	125%
Restrictive Condition	350%	250%	200%	150%	125%
Restriction Breached?	No	Yes	No	Yes	No

Workings

W1
Current operating profit before management service charge = $60,000,000 – ($12,000,000 + 22,000,000 + 4,000,000) = $22,000,000. This amount will grow by 8% every year.

W2

Year	1	2	3	4	5
Outstanding loan at the start of the year ($000's)	65,000	62,000	59,000	56,000	53,000
Interest ($000's)	5,850	5,580	5,310	5,040	4,770

(c) Implications
Based on the calculations in part (b) above, the restrictive covenant is due to be breached in years two and four. In years three and five, it has just been met, and only in year one will Tyche Co be operating well within the conditions of the restrictive covenant. This raises two main issues: firstly, Tyche Co needs to establish how the bank will react to the conditions not being met and will it put Tyche Co's business in jeopardy? Secondly, because the conditions are nearly breached in years three and five, Tyche Co needs to determine the likelihood of the revenues and costs figures being achieved. A very small deviation from the figures may cause the conditions to be breached. Sensitivity analysis and other forms of risk analysis may need to be undertaken and provisions put into place to deal with unexpected breaches in the covenant.

Possible Actions
Tyche Co can consider the following possible actions in the years where it is likely that the covenant may be breached:

(i) The directors may decide to award themselves and the other shareholders lower or no dividends. This would probably need to be negotiated and agreed.
(ii) The directors may want to ask the venture capitalist to take a higher equity stake for more funds at the outset. Both parties would need to agree to this.
(iii) Tyche Co may want to try and negotiate less onerous terms with the bank or ask it for more flexibility when applying the restrictive covenant. Given that the restrictive covenant is not likely to be breached by a significant amount, the bank will probably not want to undertake legal proceedings to close Tyche Co and would probably be open to negotiations.
(iv) Tyche Co may decide to pay off more of the loan each year from its cash reserves, if it has enough funds, in order to reduce the year-end outstanding debt.

(Note: only two actions needed)

5 **(Solution note: Question 5 can be answered in a variety of ways and the suggested answer below is indicative. Credit will be given for reasonable answers considering alternatives or additions to the explanations and discussion below)**

(a) A triple bottom line (TBL) report provides a quantitative summary of a corporation's performance in terms of its economic or financial impact, its impact on environmental quality and its impact on social performance. The principle of TBL reporting is that a corporation's true performance must be measured in terms of a balance between economic (profits), environmental (planet) and social (people) factors; with no one factor growing at the expense of the others.

A corporation's sustainable development is about how these three factors can grow and be combined so that a corporation is building a reputation as being a good citizen. The contention is that a corporation that accommodates the pressures of all the three factors will enhance shareholder value by addressing the needs of its stakeholders.

Whereas TBL reporting is a quantitative summary of the corporation performance in the three factors over a previous time period, say a year, sustainable development tends to be forward looking and qualitative. Therefore TBL provides the measurement tool to assess a corporation's performance against its stated aims.

Each factor can be assessed or measured using a number of proxies. The economic impact can be measured by considering proxies such as operating profits, dependence on imports and the extent to which the local economy is supported by purchasing locally produced goods and services. Social impact can be measured by considering proxies such as working conditions, fair pay, using appropriate labour force (not child labour), ethical investments, and maintenance of appropriate food standards. Environmental impact can be measured by considering proxies such as ecological footprint, emissions to air, water and soil, use of energy and water, investments in renewable resources.

(b) An assessment by the management of a corporation's performance in the three factors – economic, environmental and social, that make up the TBL report will result in an improvement in the financial position, if long-term shareholder value is increased as a result of the report being produced. In this case, the benefits that accrue from the assessment and production of a TBL report must exceed the costs of undertaking the report. It is likely to be the case that the costs of producing the report are relatively easy to measure but the financial benefits may be more difficult to measure and may take place over a longer time period. Some examples of the ways in which Kengai Co may benefit financially are explained below.

Focusing on and reporting the company's environmental and social impact may build and enhance its reputation. Increasing reputation may increase the long-term revenue of Kengai Co. On the other hand, if Kengai Co does not follow (or even try to lead) its competitors in this area then the loss of reputation may damage its revenues stream and lower its corporate value.

Consideration and improvement of working standards and consulting employees as part of this process, when assessing social factors, may help in retaining and attracting high performing, high calibre employees. This will benefit Kengai Co in the long term because of increased employee motivation and performance. Employee involvement may also help reduce the costs related to the company's risk management activity and thus have a direct cost reduction impact.

Improvement of due diligence procedures as part of the economic factor assessment may help limit direct legal costs and indirect costs incurred in maintaining stakeholder relationships. Communication with stakeholders and thus improving the quality of reporting may result in improvements in governance procedures. This in turn would lead to a reduction of the costs related to risk management.

Assessing and improving the environmental factor impact in the TBL report may result in Kengai Co making efforts to reduce its carbon footprint by placing less reliance on exports and developing local expertise in producing the inputs it needs. This may reduce the risk of supplier related problems and alleviate problems related to possible inventory shortfalls. It may also improve Kengai Co's reputation, leading to long-term financial benefits.

Monitoring and reporting on the performance of employees and managers as part of the assessment of economic and social factors may help identify areas where work can be done more effectively and efficiently. It may help managers reconsider business processes and question areas where improvements can be made.

In all the above examples, the result of the assessment required in producing the TBL report and comparing the corporation's progress in relation to its aim of becoming a sustainable organisation will create opportunities which senior managers can develop into financial benefits. The extent to which these opportunities are successfully developed depends on the quality of assessment and the organisation's ability to enable change to happen.

				Marks
1	(i)	Estimated future rates based on purchasing power parity		1
		Sales revenue, variable costs, component cost and fixed costs (in GR)		4
		Taxable profits and taxation		2
		Investment, terminal value and working capital		2
		Cash flows in GR		1
		Cash flows in $		1
		Discount rate of all-equity financed project		2
		Base case PVs and NPV		2
		PV of additional contribution, additional tax and opportunity cost		4
		PV of tax shield and subsidy benefits		4
		Closure costs and benefits		1
		Initial comments and conclusion		1–2
		Assumptions and sensitivity analysis		2–3
			Max	27
	(ii)	Implications of change of government		2–3
		Other business factors (1 to 2 marks per factor)		5–6
			Max	8
		Professional Marks		
		Report format		1
		Layout, presentation and structure		3
			Total	4
			Total	**39**
2	(a)	Discussion of the main advantage		2
		Discussion of the main disadvantage		2
				4
	(b)	Recommendation to go short if futures are used and purchase puts if options are used		1
		Calculation of number of contracts and remaining basis		2
		Futures contracts calculations		4
		Options contracts calculations		4
		Collar approach and calculations		4
		Supporting comments and conclusion		2–3
			Max	17
	(c)	Explanation of basis risk		2–3
		Effect of basis risk on recommendation made		2–3
			Max	4
			Total	**25**

				Marks
3	(a)	Calculation of company specific yield curve		1
		Calculation bond value based on credit rating of A		1
		Calculation of percentage fall in the value of the bond		1
				3
	(b)	Calculation of bond value based on 5% coupon		1
		Calculation of new coupon rate		4
		Advice on which type of bond to issue		2–3
			Max	7
	(c)	For each of the four criteria –		
		2 marks for explanation and suggestion of factors		8
			Total	**18**
4	(a)	1–2 marks for each point discussed	Max	4
	(b)	Calculations to get to profit before tax for the five years		3
		Calculations to get to the reserves figures for each year		2
		Calculation of the equity amount each year		1
		Calculation of debt to equity ratio for year and		
		Conclusion when the covenant is breached and when it is not		3
				9
	(c)	3 marks for discussion of implications		
		2 marks for actions Tyche Co may take		5
			Total	**18**
5	(a)	Explaining TBL and relation to sustainability		4–5
		Examples of proxies (up to 2 marks for proxies per TBL factor)		4–5
			Max	8
	(b)	Discussion of long-term shareholder wealth maximisation and benefits exceeding costs		2
		2–3 marks per well-discussed example focusing on financial impact		8–9
			Max	10
			Total	**18**

Answers

1 REPORT TO THE BOARD OF DIRECTORS, NENTE CO

IMPACT OF THE TAKEOVER PROPOSAL FROM MIJE CO AND PRODUCTION RIGHTS OF THE FOLLOW-ON PRODUCT

The report considers the value of the takeover to Nente Co and Mije Co shareholders based on a cash offer and on a share-for-share offer. It discusses the possible reaction of each group of shareholders to the two offers and how best to utilise the follow-on product opportunity. The significant assumptions made in compiling the report are also explained.

The appendices to the report show the detailed calculations in estimating the equity value of Nente Co, the value to Nente Co and Mije Co shareholders of acquiring Nente Co by cash and by a share-for-share exchange, and the value to Nente Co of the exclusive rights to the follow-on product. The results of the calculation are summarised below:

Estimated price of a Nente Co share before the takeover offer and follow-on product £2·90/share (appendix i)

Estimated increase in share price	Nente Co	Mije Co
Cash offer (appendix ii)	1·7%	9·4%
Share-for-share offer (appendix ii)	17·9%	6·9%

Estimate of the value per share of the follow-on product to Nente Co 8·7% (appendix iii)

It is unlikely that Nente Co shareholders would accept the cash offer because it is little more than the estimated price of a Nente Co share before the takeover offer. However, the share-for-share offer gives a larger increase in value of a share of 17·9%. Given that the normal premium on acquisitions ranges from 20% to 40%, this is closer to what Nente Co shareholders would find acceptable. It is also greater than the additional value from the follow-on product. Therefore, based on the financial figures, Nente Co's shareholders would find the offer of a takeover on a share-for-share exchange basis the most attractive option. The other options considered here yield lower expected percentage increase in share price.

Mije Co shareholders would prefer the cash offer so that they can maximise the price of their shares and also not dilute their shareholding, but they would probably accept either option because the price of their shares increases. However, Mije Co shareholders would probably assess whether or not to accept the acquisition proposal by comparing it with other opportunities that the company has available to it and whether this is the best way to utilise its spare cash flows.

The calculations and analysis in each case is made on a number of assumptions. For example, in order to calculate the estimated price of a Nente Co share, the free cash flow valuation model is used. For this, the growth rate, the cost of capital and effective time period when the growth rate will occur (perpetuity in this instance) are all estimates or based on assumptions. For the takeover offer, the synergy savings and P/E ratio value are both assumptions. For the value of the follow-on product and the related option, the option variables are estimates and it is assumed that they would not change during the period before the decision. The value of the option is based on the possibility that the option will only be exercised at the end of the two years, although it seems that the decision can be made any time within the two years.

The follow-on product is initially treated separately from the takeover, but Nente Co may ask Mije Co to take the value of the follow-on product into consideration in its offer. The value of the rights that allow Nente Co to delay making a decision are themselves worth $603,592 (appendix iii) and add just over 25c or 8·7% to the value of a Nente Co share. If Mije Co can be convinced to increase their offer to match this or the rights could be sold before the takeover, then the return for Nente Co's shareholders would be much higher at 26·6% (17·9% + 8·7%).

In conclusion, the most favourable outcome for Nente Co shareholders would be to accept the share-for-share offer, and try to convince Mije Co to take the value of the follow-on product into consideration. Prior to accepting the offer Nente Co shareholders would need to be assured of the accuracy of the results produced by the computations in the appendices.

Report compiled by:

Date:

(Note: credit will be given for alternative relevant discussion and suggestions)

APPENDICES

Appendix i: Estimate of Nente Co Equity Value Based on Free Cash Flows

Company value = Free cash flows (FCF) x (1 + growth rate (g))/(cost of capital (k) – g)

k = 11%

Past g = (latest profit before interest and tax (PBIT)/earliest PBIT)$^{1/\text{no. of years of growth}}$ – 1

Past g = $(1{,}230/970)^{1/3} - 1 = 0{\cdot}0824$

Future g = ¼ x 0·0824 = 0·0206

FCF Calculation
FCF = PBIT + non cash flows – cash investment – tax
FCF = $1,230,000 + $1,206,000 – $1,010,000 – ($1,230,000 x 20%) = $1,180,000

Company value = $1,180,000 x 1·0206/(0·11 – 0·0206) = $13,471,000
Equity value = $13,471,000 – $6,500,000 = $6,971,000
Per share = $6,971,000/2,400,000 shares = $2·90

Appendix ii: Estimated Returns to Nente Co and Mije Co Shareholders

Cash Offer
Gain in value to a Nente Co share = ($2·95 – $2·90)/$2·90 = 1·7%

Additional earnings after acquisition = $620,000 + $150,000 = $770,000	
Additional EPS created from acquisition = $770,000/10,000,000 = 7·7c/share	
Increase in share price based on P/E of 15 = 7·7c x 15 = $1·16	
Additional value created = $1·16 x 10,000,000 =	$11,600,000
Less: paid for Nente Co acquisition = ($2·95 x 2,400,000 shares)	$(7,080,000)
Value added for Mije shareholders =	$4,520,000
Gain in value to a Mije Co share = $4,520,000/10,000,000 =	45·2c
or 45·2c/480c =	9·4%

Share-for-share Offer
Earnings combined company = $620,000 + $150,000 + $3,200,000 = $3,970,000
Shares in combined company = 10,000,000 + 2,400,000 x 2/3 = 11,600,000
EPS = 34·2c/share [$3,970,000/11,600,000]
Expected share price = 34·2c x 15 = 513c or $5·13/share

Three Nente Co shares = $2·90 x 3 = $8·70

Gain in value to a Mije Co share = ($5·13 – $4·80)/$4·80 =	6·9%
Gain in value to a Nente Co share = ($10·26 – $8·70)/$8·70 =	17·9%

Appendix iii: Increase in Value of Follow-On Product

Present value of the positive cash flows	= $2,434,000
Present value of the cash outflow	= $(2,029,000)
Net present value of the new product	= $405,000

Based on conventional NPV, without considering the value of the option to delay the decision, the project would increase the value of the company by $405,000.

Considering the value of the option to delay the decision

Price of asset (PV of future positive cash flows)	=	$2,434,000
Exercise price (initial cost of project, not discounted)	=	$2,500,000
Time to expiry of option	=	2 years
Risk free rate (estimate)	=	3·2%
Volatility	=	42%

$d_1 = [\ln(2{,}434/2{,}500) + (0{\cdot}032 + 0{\cdot}5 \times 0{\cdot}42^2) \times 2]/(0{\cdot}42 \times 2^{1/2}) = 0{\cdot}359$
$d_2 = 0{\cdot}359 - (0{\cdot}42 \times 2^{1/2}) = -0{\cdot}235$

N(d1) = 0·5 + (0·1368 + 0·9 x (0·1406 – 0·1368)) = 0·6402
N(d2) = 0·5 – (0·0910 + 0·5 x (0·0948 – 0·0910)) = 0·4071

Value of option to delay the decision = 2,434,000 x 0·6402 – 2,500,000 x 0·4071 x $e^{-(0{\cdot}032 \times 2)}$
= $1,558,247 – $954,655 = $603,592

The project increases the value of the company by $603,592 or 25·1c per share ($603,592/2,400,000 shares). In percentage terms this is an increase of about 8·7% (25·1c/290c).

2 (a) Forecast financial position

Amounts in $'000	Current	Proposal 1	Proposal 2	Proposal 3
Non-current assets	282,000	282,000	302,000	257,000
Current assets	66,000	64,720	67,720	63,682
Total assets	348,000	346,720	369,720	320,682
Current liabilities	37,000	37,000	37,000	37,000
Non-current liabilities	140,000	160,000	160,000	113,000
Total liabilities	177,000	197,000	197,000	150,000
Share capital (40c/share)	48,000	45,500	48,000	48,000
Retained earnings	123,000	104,220	124,720	122,682
Total equity	171,000	149,720	172,720	170,682
Total liabilities and capital	348,000	346,720	369,720	320,682

Adjustments to forecast earnings

Amounts in $'000	Current	Proposal 1	Proposal 2	Proposal 3
Initial profit after tax	26,000	26,000	26,000	26,000
Interest payable on additional borrowing ($20m x 6% x (1 – 0·2))		(960)	(960)	
Additional interest payable on extra coupon ($160m x 0·25% x (1 – 0·2))		(320)	(320)	
Interest saved on less borrowing ($27m x 6% x (1 – 0·2))				1,296
Interest saved on lower coupon ($113m x 0·15% x (1 – 0·2))				136
Return on additional investment ($20m x 15%)			3,000	
Return lost on less investment ($25m x 15%)				(3,750)
Profit on sale of non-current assets				2,000
Adjusted profit after tax	26,000	24,720	27,720	25,682

	Current	Proposal 1	Proposal 2	Proposal 3
Gearing % (non-current liabilities/equity)	81·9%	106·9%	92·6%	66·2%
Number of shares ('000)	120,000	113,750	120,000	120,000
Earnings per share (adjusted profit after tax/number of shares)	21·67c	21·73c	23·10c	21·40c

Note: *Gearing defined as non-current liabilities/(non-current liabilities + equity) and/or using market value of equity is acceptable as well.*

The profit from the sale of the assets for proposal 3, of $2,000,000, is assumed to be after tax. Answers which consider the profit to be before tax, and therefore only take into account $1,600,000 as the net profit, will receive full credit.

Tutorial Note *(Explanations are not required for the answer but are included to explain the approach taken)*

Explanations of the financial position based on the three proposals

Proposal 1
Debt is increased by $20m and share capital reduced by the same amount as follows: from par value = $20m x 40c/320c = $2·5m; from retained earnings = $20m x 280c/320c = $17·5m.

Additional interest payable totaling $1,280,000 ($960,000 + $320,000) is taken off retained earnings due to reduction in profit after tax and taken off current assets because presumably it is paid from cash. Note that an alternative answer would be to add the additional interest payable to current liabilities.

Proposal 2
Debt and non-current assets are increased by $20m.

Additional interest payable as above, plus the additional investment of $20 million will generate a rate of return of 15%, which is $3,000,000 income. Net impact is $1,720,000 income which is added to retained earnings as an addition to profit after tax and added to current assets as a cash income (presumably).

Proposal 3
Net non-current assets are reduced by the $25 million, their value at disposal. Since they were sold for $27 million, this is how much the non-current liabilities are reduced by and the profit of $2 million is included in the retained earnings.

Interest saved totals $1,432,000 ($1,296,000 + $136,000). The reduction in investment of $25 million will lose $3,750,000, at a rate of return of 15%. Net impact is $2,318,000 loss which is subtracted from earnings as a reduction from profit after tax and deducted from current assets as a cash expense (presumably). Overall therefore the profit is reduced by $318,000 [$2,000,000 – $2,318,000].

If the profit from the sale of the asset is assumed to be $1,600,000 ($2,000,000 less tax), then the statement of financial position, EPS and gearing figures will all change to reflect this.

Discussion

Proposals 1 and 3 appear to produce opposite results to each other. Proposal 1 would lead to a small increase in the earnings per share (EPS) due to a reduction in the number of shares although profits would decrease by approximately 5%, due to the increase in the amount of interest payable as a result of increased borrowings. However, the level of gearing would increase substantially (by about 30%).

With proposal 3, although the overall profits would fall, because of the lost earnings due to downsizing being larger than the gain in interest saved and profit made on the sale of assets, this is less than proposal 1 (1·2%). Gearing would reduce substantially (19·2%).

Proposal 2 would give a significant boost in the EPS from 21·67c/share to 23·10c/share, which the other two proposals do not. This is mainly due to increase in earnings through extra investment. However, the amount of gearing would increase by more than 13%.

Overall proposal 1 appears to be the least attractive option. The choice between proposals 2 and 3 would be between whether the company would prefer larger EPS or less gearing. This would depend on factors such as the capital structure of the competitors, the reaction of the equity market to the proposals, the implications of the change in the risk profile of the company and the resultant impact on the cost of capital. Ennea Co should also bear in mind that the above are estimates and the actual results will probably differ from the forecasts.

(Note: credit will be given for alternative relevant comments and suggestions)

(b) Asset securitisation in this case would involve taking the future incomes from the leases that Ennea Co makes and converting them into assets. These assets are sold as bonds now and the future income from lease interest will be used to pay coupons on the bonds. Effectively Ennea Co foregoes the future lease income and receives money from sale of the assets today.

The income from the various leases would be aggregated and pooled, and new securities (bonds) issued based on these. The tangible benefit from securitisation occurs when the pooled assets are divided into tranches and tranches are credit rated. The higher rated tranches would carry less risk and have less return, compared to lower rated tranches. If default occurs, the income of the lower tranches is reduced first, before the impact of increasing defaults move to the higher rated tranches. This allows an asset of low liquidity to be converted into securities which carry higher liquidity.

Ennea Co would face a number of barriers in undertaking such a process. Securitisation is an expensive process due to management costs, legal fees and ongoing administrative costs. The value of assets that Ennea Co wants to sell is small and therefore these costs would take up a significant proportion of the income. High cost implications mean that securitisation is not feasible for small asset pools.

Normally asset pools would not offer the full value of the asset as securities. For example, only 90% of the asset value would be converted into securities, leaving the remaining 10% as a buffer against possible default. This method of credit enhancement would help to credit-rate the tranches at higher levels and help their marketability. However, Ennea Co would not be able to take advantage of the full asset value if it proceeds with the asset securitisation.

(Note: credit will be given for alternative relevant comments and suggestions)

3 (a) Gross amounts of annual interest receivable by Sembilan Co from Ratus Bank based on year 1 spot rate and years 2, 3 and 4 forward rates:

Year 1 0·025 x $320m = $8m
Year 2 0·037 x $320m = $11·84m
Year 3 0·043 x $320m = $13·76m
Year 4 0·047 x $320m = $15·04m

Gross amount of annual interest payable by Sembilan Co to Ratus Bank:

3·76¼% x $320m = $12·04m

At the start of the swap, Sembilan Co will expect to receive or (pay) the following net amounts at the end of each of the next four years:

Year 1: $8m – $12·04m = $(4·04m) payment
Year 2: $11·84m – $12·04m = $(0·20m) payment
Year 3: $13·76m – $12·04m = $1·72m receipt
Year 4: $15·04m – $12·04m = $3m receipt

Tutorial Note: *At the commencement of the swap contract the net present value of the net annual flows, discounted at the yield curve rates, is zero.*

The reason the equivalent fixed rate of 3·76¼% is less than the 3·8% four-year yield curve rate, is because the 3·8% rate reflects the zero-coupon rate with only one payment made in year four. Here the bond pays coupons at different time periods when the yield curve rates are lower. Therefore the fixed rate is lower.

(b) After taking the swap, Sembilan Co's net effect is as follows:

	% Impact	Yield Interest 3%	Yield Interest 4%
Borrow at yield interest + 60bp	(Yield+0·6)%	\$(11·52m)	\$(14·72m)
Receive yield	Yield	\$9·6m	\$12·8m
Pay fixed 3·76¼%	(3·76¼)%	\$(12·04m)	\$(12·04m)
Fee 20bp	(0·2)%	\$(0·64m)	\$(0·64m)
Net Cost	(4·56¼)%	\$(14·6m)	\$(14·6m)

The receipt and payment based on the yield curve cancels out interest rate fluctuations, fixing the rate at 3·76¼% + 0·6% + 0·2% = 4·56¼%

(c) Reducing the amount of debt by issuing equity and using the cash raised from this to reduce the amount borrowed changes the capital structure of a company and Sembilan Co needs to consider all the possible implications of this.

As the proportion of debt increases in a company's financial structure, the level of financial distress increases and with it the associated costs. Companies with high levels of financial distress would find it more costly to contract with their stakeholders. For example, they may have to pay higher wages to attract the right calibre of employees, give customers longer credit periods or larger discounts, and may have to accept supplies on more onerous terms. Furthermore, restrictive covenants may make it more difficult to borrow funds (debt and equity) for future projects. On the other hand, because interest is payable before tax, larger amounts of debt will give companies greater taxation benefits, known as the tax shield. Presumably, Sembilan Co has judged the balance between the levels of equity and debt finance, such that the positive and negative effects of gearing result in minimising the required rate of return and maximising the value of the company.

By replacing debt with equity the balance may no longer be optimal and therefore the value of Sembilan Co may not be maximised. However, reducing the amount of debt would result in a higher credit rating for the company and reduce the scale of restrictive covenants. Having greater equity would also increase the company's debt capacity. This may enable the company to raise additional finance and undertake future profitable projects more easily. Less financial distress may also reduce the costs of contracting with stakeholders.

The process of changing the financial structure can be expensive. Sembilan Co needs to determine the costs associated with early redemption of debt. The contractual clauses of the bond should indicate the level and amount of early redemption penalties. Issuing new equity can be expensive especially if the shares are offered to new shareholders, such as costs associated with underwriting the issue and communicating or negotiating the share price. Even raising funds by issuing rights can be expensive.

As well as this, Sembilan Co needs to determine the extent to which the current shareholders will be able to take up the rights and the amount of discount that needs to be given on the rights issue to ensure 100% take up. The impact on the current share price from the issue of rights needs to be considered as well. Studies on rights issues seem to indicate that the markets view the issue of rights as a positive signal and the share price does not reduce to the expected theoretical ex-rights price. However, this is mainly because the markets expect the funds raised to be used on new, profitable projects. Using funds to reduce the debt amount may not be viewed so positively.

Sembilan Co may also have to provide information and justification to the market because both the existing shareholders and any new shareholders will need to be assured that the company is not benefiting one group at the expense of the other. If sufficient information is not provided then either shareholder group may discount the share price due to information asymmetry. However, providing too much information may reduce the competitive position of the company.

(Note: credit will be given for alternative relevant comments and suggestions)

4 (a) Use Elfu Co's information to estimate the component project's asset beta. Then based on Tisa Co's capital structure, estimate the component project's equity beta and weighted average cost of capital. Assume that the beta of debt is zero.

Elfu Co MVe = \$1·20 x 400m shares = \$480m
Elfu Co MVd = \$96m

Elfu Co portfolio asset beta =
1·40 x \$480m/(\$480m + \$96m x (1 – 0·25)) = 1·217
Elfu Co asset beta of other activities =
1·25 x \$360m/(\$360m + \$76·8m x (1 – 0·25)) = 1·078

1·217 = component asset beta x 0·25 + 1·078 x 0·75
Component asset beta = [1·217 – (1·078 x 0·75)]/0·25 = 1·634

Component equity beta based on Tisa Co capital structure =
1·634 x [(\$18m + \$3·6m x 0·75)/\$18m] = 1·879
Using CAPM, component Ke = 3·5% + 1·879 x 5·8% = 14·40%
Component WACC = (14·40% x \$18m + 4·5% x \$3·6m)/(\$18m + \$3·6m) = 12·75%

(b) **Process Omega**

Year	**0**	**1**	**2**	**3**	**4**
Net cash flows ($000)	(3,800)	1,220	1,153	1,386	3,829
PV 12·75% ($000)	(3,800)	1,082	907	967	2,369
NPV ($000)	1,525				
PV 30%	(3,800)	938	682	631	1,341
NPV ($000)	(208)				

Internal rate of return is approximately 27·3%

Modified internal rate of return (MIRR) is approximately 22·7% ($[(5,325/3,800)^{1/4} \times (1{\cdot}1275)] - 1$)

Alternatively:

MIRR can be calculated as follows:

Year	**Cashflows ($000)**	**Multiplier**	**Re-invested amount ($000)**
1	1,220	$1{\cdot}1275^3$	1,749
2	1,153	$1{\cdot}1275^2$	1,466
3	1,386	1·1275	1,563
4	3,829	1	3,829

Total re-invested amount ($000) = 8,607
MIRR = $(8,607/3,800)^{1/4} - 1$ = 22·7%

The internal rate of return (IRR) assumes that positive cash flows in earlier years are reinvested at the IRR and therefore process Omega, which has higher initial cash flows when compared to process Zeta, gives a slightly higher IRR. The modified internal rate of return (MIRR) assumes that positive cash flows are reinvested at the cost of capital. This is a more reasonable assumption and produces a result consistent with the net present value. Hence, process Zeta should be adopted, although the difference is not significant.

[**Note:** *Using 13% instead of 12·75% as the cost of capital is acceptable*]

(c) 99% confidence level requires the value at risk (VAR) to be within 2·33 standard deviations from the mean, based on a single tail measure.

Annual VAR = 2·33 x $800,000 = $1,864,000
Five year VAR = $1,864,000 x $5^{1/2}$ approx. = $4,168,000

The figures mean that Elfu Co can be 99% confident that the cash flows will not fall by more than $1,864,000 in any one year and $4,168,000 in total over five years from the average returns. Therefore the company can be 99% certain that the returns will be $336,000 or more every year [$2,200,000 – $1,864,000]. And it can be 99% certain that the returns will be $6,832,000 or more in total over the five-year period [$11,000,000 – $4,168,000]. There is a 1% chance that the returns will be less than $336,000 each year or $6,832,000 over the five-year period.

5 **(Solution note: The following answer for question 5(a) is indicative. Credit will be given for alternative suggestions of risks and issues, and their management or control.)**

(a) Kilenc Co needs to consider a number of risks and issues when making the decision about whether or not to set up a subsidiary company in Lanosia. It should then consider how these may be managed or controlled.

Key Risks/Issues

Kilenc Co needs to assess the impact on its current exports to Lanosia and the nearby countries if the subsidiary is set up. Presumably, products are currently exported to these countries and if these exports stop, then there may be a negative impact on the employees and facilities currently employed in this area. Related to this may be the risk of loss of reputation if the move results in redundancies. Furthermore, Kilenc Co should consider how the subsidiary and its products would be seen in Lanosia and the nearby countries. For example, would the locally made products be perceived as being of comparative quality as the imported products?

The recession in Lanosia may have a negative impact on the market for the products. The cost of setting up the subsidiary company needs to be compared with the benefits from extra sales revenue and reduced costs. There is a risk that the perceived benefits may be less than predicted or the establishment of a subsidiary may create opportunities in the future once the country recovers from the recession.

Currently the government offers support for companies involved in the pharmaceutical industry. Kilenc Co may find it difficult to set up the subsidiary if it is viewed as impeding the development of the local industry by the government. For example, the government may impose restrictions or increase the taxes the subsidiary may have to pay. On the other hand, the subsidiary may be viewed as supporting the economy and the growth of the pharmaceutical industry, especially since 40% of the shares and 50% of the Board of Directors would be in local hands. The government may even offer the same support as it currently offers the other local companies.

Kilenc Co wants to finance the subsidiary through a mixture of equity and debt. The implications of raising equity finance are discussed in part (b) of the question. However, the risks surrounding debt finance needs further discussion. Raising debt finance in Lanosia would match the income generated in Lanosia with debt interest payments, but the company needs to consider whether or not it would be possible to borrow the money. Given that the government has had to finance the banks may mean that the availability of funds to borrow may be limited. Also interest rates are low at the moment but inflation is high, this may result in pressure on the government to raise interest rates in the future. The consequences of this may be that the borrowing costs increase for Kilenc Co.

The composition of the Board of Directors and the large proportion of the subsidiary's equity held by minority shareholders may create agency issues and risks. Kilenc Co may find that the subsidiary's Board may make decisions which are not in the interests of the parent company, or that the shareholders attempt to move the subsidiary in a direction which is not in the interests of the parent company. On the other hand, the subsidiary's Board may feel that the parent company is imposing too many restrictions on its ability to operate independently and the minority shareholders may feel that their interests are not being considered by the parent company.

Kilenc Co needs to consider the cultural issues when setting up a subsidiary in another country. These may range from cultural issues of different nationalities and doing business in the country to cultural issues within the organisation. Communication of how the company is organised and understanding of cultural issues is very important in this case. The balance between independent autonomy and central control needs to be established and agreed.

Risks such as foreign exchange exposure, product health and safety compliance, employee health and safety regulations and physical risks need to be considered and assessed. For example, foreign exchange exposures arising from exporting the products to nearby countries need to be assessed. The legal requirements around product health and safety and employee health and safety need to be understood and complied with. Risks of physical damage such as from floods or fires on the assets of the business need to be established.

Mitigating the Risks and Issues

A full analysis of the financial costs and benefits should be undertaken to establish the viability of setting up the subsidiary. Sensitivity and probability analysis should be undertaken to assess the impact and possibility of falling revenues and rising costs. Analysis of real options should be undertaken to establish the value of possible follow-on projects.

Effective marketing communication such as media advertising should be conducted on the products produced by the subsidiary to ensure that the customers' perceptions of the new products do not change. This could be supported by retaining the packaging of the products. Internal and external communication should explain the consequences of any negative impact of the move to Lanosia to minimise any reputational damage. Where possible, employees should be redeployed to other divisions, in order to minimise any negative disruption.

Negotiations with the Lanosian government should be undertaken regularly during the process of setting up the subsidiary to minimise any restrictions and to maximise any benefits such as favourable tax rates. Where necessary and possible, these may be augmented with appropriate insurance and legal advice. Continuing lobbying may also be necessary after the subsidiary has been established to reduce the possibility of new rules and regulations which may be detrimental to the subsidiary's business.

An economic analysis may be conducted on the likely movements in inflation and interest rates. Kilenc Co may also want to look into using fixed rate debt for its long-term financing needs, or use swaps to change from variable rates to fixed rates. The costs of such activity need to be taken into account.

Clear corporate governance mechanisms need to be negotiated and agreed on, to strike a balance between central control and subsidiary autonomy. The negotiations should involve the major parties and legal advice may be sought where necessary. These mechanisms should be clearly communicated to the major parties.

The subsidiary organisation should be set up to take account of cultural differences where possible. Induction sessions for employees and staff handbooks can be used to communicate the culture of the organisation and how to work within the organisation.

Foreign exchange exposure, health and safety regulation and risk of physical loss can be managed by a combination of hedging, insurance and legal advice.

(b) Dark pool trading systems allow share orders to be placed and matched without the traders' interests being declared publicly on the normal stock exchange. Therefore the price of these trades is determined anonymously and the trade is only declared publicly after it has been agreed. Large volume trades which use dark pool trading systems prevent signals reaching the markets in order to minimise large fluctuations in the share price or the markets moving against them.

The main argument put forward in support of dark pool trading systems is that by preventing large movements in the share price due to volume sales, the markets' artificial price volatility would be reduced and the markets maintain their efficiency. The contrary arguments suggest that in fact market efficiency is reduced by dark pool trading systems because such trades do not contribute to the price changes. Furthermore, because most of the individuals who use the markets to trade equity shares are not aware of the trade, transparency is reduced. This, in turn, reduces the liquidity in the markets and therefore may compromise their efficiency. The ultimate danger is that the lack of transparency and liquidity may result in an uncontrolled spread of risks similar to what led to the recent global financial crisis.

It is unlikely that the dark pool trading systems would have an impact on Kilenc Co's subsidiary company because the subsidiary's share price would be based on Kilenc Co's share price and would not be affected by the stock market in Lanosia. Market efficiency in general in Lanosia would probably be much more important.

			Marks
1	**Appendix i**		
	Based on PBIT, calculation of the growth rate		2
	Calculation of free cash flows		2
	Calculation of company value, equity value and value of each share		3
			7
	Appendix ii		
	Cash offer		
	Additional value created for Mije Co shareholders		3
	Value created per share for Nente Co shareholders		1
	Share-for-share offer		
	Expected share price for the combined company		2
	Value created for Nente Co share		1
	Value created for Mije Co share		1
			8
	Appendix iii		
	PV of underlying asset		1
	Value of exercise price		1
	$N(d_1)$		2
	$N(d_2)$		2
	Value of call		1
	Value added to Nente Co share		1
			8
	Discussion		
	Nente Co shareholders		2–3
	Mije Co shareholders		1–2
	Assumptions made		2–3
	Use of value of follow-on product		2–3
		Max	8
	Professional Marks		
	Report format		1
	Structure and presentation of the report		3
			4
		Total	**35**

				Marks
2	**(a)**	Financial position calculations: proposal 1		2
		Financial position calculations: proposal 2		2
		Financial position calculations: proposal 3		2
		Adjustments to forecast earnings		
		Interest payable on additional borrowing and higher coupon		2
		Interest saved lower borrowing and lower coupon		1
		Return on additional investment		1
		Return lost on less investment and profit on sale of non-current assets		1
		Gearing and EPS calculations		4
		Discussion of the results of the proposals		2–3
		Discussion of the implications (e.g. risk, market reaction, etc)		2–3
			Max	20
	(b)	Explanation of the process		2–3
		Key barriers in undertaking the process		2–3
			Max	5
			Total	**25**
3	**(a)**	Gross amount receivable by Sembilan Co		1
		Gross amounts payable by Sembilan Co		1
		Net amounts receivable or payable every year		2
		Explanation of why fixed rate is less than the four-year yield curve rate		2
				6
	(b)	Demonstration of impact of interest rate changes		4
		Explanation and conclusion		1
				5
	(c)	1–2 marks per relevant discussion point	Max	9
			Total	**20**
4	**(a)**	Reasoning behind cost of capital calculation		2
		Calculation of component asset beta		3
		Calculation of component equity beta, and Ke and WACC		3
				8
	(b)	Calculation of IRR for Process Omega		3
		Calculation of MIRR for Process Omega		1
		Recommendation and explanation of the recommendation		4
				8
	(c)	Annual and five-year VAR		2
		Explanation		2
				4
			Total	**20**

			Marks
5	**(a)**	Discussion of key risks or issues (2–3 marks for each)	8–10
		Suggestions for management or control of the risk or issue (1–2 marks each)	6–8
		Max	15
	(b)	Explanation of dark pool trading systems	2–3
		Consequences and how these would affect Kilenc Co	2–3
		Max	5
		Total	**20**

Answers

1 (a) Before implementing the proposal

Cost of equity = 4% + 1·1 x 6% = 10·6%
Cost of debt = 4% + 0·9% = 4·9%

Market value of debt (MV_d):
Per \$100: $5·2 x $1·049^{-1}$ + \$5·2 x $1·049^{-2}$ + \$105·2 x $1·049^{-3}$ = \$100·82
Total value = \$42,000,000 x \$100·82/\$100 = \$42,344,400

Market value of equity (MV_e):
As share price is not given, use the free cash flow growth model to estimate this. The question states that the free cash flow to equity model provides a reasonable estimate of the current market value of the company.

Assumption 1: *Estimate growth rate using the rb model. The assumption here is that free cash flows to equity which are retained will be invested to yield at least at the rate of return required by the company's shareholders. This is the estimate of how much the free cash flows to equity will grow by each year.*

r = 10·6% and b = 0·4, therefore g is estimated at 10·6% x 0·4 = 4·24%

MV_e = 2,600 x 1·0424/(0·106 – 0·0424) approximately = \$42,614,000

The proportion of MV_e to MV_d is approximately 50:50

Therefore, cost of capital:
10·6% x 0·5 + 4·9% x 0·5 x 0·8 = 7·3%

After implementing the proposal

Coeden Co, asset beta estimate
1·1 x 0·5/(0·5 + 0·5 x 0·8) = 0·61

Asset beta, hotel services only

Assumption 2: *The question does not provide an asset beta for hotel services only, which is the approximate measure of Coeden Co's business risk once the properties are sold. Assume that Coeden Co's asset beta is a weighted average of the property companies' average beta and hotel services beta.*

Asset beta of hotel services only:
0·61 = Asset beta (hotel services) x 60% + 0·4 x 40%
Asset beta (hotel services only) approximately = 0·75

Coeden Co, hotel services only, estimate of equity beta:

MV_e = \$42,614,000 (Based on the assumption stated in the question)

MV_d = Per \$100: \$5·2 x $1·046^{-1}$ + \$5·2 x $1·046^{-2}$ + \$105·2 x $1·046^{-3}$ = \$101·65
Total value = \$12,600,000 x \$101·65/\$100 = \$12,807,900 say \$12,808,000

0·75 = equity beta x 42,614/(42,614 + 12,808 x 0·8)

0·75 = equity beta x 0·806

Equity beta = 0·93

Coeden Co, hotel services only, weighted average cost of capital

Cost of equity = 4% + 0·93 x 6% = 9·6%

Cost of capital = 9·6% x 0·769 + 4·6% x 0·231 x 0·8 = 8·2%

Comment:

	Before proposal implementation	**After proposal implementation**
Cost of equity	10·6%	9·6%
WACC	7·3%	8·2%

Implementing the proposal would increase the asset beta of Coeden Co because the hotel services industry on its own has a higher business risk than a business which owns its own hotels as well. However, the equity beta and cost of equity both decrease because of the fall in the level of debt and the consequent reduction in the company's financial risk. The company's cost of capital increases because the lower debt level reduces the extent to which the weighted average cost of capital can be reduced due to the lower cost of debt. Hence the board of directors is not correct in assuming that the lower level of debt will reduce the company's cost of capital.

(b) It is unlikely that the market value of equity would remain unchanged because of the change in the growth rate of free cash flows and sales revenue, and the change in the risk situation due to the changes in the business and financial risks of the new business.

In estimating the asset beta of Coeden Co as offering hotel services only, no account is taken of the changes in business risk due to renting rather than owning the hotels. A revised asset beta may need to be estimated due to changes in the business risk.

The market value of equity is used to estimate the equity beta and the cost of equity of the business after the implementation of the proposal. But the market value of equity is dependent on the cost of equity, which is, in turn, dependent on the equity beta. Therefore, neither the cost of equity nor the market value of equity is independent of each other and they both will change as a result of the change in business strategy.

(c) Demerger
A demerger would involve the company splitting into two (or more) parts, with each part becoming a separate, independent company. The shareholders would then hold shares in each separate, independent company. Each company would most probably have its own separate management team. On the other hand, selling the hotel properties outright would be termed as a divestment, where a company would sell part of its assets.

Benefits
There are a number of possible benefits in pursuing a demerger option for Coeden Co and its shareholders. The management teams would be able to focus on creating value for each company separately, and create a unique financial structure that is suitable for each company. The full value of each company would become apparent as a result. Coeden Co's shareholders may have invested in the company specifically for its risk profile and selling the properties may imbalance their portfolios. With a demerger, the portfolio diversification remains unchanged. Communication may be stronger between the two management teams with a demerger. Since Coeden Co trades heavily on its brand name, the quality and maintenance of the hotel properties is critical, and good communication links will help ensure that these are safeguarded. However, responsibility for maintenance of the properties will need to be negotiated. Selling a lot of properties all at once may flood the market and lower the value that can be obtained for each hotel property.

Drawbacks
A number of possible drawbacks for Coeden Co and its shareholders may occur if it pursues the option to demerge. The demerger may be an expensive process to undertake and may result in a decline in the value of the companies overall. The bond holders may not agree to 70% of the long-term loans to be transferred to a property company and may ask for the terms of the loans to be re-negotiated. The new property company would need to raise the extra finance to pay the cash for the remaining property values. Coeden Co may not have the expertise amongst its management staff to manage a property company or to recruit an appropriate management team. Overall, the main drawbacks revolve around the additional costs that would probably need to be incurred if the demerger option is pursued.

[Note: Credit will be given for alternative, valid points]

2 Report to the Treasury Division, Lignum Co

Discussion and recommendations for managing the foreign exchange exposure

The report discusses and makes recommendations on how the treasury division may manage the foreign exchange exposure it faces under three unrelated circumstances or cases.

The appendices to the report show the detailed calculations to support the discussion around case one (see appendix I) and around case two (see appendix II).

Foreign exchange exposures
With case one, Lignum Co faces a possible exposure due to the receipt it is expecting in four months in a foreign currency, and the possibility that the exchange rates may move against it between now and in four months time. This is known as transactions exposure. With case two, the exposure is in the form of translation exposure, where a subsidiary's assets are being translated from the subsidiary's local currency into Euro. The local currency is facing an imminent depreciation of 20%. Finally in the third case, the present value of future sales of a locally produced and sold good is being eroded because of overseas products being sold for a relatively cheaper price. The case seems to indicate that because the US$ has depreciated against the Euro, it is possible to sell the goods at the same dollar price but at a lower Euro price. This is known as economic exposure.

Hedging strategies

Case one
Transactions exposure, as faced by Lignum Co in situation one, lasts for a short while and is easier to manage by means of derivative products or more conventional means. Here Lignum Co has access to two derivative products: an OTC forward rate and OTC option. Using the forward rate gives a higher return of €963,988, compared to options where the return is €936,715 (see appendix I). However, with the forward rate, Lignum Co is locked into a fixed rate (ZP145·23 per €1) whether the foreign exchange rates move in its favour or against it. With the options, the company has a choice and if the rate moves in its favour, that is if the Zupeso appreciates against the Euro, then the option can be allowed to lapse. Lignum Co needs to decide whether it is happy receiving €963,988, no matter what happens to the exchange rate over the four months or whether it is happy to receive at least €936,715 if the ZP weakens against the €, but with a possibility of higher gains if the Zupeso strengthens.

Lignum Co should also explore alternative strategies to derivative hedging. For example, money markets, leading and lagging, and maintaining a Zupeso account may be possibilities. If information on the investment rate in Zupesos could be obtained, then a money market hedge could be considered. Maintaining a Zupeso account may enable Lignum Co to offset any natural hedges and only convert currency periodically to minimise transaction costs.

Case two
Hedging translation risk may not be necessary if the stock market in which Lignum Co's shares are traded is efficient. Translation of currency is an accounting entry where subsidiary accounts are incorporated into the group accounts. No physical cash flows in or out of the company. In such cases, spending money to hedge such risk means that the group loses money overall, reducing the cash flows attributable to shareholders. However, translation losses may be viewed negatively by the equity holders and may impact some analytical trends and ratios negatively. In these circumstances, Lignum Co may decide to hedge the risk.

The most efficient way to hedge translation exposure is to match the assets and liabilities. In Namel Co's case the assets are more exposed to the Maram Ringit compared to the liabilities, hence the weakening of the Maram Ringit from MR35 per €1 to MR42 per €1 would make the assets lose more (accounting) value than the liabilities by €1,018,000 (see appendix II). If the exposure for the assets and liabilities were matched more closely, for example by converting non-current liabilities from loans in Euro to loans in MR, translation exposure would be reduced.

Case three
Economic exposure, which is not part of transactions exposure, is long-term in nature and therefore more difficult to manage. There are for example, few derivatives which are offered over a long period, with the possible exception of swaps. A further issue is that economic exposure may cause a substantial negative impact to a company's cash flows and value over the long period of time. In this situation, if the US$ continues to remain weak against the Euro, then Lignum Co will find it difficult to maintain a sustained advantage against its American competitor. A strategic, long-term viewpoint needs to be undertaken to manage risk of this nature, such as locating production in countries with favourable exchange rates and cheaper raw material and labour inputs or setting up a subsidiary company in the USA to create a natural hedge for the majority of the US$ cash flows.

In conclusion, the report examined, discussed and made recommendations on managing foreign exchange exposure in each of the three cases.

Report compiled by:

Date:

APPENDICES

Appendix I: Financial impact of derivative products offered by Medes Bank (case one)

Using forward rate
Forward rate = 142 x (1 + (0·085 + 0·0025)/3)/(1 + (0·022 – 0·0030)/3) = 145·23
Income in Euro fixed at ZP145·23 = ZP140,000,000/145·23 = €963,988

Using OTC options
Purchase call options to cover for the ZP rate depreciating
Gross income from option = ZP140,000,000/142 = €985,915

Cost
€985,915 x ZP7 = ZP6,901,405
In € = ZP6,901,405/142 = €48,601
€48,601 x (1 + 0·037/3) = €49,200
(Use borrowing rate on the assumption that extra funds to pay costs need to borrowed initially; investing rate can be used if that is the stated preference)

Net income = €985,915 – €49,200 = €936,715

Appendix II: Financial impact of the devaluation of the Maram Ringit (case two)

MR devalued rate = MR35 x 1·20 = MR42 per €1

	MR '000	Exposed?	€ '000 at current rate MR35 per €1	€ '000 at devalued rate MR42 per €1
Non-current assets	179,574	Yes	5,131	4,276
Current assets	146,622	60%	2,514	2,095
Non-current liabilities	(132,237)	20%	(756)	(630)
Current liabilities	(91,171)	30%	(781)	(651)
Share capital and reserves	102,788		6,108	5,090

Translation loss = €6,108,000 – €5,090,000 = €1,018,000

3 (a) Number of Sigra Co shares = 4,400,000/0·4 = 11,000,000 shares
Sigra Co earnings per share (EPS) = $4,950,000/11,000,000 shares = 45c/share
Sigra Co price to earnings (PE) ratio = $3·6/$0·45 = 8

Dentro PE ratio = 8 x 1·125 = 9
Dentro Co shares = $500,000/0·4 = 1,250,000 shares
Dentro Co EPS = $625,000/1,250,000 = 50c/share
Estimate of Dentro Co value per share = $0·5 x 9 = $4·50/share

Cash offer
Dentro share percentage gain under cash offer
$0·50/$4·50 x 100% = 11·1%

Share-for-share exchange

Equity value of Sigra Co = 11,000,000 x $3·60 =	$39,600,000
Equity value of Dentro Co = 1,250,000 x $4·50 =	$5,625,000
Synergy savings = 30% x $5,625,000 =	$1,688,000
Total equity value of combined company	$46,913,000
Number of shares for share-for-share exchange 11,000,000 + [1,250,000 x 3/2] =	12,875,000
Expected share price of combined company	$3·644/share

Dentro share percentage gain under share-for-share offer
[($3·644 x 3 – $4·50 x 2)/2]/$4·50 x 100% = 21·5%

Bond offer
Rate of return

$\$104 = \$6 \times (1 + r)^{-1} + \$6 \times (1 + r)^{-2} + \$106 \times (1 + r)^{-3}$

If r is 5%, price is $102·72
If r is 4%, price is $105·55

r is approximately = 4% + (105·55 – 104)/(105·55 – 102·72) x 1% = 4·55%

Price of new bond =
$\$2 \times 1{\cdot}0455^{-1} + \$2 \times 1{\cdot}0455^{-2} + \$102 \times 1{\cdot}0455^{-3} = \$93{\cdot}00$

Value per share = $93·00/16 = $5·81/share

Dentro share percentage gain under bond offer
Bond offer: ($5·81 – $4·50)/$4·50 x 100% = 29·1%

Comments
An initial comparison is made between the cash and the share-for-share offers. Although the share-for-share exchange gives a higher return compared to the cash offer, Dentro Co's shareholders may prefer the cash offer as the gains in the share price are dependent on the synergy gains being achieved. However, purchase for cash may mean that the shareholders face an immediate tax burden. Sigra Co's shareholders would probably prefer the cash option because the premium would only take $625,000 of the synergy benefits ($0·50 x 1,250,000 shares), whereas a share-for-share exchange would result in approximately $1,209,000 of the synergy benefits being given to the Dentro Co shareholders (21·5% x $4·50 x 1,250,000 shares).

The bond offer provides an alternative which may be acceptable to both sets of shareholders. Dentro Co's shareholders receive the highest return for this and Sigra Co's shareholders may be pleased that a large proportion of the payment is deferred for three years. In present value terms, however, a very high proportion of the projected synergy benefits are given to Dentro Co's shareholders (29·1% x $4·50 x $1,250,000 = $1,637,000).

(b) The regulatory framework within the European Union, the EU takeovers directive, will be used to discuss the proposals. However it is acceptable for candidates to refer to other directives and discuss the proposals on that basis.

Proposal 1
With regards to the first proposal, the directive gives the bidder squeeze-out rights, where the bidder can force minority shareholders to sell their shares. However, the limits set for squeeze-out rights are generally high (UK: 90%; Belgium, France, Germany and the Netherlands: 95%; Ireland 80%). It is likely therefore that Sigra Co will need a very large proportion of Dentro Co's shareholders to agree to the acquisition before they can force the rest of Dentro Co's shareholders to sell their shares. Dentro Co's minority shareholders may also require Sigra Co to purchase their shares, known as sell-out rights.

Proposal 2
With regards to the second proposal, the principle of equal treatment in the directive requires that all shareholders should be treated equally. In general terms, the bidder must offer to minority shareholders the same terms as those offered to other shareholders. It could be argued here that the principle of equal treatment is contravened because later shareholders are not offered the extra 3 cents per share, even though the 30% is less than a majority shareholding. It is highly unlikely that Sigra Co will be allowed to offer these terms.

4 **(a)** **PDur05**

Annual sales revenue = \$14 x 300,000 units = \$4,200,000
Annual costs = \$3,230,000
Annual cash flows = \$970,000

Net present value of PDur05 =
(\$2,500,000) + (\$1,200,000 x $1{\cdot}11^{-1}$) + (\$1,400,000 x $1{\cdot}11^{-2}$) + \$970,000 x 7·191 x $1{\cdot}11^{-3}$
= (\$2,500,000) + (\$1,081,000) + (\$1,136,000) + \$5,100,000
= \$383,000

In order for the net present value to fall to nil, the PV of the project's annual cash flows needs to equal to:
\$2,500,000 + \$1,081,000 + \$1,136,000 = \$4,717,000

Annual cash flows need to reduce to: \$4,717,000/(7·191 x $1{\cdot}11^{-3}$) = \$897,110
Sales revenue would reduce to: \$897,110 + \$3,230,000 = \$4,127,110
Selling price would fall to: \$4,127,110/300,000 units = \$13·76
Percentage fall = (\$14·00 – \$13·76)/\$14 x 100% = 1·7%

[Note: The estimate of the annual cash flows will differ if tables are used rather than a calculator. This is acceptable and will be allowed for when marking]

Comment: The net present value of the project is very sensitive to changes in the selling price of the product. A small fall in the selling price would reduce the net present value to nil or negative and make the project not worthwhile.

(b) A multi-period capital rationing model would use linear programming and is formulated as follows:

If:
Y1 = investment in project PDur01; Y2 = investment in project PDur02; Y3 = investment in project PDur03; Y4 = investment in project PDur04; and Y5 = investment in project PDur05

Then the objective is to maximise
464Y1 + 244Y2 + 352Y3 + 320Y4 + 383Y5

Given the following constraints
Constraint year 1: 4,000Y1 + 800Y2 + 3,200Y3 + 3,900Y4 + 2,500Y5 ≤ 9,000
Constraint year 2: 1,100Y1 + 2,800Y2 + 3,562Y3 + 0Y4 + 1,200Y5 ≤ 6,000
Constraint year 3: 2,400Y1 + 3,200Y2 + 0Y3 + 200Y4 + 1,400Y5 ≤ 5,000

And where Y1, Y2, Y3, Y4, Y5 ≥ 0

(c) **Category 1:** Total Final Value. This is the maximum net present value that can be earned within the three-year constraints of capital expenditure, by undertaking whole, part or none of the five projects. This amount is less than the total net present value of all five projects if there were no constraints.

Category 2: Adjustable Final Values. These are the proportions of projects undertaken within the constraints to maximise the net present value. In this case, all of project PDur05, 95·8% of project PDur01, 73·2% of project PDur03 and 40·7% of project PDur02 will be undertaken.

Category 3: Constraints utilised, slack. This indicates to what extent the constraint limits are used and whether any investment funds will remain unused. The figures indicate that, in order to achieve maximum net present value, all the funds in all three years are used up and no funds remain unused.

(d) **(i)** Normally, positive net present value projects should be accepted as they add to the value of the company by generating returns in excess of the required rate of return (the discount rate). However, in this case, Arbore Co seems to be employing soft capital rationing by setting internal limits on capital available for each department, possibly due to capital budget limits placed by the company on the amounts it wants to borrow or can borrow. In the latter case, the company faces limited access to capital from external sources, for example, because of restrictions in bank lending, costs related to the issue of new capital and lending to the company being perceived as too risky. This is known as hard capital rationing and can lead to soft capital rationing.

(ii) A capital investment monitoring system (CIMS) monitors how an investment project is progressing once it has been implemented. Initially the CIMS will set a plan and budget of how the project is to proceed. It sets milestones for what needs to be achieved and by when. It also considers the possible risks, both internal and external, which may affect the project. CIMS then ensures that the project is progressing according to the plan and budget. It also sets up contingency plans for dealing with the identified risks.

The benefits, to Arbore Co, of CIMS are that it tries to ensure, as much as possible, that the project meets what is expected of it in terms of revenues and expenses. Also that the project is completed on time and risk factors that are identified remain valid. A critical path of linked activities which make up the project will be identified. The departments undertaking the projects will be proactive, rather than reactive, towards the management of risk, and therefore possibly be able to reduce costs by having a better plan. CIMS can also be used as a communication device between managers charged with managing the project and the monitoring team. Finally CIMS would be able to re-assess and change the assumptions made of the project, if changes in the external environment warrant it.

5 **(Solution note: The following answer for question 5 is indicative. Credit will be given for alternative, valid points.)**

(a) The role of the IMF is to oversee the global financial systems, in particular to stabilise international exchange rates, help countries to achieve balance of payments and facilitate in the country's development through influencing the economic policies of the country in question. Where necessary, it offers temporary loans, from member states' deposits, to countries facing severe financial and economic difficulties. These temporary loans are often offered with different levels of conditions or austerity measures.

The IMF believes that in order to regain control of the balance of payments, the country should take action to reduce the level of demand for goods and services. To achieve this, the IMF often requires countries to adopt strict austerity measures such as reducing public spending and increased taxation, as conditions of the loan. It believes these conditions will help control the inflationary pressures on the economy, and reduce the demand for goods and services. As a result, this will help the country to move away from a position of a trade deficit and achieve control of its balance of payments.

However, these deflationary pressures may cause standards of living to fall and unemployment to rise. The IMF regards these as short-term hardships necessary to help countries sort out their balance of payment difficulties and international debt problems. The IMF has faced a number of criticisms for the conditions it has imposed, including the accusation that its policies impact more negatively on people with lower or mid-range incomes, hinder long-term development and growth, and possibly result in a continuous downward spiral of economic activity.

Strom Co trades throughout Europe and economic activity in these countries has been curtailed in the last few years due, initially, to the banking crisis, and then due to the austerity measures that governments have adopted. For retailers, this could pose two possible problems. First, with limited growth and higher taxes, people would have less money to spend. Secondly, increasing levels of unemployment would also limit disposable incomes. It is possible that customers may have curtailed their expenditure on clothes and clothing accessories in order to meet other needs.

Some companies may have to spend proportionally more on marketing and possibly offer more discounts and other customer incentives in order to stay competitive and to maintain market share. This additional cost would hit their profit levels negatively. From the details in the question, it is evident that in 2011 profits have reduced by more than the fall in sales revenue. This could be due to Strom Co being forced to spend more on activities such as marketing or it could be because it has found it difficult to reduce its cost base. More analysis is needed to determine the exact cause.

It seems that Strom Co is trying to address the problem of declining profitability by trimming its costs. In circumstances where sales revenues are declining, and it is not possible to stabilise or increase these, then cost structures may need to be altered in order to make reasonable profits.

(b) The low-price clothing retailers might have benefited from the austerity measures because of a switch by customers from mid-price clothes to low-price clothes. If, due to the austerity measures, people have less money to spend, and if, as stated above, austerity measures impact the mid-income and low-income earners more negatively, then it is possible that their buying preferences change from mid-price to low-price clothes. This would be especially true if there is limited brand loyalty and customers perceive that the low-price items are of a similar quality or provide a better value-for-money.

On the other hand, it is possible that brand loyalty is more significant with high-price clothes, making switching to mid-price clothes difficult. Customers who buy these clothes may prefer not to switch and would rather spend less elsewhere. It is also possible that the austerity measures did not affect the population who buy high-price clothes to the same extent as other groups of the population. Or it may be that this population group is more resilient to the austerity measures imposed by the government, especially if the assertion is true that the IMF conditions affect people who are in the low or mid income categories more than the people in the high income category.

(c) The obvious risk in reducing resources allocated to the quality control functions would be that some inspections would be reduced. This may result in defective goods being sold. The costs related to processing returns of defective goods may outweigh the savings made. Reduction in monitoring the working conditions of employees of the clothing suppliers may encourage them to retain their questionable employment practices. This may compromise the company's ethical stance and standards.

The less obvious, but more significant, risk is the impact that unethical labour practices and working conditions may have on the reputation of the company and its products. Potentially, lower quality and defective clothes could seriously harm the company's reputation and result in lower sales revenue over a long period of time. Once damaged, such reputation would be hard to rebuild. The damage in reputation of the company regarding its ethical stance could also be potentially disastrous. Different stakeholder groups could react in negative ways, for example, customers may switch their custom, investors may sell their shares and the press may run negative campaigns against the company. The consequences of such damage could be long term and sometimes permanent.

Strom Co will need to review where and how resources are allocated in order to decrease or minimise the detrimental impact of a reduction in the quality control costs. For example, savings could be made by eliminating duplicated quality control processes or eliminating processes that are not necessary. Strom Co should also evaluate whether alternative, less resource intensive processes and procedures can be implemented without compromising the quality control and monitoring of working conditions. Experts should be used to undertake the assessment. Critical processes and procedures should be retained even if they require significant resources. The risk of making errors in the assessment should be evaluated and discussed at a senior level to ensure that Strom Co is comfortable with undertaking the likely risk.

			Marks
1	(a)	**Prior to implementation of proposal**	
		Cost of equity	1
		Cost of debt	1
		Market value of debt	2
		Market value of equity	3
		Cost of capital	1
		After implementing the proposal	
		Coeden Co's current asset beta	1
		Asset beta of hotel services business only	2
		Equity beta of hotel services business only	2
		Cost of equity	1
		Cost of capital	1
		Assumptions (1–2 marks per explained assumption, award marks for reasonable alternative assumptions not in the model answer)	2–3
		Comments	2–3
		Max	20
	(b)	Discussion (1–2 marks per point)	5
	(c)	Explanation of a demerger	2–3
		Benefits	2–3
		Drawbacks	2–3
		Max	8
		Total	**33**

		Marks
2	**Requirement (i)**	
	1 mark per exposure explained	3
	Requirement (ii)	
	Calculation of forward rate	1
	Calculation of income using the forward rate	1
	Calculation of cash flows using option contracts	3
	Discussion of relative merits of forwards and options	2–3
	Discussion of alternative hedging possibilities and conclusion	1–2
	Max	9
	Requirement (iii)	
	Calculation of devalued rate	1
	Calculation of translation loss	3
	Discussion of whether risk of translation loss should be managed	2–3
	Discussion of how risk of translation loss should be managed	1–2
	Max	8
	Requirement (iv)	
	1 mark per point	3
	Professional Marks	
	Structure and presentation of report	4
	Total	**27**

			Marks
3	**(a)**	Calculation of Sigra Co PE ratio	2
		Calculation of Dentro Co share value	2
		Calculation of percentage gain under cash offer	1
		Total equity value of combined company	1
		Estimate of per share value of combined company	1
		Calculation of percentage gain under share-for-share offer	1
		Estimation of bond required rate of return	2
		Estimation of value per share under bond offer	2
		Calculation of percentage gain under bond offer	1
		Comments (1 mark per relevant point)	3–4
		Max	16
	(b)	Discussion of each proposal (2 marks per proposal)	4
		Total	**20**
4	**(a)**	Calculation of project PDur05 net present value	2
		Calculation of percentage fall of selling price	3
		Comment	1
			6
	(b)	Formulation of objective function	1
		Formulation of constraints	2
			3
	(c)	Category 1	1
		Category 2	2
		Category 3	2
			5
	(d)	**(i)** Explaining the need for capital rationing	2
		(ii) Explanation of the features of a capital investment monitoring system	1–2
		Benefits of maintaining a capital investment monitoring system (1 mark per benefit)	2–3
		Max	4
		Total	**20**
5	**(a)**	Explanation of the role and aims of the IMF	5–6
		Reasons for austerity measures affecting Strom Co negatively	4–5
		Max	10
	(b)	Suggestion(s) for not affecting low-price retailers	2
		Suggestion(s) for not affecting high-price retailers	2
			4
	(c)	Discussion of the risks	3–4
		Reduction of the detrimental impact	2–3
		Max	6
		Total	**20**

Answers

1 (a) Report to the Board of Directors, Mlima Co

Initial public listing: price range and implications

This report considers a range of values of Mlima Co and possible share price, based on 100 million issued shares in preparation of the initial public listing. The assumptions made in determining the value range and the likelihood of the unsecured bond holders accepting the 10% equity-for-debt swap offer are discussed. Alternative reasons for the listing and reasons for issuing the share at a discount are evaluated.

Mlima Co cost of capital explanation

Ziwa Co's ungeared cost of equity represents the return Ziwa Co's shareholders would require if Ziwa Co was financed entirely by equity and had no debt. The return would compensate them for the business risk undertaken by the company.

This required rate of return would compensate Mlima Co's shareholders as well because, since both companies are in the same industry, they face the same business risk. This rate is then used as Mlima Co's cost of capital because of the assumption that Mlima Co will not issue any debt and faces no financial risk. Therefore its cost of equity (ungeared) is its cost of capital.

Mlima Co Estimated Value

Based on a cost of capital of 11% (appendix 1), the value of Mlima Co is estimated at $564·3m (appendix 2), prior to considering the impact of the Bahari project. The value of the Bahari project, without taking into account the benefits of the tax shield and the subsidies, does not exceed the initial investment. With the benefits of the tax shield and subsidies, it is estimated that the project will generate a positive net present value of $21·5m (appendix 3). Taking the Bahari project into account gives a value for Mlima Co at just under $586m.

Possible share price (100m shares)	Without the Bahari project	With the Bahari project
At full value	$5·64/share	$5·86/share
With 20% discount	$4·51/share	$4·69/share

Unsecured bond holders (equity-for-debt swap)

The current value of the unsecured bond is estimated at $56·8m (appendix 4) and if the unsecured bond holders are to be offered a 10% equity stake in Mlima Co post-listing, then only the share price at $5·86 would be acceptable to them. If the listing is made at the lowest price of $4·51/share, then they would need to be offered around a 12·6% equity stake ($56·8m/$4·51 = 12·594m).

The value of the bond is based on a flat yield curve (or yield to maturity) of 7%, which is the rate at which Mlima Co can borrow funds and therefore its current yield. A more accurate method would be to assess the yield curve based on future risk-free rates and credit spreads for the company.

Assumptions

The main assumptions made are around the accuracy of the information used in estimating the values of the company and the project. For example, the value of the company is based on assumptions of future growth rates, profit margins, future tax rates and capital investment. The basis for estimating the future growth rates and profit margins on past performance may not be accurate. With the Bahari project, for example, projections of future cash flows are made for 15 years and the variability of these has been estimated. Again, the reasonableness of these estimates needs to be assessed. Are they, for example, based on past experience and/or have professional experts judged the values?

The cost of capital is estimated based on a competitor's ungeared cost of equity, on the basis that Mlima Co is in a similar line of business and therefore faces similar business risk. The financial risk element has been removed since it has been stated that Mlima Co is not looking to raise extra debt finance. However, it is possible that the business risks faced by Mlima Co and that faced by Ziwa Co, the competitor, are not the same. Accepting the Bahari project would also change the risk profile of Mlima Co and therefore its discount rate.

The values are based on the Bahari government fulfilling the subsidised loan concession it has offered. Mlima Co needs to consider the likelihood of this concession continuing for the entire 15 years and whether a change of government may jeopardise the agreement. The political and other risks need to be assessed and their impact assessed.

It has been assumed that the underwriting and other costs involved with the new listing are not significant or have been catered for, before the assessment of the cash flows. This assumption needs to be reviewed and its accuracy assessed.

Reasons for the public listing

The main reason given for the public listing is to use the funds raised to eliminate the debt in the company. There are other reasons why a company may undertake a public listing. These include: gaining a higher reputation by being listed on a recognised stock exchange and therefore reducing the costs of contracting with stakeholders; being able to raise funds more easily in the future to undertake new projects; the listing will provide the current owners with a value for their equity stake; and the listing may enable the current owners to sell their equity stakes and gain from the value in the organisation.

Issuing shares at a discount

Issuing only 20% of the share capital to the public at the initial listing would make them minority shareholders effectively. As such, their ability to influence the decision-making process in the company would be severely curtailed, since even if all the new investors voted as a bloc against a decision, they would not be able to overturn it. The discounted share price would reflect the additional risk of investing in a company as a minority shareholder. In this case, the position of the unsecured bond holders is important. If the unsecured bond holders, holding between 10% and 12·6% of the share capital in an equity-for-debt swap, are included with the new investors, then the equity stake rises to 30%–32·6%. In such a case, shareholders, as a bloc, would have a significant influence on the company's decisions. The question that should be asked is whether the current unsecured bond holders are more closely aligned to the interests of the current owners or to the interests of the new investors.

The second reason for issuing shares at a discount is to ensure that they do all get sold and as a reward for the underwriters. Research suggests that, normally, for new listings, shares are issued at a discount and the price of such shares rises immediately after launch.

Conclusion

The report and the calculations in the appendices suggest a price range for the listing of between $4·51 and $5·86 per share, depending on whether or not the Bahari project is undertaken, the discount at which the shares are issued and the assumptions made. It is recommended that Mlima Co should consult its underwriters and potential investors about the possible price they would be willing to pay before making a final decision (known as book-building).

If 20 million shares are offered to the public for $4·51 each, this will result in total funds raised of just over $90 million. If then $80 million are spent in paying for the secured bond, just over $10 million liquid funds remain. Therefore, Mlima Co needs to consider whether issuing the shares at a discount would ensure sufficient liquid funds are available for it to continue its normal business. In addition to this, the Bahari investment may result in a change in the desired capital structure of the company and have an impact on the cost of capital. Finally, being listed will result in additional listing costs and annual costs related to additional reporting requirements.

These factors should be balanced against the benefits of undertaking the new listing before a final decision is made.

Report compiled by:

Date:

APPENDICES

Appendix 1: Mlima Co, cost of capital

Ziwa Co
MV debt = $1,700m x 1·05 = $1,785m
MV equity = 200m x $7 = $1,400m

Ziwa Co, ungeared Ke
$Ke_g = Ke_u + (1 - t)(Ke_u - K_d) D/E$
$16{\cdot}83\% = Ke_u + 0{\cdot}75 \times (Ke_u - 4{\cdot}76\%) \times 1{,}785/1{,}400$
$16{\cdot}83\% + 4{\cdot}55\% = 1{\cdot}9563 \times Ke_u$
$Ke_u = 10{\cdot}93\%$ (say 11%)

Appendix 2: Mlima Co, estimate of value prior to Bahari project

Value based on future free cash flows

Historic mean sales revenue growth = $(389{\cdot}1/344{\cdot}7)^{1/2} - 1 = 0{\cdot}625$ or 6·25%
Next four years annual growth rate of sales revenue = 120% of 6·25% = 7·5%
Thereafter 3·5% of cash flows per annum
Operating profit margin (approx) = 58·4/389·1 = 54·9/366·3 = 51·7/344·7 = 15%

Year (in $ millions)	1	2	3	4
Sales revenue	418·3	449·7	483·4	519·7
Operating profit	62·7	67·5	72·5	78·0
Less taxation (25%)	(15·7)	(16·9)	(18·1)	(19·5)
Less additional capital investment (30c per $1 change in sales revenue)	(8·8)	(9·4)	(10·1)	(10·9)
Free cash flows	38·2	41·2	44·3	47·6
PV of free cash flows (11%)	34·4	33·4	32·4	31·4

PV first four years	$131·6m
PV after four years $(47{\cdot}6 \times 1{\cdot}035)/(0{\cdot}11 - 0{\cdot}035) \times 1{\cdot}11^{-4}$	$432·7m
Value of company	$564·3m

Appendix 3: Value of the Bahari project

Base case present value

Year	Free Cash flows (in $ millions)	PV (11%) (in $ millions)
1	4·0	3·6
2	8·0	6·5
3	16·0	11·7
4	18·4	12·1
5	21·2	12·6
6 to 15	21·2	**74·0
Total		120·5

** The free cash flows in years 6 to 15 are an annuity for 10 years at 11%, then discounted back for five years: 21·2 x 5·889 x 0·593 = 74·0

PV of the tax shield and subsidy
Annuity factor (7%, 15 years) = 9·108
Annual tax shield benefit interest paid = 3% x $150m x 25% = $1·1m
Subsidy benefit = 4% x $150m x (1 – 25%) = $4·5m
PV of tax shield and subsidy benefit = 5·6 x 9·108 = $51·0m

Adjusted present value = $120·5m + $51·0m – $150·0m = $21·5m

Appendix 4: Estimated value of the unsecured bond

Assume a flat yield or yield to maturity of 7%

Annual coupon interest = $5·2m (13% x $40m)
10-year annuity at 7% = 7·024; Discount factor (10 years, 7%) = 0·508

Bond value = $5·2m x 7·024 + $40m x 0·508 = $56·8m

(b) It is likely that Mlima Co's actions will be scrutinised more closely in the run up to the listing and once it has been listed. In both the situations, the company should consider the action it should take based on its ethical and accountability code. Most major corporations now publicise such codes of behaviour and would consult these in cases of ethical and/or accountability difficulties.

With the first situation concerning the relocation of the farmers, Mlima Co would consult its ethical code to judge how far its responsibility lay. It may take the view that the matter is between the farmers and the government, and it is not directly or indirectly responsible for the situation. In any case, it is likely that the mining rights will be assigned to another company, should Mlima Co decide to walk away from the deal. It is unlikely that, even if Mlima Co did not agree to the offer, the plight of the farmers would cease.

Instead, Mlima Co may decide to try to influence the government with respect to the farmers by urging the government to keep the community together and offer the farmers better land. Mlima Co may also decide to offer jobs and training to farmers who decide not to leave.

With the second situation concerning the Bahari president and Mlima Co's CEO, whilst it would make good business sense to forge strong relationships as a means of competitive advantage, Mlima Co should ensure that the negotiation was transparent and did not involve any bribery or illegal practice. If both the company and Bahari government can demonstrate that they acted in the best interests of the company and the country respectively, and individuals did not benefit as a result, then this should not be seen in a negative light.

Mlima Co needs to establish a clear strategy of how it would respond to public scrutiny of either issue. This may include actions such as demonstrating that it is acting according to its ethical code, pre-empting media scrutiny by releasing press statements, and using its influence to ensure the best and correct outcome in each case for the stakeholders concerned.

(Note: Credit will be given for alternative, relevant approaches to the calculations, comments and suggestions/ recommendations)

2 **(a)** An acquisition creates synergy benefits when the value of the combined entity is more than the sum of the two companies' values. Synergies can be separated into three types: revenue synergies which result in higher revenues for the combined entity, higher return on equity and a longer period when the company is able to maintain competitive advantage; cost synergies which result mainly from reducing duplication of functions and related costs, and from taking advantage of economies of scale; financial synergies which result from financing aspects such as the transfer of funds between group companies to where it can be utilised best, or from increasing debt capacity.

In this scenario, the following synergy benefits may arise from the two companies coming together. Financial synergies may be available because Strand Co does not have the funds to innovate new products. On the other hand, Hav Co has cash reserves available. It may be possible to identify and quantify this synergy based on the projects that can be undertaken after the acquisition, but would have been rejected before, and their corresponding net present value. Furthermore, as the company increases in size, the debt capacity of the combined company may increase, giving it additional access to finance. Finally, the acquisition may result in a decrease in the cost of capital of the combined company.

Cost synergies may arise from the larger company being able to negotiate better terms and lower costs from their suppliers. And there may be duplication of functional areas such as in research and development and head office which could be reduced and costs saved. These types of synergies are easier to identify and quantify but would be more short-lived. Therefore, if the markets are going to be positive about the acquisition, Hav Co will need to show where more long-term synergies are coming from as well as these.

Revenue synergies are perhaps where the greatest potential for growth comes from but are also more difficult to identify, quantify and enact. Good post-acquisition planning is essential for these synergies to be realised but they can be substantial and long-lasting. In this case, Hav Co's management can help market Strand Co's products more effectively by using their sales and marketing talents resulting in higher revenues and longer competitive advantage. Research and development activity can be combined to create new products using the technologies in place in both companies, and possibly bringing innovative products to market quicker. The services of the scientists from Strand Co will be retained to drive innovation forward, but these need to be nurtured with care since they had complete autonomy when they were the owners of Strand Co.

The main challenge in ensuring long-lasting benefits is not only ensuring accurate identification of potential synergies but putting into place integration processes and systems to gain full benefit from them. This is probably the greater challenge for management, and, when poorly done, can result in failure to realise the full value of the acquisition. Hav Co needs to be aware of this and make adequate provisions for it.

(Note: Credit will be given for alternative relevant comments and suggestions)

(b) Maximum premium based on excess earnings method

This method is similar to the EVA® method which measures excess return.

Average pre-tax earnings: (397 + 370 + 352)/3 = $373·0m
Average capital employed: [(882 + 210 – 209) + (838 + 208 – 180) + (801 + 198 – 140)]/3 = $869·3m
Excess annual value/annual premium = 373m – (20% x $869·3m) = $199·1m
After-tax annual premium = $199·1m x 0·8 = $159·3m
PV of annual premium (assume perpetuity) = $159·3m/0·07 = $2,275·7m

According to this method, the maximum premium payable is $2,275·7m in total.

Maximum premium based on price-to-earnings (PE) ratio method

Strand Co estimated PE ratio = 16·4 x 1·10 = 18·0
Strand Co profit after tax: $397m x 0·8 = $317·6m
Hav Co profit after tax = $1,980m x 0·8 =$1,584·0m

Hav Co, current value = $9·24 x 2,400 shares = $22,176·0m
Strand Co, current value = $317·6m x 18·0 = $5,716·8m

Combined company value = ($1,584m + $317·6m + $140·0m) x 14·5 = $29,603·2m

Maximum premium = $29,603·2m – ($22,176·0m + $5,716·8) = $1,710·4m

(c) Strand Co, current value per share = $5,716·8m/1,200m shares = $4·76 per share

Maximum premium % based on PE ratio = $1,710·4m/$5,716·8m x 100% = 29·9%
Maximum premium % based on excess earnings = $2,275·7m/$5,716·8m x 100% = 39·8%

Cash offer: premium (%)
($5·72 – $4·76)/$4·76 x 100% = 20·2%

Cash and share offer: premium (%)
1 Hav Co share for 2 Strand Co shares
Hav Co share price = $9·24
Per Strand Co share = $4·62
Cash payment per share= $1·33
Total return = $1·33 + $4·62 = $5·95
Premium percentage = ($5·95 – $4·76)/$4·76 x 100% = 25·0%

Cash and bond offer: premium (%)
Each share has a nominal value of $0·25, therefore $5 is $5/$0·25 = 20 shares
Bond value = $100/20 shares = $5 per share
Cash payment = $1·25 per share
Total = $6·25 per share
Premium percentage = ($6·25 – $4·76)/$4·76 = 31·3%

On the basis of the calculations, the cash together with bond offer yields the highest return; in addition to the value calculated above, the bonds can be converted to 12 Hav Co shares, giving them a price per share of $8·33 ($100/12). This price is below Hav Co's current share price of $9·24, and therefore the conversion option is already in-the-money. It is probable that the share price will increase in the 10-year period and therefore the value of the convertible bond should increase. A bond also earns a small coupon interest of $3 per $100 a year. The 31·3% return is the closest to the maximum premium based on the excess earnings method and more than the maximum premium based on the PE ratio method. It would seem that this payment option transfers more value to the owners of Strand Co than the value created based on the PE ratio method.

However, with this option Strand Co shareholders only receive an initial cash payment of $1·25 per share compared to $1·33 per share and $5·72 per share for the other methods. This may make it the more attractive option for the Hav Co shareholders as well, and although their shareholding will be diluted most under this option, it will not happen for some time.

The cash and share offer gives a return in between the pure cash and the cash and bonds offers. Although the return is lower, Strand Co's shareholders become owners of Hav Co and have the option to sell their equity immediately. However, the share price may fall between now and when the payment for the acquisition is made. If this happens, then the return to Strand Co's shareholders will be lower.

The pure cash offer gives an immediate and definite return to Strand Co's shareholders, but is also the lowest offer and may also put a significant burden on Hav Co having to fund so much cash, possibly through increased debt.

It is likely that Strand Co's shareholder/managers, who will continue to work within Hav Co, will accept the mixed cash and bond offer. They, therefore, get to maximise their current return and also potentially gain when the bonds are converted into shares. Different impacts on shareholders' personal taxation situations due to the different payment methods might also influence the choice of method.

3 (a) Only the transactions resulting in cash flows between Kenduri Co and Lakama Co are considered for hedging. Other transactions are not considered.

Net flow in US$: US$4·5m payment – US$2·1m receipt = US$2·4m payment

Hedge the US$ exposure using the forward market, the money market and options.

Forward market
US$ hedge: 2,400,000/1·5996 = £1,500,375 payment

Money market
US$ hedge
Invest in US$: 2,400,000/(1 + 0·031/4) = US$2,381,543
Convert into £ at spot: US$2,381,543/1·5938 = £1,494,255
Borrow in £: £1,494,255 x (1 + 0·040/4) = £1,509,198

(Note: Full credit will be given to candidates who use the investing rate of 2·8% instead of the borrowing rate of 4%, where this approach has been explained and justified)

The forward market is preferred due to lower payment costs.

Options
Kenduri Co would purchase Sterling three-month put options to protect itself against a strengthening US$ to £.

Exercise price: $1·60/£1
£ payment = 2,400,000/1·60 = 1,500,000 or 24 contracts
24 put options purchased

Premium payable = 24 x 0·0208 x 62,500 = US$31,200
Premium in £ = 31,200/1·5938 = £19,576
Total payments = £1,500,000 + £19,576 = £1,519,576

Exercise price: $1·62/£1
£ payment = 2,400,000/1·62 = 1,481,481 or 23·7 contracts
23 put options purchased

£ payment = 23 x 62,500 = £1,437,500
Premium payable = 23 x 0·0342 x 62,500 = US$49,163
Premium in £ = 49,163/1·5938 = £30,846
Amount not hedged = US$2,400,000 – (23 x 62,500 x 1·62) = US$71,250
Use forwards to hedge amount not hedged = US$71,250/1·5996 = £44,542
Total payments = 1,437,500 + 30,846 + 44,542 = £1,512,888

Both these hedges are worse than the hedge using forward or money markets. This is due to the premiums payable to let the option lapse if the prices move in Kenduri Co's favour. Options have an advantage over forwards and money markets because the prices are not fixed and the option buyer can let the option lapse if the rates move favourably. Hence options have an unlimited upside but a limited downside. With forwards and money markets, Kenduri Co cannot take advantage of the US$ weakening against the £.

Conclusion

The forward market minimises the payment and is therefore recommended over the money market. However, options give Kenduri Co the choice of an unlimited upside, although the cost is higher. Therefore the choice between the forward market and the option market depends on the risk preference of the company.

(b) Based on spot mid-rates: US$1·5950/£1; CAD1·5700/£1; JPY132·75/£1

In £000

		Payments from				
		UK	USA	Canada	Japan	Total
Receipts to	UK		1,316·6	2,165·6		3,482·2
	USA	2,821·3		940·4	877·7	4,639·4
	Canada	700·6			2,038·2	2,738·8
	Japan		2,410·5			2,410·5
Total payments		3,521·9	3,727·1	3,106·0	2,915·9	
Total receipts		3,482·2	4,639·4	2,738·8	2,410·5	
Net receipt/(payment)		(39·7)	912·3	(367·2)	(505·4)	

Each of Kenduri Co, Jaia Co and Gochiso Co will make payments of £ equivalent to the amount given above to Lakama Co.

Multilateral netting involves minimising the number of transactions taking place through each country's banks. This would limit the fees that these banks would receive for undertaking the transactions and therefore governments who do not allow multilateral netting want to maximise the fees their local banks receive. On the other hand, some countries allow multilateral netting in the belief that this would make companies more willing to operate from those countries and any banking fees lost would be more than compensated by the extra business these companies and their subsidiaries bring into the country.

(c) Gamma measures the rate of change of the delta of an option. Deltas range from near 0 for a long call option which is deep out-of-money, where the price of the option is insensitive to changes in the price of an underlying asset, to near 1 for a long call option which is deep in-the-money, where the price of the option moves in line and largely to the same extent as the price of the underlying asset. When the long call option is at-the-money, the delta is 0·5 but also changes rapidly. Hence, the gamma is highest for a long call option which is at-the-money. The gamma is also higher when the option is closer to expiry. It would seem, therefore, that the option is probably trading near at-the-money and has a relatively short time period before it expires.

4 (a) As a high growth company, Limni Co probably requires the cash flows it generates annually for investing in new projects and has therefore not paid any dividends. This is a common practice amongst high-growth companies, many of which declare that they have no intention of paying any dividends. The shareholder clientele in such companies expects to be rewarded by growth in equity value as a result of the investment policy of the company.

Capital structure theory would suggest that because of the benefit of the tax shield on interest payments, companies should have a mix of equity and debt in their capital structure. Furthermore, the pecking order proposition would suggest that companies tend to use internally generated funds before going to markets to raise debt capital initially and finally equity capital. The agency effects of having to provide extra information to the markets and where one investor group benefits at the expense of another have been cited as the main deterrents to companies seeking external sources of finance. To a certain extent, this seems to be the case with Limni Co in using internal finance first, but the pecking order proposition seems to be contradicted in that it seeks to go straight to the equity market and undertake rights issues thereafter. Perhaps the explanation for this can be gained from looking at the balance of business and financial risk. Since Limni Co operates in a rapidly changing industry, it probably faces significant business risk and therefore cannot afford to undertake high financial risk, which a capital structure containing significant levels of debt would entail. This, together with agency costs related to restrictive covenants, may have determined Limni Co's financing policy.

Risk management theory suggests that managing the volatility of cash flows enables a company to plan its investment strategy better. Since Limni Co uses internally generated funds to finance its projects, it needs to be certain that funds will be available when needed for the future projects, and therefore managing its cash flows will enable this. Moreover, because Limni Co faces high business risk, managing the risk that the company's managers cannot control through their actions, may be even more necessary.

The change to making dividend payments or undertaking share buybacks will affect all three policies. The company's clientele may change and this may cause share price fluctuations. However, since the recommendation for the change is being led by the shareholders, significant share price fluctuations may not happen. Limni Co's financing policy may change because having reduced internal funds means it may have to access debt markets and therefore have to look at its balance between business and financial risk. The change to Limni Co's financial structure may result in a change in its risk management policy, because it may be necessary to manage interest rate risk as well.

(Note: Credit will be given for alternative relevant comments)

(b) In the case of company Theta, dividends are growing but not at a stable rate. In fact company Theta is paying out $0·40 in dividends for every $1 in earnings, and has a fixed dividend cover ratio of 2·50. This would be confusing for the shareholders, as they would not know how much dividend they would receive from year to year. Although profits have risen over the past five years, if profits do fall, company Theta may reduce dividends and therefore send the wrong signals to shareholders and investors. This may cause unnecessary fluctuations of the share price or result in a depressed share price.

In the case of company Omega, annual dividends are growing at a stable rate of approximately 5% per year, while the company's earnings are growing steadily at around 3% per year, resulting in an increasing payout ratio. Also a high proportion of earnings are paid out as dividends, increasing from 60% in 2009 to almost 65% in 2013. This would indicate a company operating in a mature industry, signaling that there are few new projects to invest in and therefore reducing the retention rate. Such an investment would be attractive to investors requiring high levels of dividend returns from their investments.

In the case of company Kappa, although a lower proportion of earnings is paid out as dividends (from about 20% in 2009 to about 27% in 2013), they are growing at a higher but stable rate of 29%–30% per year. The company's earnings are growing rapidly but erratically, ranging between 3% and 35% between 2009 and 2013. This probably indicates a growing company, possibly similar to Limni Co itself, where perhaps returns to investors having been coming from share price growth, but one where dividends are becoming more prominent. Such an investment would be attractive to investors requiring lower levels of dividend returns, but higher capital returns from their investments.

Due to company Theta's confusing dividend policy, which may lead to erratic dividend payouts and a depressed share price, Limni Co would probably not want to invest in that company. The choice between company Omega and company Kappa would depend on how Limni Co wants to receive its return from the investment, maybe taking into account factors such as taxation implications, and the period of time it wishes to invest for, in terms of when the returns from an investment will be maximised and when it will need the funds for future projects.

(Note: Credit will be given for alternative relevant comments)

(c) **Limni Co, current dividend capacity**

	$000
Profit before tax (23% x $600,000,000)	138,000
Tax (26% x $138,000,000)	(35,880)
Profit after tax	102,120
Add back depreciation (25% x $220,000,000)	55,000
Less investment in assets	(67,000)
Remittances from overseas subsidiaries	15,000
Additional tax on remittances (6% x $15,000,000)	(900)
Dividend capacity	104,220

Increase in dividend capacity = 10% x $104,220,000 = $10,422,000
Gross up for tax = $10,422,000/0·94 = $11,087,234
Percentage increase in remittances from overseas subsidiaries = 73·9% [$11,087,234/$15,000,000]

Dividend repatriations need to increase by 73·9% from Limni Co's international subsidiaries in order to increase the dividend capacity by 10%. Limni Co would need to consider whether or not it is feasible for its subsidiaries to increase their repatriations to such an extent, and the impact this will have on the motivation of the subsidiaries' managers and on the subsidiaries' ability to operate as normal.

(d) The main benefit of a share buyback scheme to investors is that it helps to control transaction costs and manage tax liabilities. With the share buyback scheme, the shareholders can choose whether or not to sell their shares back to the company. In this way they can manage the amount of cash they receive. On the other hand, with dividend payments, and especially large special dividends, this choice is lost, and may result in a high tax bill. If the shareholder chooses to re-invest the funds, it will result in transaction costs. An added benefit is that, as the share capital is reduced, the earnings per share and the share price may increase. Finally, share buybacks are normally viewed as positive signals by markets and may result in an even higher share price.

(Note: Credit will be given for alternative relevant comments)

				Marks
1	**(a)**	**(i)**	Explanation of Mlima Co's cost of capital based on Ziwa Co's ungeared cost of equity	3
			Ziwa Co, cost of ungeared equity	4
				7
		(ii)	Sales revenue growth rates	1
			Operating profit rate	1
			Estimate of free cash flows and PV of free cash flows for years 1 to 4	4
			PV of free cash flows after year 4	2
			Base case Bahari project value	2
			Annual tax shield benefit	1
			Annual subsidy benefit	1
			PV of the tax shield and subsidy benefits	1
			Value of the Bahari project	1
				14
		(iii)	Calculation of unsecured bond value	2
			Comment	2
			Limitation	1
				5
		(iv)	Comments on the range of values/prices with and without the project, and concluding statement	4–5
			Discussion of assumptions	3–4
			Explanation for additional reasons for listing	2–3
			Assessment of reasons for discounted share price	2–3
			Max	**12**
		Professional marks		
		Report format		1
		Structure and presentation of the report		3
				4
	(b)	Discussion of relocation of farmers		4–5
		Discussion of relationship between Bahari president and Mlima Co CEO		4–5
			Max	**8**
			Total	**50**

			Marks
2	**(a)**	Distinguish between the different synergies	1–2
		Discuss possible financial synergy sources	2–3
		Discuss possible cost synergy sources	1–2
		Discuss possible revenue synergy sources	3–4
		Concluding comments	1–2
		Max	9
	(b)	Average earnings and capital employed	1
		After-tax annual premium	1
		PV of premium (excess earnings method)	1
		Hav Co and Strand Co values	1
		Combined company value	1
		Value created/premium (PE method)	1
			6
	(c)	Strand Co, value per share	1
		Cash offer premium (%)	1
		Cash and share offer premium (%)	2
		Cash and bond offer premium (%)	2
		Explanation and justification	4–5
		Max	10
		Total	**25**
3	**(a)**	Calculation of net US$ amount	1
		Calculation of forward market US$ amount	1
		Calculation of US$ money market amount	2
		Calculation of one put option amount (1·60 or 1·62)	3
		Calculation of the second put option amount or if the preferred exercise price choice is explained	2
		Advice and recommendation	3–4
		Max	12
	(b)	Mid spot rates calculation	1
		Calculation of the £ equivalent amounts of US$, CAD and JPY	4
		Calculation of the net receipt/payment	2
		Explanation of government reaction to multilateral hedging	3
			10
	(c)	1 mark per relevant point	3
		Total	**25**

			Marks
4	(a)	Discussion of dividend policy	1–2
		Discussion of financing policy	3–4
		Discussion of risk management policy	1–2
		Effect of dividends and share buybacks on the policies	2–3
		Max	8
	(b)	2 marks per evaluation of each of the three companies	6
		Discussion of which company to invest in	2
			8
	(c)	Calculation of initial dividend capacity	3
		Calculation of new repatriation amount	2
		Comment	1–2
		Max	6
	(d)	1 mark per relevant point	3
		Total	**25**

Answers

1 **(a)** The World Trade Organisation (WTO) was set up to continue to implement the General Agreement on Tariffs and Trade (GATT), and its main aims are to reduce the barriers to international trade. It does this by seeking to prevent protectionist measures such as tariffs, quotas and other import restrictions. It also acts as a forum for negotiation and offering settlement processes to resolve disputes between countries.

The WTO encourages free trade by applying the most favoured nation principle between its members, where reduction in tariffs offered to one country by another should be offered to all members.

Whereas the WTO has had notable success, some protectionist measures between groups of countries are nevertheless allowed and some protectionist measures, especially non-tariff based ones, have been harder to identify and control.

Mehgam could benefit from reducing protectionist measures because its actions would make other nations reduce their protectionist measures against it. Normally countries retaliate against each other when they impose protectionist measures. A reduction in these may allow Mehgam to benefit from increased trade and economic growth. Such a policy may also allow Mehgam to specialise and gain competitive advantage in certain products and services, and compete more effectively globally. Its actions may also gain political capital and more influence worldwide.

Possible drawbacks of reducing protectionist policies mainly revolve around the need to protect certain industries. It may be that these industries are developing and in time would be competitive on a global scale. However, inaction to protect them now would damage their development irreparably. Protection could also be given to old, declining industries, which, if not protected, would fail too quickly due to international competition, and would create large scale unemployment making such inaction politically unacceptable. Certain protectionist policies are designed to prevent 'dumping' of goods at a very cheap price, which hurt local producers.

(Note: Credit will be given for alternative relevant discussion)

(b) **Report to the Board of Directors (BoD), Chmura Co**

This report recommends whether or not Chmura Co should invest in a food packaging project in Mehgam, following Mehgam reducing its protectionist measures. It initially considers the value of the project without taking into account the offer made by Bulud Co to purchase the project after two years. Following this, Bulud Co's offer is considered. The report concludes by recommending a course of action for the BoD to consider further.

Estimated value of the Mehgam project and initial recommendation
The initial net present value of the project is negative at approximately $(451,000) [see Appendix 1]. This would suggest that Chmura Co should not undertake the project.

Bulud Co's offer is considered to be a real option for Mehgam Co. Since it is an offer to sell the project as an abandonment option, a put option value is calculated based on the finance director's assessment of the standard deviation and using the Black-Scholes option pricing (BSOP) model. The value of the put option is added to the initial net present value of the project without the option, to give the value of the project. Although Chmura Co will not actually obtain any immediate cash flow from Bulud Co's offer, the real option computation indicates that the project is worth pursuing because the volatility may result in increases in future cash flows.

After taking account of Bulud Co's offer and the finance director's assessment, the net present value of the project is positive at approximately $2,993,000 [see Appendix 2]. This would suggest that Chmura Co should undertake the project.

Assumptions
It is assumed that all the figures relating to variables such as revenues, costs, taxation, initial investments and their recovery, inflation figures and cost of capital are accurate. There is considerable uncertainty surrounding the accuracy of these, and in addition to the assessments of value conducted in appendices one and two, sensitivity analysis and scenario analysis are probably needed to assess the impact of these uncertainties.

It is assumed that future exchange rates will reflect the differential in inflation rates between the two countries. It is, however, unlikely that exchange rates will move fully in line with the inflation rate differentials.

It is assumed that the value of the land and buildings at the end of the project is a relevant cost, as it is equivalent to an opportunity benefit, even if the land and buildings are retained by Chmura Co.

It is assumed that Chmura Co will be given and will utilise the full benefit of the bi-lateral tax treaty and therefore will not pay any additional tax in the country where it is based.

It is assumed that the short-dated $ treasury bills are equivalent to the risk-free rate of return required for the BSOP model. And it is assumed that the finance director's assessment of the 35% standard deviation of cash flows is accurate.

It is assumed that Bulud Co will fulfil its offer to buy the project in two years' time and there is no uncertainty surrounding this. Chmura Co may want to consider making the offer more binding through a legal contract.

The BSOP model makes several assumptions such as perfect markets, constant interest rates and lognormal distribution of asset prices. It also assumes that volatility can be assessed and stays constant throughout the life of the project, and that the underlying asset can be traded. Neither of these assumptions would necessarily apply to real options. Therefore the BoD needs to treat the value obtained as indicative rather than definitive.

Additional business risks
Before taking the final decision on whether or not to proceed with the project, Chmura Co needs to take into consideration additional risks, including business risks, and where possible mitigate these as much as possible. The main business risks are as follows:

Investing in Mehgam may result in political risks. For example, the current government may be unstable and if there is a change of government, the new government may impose restrictions, such as limiting the amount of remittances which can be made to the parent company. Chmura Co needs to assess the likelihood of such restrictions being imposed in the future and consider alternative ways of limiting the negative impact of such restrictions.

Chmura Co will want to gain assurance that the countries to which it will sell the packaged food batches remain economically stable and that the physical infrastructure such as railways, roads and shipping channels are maintained in good repair. Chmura Co will want to ensure that it will be able to export the special packaging material into Mehgam. Finally, it will need to assess the likelihood of substantial protectionist measures being lifted and not re-imposed in the future.

As much as possible, Chmura Co will want to ensure that fiscal risks such as imposition of new taxes and limits on expenses allowable for taxation purposes do not change. Currently, the taxes paid in Mehgam are higher than in Chmura Co's host country, and even though the bi-lateral tax treaty exists between the countries, Chmura Co will be keen to ensure that the tax rate does not change disadvantageously.

Chmura Co will also want to protect itself, as much as possible, against adverse changes in regulations. It will want to form the best business structure, such as a subsidiary company, joint venture or branch, to undertake the project. Also, it will want to familiarise itself on regulations such as employee health and safety law, employment law and any legal restrictions around land ownership.

Risks related to the differences in cultures between the host country, Mehgam, and the countries where the batches will be exported to would be a major concern to Chmura Co. For example, the product mix in the batches which are suitable for the home market may not be suitable for Mehgam or where the batches are exported. It may contain foods which would not be saleable in different countries and therefore standard batches may not be acceptable to the customers. Chmura Co will also need to consider the cultural differences and needs of employees and suppliers.

The risk of the loss of reputation through operational errors would need to be assessed and mitigated. For example, in setting up sound internal controls, segregation of duties is necessary. However, personal relationships between employees in Mehgam may mean that what would be acceptable in another country may not be satisfactory in Mehgam. Other areas where Chmura Co will need to focus on are the quality control procedures to ensure that the quality of the food batches is similar to the quality in the host country.

Recommendation
With Bulud Co's offer, it is recommended that the BoD proceed with the project, as long as the BoD is satisfied that the offer is reliable, the sensitivity analysis/scenario analysis indicates that any negative impact of uncertainty is acceptable and the business risks have been considered and mitigated as much as possible.

If Bulud Co's offer is not considered, then the project gives a marginal negative net present value, although the results of the sensitivity analysis need to be considered. It is recommended that, if only these results are taken into consideration, the BoD should not proceed with the project. However, this decision is marginal and there may be other valid reasons for progressing with the project such as possibilities of follow-on projects in Mehgam.

Report compiled by:

Date:

APPENDICES

Appendix 1: Estimated value of the Mehgam project excluding the Bulud Co offer

(Cash flows in MP, millions)

Year	1	2	3	4	5
Sales revenue (w2)	1,209·6	1,905·1	4,000·8	3,640·7	2,205·4
Production and selling costs (w3)	(511·5)	(844·0)	(1,856·7)	(1,770·1)	(1,123·3)
Special packaging costs (w4)	(160·1)	(267·0)	(593·7)	(572·0)	(366·9)
Training and development costs	(409·2)	(168·8)	0	0	0
Tax allowable depreciation	(125)	(125)	(125)	(125)	(125)
Balancing allowance					(125)
Taxable profits/(loss)	3·8	500·3	1,425·4	1,173·6	465·2
Taxation (25%)	(1·0)	(125·1)	(356·4)	(293·4)	(116·3)
Add back depreciation	125	125	125	125	250
Cash flows (MP, millions)	127·8	500·2	1,194·0	1,005·2	598·9

(All amounts in $, 000s)

Year	1	2	3	4	5
Exchange rate (w1)	76·24	80·72	85·47	90·50	95·82
Cash flows ($ 000s)	1,676·3	6,196·7	13,969·8	11,107·2	6,250·3
Discount factor for 12%	0·893	0·797	0·712	0·636	0·567
Present values ($ 000s)	1,496·9	4,938·8	9,946·5	7,064·2	3,543·9

Present value of cash flows approx. = $26,990,000

PV of value of land, buildings and machinery in year 5 = (80% x MP1,250m + MP500m)/95·82 x 0·567 approx. = $8,876,000
PV of working capital = MP200m/95·82 x 0·567 approx. = $1,183,000

Cost of initial investment in $ = (MP2,500 million + MP200 million)/72 = $37,500,000

NPV of project = $26,990,000 + $8,876,000 + $1,183,000 – $37,500,000 = $(451,000)

Workings

1. Exchange rates

Year	1	2	3	4	5
MP/$1	72 x 1·08/1·02 = 76·24	76·24 x 1·08/1·02 = 80·72	80·72 x 1·08/1·02 = 85·47	85·47 x 1·08/1·02 = 90·50	90·50 x 1·08/1·02 = 95·82

2. Sales revenue (MP million)

Year	1	2	3	4	5
	10,000 x 115,200 x 1·05 = 1,209·6	15,000 x 115,200 x $1{\cdot}05^2$ = 1,905·1	30,000 x 115,200 x $1{\cdot}05^3$ = 4,000·8	26,000 x 115,200 x $1{\cdot}05^4$ = 3,640·7	15,000 x 115,200 x $1{\cdot}05^5$ = 2,205·4

3. Production and selling (MP million)

Year	1	2	3	4	5
	10,000 x 46,500 x 1·1 = 511·5	15,000 x 46,500 x $1{\cdot}1^2$ = 844·0	30,000 x 46,500 x $1{\cdot}1^3$ = 1,856·7	26,000 x 46,500 x $1{\cdot}1^4$ = 1,770·1	15,000 x 46,500 x $1{\cdot}1^5$ = 1,123·3

4. Special packaging (MP million)

Year	1	2	3	4	5
	10,000 x 200 x 76·24 x 1·05 = 160·1	15,000 x 200 x 80·72 x $1{\cdot}05^2$ = 267·0	30,000 x 200 x 85·47 x $1{\cdot}05^3$ = 593·7	26,000 x 200 x 90·50 x $1{\cdot}05^4$ = 572·0	15,000 x 200 x 95·82 x $1{\cdot}05^5$ = 366·9

Appendix 2: Estimated value of the Mehgam project including the Bulud Co offer

Present value of underlying asset (Pa) = $30,613,600 (approximately)
(This is the sum of the present values of the cash flows foregone in years 3, 4 and 5)

Price offered by Bulud Co (Pe) = $28,000,000

Risk free rate of interest (r) = 4% (assume government treasury bills are valid approximation of the risk free rate of return)

Volatility of underlying asset (s) = 35%

Time to expiry of option (t) = 2 years

d_1 = [ln(30,613·6/28,000) + (0·04 + 0·5 x $0{\cdot}35^2$) x 2]/[0·35 x $2^{1/2}$] = 0·589
d_2 = 0·589 – 0·35 x $2^{1/2}$ = 0·094

N(d_1) = 0·5 + 0·2220 = 0·7220
N(d_2) = 0·5 + 0·0375 = 0·5375

Call value = $30,613,600 x 0·7220 – $28,000,000 x 0·5375 x $e^{-0{\cdot}04 \times 2}$ = approx. $8,210,000
Put value = $8,210,000 – $30,613,600 + $28,000,000 x $e^{-0{\cdot}04 \times 2}$ = approx. $3,444,000

Net present value of the project with put option = $3,444,000 – $451,000 = approx. $2,993,000

(Note: Credit will be given for relevant discussion and recommendation)

2 (a) Using forward rate agreements (FRAs)

FRA rate 4·82% (3–7), since the investment will take place in three months' time for a period of four months.

If interest rates increase by 0·9% to 4·99%

Investment return = 4·79% x 4/12 x $48,000,000 =	$766,400
Payment to Voblaka bank = (4·99% – 4·82%) x $48,000,000 x 4/12 =	$(27,200)
Net receipt =	$739,200
Effective annual interest rate = 739,200/48,000,000 x 12/4 =	4·62%

If interest rates decrease by 0·9% to 3·19%

Investment return = 2·99% x 4/12 x $48,000,000 =	$478,400
Receipt from Voblaka Bank = (4·82% – 3·19%) x $48,000,000 x 4/12 =	$260,800
Net receipt =	$739,200
Effective annual interest rate (as above)	4·62%

Using futures

Need to hedge against a fall in interest rates, therefore go long in the futures market. Awan Co needs March contracts as the investment will be made on 1 February.

No. of contracts needed = $48,000,000/$2,000,000 x 4 months/3 months = 32 contracts.

Basis
Current price (on 1/11) – futures price = total basis
(100 – 4·09) – 94·76 = 1·15
Unexpired basis = 2/5 x 1·15 = 0·46

If interest rates increase by 0·9% to 4·99%

Investment return (from above) =	$766,400
Expected futures price = 100 – 4·99 – 0·46 = 94·55	
Loss on the futures market = (0·9455 – 0·9476) x $2,000,000 x 3/12 x 32 =	$(33,600)
Net return =	$732,800
Effective annual interest rate = $732,800/$48,000,000 x 12/4 =	4·58%

If interest rates decrease by 0·9% to 3·19%

Investment return (from above) =	$478,400
Expected futures price = 100 – 3·19 – 0·46 = 96·35	
Gain on the futures market = (0·9635 – 0·9476) x $2,000,000 x 3/12 x 32 =	$254,400
Net return =	$732,800
Effective annual interest rate (as above) =	4·58%

Using options on futures

Need to hedge against a fall in interest rates, therefore buy call options. As before, Awan Co needs 32 March call option contracts ($48,000,000/$2,000,000 x 4 months/3 months).

If interest rates increase by 0·9% to 4·99%

Exercise price	94·50	95·00
Futures price	94·55	94·55
Exercise ?	Yes	No
Gain in basis points	5	0
Underlying investment return (from above)	$766,400	$766,400
Gain on options (0·0005 x 2,000,000 x 3/12 x 32, 0)	$8,000	$0
Premium		
0·00432 x $2,000,000 x 3/12 x 32	$(69,120)	
0·00121 x $2,000,000 x 3/12 x 32		$(19,360)
Net return	$705,280	$747,040
Effective interest rate	4·41%	4·67%

If interest rates decrease by 0·9% to 3·19%

Exercise price	94·50	95·00
Futures price	96·35	96·35
Exercise ?	Yes	Yes
Gain in basis points	185	135
Underlying investment return (from above)	$478,400	$478,400
Gain on options		
(0·0185 x 2,000,000 x 3/12 x 32)	$296,000	
(0·0135 x 2,000,000 x 3/12 x 32)		$216,000
Premium		
As above	$(69,120)	
As above		$(19,360)
Net return	$705,280	$675,040
Effective interest rate	4·41%	4·22%

Discussion

The FRA offer from Voblaka Bank gives a slightly higher return compared to the futures market; however, Awan Co faces a credit risk with over-the-counter products like the FRA, where Voblaka Bank may default on any money owing to Awan Co if interest rates should fall. The March call option at the exercise price of 94·50 seems to fix the rate of return at 4·41%, which is lower than the return on the futures market and should therefore be rejected. The March call option at the exercise price of 95·00 gives a higher return compared to the FRA and the futures if interest rates increase, but does not perform as well if the interest rates fall. If Awan Co takes the view that it is more important to be protected against a likely fall in interest rates, then that option should also be rejected. The choice between the FRA and the futures depends on Awan Co's attitude to risk and return, the FRA gives a small, higher return, but carries a credit risk. If the view is that the credit risk is small and it is unlikely that Voblaka Bank will default on its obligation, then the FRA should be chosen as the hedge instrument.

(b) The delta value measures the extent to which the value of a derivative instrument, such as an option, changes as the value of its underlying asset changes. For example, a delta of 0·8 would mean that a company would need to purchase 1·25 option contracts (1/0·8) to hedge against a rise in price of an underlying asset of that contract size, known as the hedge ratio. This is because the delta indicates that when the underlying asset increases in value by $1, the value of the equivalent option contract will increase by only $0·80.

The option delta is equal to $N(d_1)$ from the Black-Scholes Option Pricing (BSOP) formula. This means that the delta is constantly changing when the volatility or time to expiry change. Therefore even when the delta and hedge ratio are used to determine the number of option contracts needed, this number needs to be updated periodically to reflect the new delta.

3 (a) Combined company, cost of capital

Asset beta
(1·2 x 480 + 0·9 x 1,218)/(480 + 1,218) = 0·985
Equity beta
0·985 x (60 + 40 x 0·8)/60 = 1·51
Cost of equity
2% + 1·51 x 7% = 12·57%

Cost of capital
12·57% x 0·6 + 4·55% x 0·8 x 0·4 = 9·00%

Combined company equity value
Years 1 to 4 ($ millions)

Year	1	2	3	4
Free cash flows before synergy (growing at 5%)	226·80	238·14	250·05	262·55
Synergies	20·00	20·00	20·00	20·00
Free cash flows	246·80	258·14	270·05	282·55
PV of free cash flows at 9%	226·42	217·27	208·53	200·17

(Note: the present value (PV) figures are slightly different if discount table factors are used, instead of formulae. Full credit will be given if discount tables are used to calculate PV figures.)

Total PV of cash flows (years 1 to 4) = $852·39 million
Total PV of cash flows (years 5 to perpetuity) = 262·55 x 1·0225 /(0·09 – 0·0225) x $1{\cdot}09^{-4}$ = $2,817·51 million

Total value to firm = $3,669·90 million

Value attributable to equity holders = $3,669·90 million x 0·6 = $2,201·94 million

Additional value created from the combined company = $2,201·94 million – ($1,218 million + $480 million) = $2,201·94 million – $1,698·00 million = $503·94 million (or 29·7%)

Although the equity beta and therefore the risk of the combined company is more than Makonis Co on its own, probably due to Nuvola Co's higher business risk (reflected by the higher asset beta), overall the benefits from growth in excess of the risk free rate and additional synergies have led to an increase in the value of combined company of just under 30% when compared to the individual companies' values.

However, a number of assumptions have been made in obtaining the valuation, for example:

– The assumption of growth of cash flows in perpetuity and whether this is realistic or not;
– Whether the calculation of the combined company's asset beta when based on the weighted average of market values is based on good evidence or not;
– It has been assumed that the figures such as growth rates, tax rates, free cash flows, risk free rate of return, risk premium, and so on are accurate and do not change in the future.

In all these circumstances, it may be appropriate to undertake sensitivity analysis to determine how changes in the variables would impact on the value of the combined company, and whether the large increase in value is justified.

(b) Value of Nuvola equity = $2·40 x 200m shares = $480m
30% premium: 1·3 x $480m = $624m
50% premium: 1·5 x $480m = $720m

New number of shares = 210m + ½ x 200m = 310m

Loss in value per share of combined company, if 50% premium paid instead of 30% premium = ($720m – $624m)/310m shares = $0·31/share.

This represents a drop in value of approx. 5·3% on original value of a Makonis Co share ($0·31/$5·80).

(c) The amount of cash required will increase substantially, by about $96 million, if Makonis Co agrees to the demands made by Nuvola Co's equity holders and pays the 50% premium. Makonis Co needs to determine how it is going to acquire the additional funds and the implications from this. For example, it could borrow the money required for the additional funds, but taking on more debt may affect the cost of capital and therefore the value of the company. It could raise the funds by issuing more equity shares, but this may not be viewed in a positive light by the current equity holders.

Makonis Co may decide to offer a higher proportion of its shares in the share-for-share exchange instead of paying cash for the additional premium. However, this will affect its equity holders and dilute their equity holding further. Even the current proposal to issue 100 million new shares will mean that Nuvola Co's equity holders will own just under 1/3 of the combined company and Makonis Co's shareholders would own just over 2/3 of the combined company.

Makonis Co should also consider what Nuvola Co's equity holders would prefer. They may prefer less cash and more equity due to their personal tax circumstances, but, in most cases, cash is preferred by the target firm's equity holders.

4 (a) Current and non-current liabilities = $387m + $95m = $482m

Sale of assets of supermarkets division
Proportion of assets to supermarkets division
Non-current assets = 70% x $550m = $385m; Current assets = 70% x $122m = $85·4m

Sale of assets = $385m x 1·15 + $85·4m x 0·80 = $511·07m

Sale of supermarkets division as a going concern
Profit after tax attributable to the supermarkets division: $166m/2 = $83m

Estimate of value of supermarkets division based on the PE ratio of supermarket industry: $83 x 7 = $581m

Although both options generate sufficient funds to pay for the liabilities, the sale of the supermarkets division as a going concern would generate higher cash flows and the spare cash of $99m [$581m – $482m] can be used by Nubo Co for future investments. This is based on the assumption that the value based on the industries' PE ratios is accurate.

Proportion of assets remaining within Nubo Co
30% x ($550m + $122m) = $201·6m

Add extra cash generated from the sale of $99m

Maximum debt capacity = $300·6m

Total additional funds available to Nubo Co for new investments = $300·6m + $99m = $399·6m

(b) A demerger would involve splitting Nubo Co into two separate companies which would then operate independently of each other. The equity holders in Nubo Co would continue to have an equity stake in both companies.

Normally demergers are undertaken to ensure that each company's equity values are fair. For example, the value of the aircraft parts production division based on the PE ratio gives a value of $996m (12 x $83m) and the value of the supermarkets division as $581m. If the current company's value is less than the combined values of $1,577m, then a demerger may be beneficial. However, the management and shareholders of the new supermarkets company may not be keen to take over all the debt.

Nubo Co's equity holders may view the demerger more favourably than the sale of the supermarkets division. At present their equity investment is diversified between the aircraft parts production and supermarkets. If the supermarkets division is sold, then the level of their diversification may be affected. With the demerger, since the equity holders will retain an equity stake in both companies, the benefit of diversification is retained.

However, the extra $99m cash generated from the sale will be lost in the case of a demerger. Furthermore, if the new aircraft parts production company can only borrow 100% of its asset value, then its borrowing capacity and additional funds available to it for new investments will be limited to $201·6m instead of $399·6m.

(c) With a Mudaraba contract, the profits which Pilvi Co makes from the joint venture would be shared according to a pre-agreed arrangement when the contract is constructed between Pilvi Co and Ulap Bank. Losses, however, would be borne solely by Ulap Bank as the provider of the finance, although provisions can be made where losses can be written off against future profits. Ulap Bank would not be involved in the executive decision-making process. In effect, Ulap Bank's role in the relationship would be similar to an equity holder, holding a small number of shares in a large organisation.

With a Musharaka contract, the profits which Pilvi Co makes from the joint venture would still be shared according to a pre-agreed arrangement similar to a Mudaraba contract, but losses would also be shared according to the capital or other assets and services contributed by both the parties involved in the arrangement. Therefore a value could be put to the contribution-in-kind made by Pilvi Co and any losses would be shared by Ulap Bank and Pilvi Co accordingly. Within a Musharaka contract, Ulap Bank can also take the role of an active partner and participate in the executive decision-making process. In effect, the role adopted by Ulap Bank would be similar to that of a venture capitalist.

With the Mudaraba contract, Pilvi Co would essentially be an agent to Ulap Bank, and many of the agency issues facing corporations would apply to the arrangement, where Pilvi Co can maximise its own benefit at the expense of Ulap Bank. Pilvi Co may also have a propensity to undertake excessive risk because it is essentially holding a long call option with an unlimited upside and a limited downside.

Ulap Bank may prefer the Musharaka contract in this case, because it may be of the opinion that it needs to be involved with the project and monitor performance closely due to the inherent risk and uncertainty of the venture, and also to ensure that the revenues, expenditure and time schedules are maintained within initially agreed parameters. In this way, it may be able to monitor and control agency related issues more effectively and control Pilvi Co's risky actions and decisions. Being closely involved with the venture would change both Pilvi Co's and Ulap Bank's roles and make them more like stakeholders rather than principals and agents, with a more equitable distribution of power between the two parties.

Nubo Co's concerns would mainly revolve around whether it can work with Ulap Bank and the extra time and cost which would need to be incurred before the joint venture can start. If Pilvi Co had not approached Ulap Bank for funding, the relationship between Nubo Co and Pilvi Co would be less complex within the joint venture. Although difficulties may arise about percentage ownership and profit sharing, these may be resolved through negotiation and having tight specific contracts. The day-to-day running, management and decision-making process could be resolved through negotiation and consensus. Therefore having a third party involved in all aspects of the joint venture complicates matters.

Nubo Co may feel that it was not properly consulted about the arrangements between Pilvi Co and Ulap Bank, and Pilvi Co would need to discuss the involvement of Ulap Bank with Nubo Co and gets its agreement prior to formalising any arrangements. This is to ensure a high level of trust continues to exist between the parties, otherwise the venture may fail.

Nubo Co may want clear agreements on ownership and profit-sharing. They would want to ensure that the contract clearly distinguishes them as not being part of the Musharaka arrangement which exists between Pilvi Co and Ulap Bank. Hence negotiation and construction of the contracts may need more time and may become more expensive.

Nubo Co may have felt that it could work with Pilvi Co on a day-to-day basis and could resolve tough decisions in a reasonable manner. It may not feel the same about Ulap Bank initially. Clear parameters would need to be set up on how executive decision making will be conducted by the three parties. Therefore, the integration process of bringing a third partner into the joint venture needs to be handled with care and may take time and cost more money.

The above issues would indicate that the relationship between the three parties is closer to that of stakeholders, with different levels of power and influence, at different times, as opposed to a principal–agent relationship. This would create an environment which would need ongoing negotiation and a need for consensus, which may make the joint venture hard work. Additionally, it would possibly be more difficult and time consuming to accomplish the aims of the joint venture.

(Note: Credit will be given for alternative relevant comments and suggestions for parts (b) and (c) of the question)

					Marks
1	**(a)**	Role of the World Trade Organisation			4–5
		Benefits of reducing protectionist measures			2–3
		Drawbacks of reducing protectionist measures			2–3
				Max	9
	(b)	**(i)**	Future exchange rates predicted on inflation rate differential		1
			Sales revenue		1
			Production and selling costs		1
			Special packaging costs		2
			Training and development costs		1
			Correct treatment of tax and tax allowable depreciation		2
			Years 1 to 5 cash flows in $ and present values of cash flows		2
			Ignoring initial investigation cost and additional taxation in Chmura Co host country		1
			Correct treatment of land, buildings, machinery and working capital		2
			Net present value of the project		1
					14
		(ii)	Inputting correct values for the variables		2
			Calculation of d_1 and d_2		2
			Establishing $N(d_1)$ and $N(d_2)$		2
			Call value		1
			Put value		1
			Value of the project		1
					9
		(iii)	Estimated value and initial recommendation		2–3
			Up to 2 marks per assumption discussed		5–6
			Up to 2 marks per additional business risk discussed		5–6
			Overarching recommendation(s)		1–2
				Max	14
		Professional marks			
		Report format			1
		Structure and presentation of the report			3
					4
				Total	**50**
2	**(a)**	Calculation of impact of FRA for interest rate increase and decrease			2
		Decision to go long on futures			1
		Selection of March futures and options			1
		Unexpired basis calculation			1
		Impact of interest rates increase/decrease with futures			3
		Decision to buy call options			1
		Impact of interest rates increase/decrease with options			5
		Discussion (up to 2 marks available for general explanation of the products' features)			5–6
				Max	19
	(b)	1–2 marks per well explained point		Max	6
				Total	**25**

				Marks
3	(a)	Market values of Makonis Co and Nuvola Co		1
		Combined company asset beta		1
		Combined company equity beta		1
		Combined company: cost of capital		1
		Combined company value: years 1 to 4		3
		Combined company value: years 5 to perpetuity		1
		Combined company value: value to equity holders and additional value		2
		Comment and discussion of assumptions		3–4
			Max	**13**
	(b)	Payment if 30% premium paid		1
		Payment if 50% premium paid		1
		Estimate of impact on Makonis Co's equity holders		3
				5
	(c)	Additional amount payable if 50% premium paid instead of 30% premium		1
		Discussion of how Makonis Co would finance the additional premium		6
				7
			Total	**25**
4	(a)	Sale of supermarkets division's assets		1
		Sale of supermarkets division as going concern		1
		Advice		2
		Extra cash after liabilities are paid		1
		Maximum debt which can be borrowed		1
		Additional funds available to Nubo Co		1
				7
	(b)	1–2 marks per relevant point	Max	**6**
	(c)	Discussion of why Ulap Bank might prefer a Musharaka contract		6–7
		Discussion of the key concerns of the joint venture relationship		5–6
			Max	12
			Total	**25**

Answers

1 (a) The foreign exchange exposure of the dollar payment due in four months can be hedged using the following derivative products:

Forward rate offered by Pecunia Bank;
Exchange-traded futures contracts; and
Exchange-traded options contracts

Using the forward rate
Payment in Swiss Francs = US$5,060,000/1·0677 = CHF4,739,159

Using futures contract
Since a dollar payment needs to be made in four months' time, CMC Co needs to hedge against Swiss Francs weakening. Hence, the company should go short and the six-month futures contract is undertaken. It is assumed that the basis differential will narrow in proportion to time.

Predicted futures rate = 1·0647 + [(1·0659 – 1·0647) x 1/3] = 1·0651

[*Alternatively, can predict futures rate based on spot rate: 1·0635 + [(1·0659 – 1·0635) x 4/6] = 1·0651*]

Expected payment = US$5,060,000/1·0651 = CHF4,750,728
No. of contracts sold = CHF4,750,728/CHF125,000 = approx. 38 contracts

Using options contracts
Since a dollar payment needs to be made in four months' time, CMC Co needs to hedge against Swiss Francs weakening. Hence, the company should purchase six-month put options.

Exercise price US$1·06/CHF1
Payment = US$5,060,000/1·06 = CHF4,773,585
Buy 4,773,585/125,000 = 38·19 put contracts, say 38 contracts

CHF payment = CHF4,750,000
Premium payable = 38 x 125,000 x 0·0216 = US$102,600
In CHF = 102,600/1·0635 = CHF96,474
Amount not hedged = US$5,060,000 – (38 x 125,000 x 1·06) = US$25,000
Use forward contracts to hedge this = US$25,000/1·0677 = CHF23,415
Total payment = CHF4,750,000 + CHF96,474 + CHF23,415 = CHF4,869,889

Exercise price US$1·07/CHF1
Payment = US$5,060,000/1·07 = CHF4,728,972
Buy 4,728,972/125,000 = 37·83 put contracts, say 38 contracts (but this is an over-hedge)

CHF payment = CHF4,750,000
Premium payable = 38 x 125,000 x 0·0263 = US$124,925
In CHF = 124,925/1·0635 = CHF117,466
Amount over-hedged = US$5,060,000 – (38 x 125,000 x 1·07) = US$22,500
Using forward contracts to show benefit of this = US$22,500/1·0677 = CHF21,073
Total payment = CHF4,750,000 + CHF117,466 – CHF21,073 = CHF4,846,393

Advice
Forward contracts minimise the payment and option contracts would maximise the payment, with the payment arising from the futures contracts in between these two. With the option contracts, the exercise price of US$1·07/CHF1 gives the lower cost. Although transaction costs are ignored, it should be noted that with exchange-traded futures contracts, margins are required and the contracts are marked-to-market daily.

It would therefore seem that the futures contracts and the option contract with an exercise price of US$1·06/CHF1 should be rejected. The choice between forward contracts and the 1·07 options depends on CMC Co's attitude to risk. The forward rate is binding, whereas option contracts give the company the choice to let the option contract lapse if the CHF strengthens against the US$. Observing the rates of inflation between the two countries and the exchange-traded derivatives this is likely to be the case, but it is not definite. Moreover, the option rates need to move in favour considerably before the option is beneficial to CMC Co, due to the high premium payable.

It would therefore seem that forward markets should be selected to minimise the amount of payment, but CMC Co should also bear in mind that the risk of default is higher with forward contracts compared with exchange-traded contracts.

(b)

	CMC Co	**Counterparty**	**Interest rate differential**
Fixed rate	2·2%	3·8%	1·6%
Floating rate	Yield rate + 0·4%	Yield rate + 0·8%	0·4%

CMC Co has a comparative advantage in borrowing at the fixed rate and the counterparty has a comparative advantage in borrowing at the floating rate. Total possible benefit before Pecunia Bank's fee is 1·2%, which if shared equally results in a benefit of 0·6% each, for both CMC Co and the counterparty.

	CMC Co	Counterparty
CMC Co borrows at	2·2%	
Counterparty borrows at		Yield rate + 0·8%
Advantage	60 basis points	60 basis points
Net result	Yield rate – 0·2%	3·2%
SWAP		
Counterparty receives		Yield rate
CMC Co pays	Yield rate	
Counterparty pays		2·4%
CMC Co receives	2·4%	

After paying the 20 basis point fee, CMC Co will effectively pay interest at the yield curve rate and benefit by 40 basis points or 0·4%, and the counterparty will pay interest at 3·4% and benefit by 40 basis points or 0·4% as well.

[**Note:** *Full marks will be given where the question is answered by estimating the arbitrage gain of 1·2% and deducting the fees of 0·4%, without constructing the above table*]

(c) Annuity factor, 4 years, 2% = 3·808
Equal annual amounts repayable per year = CHF60,000,000/3·808 = CHF15,756,303

Macaulay duration
(15,756,303 x 0·980 x 1 year +
15,756,303 x 0·961 x 2 years +
15,756,303 x 0·942 x 3 years +
15,756,303 x 0·924 x 4 years)/60,000,000
= 2·47 years

Modified duration = 2·47/1·02 = 2·42 years

The equation linking modified duration (D), and the relationship between the change in interest rates (Δi) and change in price or value of a bond or loan (ΔP) is given as follows:

$\Delta P = [-D \times \Delta i \times P]$
(P is the current value of a loan or bond and is a constant)

The size of the modified duration will determine how much the value of a bond or loan will change when there is a change in interest rates. A higher modified duration means that the fluctuations in the value of a bond or loan will be greater, hence the value of 2·42 means that the value of the loan or bond will change by 2·42 times the change in interest rates multiplied by the original value of the bond or loan.

The relationship is only an approximation because duration assumes that the relationship between the change in interest rates and the corresponding change in the value of the bond or loan is linear. In fact, the relationship between interest rates and bond price is in the form of a curve which is convex to the origin (i.e. non-linear). Therefore duration can only provide a reasonable estimation of the change in the value of a bond or loan due to changes in interest rates, when those interest rate changes are small.

(d) MEMORANDUM

From:

To: The Board of Directors, CMC Co

Date: xx/xx/xxxx

Subject: Discussion of the proposal to manage foreign exchange and interest rate exposures, and the proposal to move operations to four branches and consequential agency issues

This memo discusses the proposal of whether or not CMC Co should undertake the management of foreign exchange and interest rate exposure, and the agency issues resulting from the proposal to locate branches internationally and how these issues may be mitigated. Each proposal will be considered in turn.

(i) Proposal One: Management of foreign exchange and interest rate exposure

The non-executive directors are correct if CMC Co is in a situation where markets are perfect and efficient, where information is freely available and where securities are priced correctly. In this circumstance, risk management or hedging would not add value and if shareholders hold well diversified portfolios, unsystematic risk will be largely eliminated. The position against hedging states that in such cases companies would not increase shareholder value by hedging or eliminating risk because there will be no further reduction in unsystematic risk. Furthermore, the cost of reducing any systematic risk will equal or be greater than the benefit derived from such risk reduction. Shareholders would not gain from risk management or hedging; in fact, if the costs exceed the benefits, then hedging may result in a reduction in shareholder value.

Risk management or hedging may result in increasing corporate (and therefore shareholder) value if market imperfections exist, and in these situations, reducing the volatility of a company's earnings will result in higher cash inflows. Proponents of hedging cite three main situations where reduction in volatility or risk may increase cash flows –

in situations: where the rate of tax is increasing; where a firm could face significant financial distress costs due to high volatility in earnings; and where stable earnings increases certainty and the ability to plan for the future, thus resulting in stable investment policies by the firm.

Active hedging may also reduce agency costs. For example, unlike shareholders, managers and employees of the company may not hold diversified portfolios. Hedging allows the risks faced by managers and employees to be reduced. Additionally, hedging may allow managers to be less concerned about market movements which are not within their control and instead allow them to focus on business issues over which they can exercise control. This seems to be what the purchasing director is contending. On the other hand, the finance director seems to be more interested in increasing his personal benefits and not necessarily in increasing the value of CMC Co.

A consistent hedging strategy or policy may be used as a signalling tool to reduce the conflict of interest between bondholders and shareholders, and thus reduce restrictive covenants.

It is also suggested that until recently CMC Co had no intention of hedging and communicated this in its annual report. It is likely that shareholders will therefore have created their own risk management policies. A strategic change in the policy may have a negative impact on the shareholders and the clientele impact of this will need to be taken into account.

The case of whether to hedge or not is not clear cut and CMC Co should consider all the above factors and be clear about why it is intending to change its strategy before coming to a conclusion. Any intended change in policy should be communicated to the shareholders. Shareholders can also benefit from risk management because the risk profile of the company may change, resulting in a reduced cost of capital.

(ii) Proposal Two: International branches, agency issues and their mitigation

Principal–agent relationships can be observed within an organisation between different stakeholder groups. With the proposed branches located in different countries, the principal–agent relationship will be between the directors and senior management at CMC Co in Switzerland, and the managers of the individual branches. Agency issues can arise where the motivations of the branch managers, who are interested in the performance of their individual branches, diverge from the management at CMC Co headquarters, who are interested in the performance of the whole organisation.

These issues may arise because branch managers are not aware of, or appreciate the importance of, the key factors at corporate level. They may also arise because of differences in cultures and divergent backgrounds.

Mitigation mechanisms involve monitoring, compensation and communication policies. All of these mechanisms need to work in a complementary fashion in order to achieve goal congruence, much like the mechanisms in any principal–agent relationship.

Monitoring policies would involve ensuring that key aims and strategies are agreed between all parties before implementation, and results monitored to ensure adherence with the original agreements. Where there are differences, for example due to external factors, new targets need to be agreed. Where deviations are noticed, these should be communicated quickly.

Compensation packages should ensure that reward is based on achievement of organisational value and therefore there is every incentive for the branch managers to act in the best interests of the corporation as a whole.

Communication should be two-way, in that branch managers should be made fully aware of the organisational objectives, and any changes to these, and how the branch contributes to these, in order to ensure their acceptance of the objectives. Furthermore, the management at CMC Co headquarters should be fully aware of cultural and educational differences in the countries where the branches are to be set up and fully plan for how organisational objectives may nevertheless be achieved within these differences.

(Note: Credit will be given for alternative, relevant approaches to the calculations, comments and suggestions/recommendations)

2 (a) All figures are in $ million

Year	0	1	2	3	4
Sales revenue (inflated, 8% p.a.)		24·87	42·69	61·81	36·92
Costs (inflated, 4% p.a.)		(14·37)	(23·75)	(33·12)	(19·05)
Incremental profit		10·50	18·94	28·69	17·87
Tax (W1)		(0·50)	(3·39)	(5·44)	(3·47)
Working capital (W2)	(4·97)	(3·57)	(3·82)	4·98	7·38
Investment/sale of machinery	(38·00)				4·00
Cash flows	(42·97)	6·43	11·73	28·23	25·78
Discount factors (12%, W3)	1	0·893	0·797	0·712	0·636
Present values	(42·97)	5·74	9·35	20·10	16·40

Base case net present value is approximately $8·62 million.

W1 All figures are in $ million

Year	0	1	2	3	4
Incremental profit		10·50	18·94	28·69	17·87
Capital allowances		8·00	2·00	1·50	0·50
Taxable profit		2·50	16·94	27·19	17·37
Tax (20%)		0·50	3·39	5·44	3·47

W2 All figures are in $ million

Year	0	1	2	3	4
Working capital (20% of sales revenue)		4·97	8·54	12·36	7·38
Working capital required/(released)	4·97	3·57	3·82	(4·98)	(7·38)

W3 Lintu Co asset beta = 1·5 x $128m/($128m + $31·96m x 0·8) approx. = 1·25
All-equity financed discount rate = 2% + 1·25 x 8% = 12%

Financing side effects

	$'000
Issue costs 2/98 x $42,970,000	(876·94)
Tax shield Annual tax relief = ($42,970,000 x 60% x 0·015 x 20%) + ($42,970,000 x 40% x 0·04 x 20%) = 77·35 + 137·50 = 214·85	
The present value of the tax relief annuity = 214·85 x 3·63	779·91
Annual subsidy benefit $42,970,000 x 60% x 0·025 x 80% = 515·64	
The present value of the subsidy benefit annuity = 515·64 x 3·63	1,871·77
Total benefit of financing side effects	1,774·74

Financing the project entirely by debt would add just under $1·78 million to the value of the project, or approximately, an additional 20% to the all-equity financed project.

The adjusted present value (APV) of the project is just under $10·4 million and therefore it should be accepted.

Note: *In calculating the present values of the tax shield and subsidy benefits, the annuity factor used is based on 4% to reflect the normal borrowing/default risk of the company.*

Alternatively, 2% or 2·5% could be used depending on the assumptions made. Credit will be given where these are used to estimate the annuity factor, where the assumption is explained.

(b) Corrections made to the original net present value

The approach taken to exclude depreciation from the net present value computation is correct, but capital allowances need to be taken away from profit estimates before tax is calculated, reducing the profits on which tax is payable.

Interest is not normally included in the net present value calculations. Instead, it is normally imputed within the cost of capital or discount rate. In this case, it is included in the financing side effects.

Cash flows are inflated and the nominal rate based on Lintu Co's all-equity financed rate is used (see below). Where different cash flows are subject to different rates of inflation, applying a real rate to non-inflated amounts would not give an accurate answer.

The impact of the working capital requirement is included in the estimate as, although all the working capital is recovered at the end of the project, the flows of working capital are subject to different discount rates when their present values are calculated.

Approach taken

The value of the project is initially assessed considering only the business risk involved in undertaking the project. The discount rate used is based on Lintu Co's asset beta which measures only the business risk of that company. Since Lintu Co is in the same line of business as the project, it is deemed appropriate to use its discount rate, instead of 11% that Burung Co uses normally.

The impact of debt financing and the subsidy benefit are then considered. In this way, Burung Co can assess the value created from its investment activity and then the additional value created from the manner in which the project is financed.

Assumptions made

It is assumed that all figures used are accurate and any estimates made are reasonable. Burung Co may want to consider undertaking a sensitivity analysis to assess this.

It is assumed that the initial working capital required will form part of the funds borrowed but that the subsequent working capital requirements will be available from the funds generated by the project. The validity of this assumption needs to be assessed since the working capital requirements at the start of years 2 and 3 are substantial.

It is assumed that Lintu Co's asset beta and all-equity financed discount rate represent the business risk of the project. The validity of this assumption also needs to be assessed. For example, Lintu Co's entire business may not be similar to the project, and it may undertake other lines of business. In this case, the asset beta would need to be adjusted so that just the project's business risk is considered.

(Note: Credit will be given for alternative, relevant explanations)

3 (a) Vogel Co may have switched from a strategy of organic growth to one of growth by acquisition, if it was of the opinion that such a change would result in increasing the value for the shareholders.

Acquiring a company to gain access to new products, markets, technologies and expertise may be quicker and less costly. Horizontal acquisitions may help Vogel Co eliminate key competitors and enable it to take advantage of economies of scale. Vertical acquisitions may help Vogel Co to secure the supply chain and maximise returns from its value chain.

Organic growth may take a long time, can be expensive and may result in little competitive advantage being established due to the time taken. Also organic growth, especially into a new area, would need managers to gain knowledge and expertise of an area or function, which they not currently familiar with. Furthermore, in a saturated market, there may be little opportunity for organic growth.

(Note: Credit will be given for alternative relevant comments)

(b) Vogel Co can take the following actions to reduce the risk that the acquisition of Tori Co fails to increase shareholder value.

Since Vogel Co has pursued an aggressive policy of acquisitions, it needs to determine whether or not this has been too aggressive and detailed assessments have been undertaken. Vogel Co should ensure that the valuation is based on reasonable input figures and that proper due diligence of the perceived benefits is undertaken prior to the offer being made. Often it is difficult to get an accurate picture of the target when looking at it from the outside. Vogel Co needs to ensure that it has sufficient data and information to enable a thorough and sufficient analysis to be undertaken.

The sources of synergy need to be properly assessed to ensure that they are achievable and what actions Vogel Co needs to undertake to ensure their achievement. This is especially so for the revenue-based synergies. An assessment of the impact of the acquisition on the risk of the combined company needs to be undertaken to ensure that the acquisition is not considered in isolation but as part of the whole company.

The Board of Directors of Vogel Co needs to ensure that there are good reasons to undertake the acquisition, and that the acquisition should result in an increase in value for the shareholders. Research studies into mergers and acquisitions have found that often companies are acquired not for the shareholders' benefit, but for the benefit or self-interest of the acquiring company's management. The non-executive directors should play a crucial role in ensuring that acquisitions are made to enhance the value for the shareholders. A post-completion audit may help to identify the reasons behind why so many of Vogel Co's acquisitions have failed to create value. Once these reasons have been identified, strategies need to be put in place to prevent their repetition in future acquisitions.

Procedures need to be established to ensure that the acquisition is not overpaid. Vogel Co should determine the maximum premium it is willing to pay and not go beyond that figure. Research indicates that often too much is paid to acquire a company and the resultant synergy benefits are not sufficient to cover the premium paid. Often this is the result of the management of the acquiring company wanting to complete the deal at any cost, because not completing the deal may be perceived as damaging to both their own, and their company's, reputation. The acquiring company's management may also want to show that the costs related to undertaking due diligence and initial negotiation have not been wasted. Vogel Co and its management need to guard against this and maybe formal procedures need to be established which allow managers to step back without loss of personal reputation.

Vogel Co needs to ensure that it has proper procedures in place to integrate the staff and systems of the target company effectively, and also to recognise that such integration takes time. Vogel Co may decide instead to give the target company a large degree of autonomy and thus make integration less necessary; however, this may result in a reduction in synergy benefits. Vogel Co should also have strategies which allow it sufficient flexibility when undertaking integration so that it is able to respond to changing circumstances or respond to inaccurate information prior to the acquisition. Vogel Co should also be mindful that its own and the acquired company's staff and management need to integrate and ensure a good working relationship between them.

(**Note:** *The above answer covers more areas than would be needed to achieve full marks for the part. Credit will be given for alternative relevant comments*)

(c) Approach taken

The maximum premium payable is equal to the maximum additional benefit created from the acquisition of Tori Co, with no increase in value for the shareholders of Vogel Co (although the shareholders of Vogel Co would probably not approve of the acquisition if they do not gain from it).

The additional benefit can be estimated as the sum of the cash gained (or lost) from selling the assets of Department C, spinning off Department B and integrating Department A, less the sum of the values of Vogel Co and Tori Co as separate companies.

Estimation
Cash gained from selling the assets of Department C = (20% x $98·2m) + (20% x $46·5m x 0·9) – ($20·2 + $3m) = $19·64m + $8·37m – $23·2m = $4·81m

Value created from spinning off Department B into Ndege Co

Free cash flow of Ndege Co	**$ million**
Current share of PBDIT (0·4 x $37·4m)	14·96
Less: attributable to Department C (10%)	(1·50)
Less: tax allowable depreciation (0·4 x 98·2 x 0·10)	(3·93)
Profits before tax	9·53
Tax (20%)	(1·91)
Free cash flows	7·62

Value of Ndege Co =
Present value of cash flow in year 1: $7·62m x 1·2 x $1·1^{-1}$ = $8·31m
Add: present value of cash flows from year 2 onwards:
($9·14m x 1·052)/(0·1 – 0·052) x $1·1^{-1}$ = $182·11m
Less debt = $40m
Value to shareholders of Ndege Co = $150·42m

Vogel Co's current value = $3 x 380m = $1,140m
Vogel Co, profit after tax = $158·2m x 0·8 = $126·56m
Vogel Co, PE ratio before acquisition = $1,140·0m/$126·56m = 9·01 say 9
Vogel Co, PE ratio after acquisition = 9 x 1·15 = 10·35
Tori Co, PE ratio before acquisition = 9 x 1·25 = 11·25

Tori Co's current value = 11·25 x ($23·0 x 0·8) = $207·0m

Value created from combined company
($126·56m + 0·5 x $23·0m x 0·8 + $7m) x 10·35 = $1,477·57

Maximum premium = ($1,477·57m + $150·42m + $4·81m) – ($1,140m + $207·0m) = $285·80m

Assumptions
Based on the calculations given above, it is estimated that the value created will be 64·9% or $285·80m.

However, Vogel Co needs to assess whether the numbers it has used in the calculations and the assumptions it has made are reasonable. For example, Ndege Co's future cash flows seem to be growing without any additional investment in assets and Vogel Co needs to establish whether or not this is reasonable. It also needs to establish how the increase in its PE ratio was determined after acquisition. Perhaps sensitivity analysis would be useful to show the impact on value changes, if these figures are changed. Given its poor record in generating value previously, Vogel Co needs to pay particular attention to these figures.

4 (a) With conventional investment decisions, it is assumed that once a decision is made, it has to be taken immediately and carried to its conclusion. These decisions are normally made through conventional assessments using methods such as net present value. Assessing projects through option pricing may aid the investment decision making process.

Where there is uncertainty with regard to the investment decision and where a company has flexibility in its decision making, valuing projects using options can be particularly useful. For example, situations may exist where a company does not have to make a decision on a now-or-never basis, or where it can abandon a decision, which has been made, at some future point, or where it has an opportunity for further expansion as a result of the original decision. In such situations, using option pricing formulae, which incorporate the uncertainty surrounding a project and the time before a decision has to be made, can determine a value attached to this flexibility. This value can be added to the conventional net present value computation to give a more accurate assessment of the project's value.

In the situation which Faoilean Co is considering, the initial exploration rights may give it the opportunity to delay the decision of whether to undertake the extraction of oil and gas to a later date. In that time, using previous knowledge and experience, it can estimate the quantity of oil and gas which is present more accurately. It can also use its knowledge to assess the variability of the likely quantity. Faoilean Co may be able to negotiate a longer time scale with the government of Ireland for undertaking the initial exploration, before it needs to make a final decision on whether and how much to extract.

Furthermore, Faoilean Co can explore the possibilities of it exiting the extraction project, once started, if it is proving not to be beneficial, or if world prices of oil and gas have moved against it. It could, for example, negotiate a get-out clause which gives it the right to sell the project back to the government at a later date at a pre-agreed price. Alternatively, it could build facilities in such a way that it can redeploy them to other activities, or scale the production up or down more easily and at less cost. These options give the company the opportunity to step out of a project at a future date, if uncertainties today become negative outcomes in the future.

Finally, Faoilean Co can explore whether or not applying for the rights to undertake this exploration project could give it priority in terms of future projects, perhaps due to the new knowledge or technologies it builds during the current project. These opportunities would allow it to gain competitive advantage over rivals, which, in turn, could provide it with greater opportunities in the future, but which are uncertain at present.

Faoilean Co can incorporate these uncertainties and the time before the various decisions need to be made into the option formulae to determine the additional value of the project, on top of the initial net present value calculation.

The option price formula used with investment decisions is based on the Black-Scholes Option Pricing (BSOP) model. The BSOP model makes a number of assumptions as follows:

– The underlying asset operates in perfect markets and therefore the movement of market prices cannot be predicted;
– The BSOP model uses the risk-free rate of interest. It is assumed that this is known and remains constant, which may not be the case where the time it takes for the option to expire may be long;
– The BSOP model assumes that volatility can be assessed and stays constant throughout the life of the project; again with long-term projects these assumptions may not be valid;
– The BSOP model assumes that the underlying asset can be traded freely. This is probably not accurate where the underlying asset is an investment project.

These assumptions mean that the value based around the BSOP model is indicative and not definitive.

(Note: Credit will be given for alternative relevant comments)

(b) Equity can be regarded as purchasing a call option by the equity holders on the value of a company, because they will possess a residual claim on the assets of the company. In this case, the face value of debt is equivalent to the exercise price, and the repayment term of debt as the time to expiry of the option.

If at expiry, the value of the company is greater than the face value of debt, then the option is in-the-money, otherwise if the value of the firm is less than the face value of debt, then the option is out-of-money and equity is worthless.

For example, say V is the market value of the assets in a company, E is the market value of equity, and F is the face value of debt, then,

If at expiry $V > F$ (option is in-the-money), then the option has intrinsic value to the equity holders and $E = V – F$;
Otherwise if $F > V$ (option is out-of-money), then the option has no intrinsic value and no value for the equity holders, and $E = 0$.

Prior to expiry of the debt, the call option (value to holders of equity) will also have a time value attached to it. The BSOP model can be used to assess the value of the option to the equity holders, the value of equity, which can consist of both time value and intrinsic value if the option is in-the-money, or just time value if the option is out-of-money.

Within the BSOP model, $N(d_1)$, the delta value, shows how the value of equity changes when the value of the company's assets change. $N(d_2)$ depicts the probability that the call option will be in-the-money (i.e. have intrinsic value for the equity holders).

Debt can be regarded as the debt holders writing a put option on the company's assets, where the premium is the receipt of interest when it falls due and the capital redemption. If $N(d_2)$ depicts the probability that the call option is in-the-money, then $1 – N(d_2)$ depicts the probability of default.

Therefore the BSOP model and options are useful in determining the value of equity and default risk.

Option pricing can be used to explain why companies facing severe financial distress can still have positive equity values. A company facing severe financial distress would presumably be one where the equity holders' call option is well out-of-money and therefore has no intrinsic value. However, as long as the debt on the option is not at expiry, then that call option will still have a time value attached to it. Therefore, the positive equity value reflects the time value of the option, even where the option is out-of-money, and this will diminish as the debt comes closer to expiry. The time value indicates that even though the option is currently out-of-money, there is a possibility that due to the volatility of asset values, by the time the debt reaches maturity, the company will no longer face financial distress and will be able to meet its debt obligations.

(Note: Credit will be given for alternative relevant comments)

(c) According to the BSOP model, the value of an option is dependent on five variables: the value of the underlying asset, the exercise price, the risk-free rate of interest, the implied volatility of the underlying asset, and the time to expiry of the option. These five variables are input into the BSOP formula, in order to compute the value of a call option (the value of an equivalent put option can be computed by the BSOP model and put-call parity relationship). The different risk factors determine the impact on the option value of the changes in the five variables.

The 'vega' determines the sensitivity of an option's value to a change in the implied volatility of the underlying asset. Implied volatility is what the market is implying the volatility of the underlying asset will be in the future, based on the price changes in an option. The option price may change independently of whether or not the underlying asset's value changes, due to new information being presented to the markets. Implied volatility is the result of this independent movement in the option's value, and this determines the 'vega'. The 'vega' only impacts the time value of an option and as the 'vega' increases, so will the value of the option.

(Note: Credit will be given for alternative relevant comments)

				Marks
1	**(a)**	Calculation of payment using the forward rate		1
		Going short on futures and purchasing put options		2
		Predicted futures rate based on basis reduction		1
		Futures: Expected payment and number of contracts		2
		Options calculation using either 1·06 or 1·07 rate		3
		Options calculation using the second rate (or explanation)		2
		Advice (1 to 2 marks per point)		4–5
			Max	15
	(b)	Comparative advantage and recognition of benefit as a result		2
		Initial decision to borrow fixed by CMC Co and floating by counterparty		1
		Swap impact		2
		Net benefit after bank charges		1
				6
	(c)	Calculation of annual annuity amount		1
		Calculation of Macaulay duration		2
		Calculation of modified duration		1
		Explanation		3
				7
	(d)	**(i)** 2–3 marks per point	Max	9
		(ii) Discussion of the agency issues		3–4
		Discussion of mitigation strategies and policies		4–6
			Max	9
		Professional marks		
		Memorandum format		1
		Structure and presentation of the memorandum		3
				4
			Total	**50**
2	**(a)**	Inflated incremental profit		2
		Taxation		2
		Working capital		2
		Estimate of discount rate		2
		Net present value		1
		Issue costs		1
		Tax shield benefit		2
		Subsidy benefit		1
		Adjusted present value and conclusion		2
				15
	(b)	Corrections made		4–5
		Approach taken		2–3
		Assumptions made		3–4
			Max	10
			Total	**25**

				Marks
3	(a)	1–2 marks per point	Max	4
	(b)	2–3 marks per point	Max	7
	(c)	Cash gained from sales of Department C assets		1
		Calculation of free cash flows for Ndege Co		2
		Calculation of present values of Ndege Co cash flows and value		2
		Vogel Co PE ratios before and after acquisition		2
		Tori Co PE ratio and value		1
		Value created from combining Department A with Vogel Co		1
		Maximum premium payable		1
		Approach taken		1–2
		Assumptions made		2–3
			Max	14
			Total	**25**
4	(a)	Discussion of the idea of using options in making the project investment decision		7–8
		Explanation of the assumptions		3–4
			Max	11
	(b)	Discussion of using options to value equity		4–5
		Discussion of using options to assess default risk		2–3
		Discussion of financial distress and time value of an option		2–3
			Max	9
	(c)	Explanation of why option values are determined by different risk factors		2–3
		Explanation of what determines 'vega'		2–3
			Max	5
			Total	**25**

Answers

1 **(a)** Risk diversification, especially into diverse business sectors, has often been stated as a reason for undertaking mergers and acquisitions (M&As). Like individuals holding well-diversified portfolios, a company with a number of subsidiaries in different sectors could reduce its exposure to unsystematic risk. Another possible benefit of diversification is sometimes argued to be a reduction in the volatility of cash flows, which may lead to a better credit rating and a lower cost of capital.

The argument against this states that since individual investors can undertake this level of risk diversification both quickly and cheaply themselves, there is little reason for companies to do so. Indeed, research suggests that markets do not reward this risk diversification.

Nevertheless, for Nahara Co, undertaking M&As may have beneficial outcomes, especially if the sovereign fund has its entire investment in the holding company and is not well-diversified itself. In such a situation unsystematic risk reduction can be beneficial. The case study does not state whether or not the sovereign funds are invested elsewhere and therefore a definitive conclusion cannot be reached.

If Nahara Co is able to identify undervalued companies and after purchasing the company can increase the value for the holding company overall, by increasing the value of the undervalued companies, then such M&As activity would have a beneficial impact on the funds invested. However, for this strategy to work, Nahara Co must:

(i) Possess a superior capability or knowledge in identifying bargain buys ahead of its competitor companies. To achieve this, it must have access to better information, which it can tap into quicker, and/or have superior analytical tools. Nahara Co should assess whether or not it does possess such capabilities, otherwise its claim is not valid;

(ii) Ensure that it has quick access to the necessary funds to pursue an undervalued acquisition. Even if Nahara Co possesses superior knowledge, it is unlikely that this will last for a long time before its competitors find out; therefore it needs to have the funds ready, to move quickly. Given that it has access to sovereign funds from a wealthy source, access to funds is probably not a problem;

(iii) Set a maximum ceiling for the price it is willing to pay and should not go over this amount, or the potential value created will be reduced.

If, in its assessment, Nahara Co is able to show that it meets all the above conditions, then the strategy of identifying and pursuing undervalued companies may be valid.

(b) In a similar manner to the Competition and Markets Authority in the UK, the European Union (EU) will assess significant mergers and acquisitions' (M&As) impact on competition within a country's market. It will, for example, use tests such as worldwide turnover and European turnover of the group after the M&A. It may block the M&A, if it feels that the M&A will give the company monopolistic powers or enable it to carve out a dominant position in the market so as to negatively affect consumer choice and prices.

Sometimes the EU may ask for the company to sell some of its assets to reduce its dominant position rather than not allow an M&A to proceed. It would appear that this may be the case behind the EU's concern and the reason for its suggested action.

(c) Report to the Board of Directors, Avem Co

Proposed acquisition of Fugae Co

This report evaluates whether or not it is beneficial for Avem Co to acquire Fugae Co. Initially the value of the two companies is determined separately and as a combined entity, to assess the additional value created from bringing the two companies together. Following this, the report considers how much Nahara Co and Avem Co will gain from the value created. The assumptions made to arrive at the additional value are also considered. The report concludes by considering whether or not the acquisition will be beneficial to Avem Co and to Nahara Co.

Appendix 1 shows that the additional value created from combining the two companies is approximately $451·5 million, of which $276·8 million will go to Nahara Co, as the owner of Fugae Co. This represents a premium of about 30% which is the minimum acceptable to Nahara Co. The balance of the additional value will go to Avem Co which is about $174·7 million, representing an increase in value of 1·46% [$174·7m/$12,000m].

Appendix 2 shows that accepting the project would increase Fugae Co's value as the expected net present value is positive. After taking into account Lumi Co's offer, the expected net present value is higher. Therefore, it would be beneficial for Fugae Co to take on the project and accept Lumi Co's offer, if the tourism industry does not grow as expected, as this will increase Fugae Co's value.

Assumptions

It is assumed that all the figures relating to synergy benefits, betas, growth rates, multipliers, risk adjusted cost of capital and the probabilities are accurate. There is considerable uncertainty surrounding the accuracy of these, and in addition to the probability analysis conducted in appendix 2 and the assessments of value conducted in appendix 1, a sensitivity analysis is probably needed to assess the impact of these uncertainties.

It is assumed that the rb model provides a reasonably good estimate of the growth rate, and that perpetuity is not an unreasonable assumption when assessing the value of Fugae Co.

It is assumed that the capital structure would not change substantially when the new project is taken on. Since the project is significantly smaller than the value of Fugae Co itself, this is not an unreasonable assumption.

When assessing the value of the project, the outcomes are given as occurring with discrete probabilities and the resulting cash flows from the outcomes are given with certainty. There may be more outcomes in practice than the ones given and financial impact of the outcomes may not be known with such certainty. The Black-Scholes Option Pricing model may provide an alternative and more accurate way of assessing the value of the project.

It is assumed that Fugae Co can rely on Lumi Co paying the $50m at the beginning of year two with certainty. Fugae Co may want to assess the reliability of Lumi Co's offer and whether formal contracts should be drawn up between the two companies. Furthermore, Lumi Co may be reluctant to pay the full amount of money once Fugae Co becomes a part of Avem Co.

Concluding comments
Although Nahara Co would gain more than Avem Co from the acquisition both in percentage terms and in monetary terms, both companies benefit from the acquisition. If Fugae Co were to take on the project, although it is value-neutral to the acquisition, Nahara Co could ask for an additional 30% of $12·3 million value to be transferred to it, which is about $3·7 million. Hence the return to Avem Co would reduce by a small amount, but not significantly.

As long as all the parties are satisfied that the value is reasonable despite the assumptions highlighted above, it would appear that the acquisition should proceed.

Report compiled by:

Date:

Appendices

Appendix 1: Additional value created from combining Avem Co and Fugae Co

Avem Co, current value = $7·5/share x 1,600 million shares = $12,000m
Avem Co, free cash flow to equity = $12,000 million/7·2 = $1,666·7m

The growth rate is calculated on the basis of the rb model.
Fugae Co, estimate of growth rate = 0·227 x 0·11 = 0·025 = 2·5%
Fugae Co, current value estimate = $76·5 million x 1·025/(0·11 – 0·025) = $922·5m

Combined company, estimated additional value created =
([$1,666·7m + $76·5m + $40m] x 7·5) – ($12,000m + $922·5m) = $451·5m

Gain to Nahara for selling Fugae Co, 30% x $922·5m = $276·8m

Avem Co will gain $174·7 million of the additional value created, $451·5m – $276·8m = $174·7m

Appendix 2: Value of project to Fugae Co

Appendix 2.1
Estimate of risk-adjusted cost of capital to be used to discount the project's cash flows

The project value is calculated based on its cash flows which are discounted at the project's risk adjusted cost of capital, to reflect the business risk of the project.

Reka Co's asset beta
Reka Co equity value = $4·50 x 80 million shares = $360m
Reka Co debt value = 1·05 x $340 million = $357m
Asset beta = 1·6 x $360m/($360m + $357m x 0·8) = 0·89

Project's asset beta (PAB)
0·89 = PAB x 0·15 + 0·80 x 0·85
PAB = 1·4

Fugae Co
MVe = $922·5m

MVd
Cost of debt = Risk free rate of return plus the credit spread
= 4% + 0·80% = 4·80%

Current value of a $100 bond: $5·4 x 1·048^{–1}$ + $5·4 x 1·048^{–2}$ + $5·4 x 1·048^{–3}$ + $105·4 x 1·048^{–4}$ = $102·14 per $100
MVd = 1·0214 x $380m = $388·1m

Project's risk adjusted equity beta
1·4 x ($922·5m + $388·1m x 0·8)/$922·5m = 1·87

Project's risk adjusted cost of equity
4% + 1·87 x 6% = 15·2%

Project's risk adjusted cost of capital
(15·2% x $922·5m + 4·8% x 0·8 x $388·1m)/($922·5m + $388·1m) = 11·84%, say 12%

Appendix 2.2

Estimate of expected value of the project without the offer from Lumi Co

(All amounts in $, 000s)

Year	1	2	3	4
Cash flows	3,277·6	16,134·3	36,504·7	35,683·6
Discount factor for 12%	0·893	0·797	0·712	0·636
Present values	2,926·9	12,859·0	25,991·3	22,694·8

Probabilities are assigned to possible outcomes based on whether or not the tourism market will grow. The expected net present value (PV) is computed on this basis.

PV year 1: $2,926,900

50% of PV years 1 to 4: $32,236,000

PV years 2 to 4: $61,545,100
40% PV years 2 to 4: $24,618,040

Expected present value of cash flows = [0·75 x (2,926,900 + (0·8 x 61,545,100 + 0·2 x 24,618,040))] + [0·25 x 32,236,000]
= [0·75 x (2,926,900 + 54,159,688)] + [0·25 x 32,236,000] = 42,814,941 + 8,059,000 = $50,873,941

Expected NPV of project = $50,873,941 – $42,000,000 = $8,873,941

Estimate of expected value of the project with the offer from Lumi Co

PV of $50m = $50,000,000 x 0·893 = $44,650,000

If the tourism industry does not grow as expected in the first year, then it is more beneficial for Fugae Co to exercise the offer made by Lumi Co, given that Lumi Co's offer of $44·65 million (PV of $50 million) is greater than the PV of the years two to four cash flows ($30·8 million approximately) for that outcome. This figure is then incorporated into the expected net present value calculations.

50% of year 1 PV: $1,463,450

Expected present value of project =
[0·75 x (2,926,900 + 54,159,688)] + [0·25 x (1,463,450 + 44,650,000)] = 42,814,941 + 11,528,363 = $54,343,304

Expected NPV of project = $54,343,304 – $42,000,000 = $12,343,304

(Note: Credit will be given for alternative, relevant approaches to the calculations, comments and suggestions/recommendations)

2 (a) Using traded options

Need to hedge against a rise in interest rates, therefore buy put options.
Keshi Co needs 42 March put option contracts ($18,000,000/$1,000,000 x 7 months/3 months).

Expected futures price on 1 February if interest rates increase by 0·5% =
100 – (3·8 + 0·5) – 0·22 = 95·48

Expected futures price on 1 February if interest rates decrease by 0·5% =
100 – (3·8 – 0·5) – 0·22 = 96·48

If interest rates increase by 0·5% to 4·3%		
Exercise price	95·50	96·00
Futures price	95·48	95·48
Exercise?	Yes	Yes
Gain in basis points	2	52
Underlying cost of borrowing		
4·7% x 7/12 x $18,000,000	$493,500	$493,500
Gain on options		
0·0002 x $1,000,000 x 3/12 x 42	$2,100	
0·0052 x $1,000,000 x 3/12 x 42		$54,600
Premium		
0·00662 x $1,000,000 x 3/12 x 42	$69,510	
0·00902 x $1,000,000 x 3/12 x 42		$94,710
Net cost	$560,910	$533,610
Effective interest rate	5·34%	5·08%

If interest rates decrease by 0·5% to 3·3%

Exercise price	95·50	96·00
Futures price	96·48	96·48
Exercise?	No	No
Gain in basis points	0	0
Underlying cost of borrowing		
3·7% x 7/12 x $18,000,000	$388,500	$388,500
Gain on options	$0	$0
Premium	$69,510	$94,710
Net cost	$458,010	$483,210
Effective interest rate	4·36%	4·60%

Using swaps

	Keshi Co	Rozu Bank offer	Basis differential
Fixed rate	5·5%	4·6%	0·9%
Floating rate	LIBOR + 0·4%	LIBOR + 0·3%	0·1%

Prior to the swap, Keshi will borrow at LIBOR + 0·4% and swaps this rate to a fixed rate. Total possible benefit is 0·8% before Rozu Bank's charges.

Keshi Co borrows at	LIBOR + 0·4%
From swap Keshi Co receives	LIBOR
Keshi Co gets 70% of the benefit	
Advantage (70% x 0·8 – 0·10)	0·46%
Keshi Co's effective borrowing rate (after swap)	5·04%

Alternatively (Swap)

From swap Keshi Co receives	LIBOR
Keshi Co pays	4·54%
Effective borrowing rate (as above)	4·54% + 0·4% + 0·10% = 5·04%

Discussion and recommendation

Under each choice the interest rate cost to Keshi Co will be as follows:

	Doing nothing	95·50 option	96·00 option	Swap
If rates increase by 0·5%	4·7% floating; 5·5% fixed	5·34%	5·08%	5·04%
If rates decrease by 0·5%	3·7% floating; 5·5% fixed	4·36%	4·60%	5·04%

Borrowing at the floating rate and undertaking a swap effectively fixes the rate of interest at 5·04% for the loan, which is significantly lower than the market fixed rate of 5·5%.

On the other hand, doing nothing and borrowing at the floating rate minimises the interest rate at 4·7%, against the next best choice which is the swap at 5·04% if interest rates increase by 0·5%. And should interest rates decrease by 0·5%, then doing nothing and borrowing at a floating rate of 3·7% minimises cost, compared to the next best choice which is the 95·50 option.

On the face of it, doing nothing and borrowing at a floating rate seems to be the better choice if interest rates increase or decrease by a small amount, but if interest rates increase substantially then this choice will no longer result in the lowest cost.

The swap minimises the variability of the borrowing rates, while doing nothing and borrowing at a floating rate maximises the variability. If Keshi Co wants to eliminate the risk of interest rate fluctuations completely, then it should borrow at the floating rate and swap it into a fixed rate.

(b) Free cash flows and therefore shareholder value are increased when corporate costs are reduced and/or income increased. Therefore, consideration should be given to how the centralised treasury department may reduce costs and increase income.

The centralised treasury department should be able to evaluate the financing requirements of Keshi Co's group as a whole and it may be able to negotiate better rates when borrowing in bulk. The department could operate as an internal bank and undertake matching of funds. Therefore it could transfer funds from subsidiaries which have spare cash resources to ones which need them, and thus avoid going into the costly external market to raise funds. The department may be able to undertake multilateral internal netting and thereby reduce costs related to hedging activity. Experts and resources within one location could reduce duplication costs.

The concentration of experts and resources within one central department may result in a more effective decision-making environment and higher quality risk monitoring and control. Further, having access to the Keshi Co group's entire cash funds may give the company access to larger and more diverse investment markets. These factors could result in increasing the company's cash inflows, as long as the benefits from such activity outweigh the costs.

Decentralising Keshi Co's treasury function to its subsidiary companies may be beneficial in several ways. Each subsidiary company may be better placed to take local regulations, custom and practice into consideration. An example of custom and practice is the case of Suisen Co's need to use Salam contracts instead of conventional derivative products which the centralised treasury department may use as a matter of course.

Giving subsidiary companies more autonomy on how they undertake their own fund management may result in increased motivation and effort from the subsidiary's senior management and thereby increase future income. Subsidiary companies which have access to their own funds may be able to respond to opportunities quicker and establish competitive advantage more effectively.

(c) Islamic principles stipulate the need to avoid uncertainty and speculation. In the case of Salam contracts, payment for the commodity is made at the start of the contract. The buyer and seller of the commodity know the price, the quality and the quantity of the commodity and the date of future delivery with certainty. Therefore, uncertainty and speculation are avoided.

On the other hand, futures contracts are marked-to-market daily and this could lead to uncertainty in the amounts received and paid everyday. Furthermore, standardised futures contracts have fixed expiry dates and pre-determined contract sizes. This may mean that the underlying position is not hedged or covered completely, leading to limited speculative positions even where the futures contracts are used entirely for hedging purposes. Finally, only a few commodity futures contracts are offered to cover a range of different quality grades for a commodity, and therefore price movement of the futures market may not be completely in line with the price movement in the underlying asset.

(Note: Credit will be given for alternative, relevant discussion for parts (b) and (c))

3 (a) A free trade area like the European Union (EU) aims to remove barriers to trade and allow freedom of movement of production resources such as capital and labour. The EU also has an overarching common legal structure across all member countries and tries to limit any discriminatory practice against companies operating in these countries. Furthermore, the EU erects common external barriers to trade against countries which are not member states.

Riviere Co may benefit from operating within the EU in a number of ways as it currently trades within it. It should find that it is able to compete on equal terms with rival companies within the EU. Companies outside the EU may find it difficult to enter the EU markets due to barriers to trade. A common legal structure should ensure that the standards of food quality and packaging apply equally across all the member countries. Due diligence of logistic networks used to transport the food may be easier to undertake because of common compliance requirements. Having access to capital and labour within the EU may make it easier for the company to set up branches inside the EU, if it wants to. The company may also be able to access any grants which are available to companies based within the EU.

(b) Project Drugi

Internal rate of return (IRR)
10% NPV: €2,293,000 approximately

Year	Current	1	2	3	4	5
Cash flows (€000s)	(11,840)	1,230	1,680	4,350	10,240	2,200
Try 20%		0·833	0·694	0·579	0·482	0·402
	(11,840)	1,025	1,166	2,519	4,936	884

NPV = €(1,310,000)

IRR = 10% + 2,293/(2,293 + 1,310) x 10% approximately = 16·4%

Modified internal rate of return (MIRR)
Total PVs years 1 to 5 at 10% discount rate = €11,840,000 + €2,293,000 = €14,133,000

MIRR (using formula) = $[(14{,}133/11{,}840)^{1/5} \times 1{\cdot}10] - 1 = 14\%$

Alternatively:

Year	Cash flows (€000s)	Multiplier	Re-invested amount (€000s)
1	1,230	$1{\cdot}1^4$	1,801
2	1,680	$1{\cdot}1^3$	2,236
3	4,350	$1{\cdot}1^2$	5,264
4	10,240	$1{\cdot}1^1$	11,264
5	2,200	1	2,200

Total re-invested amount approx. = €22,765,000
MIRR = $(€22{,}765{,}000/€11{,}840{,}000)^{1/5} - 1 = 14\%$

Value at risk (VAR)
Based on a single tail test:

A 95% confidence level requires the annual present value VAR to be within approximately 1·645 standard deviations from the mean.

A 90% confidence level requires annual present value VAR to be within approximately 1·282 standard deviations from the mean.

(**Note:** *An approximation of standard deviations to two decimal places is acceptable*)

95%, five-year present value VAR = \$400,000 x 1·645 x $5^{0 \cdot 5}$ = approx. €1,471,000
90%, five-year present value VAR = \$400,000 x 1·282 x $5^{0 \cdot 5}$ = approx. €1,147,000

	Privi	**Drugi**
Net present value (10%)	€2,054,000	€2,293,000
Internal rate of return	17·6%	16·4%
Modified internal rate of return	13·4%	14·0%
VAR (over the project's life)		
95% confidence level	€1,103,500	€1,471,000
90% confidence level	€860,000	€1,147,000

The net present value and the modified internal rate of return both indicate that project Drugi would create more value for Riviere Co. However, the internal rate of return (IRR) for project Privi is higher. Where projects are mutually exclusive, the IRR can give an incorrect answer. This is because the IRR assumes that returns are re-invested at the internal rate of return, whereas net present value and the modified IRR assume that they are re-invested at the cost of capital (discount rate) which in this case is 10%. The cost of capital is a more realistic assumption as this is the minimum return required by investors in a company. Furthermore, the manner in which the cash flows occur will have a bearing on the IRR calculated. For example, with project Drugi, a high proportion of the cash flows occur in year four and these will be discounted by using the higher IRR compared to the cost of capital, thus reducing the value of the project faster. The IRR can give the incorrect answer in these circumstances. Therefore, based purely on cash flows, project Drugi should be accepted due to the higher net present value and modified IRR, as they give the theoretically correct answer of the value created.

The VAR provides an indication of the potential riskiness of a project. For example, if Riviere Co invests in project Drugi then it can be 95% confident that the present value will not fall by more than €1,471,000 over its life. Hence the project will still produce a positive net present value. However, there is a 5% chance that the loss could be greater than €1,471,000. With project Privi, the potential loss in value is smaller and therefore it is less risky. It should be noted that the VAR calculations indicate that the investments involve different risk. However, the cash flows are discounted at the same rate, which they should not be, since the risk differs between them.

Notwithstanding that, when risk is also taken into account, the choice between the projects is not clear cut and depends on Riviere Co's attitude to risk and return. Project Drugi gives the higher potential net present value but is riskier, whereas project Privi is less risky but gives a smaller net present value. This is before taking into account additional uncertainties such as trading in an area in which Riviere Co is not familiar. It is therefore recommended that Riviere Co should only proceed with project Drugi if it is willing to accept the higher risk and uncertainty.

(c) Possible legal risks

There are a number of possible legal risks which Riviere Co may face, for example:

– The countries where the product is sold may have different legal regulations on food preparation, quality and packaging. The company needs to ensure that the production processes and the transportation of the frozen foods comply with these regulations. It also needs to ensure that the promotional material on the packaging complies with regulations in relation to what is acceptable in each country.

– The legal regulations may be more lax in countries outside the EU but Riviere Co needs to be aware that complying only with the minimum standards may impact its image negatively overall, even if they are acceptable in the countries concerned.

– There may be import quotas in the countries concerned or the governments may give favourable terms and conditions to local companies, which may make it difficult for Riviere Co to compete.

– The legal system in some countries may not recognise the trademarks or production patents which the company holds on its packaging and production processes. This may enable competitors to copy the food and the packaging.

– Different countries may have different regulations regarding product liability from poorly prepared and/or stored food which cause harm to consumers. For example, Riviere Co may use other companies to transport its food and different supermarkets may sell its food. It needs to be aware of the potential legal claims on it and its supplier should the food prove harmful to the customers.

Possible mitigation strategies

– Riviere Co needs to undertake sufficient research of the countries' current laws and regulations to ensure that it complies with the standards required. It may even want to ensure that it exceeds the required standards to ensure that it maintains its reputation.

– Riviere Co needs to ensure that it also keeps abreast of potential changes in the law. It may also want to ensure that it complies with best practice, even if it is not the law yet. Often current best practices become enshrined in future legislation.

– Riviere Co needs to investigate the extent to which it may face difficulty in overcoming quota restrictions, less favourable trading conditions and lack of trademark and patent protection. If necessary, these should be factored into the financial analysis. It could be that Riviere Co has already taken these into account.

– Strict contracts need to be set up between Riviere Co and any agents it uses to transport and sell the food. These could be followed up by regular checks to ensure that the standards required are maintained.

– All the above will add extra costs and if these have not been included in the financial analysis, they need to be. These extra costs may mean that the project is no longer viable.

(Note: Credit will be given for alternative, relevant discussion for parts (a) and (c))

4 (a) Advantages of EVA™

The cost of capital indicates the minimum value which is required by the investors of a company and therefore any positive economic profit greater than the cost of capital times the capital employed should result in an increase in value for the investors. If the debt holders are paid a fixed return, then all the additional value created will go to the shareholders. EVA™ focuses on creating shareholder value.

Capital is needed for investment purposes to create value and EVA™ recognises this when it takes into account the capital employed.

EVA™ captures performance into a single figure, which if positive should increase shareholder value. Ratios on the other hand may require various different targets to be set.

EVA™ is based on the residual income value principle and therefore it is relatively easy to understand. An EVA™ trend would give an indication of how the company is creating value over a number of years.

Drawbacks of EVA™

EVA™ is an annual measure and therefore it is relatively easy to manipulate. Short-term projects with early redemption but low yields may be chosen to the detriment of longer term, high yield projects which may not show immediate high returns. Focusing on annual EVA™ figures may make the company's managers adopt a short-term attitude and this may be to the detriment of the company's long-term success. Paying attention to EVA™ trends instead may reduce or eliminate this drawback.

Furthermore, EVA™ is an absolute measure, making comparison between companies in different industrial sectors more difficult.

(b) EVA™ calculation: Kamala Co

	30 November 2013 $m	30 November 2014 $m
Operating profit	819	1,098
Add: Depreciation	826	1,150
Less: Economic depreciation	(990)	(1,380)
Add: Non-cash expenses	150	170
Taxation excluding finance costs: 2013: 25% x $819m and 2014 25% x $1,098m	(205)	(275)
Economic profit	600	763
Capital employed: 2013: $1,484m + $2,226m; 2014: $2,184m + $2,577m + $616m	3,710	5,377
10% x capital employed	371	538
EVA™	229	225

(c) Additional ratio trends: Kamala Co

	2012	2013	2014	
Return on capital employed	19·2%	17·2%	16·7%	Operating profit/cap employed
Asset turnover	1·01	0·85	0·79	Sales revenue/cap employed
Current ratio	0·80	0·65	0·63	Current assets/current liabilities
Current ratio without bank overdraft (o/d)	0·80	0·92	0·93	(Current assets – bank o/d)/current liabilities
Gearing with bank o/d	40%	52%	60%	(NCL + bank o/d)/(equity + NCL + bank o/d)
Kamala Co PE ratio	11·0:1	12·3:1	13·0:1	Share price/earnings per share
Kamala Co dividend yield	2·7%	2·3%	1·9%	Dividend per share/share price

Activity trends

	Construction %	Hospitals and biomedical %
Profit margin		
2012	19·0%	19·0%
2013	18·5%	23·4%
2014	18·9%	25·8%
Average annual growth: Sales revenue	23·0%	8·2%
Average annual growth: Operating profit	22·7%	26·3%

Share price and indices analysis

	%
Kamala Co: Average annual share price growth	26·2
Market index: Average annual growth	20·1
Industry index: Average annual growth	30·1

Evaluation

A number of ratios and the EVA™ figures support the CEO's assertions. The EVA™ shows that positive value is created in both years and these lend support to the increase in the share price, and this increase in the share price (26·2% annually over the last two years) has grown more rapidly than the market index (20·1% annually over the same period). There is a steady growth in the sales revenue, profit margin, earnings per share and dividend cover, together with significant investment in non-current assets, which seems to indicate a successful company. If the bank overdraft is not included, the current ratio has been improving as well.

On the other hand, there are a number of indicators which suggest that perhaps Kamala Co's investment strategy may be flawed. Even though the economic profit between 2013 and 2014 has increased, the EVA™ in 2014 is less than the EVA™ in 2013. This is because there has been a substantial investment in non-current assets. This is also evidenced by a decline in the asset turnover ratio. Less sales revenue is being generated for every $ invested, leading to a decline in the return on capital employed, even though there has been a small increase in the profit margin in the three-year period.

It may be that Kamala Co is making substantial investment now for deferred benefits in the future, but the markets do not seem to be convinced. In 2012, Kamala Co's price to earnings (PE) ratio was greater than the industry sector's average PE ratio (11·0 against 9·2), but by 2014 it was lower (13·0 against 15·3). The share price growth has also been lower at 26·2% on average annually, compared to the industry's index growth of 30·1% on average annually. Furthermore, the fall in dividend yield between 2012 and 2014 from 2·7% to 1·9% may not be viewed positively by equity holders, possibly leading to a less robust share price growth.

It seems that Kamala Co is making substantial investments in its construction activity (23% annual sales revenue growth) but significantly less in the hospitals and biomedical activity (8·2% annual sales revenue growth). However, the profit margin of the hospitals and biomedical research activity has grown faster, from 19% to 25·8%, compared to the construction activity, which has remained static at between 18·5% and 19%. The markets seemed to have viewed this in a less positive light, especially since the CEO stated that the company wanted to make further investment in an acquisition in the construction activity in 2015.

The financing strategy of Kamala Co is also problematic. There seems to be a higher reliance on debt as a finance source, with no fresh issues of equity between 2012 and 2014. Book value gearing has increased from 40% to 54% from 2012 to 2014. If the overdraft is added to gearing, based on the assumption that it is being used as a long-term source of finance, then the increase is even more pronounced, increasing from 40% to 60% in the same period. It seems that Kamala Co is continuing to carry on with this policy in the future as well, because it wants to acquire a new company using mostly debt finance, by issuing a new bond, with only a small rights issue. This may lead to an increase in financial risk, putting further pressure on the share price.

In conclusion, Kamala Co has been pursuing growth through debt finance. This may increase its financial risk but it is difficult to say for sure. If the industry's norm is to have higher debt levels, then the financial strategy may be acceptable. However, the investment strategy looks to be flawed. The company seems to be pursuing growth in the lower profit margin activity area. This may have led to the share price not increasing at the same rate as the industry sector as a whole. The investment strategy warrants a re-assessment.

(Note: credit will be given for alternative relevant comments for parts (a) and (c))

					Marks
1	**(a)**	Risk diversification			2–3
		Purchasing undervalued companies			4–5
				Max	7
	(b)	1–2 marks per point		Max	4
	(c)	**(i)**	Avem Co, current value		1
			Avem Co, free cash flows to equity		1
			Fugae Co, estimate of growth rate		2
			Fugae Co, current value estimate		2
			Combined company, estimated additional value created		2
			Gain to Nahara Co when selling Fugae Co		1
			Gain to Avem Co		1
					10
		(ii)	Reka Co asset beta		2
			Project asset beta		1
			Fugae Co's market value of debt		2
			Project's risk adjusted equity beta		1
			Project's risk adjusted cost of equity		1
			Project's risk adjusted cost of capital		1
			Annual PVs of project		1
			Different outcomes PVs (year 1, years 2 to 4, 50% and 40%)		2
			Expected NPV of project before Lumi Co offer		3
			PV of Lumi Co offer		1
			Expected NPV of project with Lumi Co's offer		3
					18
		(iii)	Presentation of benefits to each group of equity holders		
			With and without the project		2–3
			Assumptions made		3–4
			Concluding comments		1–2
				Max	7
			Note: Maximum 6 marks if no concluding comments given		
		Professional marks			
		Report format			1
		Structure and presentation of the report			3
					4
				Total	50

				Marks
2	(a)	Buy put options and number of contracts		1
		Futures prices if interest rates increase or decrease		1
		Option contracts calculations: either exercise price		3
		Option contract calculations: second exercise price (or justification for calculations of just one exercise price)		2
		Swap: Keshi Co initially borrows at the floating rate and resulting advantage		2
		Swap impact		2
		Effective borrowing rate		1
		Discussion and recommendation		3–4
			Max	15
	(b)	Discussion of why a centralised treasury department may increase value		3–4
		Discussion of reasons for decentralisation		2–3
			Max	6
	(c)	1–2 marks per point	Max	4
			Total	**25**
3	(a)	Discussion of the EU as a free trade area		2–3
		Discussion of the possible benefits to Riviere Co		2–3
			Max	5
	(b)	Calculation of internal rate of return		2
		Calculation of modified internal rate of return		2
		Determining the two standard deviations (1·645 and 1·282)		1
		Calculations of the two value at risk figures		2
		Explanation of weakness of internal rate of return and why net present value and modified internal rate of return are better		2–3
		Explanation of value at risk figures and what they indicate		2–3
		Recommendation		1–2
			Max	13
	(c)	Discussion of possible legal risks		3–4
		Discussion of how these may be mitigated		3–4
			Max	7
			Total	**25**

				Marks
4	(a)	Advantages of EVA™ (1–2 marks per point)		3–4
		Drawbacks of EVA™ (1–2 marks per point)		2–3
			Max	6
	(b)	**For 2013 and 2014**		
		Add depreciation and deduct economic depreciation to operating profit		1
		Adding back non-cash expenses		1
		Calculation of economic profit after tax		1
		EVA™ calculations		2
				5
	(c)	Additional ratio trends (1 mark per key three-year ratio trend)		4–5
		Activity trends		2–3
		Analysis of the share price against market and industry indices		1–2
		Discussion of investment strategy including impact on share price, EVA™ and PE ratios		3–4
		Discussion of financing strategy		1–2
		Other discussion points		1–2
		Concluding comments		1–2
			Max	14
		Note: Maximum 13 marks if no conclusion provided		
			Total	**25**

Answers

1 (a) Benefits of own investment as opposed to licensing

Imoni Co may be able to benefit from setting up its own plant as opposed to licensing in a number of ways. Yilandwe wants to attract foreign investment and is willing to offer a number of financial concessions to foreign investors which may not be available to local companies. The company may be able to control the quality of the components more easily, and offer better and targeted training facilities if it has direct control of the labour resources. The company may also be able to maintain the confidentiality of its products, whereas assigning the assembly rights to another company may allow that company to imitate the products more easily. Investing internationally may provide opportunities for risk diversification, especially if Imoni Co's shareholders are not well-diversified internationally themselves. Finally, direct investment may provide Imoni Co with new opportunities in the future, such as follow-on options.

Drawbacks of own investment as opposed to licensing

Direct investment in a new plant will probably require higher, upfront costs from Imoni Co compared to licensing the assembly rights to a local manufacturer. It may be able to utilise these saved costs on other projects. Imoni Co will most likely be exposed to higher risks involved with international investment such as political risks, cultural risks and legal risks. With licensing these risks may be reduced somewhat. The licensee, because it would be a local company, may understand the operational systems of doing business in Yilandwe better. It will therefore be able to get off-the-ground quicker. Imoni Co, on the other hand, will need to become familiar with the local systems and culture, which may take time and make it less efficient initially. Similarly, investing directly in Yilandwe may mean that it costs Imoni Co more to train the staff and possibly require a steeper learning curve from them. However, the scenario does say that the country has a motivated and well-educated labour force and this may mitigate this issue somewhat.

(Note: Credit will be given for alternative, relevant suggestions)

(b) Report on the proposed assembly plant in Yilandwe

This report considers whether or not it would be beneficial for Imoni Co to set up a parts assembly plant in Yilandwe. It takes account of the financial projections, presented in detail in appendices 1 and 2, discusses the assumptions made in arriving at the projections and discusses other non-financial issues which should be considered. The report concludes by giving a reasoned recommendation on the acceptability of the project.

Assumptions made in producing the financial projections

It is assumed that all the estimates such as sales revenue, costs, royalties, initial investment costs, working capital, and costs of capital and inflation figures are accurate. There is considerable uncertainty surrounding the accuracy of these and a small change in them could change the forecasts of the project quite considerably. A number of projections using sensitivity and scenario analysis may aid in the decision making process.

It is assumed that no additional tax is payable in the USA for the profits made during the first two years of the project's life when the company will not pay tax in Yilandwe either. This is especially relevant to year 2 of the project.

No details are provided on whether or not the project ends after four years. This is an assumption which is made, but the project may last beyond four years and therefore may yield a positive net present value. Additionally, even if the project ceases after four years, no details are given about the sale of the land, buildings and machinery. The residual value of these non-current assets could have a considerable bearing on the outcome of the project.

It is assumed that the increase in the transfer price of the parts sent from the USA directly increases the contribution which Imoni Co earns from the transfer. This is probably not an unreasonable assumption. However, it is also assumed that the negotiations with Yilandwe's government will be successful with respect to increasing the transfer price and the royalty fee. Imoni Co needs to assess whether or not this assumption is realistic.

The basis for using a cost of capital of 12% is not clear and an explanation is not provided about whether or not this is an accurate or reasonable figure. The underpinning basis for how it is determined may need further investigation.

Although the scenario states that the project can start almost immediately, in reality this may not be possible and Imoni Co may need to factor in possible delays.

It is assumed that future exchange rates will reflect the differential in inflation rates between the respective countries. However, it is unlikely that the exchange rates will move fully in line with the inflation rate differentials.

Other risks and issues

Investing in Yilandwe may result in significant political risks. The scenario states that the current political party is not very popular in rural areas and that the population remains generally poor. Imoni Co needs to assess how likely it is that the government may change during the time it is operating in Yilandwe and the impact of the change. For example, a new government may renege on the current government's offers and/or bring in new restrictions. Imoni Co will need to decide what to do if this happens.

Imoni Co needs to assess the likelihood that it will be allowed to increase the transfer price of the parts and the royalty fee. Whilst it may be of the opinion that currently Yilandwe may be open to such suggestions, this may depend on the interest the government may get from other companies to invest in Yilandwe. It may consider that agreeing to such demands from Imoni Co may make it obligated to other companies as well.

The financial projections are prepared on the basis that positive cash flows from Yilandwe can be remitted back to the USA. Imoni Co needs to establish that this is indeed the case and that it is likely to continue in the future.

Imoni Co needs to be careful about its ethical stance and its values, and the impact on its reputation, given that a school is being closed in order to provide it with the production facilities needed. Whilst the government is funding some of the transport costs for the children, the disruption this will cause to the children and the fact that after six months the transport costs become the parents' responsibility, may have a large, negative impact on the company's image and may be contrary to the ethical values which the company holds. The possibility of alternative venues should be explored.

Imoni Co needs to take account of cultural risks associated with setting up a business in Yilandwe. The way of doing business in Yilandwe may be very different and the employees may need substantial training to adapt to Imoni Co's way of doing business. On the other hand, the fact that the population is well educated, motivated and keen may make this process easier to achieve.

Imoni Co also needs to consider fiscal and regulatory risks. The company will need to assess the likelihood of changes in tax rates, laws and regulations, and set up strategies to mitigate eventualities which can be predicted. In addition to these, Imoni Co should also consider and mitigate as far as possible, operational risks such as the quality of the components and maintenance of transport links.

Imoni Co should assess and value alternative real options which it may have. For example, it could consider whether licensing the production of the components to a local company may be more financially viable; it could consider alternative countries to Yilandwe, which may offer more benefits; it could consider whether the project can be abandoned if circumstances change against the company; entry into Yilandwe may provide Imoni Co with other business opportunities.

Recommendation

The result from the financial projections is that the project should be accepted because it results in a positive net present value. It is recommended that the financial projections should be considered in conjunction with the assumptions, the issues and risks, and the implications of these, before a final decision is made.

There is considerable scope for further investigation and analysis. It is recommended that sensitivity and scenario analysis be undertaken to take into consideration continuing the project beyond four years and so on. The value of any alternative real options should also be considered and incorporated into the decision.

Consideration must also be given to the issues, risks and factors beyond financial considerations, such as the impact on the ethical stance of the company and the impact on its image, if the school affected is closed to accommodate it.

Report compiled by:

Date:

Appendices

Appendix 1

(All amounts in YR, millions)

Year	0	1	2	3	4
Sales revenue (w2)		18,191	66,775	111,493	60,360
Parts costs (w2)		(5,188)	(19,060)	(31,832)	(17,225)
Variable costs (w2)		(2,921)	(10,720)	(17,901)	(9,693)
Fixed costs		(5,612)	(6,437)	(7,068)	(7,760)
Royalty fee (w3)		(4,324)	(4,813)	(5,130)	(5,468)
Tax allowable depreciation		(4,500)	(4,500)	(4,500)	(4,500)
Taxable profits/(loss)		(4,354)	21,245	45,062	15,714
Tax loss carried forward				(4,354)	
				40,708	
Taxation (40%)		0	0	(16,283)	(6,286)
Add back loss carried fwd				4,354	
Add back depreciation		4,500	4,500	4,500	4,500
Cash flows after tax		146	25,745	33,279	13,928
Working capital	(9,600)	(2,112)	(1,722)	(1,316)	14,750
Land, buildings and machinery	(39,000)				
Cash flows (YR, millions)	(48,600)	(1,966)	24,023	31,963	28,678

(All amounts in $, 000s)

Year	0	1	2	3	4
Exchange rate	101·4	120·1	133·7	142·5	151·9
Remittable flows	(479,290)	(16,370)	179,678	224,302	188,795
Contribution (parts sales) ($120 + inflation per unit)		18,540	61,108	95,723	48,622
Royalty (w3)		36,000	36,000	36,000	36,000
Tax on contribution and royalty (20%)		(10,908)	(19,422)	(26,345)	(16,924)
Cash flows	(479,290)	27,262	257,364	329,680	256,493
Discount factors (12%)	1	0·893	0·797	0·712	0·636
Present values	(479,290)	24,345	205,119	234,732	163,130

Net present value project before considering the impact of the lost contribution and redundancy is approximately $148·0 million.

Lost contribution and redundancy cost

The lost contribution and redundancy costs are small compared to the net present value and would therefore have a minimal impact of reducing the net present value by $0·1 million approximately.

(Note: Full credit will be given if the assumption is made that the amounts are in $ 000s instead of $.)

Appendix 2: Workings

W1: Unit prices and costs including inflation

Year	1	2	3	4
Selling price (€)	735	772	803	835
Parts ($)	288	297	306	315
Variable costs (YR)	19,471	22,333	24,522	26,925

W2: Sales revenue and costs

In YR millions

Year	1	2	3	4
Sales revenue	150 x 735 x 165 = 18,191	480 x 772 x 180·2 = 66,775	730 x 803 x 190·2 = 111,493	360 x 835 x 200·8 = 60,360
Parts costs	150 x 288 x 120·1 = 5,188	480 x 297 x 133·7 = 19,060	730 x 306 x 142·5 = 31,832	360 x 315 x 151·9 = 17,225
Variable costs	150 x 19,471 = 2,921	480 x 22,333 = 10,720	730 x 24,522 = 17,901	360 x 26,925 = 9,693

W3: Royalty fee

$20 million x 1·8 = $36 million
This is then converted into YR at the YR/$ rate for each year: 120·1, 133·7, 142·5 and 151·9 for years 1 to 4 respectively.

(Note: Credit will be given for alternative, relevant approaches to the calculations, and to the discussion of the assumptions, risks and issues)

2 (a) A dark pool network allows shares to be traded anonymously, away from public scrutiny. No information on the trade order is revealed prior to it taking place. The price and size of the order are only revealed once the trade has taken place. Two main reasons are given for dark pool networks: first they prevent the risk of other traders moving the share price up or down; and second they often result in reduced costs because trades normally take place at the mid-price between the bid and offer; and because broker-dealers try and use their own private pools, and thereby saving exchange fees.

Chawan Co's holding in Oden Co is 27 million shares out of a total of 600 million shares, or 4·5%. If Chawan Co sold such a large holding all at once, the price of Oden Co shares may fall temporarily and significantly, and Chawan Co may not receive the value based on the current price. By utilising a dark pool network, Chawan Co may be able to keep the price of the share largely intact, and possibly save transaction costs.

Although the criticism against dark pool systems is that they prevent market efficiency by not revealing bid-offer prices before the trade, proponents argue that in fact market efficiency is maintained because a large sale of shares will not move the price down artificially and temporarily.

(b) Ratio calculations

Focus on investor and profitability ratios

Oden Co	2012	2013	2014	2015
Operating profit/sales revenue		16·2%	15·2%	10·4%
Operating profit/capital employed		22·5%	20·4%	12·7%
Earnings per share		$0·27	$0·24	$0·12
Price to earnings ratio		9·3	10·0	18·3
Gearing ratio (debt/(debt + equity))		37·6%	36·9%	37·1%
Interest cover (operating profit/finance costs)		9·5	7·5	3·5
Dividend yield	7·1%	7·2%	8·3%	6·8%

Travel and leisure (T&L) sector	2012	2013	2014	2015
Price to earnings ratio	11·9	12·2	13·0	13·8
Dividend yield	6·6%	6·6%	6·7%	6·4%

Other calculations
Oden Co, sales revenue annual growth rate average between 2013 and 2015 = $(1,185/1,342)^{1/2}$ – 1 = –6·0%.
Between 2014 and 2015 = (1,185 – 1,335)/1,335 = –11·2%.

Oden Co, average financing cost
2013: 23/(365 + 88) = 5·1%
2014: 27/(368 + 90) = 5·9%
2015: 35/(360 + 98) = 7·6%

Share price changes	2012–2013	2013–2014	2014–2015
Oden Co	19·0%	–4·0%	–8·3%
T&L sector	15·8%	–2·3%	12·1%

Oden Co

Return to shareholders (RTS)	**2013**	**2014**	**2015**
Dividend yield	7·2%	8·3%	6·8%
Share price gain	19·0%	–4·0%	–8·3%
Total	26·2%	4·3%	–1·5%
Average: 9·7%			
Required return (based on capital asset pricing model (CAPM))	13·0%	13·6%	16·0%
Average: 14·2%			

T&L sector (RTS)	**2013**	**2014**	**2015**
Dividend yield	6·6%	6·7%	6·4%
Share price gain	15·8%	–2·3%	12·1%
Total	22·4%	4·4%	18·5%
Average: 15·1%			
Required return (based on CAPM)	12·4%	13·0%	13·6%
Average: 13·0%			

(Note: The averages for Oden Co, RTS and for the T&L sector, RTS are the simple averages of the three years: 2013 to 2015)

Discussion

The following discussion compares the performance of Oden Co over time, to the T&L sector and against expectations, in terms of it being a sound investment. It also considers the wider aspects which Chawan Co should take account of and the further information which the company should consider before coming to a final decision.

In terms of Oden Co's performance between 2013 and 2015, it is clear from the calculations above, that the company is experiencing considerable financial difficulties. Profit margins have fallen and so has the earnings per share (EPS). Whereas the amount of gearing appears fairly stable, the interest cover has deteriorated. The reason for this is that borrowing costs have increased from an average of 5·1% to an average of 7·6% over the three years. The share price has decreased over the three years as well and in the last year so has the dividend yield. This would indicate that the company is unable to maintain adequate returns for its investors (please also see below).

Although Oden Co has tried to maintain a dividend yield which is higher than the sector average, its price to earnings (PE) ratio has been lower than the sector average between 2013 and 2014. It does increase significantly in 2015, but this is because of the large fall in the EPS, rather than an increase in the share price. This could be an indication that there is less confidence in the future prospects of Oden Co, compared to the rest of the T&L sector. This is further corroborated by the higher dividend yield which may indicate that the company has fewer value-creating projects planned in the future. Finally, whereas the T&L sector's average share price seems to have recovered strongly in 2015, following a small fall in 2014, Oden Co's share price has not followed suit and the decline has gathered pace in 2015. It would seem that Oden Co is a poor performer within its sector.

This view is further strengthened by comparing the actual returns to the required returns based on the capital asset pricing model (CAPM). Both the company and the T&L sector produced returns exceeding the required return in 2013 and Oden Co experienced a similar decline to the sector in 2014. However, in 2015, the T&L sector appears to have recovered but Oden Co's performance has worsened. This has resulted in Oden Co's actual average returns being significantly below the required returns between 2012 and 2015.

Taking the above into account, the initial recommendation is for Chawan Co to dispose of its investment in Oden Co. However, there are three important caveats which should be considered before the final decision is made.

The first caveat is that Chawan Co should look at the balance of its portfolio of investments. A sale of $58 million worth of equity shares within a portfolio total of $360 million may cause the portfolio to become unbalanced, and for unsystematic risk to be introduced into the portfolio. Presumably, the purpose of maintaining a balanced portfolio is to virtually eliminate unsystematic risk by ensuring that it is well diversified. Chawan Co may want to re-invest the proceeds from the sale of Oden Co (if it decides to proceed with the disposal) in other equity shares within the same sector to ensure that the portfolio remains balanced and diversified.

The second caveat is that Chawan Co may want to look into the rumours of a takeover bid of Oden Co and assess how realistic it is that this will happen. If there is a realistic chance that such a bid may happen soon, Chawan Co may want to hold onto its investment in Oden Co for the present time. This is because takeover bids are made at a premium and the return to Chawan Co may increase if Oden Co is sold during the takeover.

The third caveat is that Chawan Co may want to consider Oden Co's future prospects. The calculations above are based on past performance between 2012 and 2015 and indicate an increasingly poor performance. However, the economy is beginning to recover, albeit slowly and erratically. Chawan Co may want to consider how well placed Oden Co is to take advantage of the improving conditions compared to other companies in the same industrial sector.

If Chawan Co decides that none of the caveats materially affect Oden Co's poor performance and position, then it should dispose of its investment in Oden Co.

3 **(a)** A management buy-out (MBO) involves the purchase of a business by the management team running that business. Hence, an MBO of Okazu Co would involve the takeover of that company from Bento Co by Okazu Co's current management team. However, a management buy-in (MBI) involves purchasing a business by a management team brought in from outside the business.

The benefits of a MBO relative to a MBI to Okazu Co are that the existing management is likely to have detailed knowledge of the business and its operations. Therefore they will not need to learn about the business and its operations in a way which a new external management team may need to. It is also possible that a MBO will cause less disruption and resistance from the employees when compared to a MBI. If Bento Co wants to continue doing business with the new company after it has been disposed of, it may find it easier to work with the management team which it is more familiar with. The internal management team may be more focused and have better knowledge of where costs can be reduced and sales revenue increased, in order to increase the overall value of the company.

The drawbacks of a MBO relative to a MBI to Okazu Co may be that the existing management may lack new ideas to rejuvenate the business. A new management team, through their skills and experience acquired elsewhere, may bring fresh ideas into the business. It may be that the external management team already has the requisite level of finance in place to move quickly and more decisively, whereas the existing management team may not have the financial arrangements in place yet. It is also possible that the management of Bento Co and Okazu Co have had disagreements in the past and the two teams may not be able to work together in the future if they need to. It may be that a MBI is the only way forward for Okazu Co to succeed in the future.

(b) Annuity (8%, 4 years) = 3·312
Annuity payable per year on loan = $30,000,000/3·312 = $9,057,971

Interest payable on convertible loan, per year = $20,000,000 x 6% = $1,200,000

Annual interest on 8% bond
(All amounts in $ 000s)

Year end	**1**	**2**	**3**	**4**
Opening loan balance	30,000	23,342	16,151	8,385
Interest at 8%	2,400	1,867	1,292	671
Annuity	(9,058)	(9,058)	(9,058)	(9,058)
Closing loan balance	23,342	16,151	8,385	(2)*

*The loan outstanding in year 4 should be zero. The small negative figure is due to rounding.

Estimate of profit and retained earnings after MBO
(All amounts in $ 000s)

Year end	**1**	**2**	**3**	**4**
Operating profit	13,542	15,032	16,686	18,521
Finance costs	3,600	3,067	2,492	1,871
Profit before tax	9,942	11,965	14,194	16,650
Taxation	1,988	2,393	2,839	3,330
Profit for the year	7,954	9,572	11,355	13,320
Dividends	1,989	2,393	2,839	3,330
Retained earnings	5,965	7,179	8,516	9,990

Estimate of gearing
(All amounts in $ 000s)

Year end	**1**	**2**	**3**	**4**
Book value of equity	15,965*	23,144	31,660	41,650
Book value of debt	43,342	36,151	28,385	20,000
Gearing	73%	61%	47%	32%
Covenant	75%	60%	50%	40%
Covenant breached?	No	Yes	No	No

*The book value of equity consists of the sum of the 5,000,000 equity shares which Dofu Co and Okazu Co's senior management will each invest in the new company (total 10,000,000), issued at their nominal value of $1 each, and the retained earnings from year 1. In subsequent years the book value of equity is increased by the retained earnings from that year.

The gearing covenant is forecast to be breached in the second year only, and by a marginal amount. It is forecast to be met in all the other years. It is unlikely that Dofu Co will be too concerned about the covenant breach.

(c) **Net asset valuation**

Based on the net asset valuation method, the value of the new company is approximately:
1·3 x $40,800,000 + $12,300,000 – $7,900,000 approx. = $57,440,000

Dividend valuation model

Year	Dividend ($000s)	DF (12%)	PV ($000s)
1	1,989	0·893	1,776
2	2,393	0·797	1,907
3	2,839	0·712	2,021
4	3,330	0·636	2,118
Total			7,822

Annual dividend growth rate, years 1 to 4 = $(3,330/1,989)^{1/3} - 1$ = 18·7%
Annual dividend growth rate after year 4 = 7·5% [40% x 18·7%]
Value of dividends after year 4 = ($3,330,000 x 1·075)/(0·12 – 0·075) x 0·636 = $50,594,000 approximately

Based on the dividend valuation model, the value of new company is approximately:
$7,822,000 + $50,594,000 = $58,416,000

The $60 million asked for by Bento Co is higher than the current value of the new company's net assets and the value of the company based on the present value of future dividends based on the dividend valuation model. Although the future potential of the company represented by the dividend valuation model, rather than the current value of the assets, is probably a better estimate of the potential of the company, the price of $60 million seems excessive.

Nevertheless, both the management team and Dofu Co are expected to receive substantial dividends during the first four years and Dofu Co's 8% bond loan will be repaid within four years.

Furthermore, the dividend valuation model can produce a large variation in results if the model's variables are changed by even a small amount. Therefore, the basis for estimating the variables should be examined carefully to judge their reasonableness, and sensitivity analysis applied to the model to demonstrate the impact of the changes in the variables. The value of the future potential of the new company should also be estimated using alternative valuation methods including free cash flows and price-earnings methods.

It is therefore recommended that the MBO should not be rejected at the outset but should be considered further. It is also recommended that the management team and Dofu Co try to negotiate the sale price with Bento Co.

(Note: Credit will be given for alternative, relevant discussion for parts (a) and (c))

4 (a) Borrowing period is 6 months (11 months – 5 months)

Current borrowing cost = $34,000,000 x 6 months/12 months x 4·3% = $731,000
Borrowing cost if interest rates increase by 80 basis points (0·8%) = $34,000,000 x 6/12 x 5·1% = $867,000
Additional cost = $136,000 [$34,000,000 x 6/12 x 0·8%]

Using futures to hedge

Need to hedge against a rise in interest rates, therefore go short in the futures market.

Borrowing period is 6 months
No. of contracts needed = $34,000,000/ $1,000,000 x 6 months/3 months = 68 contracts.

Basis
Current price (on 1 June 2015) – futures price = total basis
(100 – 3·6) – 95·84 = 0·56
Unexpired basis (at beginning of November) = 2/7 x 0·56 = 0·16

Assume that interest rates increase by 0·8% (80 basis points) to 4·4%

Expected futures price = 100 – 4·4 – 0·16 = 95·44
Gain on the futures market = (95·84 – 95·44) x $25 x 68 = $68,000

Net additional cost = ($136,000 – $68,000) $68,000

Using options on futures to hedge

Need to hedge against a rise in interest rates, therefore buy put options. As before, 68 put option contracts are needed ($34,000,000/$1,000,000 x 6 months/3 months).

Assume that interest rates increase by 0·8% (80 basis points) to 4·4%

Exercise price	95·50	96·00
Futures price	95·44	95·44
Exercise ?	Yes	Yes
Gain in basis points	6	56
Gain on options		
6 x $25 x 68	$10,200	
56 x $25 x 68		$95,200
Premium		
30·4 x $25 x 68	$51,680	
50·8 x $25 x 68		$86,360
Option benefit/(cost)	$(41,480)	$8,840
Net additional cost		
($136,000 + $41,480)	$177,480	
($ 136,000 – $8,840)		$127,160

Using a collar on options to hedge

Buy put options at 95·50 for 0·304 and sell call at 96·00 for 0·223
Net premium payable = 0·081

Assume that interest rates increase by 0·8% (80 basis points) to 4·4%

	Buy put	**Sell call**
Exercise price	95·50	96·00
Futures price	95·44	95·44
Exercise?*	Yes	No

(*The put option is exercised, since by exercising the option, the option holder has the right to sell the instrument at 95·50 instead of the market price of 95·44 and gain 6 basis points per contract. The call option is not exercised, since by not exercising the option, the option holder can buy the instrument at a lower market price of 95·44 instead of the higher option exercise price of 96·00)

Gain on options	
6 x $25 x 68	$10,200
Premium payable	
8·1 x $25 x 68	$13,770
Net cost of the collar	$3,570
Net additional cost	
($136,000 + $3,570)	$139,570

Based on the assumption that interest rates increase by 80 basis points in the next five months, the futures hedge would lower the additional cost by the greatest amount and is significantly better than either of the options hedge or the collar hedge. In addition to this, futures fix the amount which Daikon Co is likely to pay, assuming that there is no basis risk. The benefits accruing from the options are lower, with the 95·50 option and the collar option actually increasing the overall cost. In each case, this is due to the high premium costs. However, if interest rates do not increase and actually reduce, then the options (and to some extent the collar) provide more flexibility because they do not have to be exercised when interest rates move in the company's favour. But the movement will need to be significant before the cost of the premium is covered.

On that basis, on balance, it is recommended that hedging using futures is the best choice as they will probably provide the most benefit to Daikon Co.

However, it is recommended that the points made in part (b) are also considered before a final conclusion is made.

(b) Mark-to-market: Daily settlements

2 June: 8 basis points (95·76 – 95·84) x $25 x 50 contracts = $10,000 loss

3 June: 10 basis points (95·66 – 95·76) x $25 x 50 contracts + 5 basis points (95·61 – 95·66) x $25 x 30 contracts = $16,250 loss

[Alternatively: 15 basis points (95·61 – 95·76) x $25 x 30 contracts + 10 basis points (95·66 – 95·76) x $25 x 20 contracts = $16,250 loss]

4 June: 8 basis points (95·74 – 95·66) x $25 x 20 contracts = $4,000 profit

Both mark-to-market and margins are used by markets to reduce (eliminate) the risk of non-payment by purchasers of the derivative products if prices move against them.

Mark-to-market closes all the open deals at the end of each day at that day's settlement price, and opens them again at the start of the following day. The notional profit or loss on the deals is then calculated and the margin account is adjusted accordingly on a daily basis. The impact on Daikon Co is that if losses are made, then the company may have to deposit extra funds with its broker if the margin account falls below the maintenance margin level. This may affect the company's ability to plan adequately and ensure it has enough funds for other activities. On the other hand, extra cash accruing from the notional profits can be withdrawn from the broker account if needed.

Each time a market-traded derivative product is opened, the purchaser needs to deposit a margin (initial margin) with the broker, which consists of funds to be kept with the broker while the position is open. As stated above, this amount may change daily and would affect Daikon Co's ability to plan for its cash requirements, but also open positions require that funds are tied up to support these positions and cannot be used for other purposes by the company.

The value of an option prior to expiry consists of time value, and may also consist of intrinsic value if the option is in-the-money. If an option is exercised prior to expiry, Daikon Co will only receive the intrinsic value attached to the option but not the time value. If the option is sold instead, whether it is in-the-money or out-of-money, Daikon Co will receive a higher value for it due to the time value. Unless options have other features, like dividends, attached to them, which are not reflected in the option value, they would not normally be exercised prior to expiry.

					Marks
1	**(a)**	Benefits			2–3
		Drawbacks			2–3
				Max	5
	(b)	**(i)**	Sales revenue		3
			Parts costs		3
			Variable costs		2
			Fixed costs		1
			Royalty fee		1
			Tax payable in Yilandwe		3
			Working capital		2
			Remittable flows ($)		1
			Contribution from parts ($)		2
			Tax on parts' contribution and royalty		1
			Impact of lost contribution and redundancy		1
			NPV of project		1
					21
		(ii)	Up to 2 marks per assumption discussed	Max	9
			2–3 marks per issue/risk discussed	Max	11
				Max	17
			Note: For (b)(ii), where points can be made either as assumptions or as issues, marks will be allocated to either area as relevant.		
		(iii)	Reasoned recommendation		3
		Professional marks for part (b)			
		Report format			1
		Structure and presentation of the report			3
					4
				Total	**50**
2	**(a)**	Explanation of a dark pool network			3–4
		Explanation of why Chawan Co may want to use one			1–2
				Max	5
	(b)	Profitability ratios			1–2
		Investor ratios			3–4
		Other ratios			1–2
		Trends and other calculations			3–4
				Max	10
		Note: Maximum 7 marks if only ratio calculations provided			
		Discussion of company performance over time			2–3
		Discussion of company performance against competitors			2–3
		Discussion of actual returns against expected returns			1–2
		Discussion of need to maintain portfolio and alternative investments			1–2
		Discussion of future trends and expectations			1–2
		Discussion of takeover rumour and action as a result			1–2
		Other relevant discussion/commentary			1–2
				Max	10
				Total	**25**

				Marks
3	**(a)**	Distinguishing between an MBI and MBO		1–2
		Discussing the relative benefits and drawbacks between the two		3–4
			Max	**5**
	(b)	Amount of annual annuity of 8% bond		1
		Annual split between interest and capital repayment of 8% bond		2
		Operating profit for first four years		1
		Finance costs		2
		Tax payable for the first four years		1
		Dividend payable for the first four years		1
		Book values of debt and of equity in years 1 to 4		2
		Gearing levels and concluding comment		2
				12
	(c)	Company value based on net asset valuation method		1
		Company value based on the dividend valuation method		3
		Discussion (1 to 2 marks per point)		4
				8
			Total	**25**
4	**(a)**	Additional interest cost		1
		Recommendation to go short if futures are used and purchase puts if options are used		1
		Calculation of number of contracts and remaining basis		2
		Futures contracts calculation		1
		Options contracts calculations		4
		Collar on options calculations		4
		Supporting comments and conclusion		2–3
			Max	**15**
	(b)	Mark-to-market calculations (1 mark each for 2, 3 and 4 June)		3
		Impact of the daily mark-to-market		2–3
		Impact of the margin requirements		2–3
		Impact of the options sold instead of being exercised		2–3
			Max	**10**
		Note: Maximum of 7 marks if no mark-to-market calculations provided		
			Total	**25**

Answers

1 (a) Both forms of unbundling involve disposing the non-core parts of the company.

The divestment through a sell-off normally involves selling part of a company as an entity or as separate assets to a third party for an agreed amount of funds or value. This value may comprise of cash and non-cash based assets. The company can then utilise the funds gained in alternative, value-enhancing activities.

The management buy-in is a particular type of sell-off which involves selling a division or part of a company to an external management team, who will take up the running of the new business and have an equity stake in the business. A management buy-in is normally undertaken when it is thought that the division or part of the company can probably be run better by a different management team compared to the current one.

(b) Report to the board of directors (BoD), CIGNO CO

This report assesses the potential value of acquiring Anatra Co for the equity holders of Cigno Co, both with and without considering the benefits of the reduction in taxation and in employee costs. The possible issues raised by reduction in taxation and in employee costs are discussed in more detail below. The assessment also discusses the estimates made and the methods used.

Assessment of value created

Cigno Co estimates that the premium payable to acquire Anatra Co largely accounts for the benefits created from the acquisition and the divestment, before considering the benefits from the tax and employee costs' saving. As a result, before these savings are considered, the estimated benefit to Cigno Co's shareholders of $128 million (see appendix 3) is marginal. Given that there are numerous estimations made and the methods used make various assumptions, as discussed below, this benefit could be smaller or larger. It would appear that without considering the additional benefits of cost and tax reductions, the acquisition is probably too risky and would probably be of limited value to Cigno Co's shareholders.

If the benefits of the taxation and employee costs saved are taken into account, the value created for the shareholders is $5,609 million (see appendix 4), and therefore significant. This would make the acquisition much more financially beneficial. It should be noted that no details are provided on the additional pre-acquisition and post-acquisition costs or on any synergy benefits that Cigno Co may derive in addition to the cost savings discussed. These should be determined and incorporated into the calculations.

Basing corporate value on the price-earnings (PE) method for the sell-off, and on the free cash flow valuation method for the absorbed business, is theoretically sound. The PE method estimates the value of the company based on its earnings and on competitor performance. With the free cash flow method, the cost of capital takes account of the risk the investors want to be compensated for and the non-committed cash flows are the funds which the business can afford to return to the investors, as long as they are estimated accurately.

However, in practice, the input factors used to calculate the organisation's value may not be accurate or it may be difficult to assess their accuracy. For example, for the free cash flow method, it is assumed that the sales growth rate, operating profit margin, the taxation rate and incremental capital investment can be determined accurately and remain constant. It is assumed that the cost of capital will remain unchanged and it is assumed that the asset beta, the cost of equity and cost of debt can be determined accurately. It is also assumed that the length of the period of growth is accurate and that the company operates in perpetuity thereafter. With the PE model, the basis for using the average competitor figures needs to be assessed, for example, have outliers been ignored; and the basis for the company's higher PE ratio needs to be justified as well. The uncertainties surrounding these estimates would suggest that the value is indicative, rather than definitive, and it would be more prudent to undertake sensitivity analysis and obtain a range of values.

Key factors to consider in relation to the redundancies and potential tax savings

It is suggested that the BoD should consider the impact of the cost-savings from redundancies and from the tax payable in relation to corporate reputation and ethical considerations.

At present, Cigno Co enjoys a good reputation and it is suggested that this may be because it has managed to avoid large scale redundancies. This reputation may now be under threat and its loss could affect Cigno Co negatively in terms of long-term loss in revenues, profits and value; and it may be difficult to measure the impact of this loss accurately.

Whilst minimising tax may be financially prudent, it may not be considered fair. For example, currently there is ongoing discussion and debate from a number of governments and other interested parties that companies should pay tax in the countries they operate and derive their profits, rather than where they are based. Whilst global political consensus in this area seems some way off, it is likely that the debate in this area will increase in the future. Companies that are seen to be operating unethically with regard to this, may damage their reputation and therefore their profits and value.

Nonetheless, given that Cigno Co is likely to derive substantial value from the acquisition, because of these savings, it should not merely disregard the potential savings. Instead it should consider public relations exercises it could undertake to minimise the loss of reputation, and perhaps meet with the government to discuss ways forward in terms of tax payments.

Conclusion

The potential value gained from acquiring and unbundling Anatra Co can be substantial if the potential cost savings are taken into account. However, given the assumptions that are made in computing the value, it is recommended that sensitivity analysis is undertaken and a range of values obtained. It is also recommended that Cigno Co should undertake public relations exercises to minimise the loss of reputation, but it should probably proceed with the acquisition, and undertake the cost saving exercise because it is likely that this will result in substantial additional value.

Report compiled by:

Date:

Appendix 1: Estimate of value created from the sell-off of the equipment manufacturing business

Average industry PE ratio = $2·40/$0·30 = 8
Anatra Co's equipment manufacturing business PE ratio = 8 x 1·2 = 9·6

Value from sell-off of equipment manufacturing business
Share of pre-tax profit = 30% x $2,490 million = $747m
After tax profit = $747 million x (1 – 0·22) = $582·7m
Value from sell-off = $582·7 million x 9·6 = $5,594m (approximately)

Appendix 2: Estimate of the combined company cost of capital

Anatra Co, asset beta = 0·68

Cigno Co, asset beta:
Equity beta =1·10
Proportion of market value of debt = 40%; Proportion of market value of equity = 60%
Asset beta = 1·10 x 0·60/(0·60 + 0·40 x 0·78) = 0·72

Combined company, asset beta
Market value of equity, Anatra Co = $3 x 7,000 million shares = $21,000m
Market value of equity, Cigno Co = 60% x $60,000 million = $36,000m
Asset beta = (0·68 x 21,000 + 0·72 x 36,000)/(21,000 + 36,000) = 0·71 (approximately)

Combined company equity beta = 0·71 x (0·6 + 0·4 x 0·78)/0·6 = 1·08

Combined company, cost of equity = 4·3% + 1·08 x 7% = 11·86%
Combined company, cost of capital = 11·86% x 0·6 + 6·00% x 0·78 x 0·4 = 8·99, say 9%

Appendix 3: Estimate of the value created for Cigno Co's equity holders from the acquisition

Anatra Co, Medical R&D value estimate:

Sales revenue growth rate = 5%
Operating profit margin = 17·25%
Tax rate = 22%
Additional capital investment = 40% of the change in sales revenue
Cost of capital = 9% (Appendix 2)
Free cash flow growth rate after four years = 3%
Current sales revenue = 70% x $21,400m = $14,980m

Cash flows, years 1 to 4 ($ millions)

Year	1	2	3	4
Sales revenue	15,729	16,515	17,341	18,208
Profit before interest and tax	2,713	2,849	2,991	3,141
Tax	597	627	658	691
Additional capital investment	300	314	330	347
Free cash flows	1,816	1,908	2,003	2,103
Present value of cash flows (9% discount)	1,666	1,606	1,547	1,490

Value, years 1 to 4: $6,309m
Value, year 5 onwards: [$2,103 x 1·03 / (0·09 – 0·03)] x $1·09^{-4}$ = $25,575m
Total value of Anatra Co's medical R&D business area = $31,884m

Total value of Anatra Co following unbundling of equipment manufacturing business and absorbing medical R&D business: $5,594m (appendix 1) + $31,884m = $37,478m (approximately)

Anatra Co, current market value of equity = $21,000m
Anatra Co, current market value of debt = $9,000m
Premium payable = $21,000m x 35% = $7,350m

Total value attributable to Anatra Co's investors = $37,350m

Value attributable to Cigno Co's shareholders from the acquisition of Anatra Co before taking into account the cash benefits of potential tax savings and redundancies = $128m

Appendix 4: Estimate of the value created from savings in tax and employment costs following possible redundancies

Cash flows, years 1 to 4 ($ millions)

Year	1	2	3	4
Cash flows (4% increase p.a.)	1,600	1,664	1,731	1,800
Present value of cash flows (9%)	1,468	1,401	1,337	1,275

Total value = $5,481m

Value attributable to Cigno Co's shareholders from the acquisition of Anatra Co after taking into account the cash benefits of potential tax savings and redundancies = $5,609m

(c) The feasibility of disposing of assets as a defence tool against a possible acquisition depends upon the type of assets sold and how the funds generated from the sale are utilised.

If the type of assets are fundamental to the continuing business then this may be viewed as disposing of the corporation's 'crown-jewels'. Such action may be construed as being against protecting the rights of shareholders (similar to the conditions discussed in part (d) below). In order for key assets to be disposed of, the takeover regulatory framework may insist on the corporation obtaining permission from the shareholders first before carrying it out.

On the other hand, the assets may be viewed as not being fundamental to the core business and may be disposed off to generate extra funds through a sell-off (see part (a) above). This may make sense if the corporation is undertaking a programme of restructuring and re-organisation.

In addition to this, the company needs to consider what it intends to do with the funds raised from the sale of assets. If the funds are used to grow the core business and therefore enhancing value, then the shareholders would see this positively and the value of the corporation will probably increase. Alternatively, if there are no profitable alternatives, the funds could be returned to the shareholders through special dividends or share buy-backs. In these circumstances, disposing of assets may be a feasible defence tactic.

However, if the funds are retained but not put to value-enhancing use or returned to shareholders, then the share price may continue to be depressed. And the corporation may still be an attractive takeover target for corporations which are in need of liquid funds. In these circumstances, disposing of assets would not be a feasible defence tactic.

(d) Each of the three conditions aims to ensure that shareholders are treated fairly and equitably.

The mandatory-bid condition through sell out rights allows remaining shareholders to exit the company at a fair price once the bidder has accumulated a certain number of shares. The amount of shares accumulated before the rule applies varies between countries. The bidder must offer the shares at the highest share price, as a minimum, which had been paid by the bidder previously. The main purpose for this condition is to ensure that the acquirer does not exploit their position of power at the expense of minority shareholders.

The principle of equal treatment condition stipulates that all shareholder groups must be offered the same terms, and that no shareholder group's terms are more or less favourable than another group's terms. The main purpose of this condition is to ensure that minority shareholders are offered the same level of benefits, as the previous shareholders from whom the controlling stake in the target company was obtained.

The squeeze-out rights condition allows the bidder to force minority shareholders to sell their stake, at a fair price, once the bidder has acquired a specific percentage of the target company's equity. The percentage varies between countries but typically ranges between 80% and 95%. The main purpose of this condition is to enable the acquirer to gain 100% stake of the target company and prevent problems arising from minority shareholders at a later date.

Note: *Credit will be given for alternative, relevant approaches to the calculations, comments and suggestions/recommendations*

2 (a) (i)

Owed by	Owed to	Local currency m	$ m
Armstrong (USA)	Horan (South (Africa)	US $12·17	12·17
Horan (South Africa)	Massie (Europe)	SA R42·65	3·97
Giffen (Denmark)	Armstrong (USA)	D Kr21·29	3·88
Massie (Europe)	Armstrong (USA)	US $19·78	19·78
Armstrong (USA)	Massie (Europe)	€1·57	2·13
Horan (South Africa)	Giffen (Denmark)	D Kr16·35	2·98
Giffen (Denmark)	Massie (Europe)	€1·55	2·11

Owed to	Owed by				
	Giffen (De) $m	Armtg (US) $m	Horan (SA) $m	Massie (Eu) $m	Total $m
Giffen (De)			2·98		2·98
Armtg (US)	3·88			19·78	23·66
Horan (SA)		12·17			12·17
Massie (Eu)	2·11	2·13	3·97		8·21
Owed by	(5·99)	(14·30)	(6·95)	(19·78)	
Owed to	2·98	23·66	12·17	8·21	
Net	(3·01)	9·36	5·22	(11·57)	

Under the terms of the arrangement, Massie, as the company with the largest debt, will pay Horan $5·22m, as the company with the smallest amount owed. Then Massie will pay Armstrong $6·35m and Giffen will pay Armstrong $3·01m.

(ii) The Armstrong Group may have problems if any of the governments of the countries where the subsidiaries are located object to multilateral netting. However, this may be unlikely here.

The new system may not be popular with the management of the subsidiaries because of the length of time before settlement (up to six months). Not only might this cause cash flow issues for the subsidiaries, but the length of time may mean that some of the subsidiaries face significant foreign exchange risks. The system may possibly have to allow for immediate settlement in certain circumstances, for example, if transactions are above a certain size or if a subsidiary will have significant cash problems if amounts are not settled immediately.

(b) Need to hedge against a fall in interest rate, therefore buy call options. Require 50 contracts (25,000,000/1,000,000) x 6/3.

As Massie is looking to invest on 30 November, December contracts are needed.

Basis
Current price (1 September) – futures price = basis
(100 – 3·6) – 95·76 = 0·64
Unexpired basis = 1/4 x 0·64 = 0·16

Option
Amount received will be (LIBOR – 0·4%) x 25,000,000 x 6/12

If interest rates increase by 0·5% to 4·1%
Expected futures price = (100 – 4·1) – 0·16 = 95·74

Exercise price	97·00	96·50
Futures price	95·74	95·74
Exercise option?	No	No
Gain in basis points	–	–
	€	€
Interest received		
(€25m x 6/12 x (4·1 – 0·4)%)	462,500	462,500
Gain on options	–	–
Premium		
(3·2 x €25 x 50)	(4,000)	
(18·2 x €25 x 50)		(22,750)
Net receipt	458,500	439,750
Effective interest rates	3·67%	3·52%

If interest rates fall by 0·5% to 3·1%
Expected futures price = (100 – 3·1) – 0·16 = 96·74

Exercise price	97·00	96·50
Futures price	96·74	96·74
Exercise option?	No	Yes
Gain in basis points	–	24
	€	€
Interest received		
(€25m x 6/12 x (3·1 – 0·4)%)	337,500	337,500
Gain on options		
(0 and 24 x €25 x 50)	–	30,000
Premium		
(3·2 x €25 x 50)	(4,000)	
(18·2 x €25 x 50)		(22,750)
Net receipt	333,500	344,750
Effective interest rates	2·67%	2·76%

Using a collar

Buy December call at 97·00 for 0·032 and sell December put at 96·50 for 0·123. Net premium received = 0·091.

If interest rates increase to 4·1%

	Buy call	**Sell put**
Exercise price	97·00	96·50
Futures price	95·74	95·74
Exercise option?	No	Yes
	€	
Interest received	462,500	
Loss on exercise		
(76 x €25 x 50)	(95,000)	
Premium		
(9·1 x €25 x 50)	11,375	
Net receipt	378,875	
Effective interest rate	3·03%	

If interest rates fall to 3·1%

	Buy call	**Sell put**
Exercise price	97·00	96·50
Futures price	96·74	96·74
Exercise option?	No	No
	€	
Interest received	337,500	
Loss on exercise	–	
Premium		
(9·1 x €25 x 50)	11,375	
Net receipt	348,875	
Effective interest rate	2·79%	

Summary

	97·00	**96·50**	**Collar**
Interest rates rise to 4·1%	3·67%	3·52%	3·03%
Interest rates fall to 3·1%	2·67%	2·76%	2·79%

The option with the 97·00 exercise price has a higher average figure than the option with the 96·50 exercise price, and can be recommended on that basis, as its worst result is only marginally worse than the 96·50 option. There is not much to choose between them. The collar gives a significantly worse result than either of the options if interest rates rise, because Massie cannot take full advantage of the increase. It is marginally the better choice if interest rates fall. The recommendation would be to choose the option with the 97·00 exercise price, unless interest rates are virtually certain to fall.

3 **(a)** **(i)** **SOFP if Gupte VC shares are purchased by Flufftort Co and cancelled**

	$m
Assets	
Non-current assets	69
Current assets excluding cash	18
Cash	–
Total assets	87
Equity and liabilities	
Share capital	40
Retained earnings	5
Total equity	45
Long-term liabilities	
Bank loan	30
Loan note	5
Total long-term liabilities	35
Current liabilities	7
Total liabilities	42
Total equity and liabilities	87

(ii) **SOFP if full refinancing takes place**

	$m
Assets	
Non-current assets	125
Current assets excluding cash	42
Cash (balancing figure)	5
Total assets	172
Equity and liabilities	
Share capital	90
Retained earnings	5
Total equity	95
Long-term liabilities	
Bank loan	65
Loan note	–
Total long-term liabilities	65
Current liabilities	12
Total liabilities	77
Total equity and liabilities	172

(iii) **Projected SOPL**

	2017 **$m**	**2018** **$m**
Operating profit	20·0	25·0
Finance cost	(6·5)	(6·5)
Profit before tax	13·5	18·5
Taxation 20%	(2·7)	(3·7)
Profit after tax	10·8	14·8
Dividends	–	–
Retained earnings	10·8	14·8

(b) **Current situation**

Initial product developments have not generated the revenues required to sustain growth. The new Easicushion chair appears to offer Flufftort Co much better prospects of commercial success. At present, however, Flufftort Co does not have the resources to make the investment required.

Purchase of Gupte VC's shares

In the worst case scenario, Gupte VC will demand repayment of its investment in a year's time. The calculations in (a) show the financial position in a year's time, assuming that there is no net investment in non-current assets or working capital, the purchase of shares is financed solely out of cash reserves and the shares are cancelled. Repayment by this method would mean that the limits set out in the covenant would be breached (45/35 = 1·29) and the bank could demand immediate repayment of the loan.

The directors can avoid this by buying some of Gupte VC's shares themselves, but this represents money which is not being put into the business. In addition, the amount of shares which the directors would have to purchase would be greater if results, and therefore reserves, were worse than expected.

Financing the investment

The calculations in (a) show that the cash flows associated with the refinancing would be enough to finance the initial investment. The ratio of equity to non-current liabilities after the refinancing would be 1·46 (95/65), in line with the current limits in the bank's covenant. However, financing for the subsequent investment required would have to come from surplus cash flows.

Shareholdings

The disposition of shareholdings will change as follows:

	Current shareholdings		Shareholdings after refinancing	
	Number in million	%	Number in million	%
Directors	27·5	55·0	42·5	47·2
Other family members	12·5	25·0	12·5	13·9
Gupte VC	10·0	20·0	30·0	33·3
Loan note holder	–	–	5·0	5·6
	50·0	100·0	90·0	100·0

Gupte VC's percentage shareholding will rise from 20% to 33·3%, enough possibly to give it extra rights over the company. The directors' percentage shareholding will fall from 55% to 47·2%, which means that collectively they no longer have control of the company. The percentage of shares held by family members who are not directors falls from 25% to around 19·5%, taking into account the conversion of the loan note. This will mean, however, that the directors can still maintain control if they can obtain the support of some of the rest of the family.

Position of finance providers

The refinancing has been agreed by the chief executive and finance director. At present, it is not clear what the views of the other directors are, or whether the $15 million contributed by directors will be raised from them in proportion to their current shareholdings. Some of the directors may not be able to, or wish to, make a significant additional investment in the company. On the other hand, if they do not, their shareholdings, and perhaps their influence within the company, will diminish. This may be a greater concern than the board collectively losing control over the company, since it may be unlikely that the other shareholders will combine to outvote the board.

The other family shareholders have not been actively involved in Flufftort Co's management out of choice, so a reduction in their percentage shareholdings may not be an issue for them. They may have welcomed the recent dividend payment as generating a return on their investment. However, as they appear to have invested for the longer term, the new investment appears to offer much better prospects in the form of a capital gain on listing or buy-out than an uncertain flow of dividends. The new investment appears only to have an upside for them in the sense that they are not being asked to contribute any extra funding towards it.

Rajiv Patel is unlikely to be happy with the proposed scheme. He is exchanging a guaranteed flow of income for an uncertain flow of future dividends sometime after 2018. On the other hand, his investment may be jeopardised by the realisation of the worst case scenario, since his debt is subordinated to the bank's debt.

The most important issue from Gupte VC's viewpoint is whether the extra investment required is likely to yield a better outcome than return of its initial investment in a year's time. The plan that no dividends would be paid until after 2018 is a disadvantage. On the other hand, the additional investment seems to offer the only prospect of realising a substantial gain either by Flufftort Co being listed or sold. The arrangement will mean that Gupte VC may be able to exercise greater influence over Flufftort Co, which may provide it with a greater sense of reassurance about how Flufftort Co is being run. The fact that Gupte VC has a director on Flufftort Co's board should also give it a clear idea of how successful the investment is likely to be.

The bank will be concerned about the possibility of Flufftort Co breaching the covenant limits and may be concerned whether Flufftort Co is ultimately able to repay the full amount without jeopardising its existence. The bank will be concerned if Flufftort Co tries to replace loan finance with overdraft finance. The refinancing provides reassurance to the bank about gearing levels and a higher rate of interest. The bank will also be pleased that the level of interest cover under the refinancing is higher and increasing (from 2·0 in 2016 to 3·1 in 2017 and 3·8 in 2018). However, it will be concerned about how Flufftort Co finances the additional investment required if cash flows from the new investment are lower than expected. In those circumstances Flufftort Co may seek to draw on its overdraft facility.

Conclusion

The key players in the refinancing are Gupte VC, the bank and the directors other than the chief executive and the finance director. If they can be persuaded, then the scheme has a good chance of being successful. However, Rajiv Patel could well raise objections. He may be pacified if he retains the loan note. This would marginally breach the current covenant limit (90/70 = 1·29), although the bank may be willing to overlook the breach as it is forecast to be temporary. Alternatively, the refinancing would mean that Flufftort Co just had enough spare cash initially to redeem the loan note, although it would be more dependent on cash surpluses after the refinancing to fund the additional investment required.

4 (a) An annual cash flow account compares the estimated cash flows receivable from the property against the liabilities within the securitisation process. The swap introduces leverage into the arrangement.

Cash flow receivable	$ million	Cash flow payable	$ million
$200 million × 11%	22·00	A-rated loan notes	
Less: Service charge	(0·20)	Pay $108 million (W1) × 11% (W2)	11·88
		B-rated loan notes	
		Pay $27 million (W1) × 12%	3·24
		C-rated loan notes	
		Pay $27 million (W1) × 13%	3·51
	21·80		18·63
		Balance to the subordinated certificates	3·17

Workings

1. Loan notes

		$million
A	$200m × 0·9 × 0·6	108
B	$200m × 0·9 × 0·15	27
C	$200m × 0·9 × 0·15	27

2. Swap

Pay fixed rate under swap	9·5%
Pay floating rate	LIBOR + 1·5%
Receive floating rate under swap	(LIBOR)
Net payment	11%

The holders of the certificates are expected to receive $3·17million on $18 million, giving them a return of 17·6%. If the cash flows are 5% lower than the non-executive director has predicted, annual revenue received will fall to $20·90 million, reducing the balance available for the subordinated certificates to $2·07 million, giving a return of 11·5% on the subordinated certificates, which is below the returns offered on the B and C-rated loan notes. The point at which the holders of the certificates will receive nothing and below which the holders of the C-rated loan notes will not receive their full income will be an annual income of $18·83 million (a return of 9·4%), which is 14·4% less than the income that the non-executive director has forecast.

(b) Benefits

The finance costs of the securitisation may be lower than the finance costs of ordinary loan capital. The cash flows from the commercial property development may be regarded as lower risk than Moonstar Co's other revenue streams. This will impact upon the rates that Moonstar Co is able to offer borrowers.

The securitisation matches the assets of the future cash flows to the liabilities to loan note holders. The non-executive director is assuming a steady stream of lease income over the next 10 years, with the development probably being close to being fully occupied over that period.

The securitisation means that Moonstar Co is no longer concerned with the risk that the level of earnings from the properties will be insufficient to pay the finance costs. Risks have effectively been transferred to the loan note holders.

Risks

Not all of the tranches may appeal to investors. The risk-return relationship on the subordinated certificates does not look very appealing, with the return quite likely to be below what is received on the C-rated loan notes. Even the C-rated loan note holders may question the relationship between the risk and return if there is continued uncertainty in the property sector.

If Moonstar Co seeks funding from other sources for other developments, transferring out a lower risk income stream means that the residual risks associated with the rest of Moonstar Co's portfolio will be higher. This may affect the availability and terms of other borrowing.

It appears that the size of the securitisation should be large enough for the costs to be bearable. However Moonstar Co may face unforeseen costs, possibly unexpected management or legal expenses.

(c) **(i)** Sukuk finance could be appropriate for the securitisation of the leasing portfolio. An asset-backed Sukuk would be the same kind of arrangement as the securitisation, where assets are transferred to a special purpose vehicle and the returns and repayments are directly financed by the income from the assets. The Sukuk holders would bear the risks and returns of the relationship.

The other type of Sukuk would be more like a sale and leaseback of the development. Here the Sukuk holders would be guaranteed a rental, so it would seem less appropriate for Moonstar Co if there is significant uncertainty about the returns from the development.

The main issue with the asset-backed Sukuk finance is whether it would be as appealing as certainly the A-tranche of the securitisation arrangement which the non-executive director has proposed. The safer income that the securitisation offers A-tranche investors may be more appealing to investors than a marginally better return from the Sukuk. There will also be costs involved in establishing and gaining approval for the Sukuk, although these costs may be less than for the securitisation arrangement described above.

(ii) A Mudaraba contract would involve the bank providing capital for Moonstar Co to invest in the development. Moonstar Co would manage the investment which the capital funded. Profits from the investment would be shared with the bank, but losses would be solely borne by the bank. A Mudaraba contract is essentially an equity partnership, so Moonstar Co might not face the threat to its credit rating which it would if it obtained ordinary loan finance for the development. A Mudaraba contract would also represent a diversification of sources of finance. It would not require the commitment to pay interest that loan finance would involve.

Moonstar Co would maintain control over the running of the project. A Mudaraba contract would offer a method of obtaining equity funding without the dilution of control which an issue of shares to external shareholders would bring. This is likely to make it appealing to Moonstar Co's directors, given their desire to maintain a dominant influence over the business.

The bank would be concerned about the uncertainties regarding the rental income from the development. Although the lack of involvement by the bank might appeal to Moonstar Co's directors, the bank might not find it so attractive. The bank might be concerned about information asymmetry – that Moonstar Co's management might be reluctant to supply the bank with the information it needs to judge how well its investment is performing.

					Marks
1	(a)	Up to 2 marks for distinguishing the two forms of unbundling			4
	(b)	(i)	Appendix 1		
			Anatra Co, Manufacturing business, PE ratio		1
			Estimate of the value created from sell-off		3
			Appendix 2		
			Cigno Co asset beta		1
			Combined company asset beta		1
			Combined company equity beta		1
			Combined company cost of capital		1
			Appendix 3		
			Sales revenue, years 1 to 4		1
			Operating profit, years 1 to 4		1
			Taxation, years 1 to 4		1
			Capital investment, years 1 to 4		1
			Value from years 1 to 4		1
			Value from year 5 onwards		1
			Value for Cigno Co shareholders before impact of savings from tax and employee cost reduction		2
			Appendix 4		
			Value created from tax and employee cost savings		1
			Value for Cigno Co shareholders after impact of savings from tax and employee cost reduction		1
					18
		(ii)	Discussion of values for the equity holders, additional costs/benefits not given		3–4
			Methods used and assumptions made		4–5
				Max	8
		(iii)	Reputation factors		1–2
			Ethical factors		1–2
			Comment on value		1–2
				Max	4
		Professional marks for part (b)			
		Report format			1
		Structure and presentation of the report			3
					4
	(c)	1 to 2 marks per point		Max	6
	(d)	Up to 2 marks for explaining the purpose of each condition			6
				Total	**50**

					Marks
2	**(a)**	**(i)**	Dollar amounts owed and owing		2
			Totals owed and owing		3
			Net amounts owed		1
			Payments and receipts		2
					8
		(ii)	1–2 marks per problem discussed	Max	**3**
	(b)		Recommendation to purchase calls		1
			Number and month of contracts		1
			Calculation of basis		1
			Options contracts calculations		4
			Collars approach and calculations		5
			Comments and conclusion		2–3
				Max	**14**
				Total	**25**
3	**(a)**	**(i)**	SOFP if shares purchased and cancelled		
			Cash and other assets		2
			Equity		1
			Liabilities		1
					4
		(ii)	SOFP if full refinancing takes place		
			Cash and other assets		2
			Equity		1
			Liabilities		1
					4
		(iii)	2017 forecast		2
			2018 forecast		2
					4
	(b)		Up to 2 marks for each well discussed point	Max	**13**
				Total	**25**
4	**(a)**		Calculation of receivable		1
			Loan note amounts attributable to the A, B and C tranches		1
			Impact of swap		2
			Calculation of interest payable on interest for tranches A, B and C-rated tranches		3
			Estimation of return to subordinated certificates		1
			Comments and calculation relating to sensitivity		3
					11
	(b)		Benefits of securitisation		3–4
			Risks associated with securitisation		2–3
				Max	**6**
	(c)	**(i)**	Explanation/discussion of suitability of Sukuk finance		2–3
			Discussion of investors' views		1–2
				Max	**4**
		(ii)	Explanation/discussion of suitability of Mudaraba contract		2–3
			Discussion of bank's views		1–2
				Max	**4**
				Total	**25**